

“A Stieltjes Pole-Package Identity for the Prime-Weighted Quantum Operator and the Riemann Hypothesis”

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Abstract:

The proof proceeds through a fixed four-step spine. Appendix D constructs the canonical whole-line operator-side transform on $\Omega = \mathbb{C} \setminus (-\infty, 0]$ and isolates the explicit prime-power package. Appendix H constructs independently the zero-side meromorphic pole package on the same domain. Appendix E proves the overlap identity between the two constructions on a common half-plane. Appendix F then applies rigidity on Ω : an off-critical zero would force an isolated pole incompatible with the holomorphic/Stieltjes structure of the operator-side transform.

Keywords:

Riemann Hypothesis; Riemann zeta function; explicit formula; prime-power spike package; resolvent trace; Stieltjes transform; meromorphic continuation; operator rigidity; whole-line Schrödinger operator; prime-weighted quantum operator.

Introduction

The Riemann Hypothesis asserts that every nontrivial zero of the Riemann zeta function has real part $1/2$. This manuscript presents a proof organized around a fixed operator-side / zero-side / interface / rigidity spine. The main purpose of the present introduction is to make the dependency structure explicit at the outset, so that the burden of verification falls on a finite list of stated analytic steps rather than on reconstruction of the argument from dispersed remarks.

The proof is carried out on the slit plane

$$\Omega = \mathbb{C} \setminus (-\infty, 0]$$

On the operator side, the central object is a canonical whole-line transform F_H constructed from the prime-weighted quantum operator by a resolvent/Laplace-transform procedure. On the zeta side, the central object is a meromorphic pole package built from the nontrivial zeros of ζ , together with the archimedean correction term. The argument is to construct these two objects independently, identify them on a nonempty overlap domain, and then use rigidity on Ω to exclude any off-critical pole.

A global convention is fixed once and for all. The arithmetic spike package is indexed by prime powers $q = p^m$, with centers at $\log q$ and frozen weight

$$w(q) = \frac{\Lambda(q)}{\sqrt{q}} = (\log p) p^{-m/2}.$$

This is the only normalization used in the RH spine. Finite-cutoff and finite-window objects are auxiliary regularizations; the closure argument is stated for the canonical whole-line transform. Appendix A supplies confining/background operator input and is not a second definition of the arithmetic spike package.

The proof proceeds in four steps.

Claim calibration. The RH implication proved later is not a fifth construction and does not depend on unrestricted normal convergence of a prime-side series on all of Ω . The load-bearing closure consists exactly of: (i) the operator-side decomposition $F_H = B + R_H$ with $R_H \in O(\Omega)$ from Appendix D; (ii) the independent zero-side meromorphic package $Z_{pole, \Omega} + H_{arch}$ from Appendix H; (iii) the local overlap identity on a nonempty half-plane $U_{\sigma_0} \subset \Omega$ from Appendix E; and (iv) the rigidity argument on Ω from Appendix F. Once Appendices D and H have produced their Ω -objects, the final contradiction follows from equality on U_{σ_0} and uniqueness on the connected slit plane Ω .

The remaining burden of verification is accordingly concentrated in a finite analytic list: the fixed-cutoff extraction of the explicit prime-power package, the holomorphic control of the operator-side remainder on Ω , the stability of the cutoff-removal passage, and the independent meromorphic construction of the zero-side pole package. In that sense, the manuscript is intended to be read as a proof organized around explicit, checkable analytic lemmas rather than as a speculative program or philosophical analogy.

We now turn to the precise construction of the operator model and the canonical arithmetic package.

Canonical proof architecture :

Appendix D

Construct the canonical operator-side transform

$$F_H \in O(\Omega), \Omega = \mathbb{C} \setminus (-\infty, 0],$$

and construct an operator-side remainder

$$R_H \in O(\Omega).$$

On the common overlap half-plane $U_{\sigma_0} \subset \Omega$, Appendix D identifies the operator transform with the canonical shifted package by

$$F_H(s) = B_{can}(s) + R_H(s), \quad s \in U_{\sigma_0}.$$

Appendix D does not assert that B_{can} is holomorphic on all of Ω . The global continuation of the package is supplied later by Appendix H through the zero-side meromorphic pole package.

Appendix H

Construct independently the zero-side meromorphic object

$$Z(s) = \sum_{\rho} \frac{m(\rho)}{s + \rho(1-\rho)}$$

plus the allowed holomorphic remainder, first on a half-plane where the Laplace transform is justified termwise, and then continue it meromorphically to Ω without, continuation comes from identifying it with F .

- on the initial half-plane, the Laplace transform yields the zero-pole package;
- this defines a meromorphic function there;
- the resulting meromorphic expression has a canonical meromorphic continuation to Ω by its explicit pole formula plus holomorphic remainder.

The key point is: Appendix H owns it's own the zero-side meromorphic package.

Appendix E

Appendix E proves only the local bridge identity on a nonempty overlap half-plane $U_{\sigma_0} \subset \Omega$:

- $F_H(s) = H_{tot}(s) - Z_{pole,\Omega}(s), \quad s \in U_{\sigma_0},$
- where $H_{tot} = R_H + H_{arch} \in O(\Omega)$

Appendix E performs no global continuation. The global extension from U_{σ_0} to Ω is carried out in Appendix F by the meromorphic identity theorem, using the global objects already constructed in Appendices D and H.

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Appendix F

We use the now-global comparison:

- F is Stieltjes/Herglotz-type on Ω ,
- $F=Z+H$ globally on Ω ,
- H is holomorphic,
- therefore any off-line zero would produce a forbidden pole of F ,
- contradiction,
- hence RH.

Scope note. Sections F.5–F.6 (RKHS/de Branges interpretation and kernel-identification targets) are optional motivation and are not used in the RH closure, which depends only on the spine above.

4. Notation and conventions (read once)

(i) The slit plane. We write $\Omega = \mathbb{C} \setminus (-\infty, 0]$. All holomorphy statements are on Ω , unless explicitly restricted to a right half-plane.

(ii) Prime-power normalization. The spike package is indexed by prime powers $q = p^m$. We use the normalization

$$\frac{\Lambda(q)}{\sqrt{q}} \text{ equivalently } (\log p) p^{-m/2},$$

and treat these as the same convention written in two indexings.

(iii) Decay vs compact support. The bump profile W_α is Schwartz (Gaussian is admissible).

Any earlier “support separation” phrasing is to be read as effective localization/decay separation; the mixed-term bounds use off-diagonal heat-kernel decay across cluster gaps. No step of the RH closure depends on literal compact support.

(iv) Gaussian normalization. We use the heat-kernel normalization

$G_t(u) = (4\pi t)^{-1/2} e^{-u^2/(4t)}$, with Fourier transform $e^{-t\xi^2}$ and we distinguish this displacement kernel from the Laplace kernel $e^{-t\lambda}$ acting on spectral parameters.

Canonical transform convention.

Throughout the RH closure spine, the symbol F_H denotes the canonical displacement/commutator transform exported by Appendix D:

$$F_H := F_H^{can}$$

It does not denote the raw multiplication-bump heat trace. The raw multiplication-bump calculation and its local-loop kernel $J_{t,L,\alpha}$ are auxiliary diagnostics only. Appendices E and F consume F_H^{can} , the transform whose one-letter package is the canonical displacement package

$$\sum_{\log q \leq R} w(q) g_{t,L}(\log q), \quad g_{t,L}(a) = \text{Tr}(S_t A_L T_a)$$

Boxed Global Conventions

GLOBAL CONVENTIONS (used throughout Sections 1–5 and Appendices A–H).

1. Hilbert space and boundary conditions.

Fix $L \in (0, \infty]$ Work on the Hilbert space

$$L := \{L^2([-L, L]) \text{ with Dirichlet boundary at } \pm L, (L < \infty), L^2(\mathbb{R}) (L = \infty).$$

When $L < \infty$, all trace objects are taken on H_L . When $L = \infty$, traces are interpreted via the specified cutoff/observable A_L (see item 3).

2. Free operator and sign conventions.

The free Hamiltonian is

$$H_0 := -\partial_u^2 \text{ (nonnegative)}.$$

The full operator is

$$H := H_0 + V,$$

where V is real-valued and chosen so that H is self-adjoint and bounded below on H_L (precise hypotheses stated in Appendix A).

3. Observable/cutoff used for traces.

Fix a bounded multiplication window M_w (with $w \geq 0$, compactly supported or rapidly decaying as specified), and define the cutoff observable

$$A_L := \Pi_L M_w \Pi_L,$$

where Π_L is the projector to $[-L, L]$ (identity if $L = \infty$ and the “ L ” is viewed as the window-scale in A_L). All heat and resolvent traces appearing in the prime-package/spike arguments are taken in the weighted sense:

$$F_L(s) := \text{Tr} (A_L (H + s)^{-1}), \Theta_L(t) := \text{Tr} (A_L e^{-tH}),$$

whenever these traces are defined.

Reader note on density normalization. Throughout Appendix D we keep the cutoff observable in the raw form $A_L := \Pi_L M_w \Pi_L$,

. The object compared with the canonical prime-power package is the density-normalized spike package defined later by $\widehat{g_{t,L}} := (2L)^{-1} g_{t,L}$ and $\widehat{P_{L,R}} := (2L)^{-1} \sum_{\log q \leq R} w(q) g_{t,L}(\log q)$

4. Domain Ω and branch convention.

Set

$$\Omega := \mathbb{C} \setminus (-\infty, 0]$$

(the slit plane). The principal branch of $\sqrt{\cdot}$ is used on Ω , and any Stieltjes/Herglotz statements refer to this choice.

5. Prime-power indexing and normalization (fixed globally).

Prime powers are denoted p^m with p prime and $m \geq 1$. The von Mangoldt weight is

$\Lambda(q) = \log p$ if $q = p^m$ and $\Lambda(q) = 0$ otherwise. **The global spike normalization is**

spike weight $w(q) = \frac{\Lambda(q)}{\sqrt{q}}$.

Equivalently, for $q = p^m$, $w(q) = (\log p) p^{-m/2}$. No other q -powers occur in the spike package.

6. Truncations and limits.

All “prime packages” are first formed at finite truncation $R < \infty$ (i.e. $\log q \leq R$), defining $H_R, F_{L,R}, \Theta_{L,R}$, etc. Limits $R \rightarrow \infty$ are taken **only** in the precise sense stated in Appendix D

(e.g. norm-resolvent on Ω -compacts, or via the meromorphic identity theorem after establishing overlap on a half-plane $U \subset \Omega$). No informal interchange of $R \rightarrow \infty$ with analytic continuation is permitted.

7. Global package convention.

The manuscript uses two related Gaussian kernels:

$$G_t(u) = (4\pi t)^{-1/2} e^{-u^2/(4t)}$$

for the displacement-side spike extraction in Appendix D, and

$$K_t(u) = e^{-t/4} G_t(u)$$

for the shifted closure package used in the Appendix E / Appendix F / Appendix H RH spine. The raw G_t -package and the shifted K_t -package are not identified with the same ss-plane pole geometry. The canonical closure package is always the shifted K_t -package.

Global normalization firewall.

Throughout Sections 5 and Appendices D, E, F, and H, every arithmetic spike package is indexed by prime powers

$$q = p^m, m \geq 1,$$

with center

$$u_q = \log q = m \log p$$

and frozen spike weight

$$w(q) = \frac{\Lambda(q)}{\sqrt{q}} = (\log p) p^{-m/2}.$$

This is the only arithmetic normalization used in the RH closure spine. In particular, no step in the D/E/F/H argument uses $\frac{\Lambda(q)}{\sqrt{q}}$ as the spike coefficient. Any occurrence of $\frac{\Lambda(q)}{\sqrt{q}}$ in the RH spine is to be read as a typographical or extraction error unless it is immediately rewritten in the prime-power form $(\log p) p^{-m/2}$.

The canonical packages are therefore

$$B_{raw,R}(t) = \sum_{\log q \leq R} \frac{\Lambda(q)}{\sqrt{q}} G_t(\log q),$$

and

$$B_{can,R}(t) = e^{-t/4} B_{raw,R}(t) = \sum_{\log q \leq R} \frac{\Lambda(q)}{\sqrt{q}} K_t(\log q), \quad K_t(u) = e^{-t/4} G_t(u).$$

The raw G_t -package and shifted K_t -package have different Laplace-side pole geometry.

The RH closure uses only the shifted package B_{can}

D.NORM (Global prime-power indexing and normalization).

Throughout the manuscript, prime-power spikes are indexed by prime powers

$$q = p^m \text{ (} p \text{ prime, } m \geq 1\text{),}$$

$$\text{and we write } u_q := \log q = m \log p, \quad \Lambda(q) := \log p.$$

The global spike weight is fixed as

$$w(q) := \frac{\Lambda(q)}{\sqrt{q}} = (\log p)/p^{\{m/2\}}.$$

This is the unique normalization used in every spike package (time domain), in every spike transform (Laplace/resolvent domain), and in the prime-power distribution μ_{pr} appearing in Appendix H.

Section 2. The Prime–Power Weighted Quantum Operator (PWQO)

2.1 Global Sign and Normalization Convention

Throughout the paper we fix the arithmetic potential as a **nonnegative multiplication operator**, and we place the minus sign at the Hamiltonian level.

For each prime power

$$q = p^m \quad (m \geq 1),$$

define its center

$$u_q := \log q,$$

and its weight

$$w(q) := \frac{\Lambda(q)}{\sqrt{q}} = (\log p) p^{-m/2} \geq 0.$$

Let W_α be a fixed nonnegative Schwartz bump centered at 0.

For a cutoff $R > 0$ define the truncated prime-power package potential by

$$V_{pp}^{(R)}(u) = \sum_{\log q \leq R} \frac{\Lambda(q)}{\sqrt{q}} W_\alpha(u - \log q) \quad q = p^m.$$

Equivalently,

$$V_{pp}^{(R)}(u) = \sum_p \sum_{m \geq 1, m \log p \leq R} (\log p) p^{-m/2} W_\alpha(u - m \log p)$$

Then the truncated PWQO is

$$H_{\varepsilon, R} = -\partial_u^2 + V_{env}(u) + \varepsilon U_{left}(u) - \sum_{\log q \leq R} \frac{\Lambda(q)}{\sqrt{q}} W_\alpha(u - \log q).$$

The minus sign at the Hamiltonian level makes each prime-power term an attractive well. The coefficient of that well is always

$$\frac{\Lambda(q)}{\sqrt{q}}$$

Throughout the RH spine, the canonical arithmetic package is the prime-power spike family with centers at $\log q$ and frozen weights w of q equal to $\Lambda(q)$ divided by square root q , for q equal to p to the m . The ambient Hilbert space is the whole-line L^2 space in the manuscript's chosen displacement variable. The operator used for R_H closure is the canonical whole-line object; finite-volume and windowed operators are auxiliary regularizations only. The baseline decomposition is: confining envelope, optional left-wall penalty, and truncated arithmetic spike package. Weighted traces involving $A_{sub L}$ are regularizing devices and are not alternate definitions of the final operator-side transform.

Any appearance elsewhere of an alternate scaling or an OCR-distorted formula is non-authoritative; the frozen arithmetic normalization used throughout the RH spine is $\Lambda(q)$ divided by square root q .

2.2 Baseline Confining Operator

We work on the Hilbert space $L^2(R, du)$

Let

$$H_0 = -\partial_u^2$$

with domain $H^2(R)$ (Friedrichs extension from (C_c^∞)).

Let V_{env} be a smooth confining envelope satisfying

- $V_{env} \in L^1_{loc}(R)$,
- $V_{env}(u) \geq -C$,
- $V_{env}(u) \rightarrow +\infty$ as $|u| \rightarrow \infty$.

Let $U_{left} \geq 0$ be a fixed left-wall penalty supported in $(-\infty, U^*)$.

Define the baseline operator

$$H_{base} = H_0 + V_{env} + \varepsilon U_{left}.$$

Appendix A proves that H_{base} is self-adjoint with compact resolvent and purely discrete real spectrum.

2.3 Definition of the Truncated PWQO

We now define the truncated prime-power weighted operator by

For a cutoff $R > 0$ define the truncated prime-power package potential by

$$V_{pp}^{(R)}(u) = \sum_{\log q \leq R} \frac{\Lambda(q)}{\sqrt{q}} W_\alpha(u - \log q) \quad q = p^m.$$

Equivalently,

$$V_{pp}^{(R)}(u) = \sum_p \sum_{m \geq 1} (\log p) p^{-m/2} W_\alpha(u - m \log p)$$

Then the truncated PWQO is

$$H_{\varepsilon, R} = -\partial_u^2 + V_{env}(u) + \varepsilon U_{left}(u) - \sum_{\log q \leq R} \frac{\Lambda(q)}{\sqrt{q}} W_\alpha(u - \log q).$$

The minus sign at the Hamiltonian level makes each prime-power term an attractive well. The coefficient of that well is always

$$\frac{\Lambda(q)}{\sqrt{q}}$$

2.4 Quadratic Form Realization

The operator $H_{\varepsilon,R}$ is realized via the closed quadratic form

$$q_R[\Psi] = \int_R (|\Psi'(u)|^2 + (V_{env}(u) + \varepsilon U_{left}(u)) |\Psi(u)|^2) du - \int_R V_{pp}^{(R)}(u) |\Psi(u)|^2 du,$$

with form domain

$$H^1(R) \cap L^2((V_{env} + U_{left}) du)$$

Since $V_{pp}^{(R)} \in L^\infty$ for each fixed R , it is form-bounded with relative bound zero. Therefore $H_{\varepsilon,R}$ is self-adjoint and lower semibounded.

2.5 Spectral Normalization (Stieltjes Compatibility)

Because the Stieltjes rigidity argument in Appendix F is carried out on the slit plane

$$\Omega = \mathbb{C} \setminus (-\infty, 0],$$

we use a single global spectral normalization for the finite-cutoff family.

By the cutoff-uniform lower-bound theorem proved in Appendix A/B, there exists a constant

$$M < \infty$$

independent of both L and R , such that every finite-window truncated operator satisfies the quadratic-form lower bound

$$H_{L,R}^{orig} \geq -M.$$

Equivalently,

$$\langle \psi, H_{L,R}^{orig} \psi \rangle \geq -M \|\psi\|_2^2$$

for all admissible test functions ψ in the finite-window form domain, uniformly in L and R.

We therefore fix once and for all the globally shifted operators

$$H_{L,R}^{\sim} := H_{L,R}^{orig} + MI$$

Then

$$H_{L,R}^{\sim} \geq 0, \quad \sigma(H_{L,R}^{\sim}) \subset [0, \infty)$$

for every finite cutoff stage (L,R).

This normalization is global: the same constant M is used for the entire cutoff family. No L-dependent or R-dependent spectral shift is introduced.

The shift does not alter the prime-power spike amplitudes, the spike centers, or the diagonal extraction mechanism. In particular, the arithmetic package remains

$$w(q) = \frac{\Lambda(q)}{\sqrt{q}}, u_q = \log q$$

The shift only places the operator spectrum in the nonnegative half-line so that the resolvent/Stieltjes transform

$$s \mapsto \text{Reg Tr}(H^{\sim} + s)^{-1}$$

is naturally holomorphic on

$$\Omega = \mathbb{C} \setminus (-\infty, 0]$$

Equivalently, the pole locations of the normalized resolvent are contained in the excluded cut

$$(-\infty, 0]$$

All later occurrences of the operator-side transform F_H refer to this globally normalized, regularized/bulk-subtracted transform. Thus the RH closure uses

$$F_H(s) := F_H^{can}(s) = L[\Theta_H^{can} - A_{class}^{can}](s)$$

Or equivalently,

$$F_H^{can}(s) = \text{Reg Tr}^{can}(H^{\sim} + s)^{-1} := L[\Theta_H^{can} - A_{class}^{can}](s).$$

The notation $RegTr^{can}(H^\sim + s)^{-1}$, when used, is only shorthand for this canonical bulk-subtracted transform. It is not the raw whole-line multiplication-bump resolvent trace.

Referee note

The purpose of this normalization is only Stieltjes compatibility. It is not a new arithmetic input and it is not a cutoff-dependent adjustment. The same global shift is applied before the ordered cutoff-removal procedure, so it cannot change the prime-power package or create/cancel any off-cut pole in Ω .

2.6 Duhamel Letter and First-Order Spike Sign

In Appendix D we define the perturbation letter

$$B_{L,R} = - \Pi_L V_{pp}^{(R)} \Pi_L,$$

so that

$$H_{L,R} = H_0 + B_{L,R}.$$

The Duhamel expansion yields

$$e^{-tH_{L,R}} = \sum_{k \geq 0} (-1)^k \int_{\Delta_k(t)} G_{t-s_k} B_{L,R} G_{s_k-s_{k-1}} \cdots B_{L,R} G_{s_1} ds.$$

At first order (k=1) we obtain

$$(-1)^1 B_{L,R} = + \Pi_L V_{pp}^{(R)} \Pi_L$$

Therefore the first spike coefficient appears with amplitude

$$+ \frac{\Lambda(q)}{\sqrt{q}}$$

This matches exactly the explicit-formula spike weight and removes any sign ambiguity.

2.7 Ambient space, baseline, and envelopes

Work on the Hilbert space $H: = L^2(R, du)$ Let

$$H_0: = -\partial_u^2$$

with domain $D(H_0) = H^2(R)$ equivalently, the Friedrichs extension of $-\partial_u^2$ from $C_c^\infty(R)$.

Fix a smooth confining envelope $V_{env}: R \rightarrow R$ satisfying:

$$V_{env} \in L_{loc}^1(R), V_{env}(u) \geq -C$$

- $V_{env}(u) \rightarrow +\infty$ as $|u| \rightarrow \infty$ (any standard confining condition sufficient for compact resolvent; see Appendix A).

Fix also a left-wall penalty $U_{left}: R \rightarrow [0, \infty)$ supported on $u \leq u_*$ (for some fixed u_*), used only to strengthen confinement on $(-\infty, u_*]$ (The exact choice is not load-bearing; Appendix A records a concrete admissible class and proves the corresponding self-adjointness and discreteness.)

2.8 Gaussian well profile

Fix $\alpha > 0$ and a nonnegative Schwartz bump W_α centered at 0. For definiteness we take

$W_\alpha(u) := \frac{1}{\sqrt{2\pi\alpha}} \exp(-\frac{u^2}{2\alpha})$, though any fixed Schwartz profile with the same regularity/decay properties is admissible for the trace expansions in Appendices B–D.

2.9 Prime-power arithmetic package and truncation

Definition (truncated prime-power package).

For a cutoff $R > 0$ define the truncated prime-power package potential by

$$V_{pp}^{(R)}(u) = \sum_{\log q \leq R} \frac{\Lambda(q)}{\sqrt{q}} W_\alpha(u - \log q) \quad q = p^m.$$

Equivalently,

$$V_{pp}^{(R)}(u) = \sum_p \sum_{m \geq 1, m \log p \leq R} (\log p) p^{-m/2} W_\alpha(u - m \log p)$$

Then the truncated PWQO is

$$H_{\varepsilon,R} = -\partial_u^2 + V_{env}(u) + \varepsilon U_{left}(u) - \sum_{\log q \leq R} \frac{\Lambda(q)}{\sqrt{q}} W_\alpha(u - \log q).$$

The minus sign at the Hamiltonian level makes each prime-power term an attractive well. The coefficient of that well is always

$$\frac{\Lambda(q)}{\sqrt{q}}$$

2.10 Definition of the truncated PWQO operator

Define the truncated PWQO operator

$$H_R = H_0 + V_{env}(u) + U_{left}u - V_{pp}^{(R)}(u).$$

We realize H_R as the self-adjoint operator associated to the closed, lower-semibounded quadratic form

$$q_R[\psi] = \int_R (|\psi'(u)|^2 + (V_{env}(u) + U_{left}(u))|\psi(u)|^2) du - \int_R V_{pp}^{(R)}(u) |\psi(u)|^2 du,$$

with form domain $H^1(R) \cap L^2((V_{env} + U_{left})du)$. Since $V_{pp}^{(R)} \in L^\infty$ is a bounded multiplication operator for each fixed R , the perturbation is form-bounded with bound 0. Consequently H_R is self-adjoint and lower-semibounded by standard form methods (Appendix A).

2.11 Spectral discreteness and heat-trace legality (for fixed R)

For each fixed R , the confining hypothesis on $V_{env} + U_{left}$ implies H_R has compact resolvent and purely discrete spectrum (Appendix A). In particular the heat operator e^{-tH_R} is trace-class for every $t > 0$, and the localized/regularized trace functionals used in Appendices B–D are well-defined.

2.12 The $R \rightarrow \infty$ limit is taken at the level of trace identities

The full arithmetic package is obtained by letting $R \rightarrow \infty$. We do **not** require an a priori strong operator limit $H_R \rightarrow H$ for the argument; instead, we take the limit at the level of the trace identities and Laplace transforms:

- Appendix D proves, for each R , a diagonal spike decomposition for the localized trace of H_R with coefficients $\frac{\Lambda(q)}{\sqrt{q}}$ and spike locations $\log q$.
- Lemma D.LIMIT establishes normal convergence of the prime-power spike package as $R \rightarrow \infty$ and the remainder shield (D.SHIELD) controls the limit of the Ω -holomorphic remainder terms.

Accordingly we define the limiting Gaussian functional $F_H(t) := \lim_{R \rightarrow \infty} \langle \mu_{H_R}, G_t \rangle$ wherever it appears; all equalities in Section 5 are justified by the D.LIMIT + D.SHIELD passage.

Remark (what is “the operator” in this paper). The rigorous object used throughout is the family H_R and its trace identities, together with the controlled $R \rightarrow \infty$ limit. This avoids ambiguity between “primes-only wells” and the prime-power package and aligns the operator-side coefficients with the explicit-formula coefficients.

Section 3. Trace identity and explicit prime encoding

With (SA), (Agmon), and (Compact) in hand, we turn to the heat trace

$$\Theta_H(t) := \text{Tr}(e^{-tH}) = \sum_{n \geq 1} e^{-t\lambda_n}, \quad t > 0.$$

Our objective is to extract from the short-time behaviour of $\Theta_H(t)$ the arithmetic prime-power data $m \log p$ and to do so **purely spectrally**, without assuming any prior correspondence with zeta zeros. The analysis proceeds in three steps: (i) regularisation and short-time asymptotics, (ii) a prime-power kernel decomposition, and (iii) a resonance formulation showing how prime-power locations are encoded.

3.1 Regularised trace and short-time expansion

On R , the free heat trace diverges. We therefore work with a standard regularisation that is both rigorous and compatible with the appendix machinery.

For $L > 0$, let H_L be the Dirichlet restriction of H to $[-L, L]$, and let $H_{0,L} = -d^2/dt^2$ be the Dirichlet free Laplacian on $[-L, L]$. Then e^{-tH_L} and $e^{-tH_{0,L}}$ are trace-class and

$$\Theta H_L(t) = \text{Tr}(e^{-tH_L}), \Theta_{0,L}(t) = \text{Tr}(e^{-tH_{0,L}}).$$

Define the **finite-volume regularised trace**

$$\Theta_{reg,L}(t) := \text{Tr}(e^{-tH_L} - e^{-tH_{0,L}}), t > 0.$$

Appendix D proves the trace-class estimate (uniform for $0 < t \leq 1$)

$$\|e^{-tH_L} - e^{-tH_{0,L}}\|_1 \leq C_\alpha t^{1/2},$$

and shows that $\Theta_{reg,L}(t)$ admits a limit as $L \rightarrow \infty$. We define

$$\Theta_{reg}(t) := \lim_{L \rightarrow \infty} \Theta_{reg,L}(t), 0 < t \leq 1.$$

Proposition 3.1 (Short-time expansion; Appendix D).

As $t \rightarrow 0^+$,

$$\Theta_{reg}(t) = \frac{1}{2\pi} \log \frac{1}{t} + B + R(t), \quad R(t) = O(t^{1/2}).$$

Here B is a finite constant depending on the fixed parameters (α, γ) and the chosen well profile; its exact value is immaterial for counting-function arguments. The coefficient $\frac{1}{2\pi}$ in front of $\log(1/t)$ is the key term later transported into the eigenvalue counting asymptotics.

3.2 Prime-power kernel decomposition (Selberg-type expansion)

Appendix D refines the remainder $R(t)$ and shows it decomposes into contributions indexed by prime powers pm . The essential point is that the heat-kernel response of a **single** Gaussian well induces a universal Gaussian-type kernel in the arithmetic variable $a = m \log p$.

“From this point onward, the prime encoding is taken on the **prime-power lattice** $u_{p,m} = m \log p$, i.e., the potential is assembled from localized wells centered at $m \log p$ with canonical weights $(\log p)p^{-m/2}$.”

Let $g_t(a)$ denote the local response kernel arising from the heat kernel of a single Gaussian well. Appendix D proves $g_t(\cdot)$ is smooth and rapidly decaying in a for each fixed t , and that for small t it has a Gaussian leading behaviour of the form

$$g_t(a) = \frac{1}{4\pi t} \exp\left(-\frac{a^2}{4t}\right) + (\text{uniformly controlled error}), 0 < t \leq 1,$$

with error terms that are $O(t^{1/2})$ after summation over prime powers (Appendix D).

Proposition 3.2 (Prime-power kernel expansion; Appendix D).

There exists a smooth, rapidly decaying kernel $g_t(a)$ such that, for $0 < t \leq 1$,

$$R(t) = \sum_{p \in P} \sum_{m \geq 1} (\log p) p^{-m/2} g_t(m \log p) + O(t^{1/2}).$$

This is the Selberg-type “spectral–prime” identity for the PWQO: prime powers contribute at the arithmetic locations $a = m \log p$ with amplitudes $(\log p) p^{-m/2}$, and no use of zeta zeros is made at this stage.

Incremental (shell) form.

For a prime cutoff P , let $H(\leq P)$ be the operator where the sum over primes is restricted to $p \leq P$, and define $\Theta_{reg}^{(\leq P)}(t)$ accordingly. Then for $P_1 < P_2$,

$$\Delta \Theta_{reg}(t) := \Theta_{reg}^{(\leq P_2)}(t) - \Theta_{reg}^{(\leq P_1)}(t)$$

isolates exactly the incremental contribution of primes $p \in (P_1, P_2]$ and is predicted by the corresponding incremental prime-power kernel sum. This “shell difference” formulation is the one directly targeted by numerical stress tests of the Appendix-D mechanism.

3.3 Resonance formulation and arithmetic frequencies

Proposition 3.2 can be recast as an explicit statement about an atomic measure supported on prime-power locations.

Define the **prime-power measure** on $(0, \infty)$ by

$$d\mu(a) := \sum_{p \in P} \sum_{m \geq 1} (\log p) p^{-m/2} \delta(a - m \log p) da.$$

Then Proposition 3.2 is equivalently the statement that

$$R(t) = \int_0^\infty g_t(a) d_\mu(a) + O(t^{1/2}), 0 < t \leq 1$$

Thus the oscillatory remainder is a smoothed probe of an arithmetic atomic measure supported precisely at the locations $a = m \log p$.

Taking Fourier transforms in the arithmetic variable a gives a resonance representation:

$$\widehat{\mu}(\xi) := \int_0^\infty e^{-i\xi a} d\mu(a) = \sum_{p \in P} \sum_{m \geq 1} (\log p) p^{-m/2} e^{-i\xi m \log p}.$$

Since g_t is smooth and rapidly decaying, $\widehat{g}_t(\xi)$ is a smooth bump-type function, and the product

$$\widehat{R}(\xi; t) \approx \widehat{g}_t(\xi) \widehat{\mu}(\xi)$$

makes explicit that the spectral remainder carries the logarithmic prime frequencies $\log p$ (and their harmonics $m \log p$) through the atomic support of μ . This is the PWQO analogue of “Selberg-type resonance”: primes appear as discrete arithmetic frequencies in a purely spectral identity.

3.4 Summary of Section 3

- The regularised trace $\Theta_{reg}(t)$ admits a controlled short-time expansion

$$\Theta_{reg}(t) = \frac{1}{2\pi} \log \frac{1}{t} + B + O(t^{1/2}),$$

supplying the key logarithmic coefficient for eigenvalue counting arguments.

The refined decomposition expresses the remainder as a prime-power kernel sum

$$R(t) = \sum_{p,m} (\log p) p^{-m/2} g_t(m \log p) + O(t^{1/2}),$$

yielding an explicit prime encoding independent of any assumed link to zeta zeros.

- The measure-and-Fourier formulation shows the remainder is a smoothed probe of an atomic measure supported at $a = m \log p$, making the arithmetic resonance structure transparent.

These results are used in two distinct ways later:

1. **Appendix D** uses them (with the full Duhamel/Feynman–Kac machinery) to justify the regularised expansion, to control the remainder, and to supply the coefficients entering the eigenvalue counting function.
 2. **Appendix F** combines the trace-side arithmetic encoding with the Exact Gaussian Identity (Appendix E) and transform-uniqueness arguments to compare spectral measures and counting functions, forming the bridge to the Ξ -side correspondence.
-
-

Section 4. Functional mapping and ξ -factorisation

This section associates to each PWQO eigenfunction a canonical entire transform and proves that **all nontrivial zeros of Ξ** are zeros of that transform (for every eigenfunction), yielding a **local ξ -factorisation** that does **not** assume RH. The construction is used later only after the counting/bijection step (Appendix F) to avoid circularity.

4.1 Mellin–Gaussian transform of eigenfunctions

Let $H = -\frac{d^2}{dt^2} + V(t)$ be the PWQO from Section 2 with discrete spectrum $\{(\lambda_n, \psi_n)\}_{n \geq 1}$. Define, for $s \in \mathbb{C}$,

$$\Phi_n(s) := \int_{-\infty}^{\infty} \psi_n(t) e^{-t^2/2} e^{st} dt. \quad (4.1)$$

This is well-defined for all s because the Gaussian weight and uniform Agmon decay control the tails (Agmon decay is one of the standing hypotheses carried from Section 2; cf.)

Lemma 4.1 (Entirety and order; Appendix C).

Each Φ_n extends to an entire function of order ≤ 1 , with bounds uniform in n . In particular, for $\sigma, y \in \mathbb{R}$,

$$\log |\Phi_n(\sigma + iy)| \leq C_0 + C_1(1 + |\sigma|) \log(2 + |\sigma|), \quad (4.2)$$

and $|\Phi_n(\sigma + iy)| \ll 1$ on vertical lines as $|y| \rightarrow \infty$. (This is the “order ≤ 1 ” package you cite from Appendix C in your current Section 4.1.)

It is often convenient to remove the Gamma factor by defining the “de-Gamma’d” normalisation

$$G_n(s) := 2^{1-\frac{s}{2}} \Gamma\left(\frac{s}{2}\right)^{-1} \Phi_n(s),$$

so that G_n is entire of finite exponential type (uniformly in n) and local bounds near Ξ -zeros are cleaner.

Remark: “We do **not** use any global divisibility $\Xi|\Phi_n$. The only factorization used is the local statement Lemma C.5.1’(Appendix C): if Ξ and Φ_n vanish at the same point, then $\Phi_n = \Xi \cdot F_{n,*}$ locally near that point.”

4.3 Uniform control away from the zero set

The local nonvanishing near zeros and the maximum-modulus principle imply uniform bounds on compacta that avoid $\Xi^{-1}(0)$:

Lemma 4.4 (Uniform control on compacta).

Let $K \subset \mathbb{C}$ be compact with $\text{dist}(K, \Xi^{-1}(0)) > 0$. Then there exists $C_K > 0$ (depending only on that distance) such that for all n ,

$$\sup_{s \in K} |\tilde{F}_n(s)| \leq C_K, \quad \inf_{s \in K} |\tilde{F}_n(s)| \geq C_K^{-1} \quad (4.6)$$

This uniformity is exactly what is invoked later when passing to limits in the measure/counting arguments.

4.4 What Section 4 does—and does not—assert (avoiding circularity)

From (4.5) we immediately have

$$\Xi(s) = 0 \Rightarrow \Phi_n(s) = 0 \text{ for every } n. \quad (4.7)$$

At this stage we do not claim that $\Phi_n\left(\frac{1}{2} + i\lambda_n\right) = 0$ “because λ_n is a zero ordinate”.

That implication is only made after Appendix F establishes the bijection/equality of counting functions. In other words:

- Section 4 gives a one-way spectral vanishing statement: Ξ -zeros force Φ -zeros for all n (no RH assumed).
- Appendix F supplies the matching/bijection $\lambda_n \leftrightarrow \gamma_n$ (or equivalent equality of measures), at which point (4.7) can be evaluated at the matched points.

Formally:

Corollary 4.5 (Spectral vanishing at matched points).

Suppose the Appendix-F bijection between $\{\lambda_n\}$ and the ordinates $\{\gamma_n\}$ of the nontrivial zeros of Ξ . Then for every n ,

$$\Phi_n\left(\frac{1}{2} + i\lambda_n\right) = 0, \quad \tilde{F}_n\left(\frac{1}{2} + i\lambda_n\right) \neq 0. \quad (4.8)$$

The nonvanishing follows from the local nonvanishing lemma used in Proposition 4.3.

4.5 Output of Section 4 for the proof chain

Section 4 contributes exactly two reusable objects:

1. the canonical map $\psi_n \mapsto \Phi_n$ (and Gn if preferred), and
2. the ξ -factorisation $\Phi_n = \Xi \tilde{F}_n$ with local nonvanishing near Ξ -zeros.

No counting arguments occur here; those enter via the EGI and determinacy mechanism in Section 5 and Appendix F.

Canonical Proof Structure for the RH Closure

The proof is organized around four independent stages: the operator track, the explicit-formula track, the interface step, and the rigidity closure. Throughout the RH closure spine, the notation

$$F_H := F_H^{can}$$

and

$$R_H := R_H^{can}$$

is used for readability. These are the canonical bulk-subtracted displacement/commutator objects exported by Appendix D. They are not raw finite-volume traces, not raw multiplication-bump heat traces, and not unregularized whole-line traces.

Let

$$\Omega = \mathcal{C} \setminus (-\infty, 0]$$

1. Operator track: Appendix D

Appendix D is independent of the explicit formula and does not use the zero-side pole package.

It constructs the canonical of

$$F_H^{can} O(\Omega),$$

and the canonical operator-side holomorphic remainder

$$R_H^{can} \in O(\Omega)$$

On a nonempty right half-plane

$$U_{\sigma_0} \subset \Omega,$$

Appendix D proves the local decomposition

$$F_H^{can}(s) = B_{can}(s) + R_H^{can}(s)$$

Here B_{can} is the shifted canonical prime-power package

$$B_{can}(t) = \sum_q w(q) K_t(\log q)$$

where

$$w(q) = \frac{\Lambda(q)}{\sqrt{q}}$$

and

$$K_t(u) = e^{-t/4} Gt(u)$$

Appendix D exports $F_H^{can} \in O(\Omega)$, $R_H^{can} \in O(\Omega)$, and the overlap decomposition on U_{σ_0} . Appendix D does not assert that the infinite prime-side package Bcan is holomorphic on all of Ω .

2. Explicit-formula track: Appendix H

Appendix H is independent of the operator construction.

It constructs the zero-side meromorphic pole package

$$Z_{pole,\Omega}(s)$$

on Ω , together with the archimedean holomorphic term

$$H_{arch} \in O(\Omega)$$

On the same overlap half-plane $U\sigma_0$, Appendix H proves the package identity

$$B_{can}$$

This identity is first justified on the Laplace-legal half-plane. The zero-side object is then represented on Ω by the explicit meromorphic pole package plus the holomorphic archimedean term.

3. Interface step: Appendix E

Appendix E performs only the local substitution step. It does not construct a new global continuation.

On U_{σ_0} , combine the Appendix D identity

$$F_H^{can}(s) = B_{can}(s) + R_H^{can}(s)$$

with the Appendix H identity

$$B_{can}(s) = H_{arch}(s) - Z_{pole,\Omega}(s)$$

Therefore, on $U\sigma_0$,

$$F_H^{can}(s) = H_{tot}(s) - Z_{pole,\Omega}(s)$$

where

$$H_{tot} := R_H^{can} + H_{arch}$$

Since both R_H^{can} and H_{arch} are holomorphic on Ω , we have

$$H_{tot} \in O(\Omega)$$

Thus Appendix E proves the local overlap identity only. The passage from this local identity to a global identity on Ω is deferred to Appendix F.

4. Rigidity closure: Appendix F

Appendix F applies the meromorphic identity theorem on Ω .

Because

$$F_H^{can} \in O(\Omega)$$

$$H_{tot} \in O(\Omega)$$

and

$$Z_{pole,\Omega}$$

is meromorphic on Ω , the local identity

$$F_H^{can}(s) = H_{tot}(s) - Z_{pole,\Omega}(s)$$

on $U_{\sigma 0}$ extends meromorphically to all of Ω .

Now suppose there is a nontrivial zero

$$\rho = \beta + i\gamma$$

with

$$\beta \neq 1/2$$

Since ζ has no real zeros in the open critical strip, $\gamma \neq 0$. Then

$$s_\rho = -\rho(1 - \rho)$$

lies in Ω , not on the excluded cut $(-\infty, 0]$. By the zero-side pole package, $Z_{pole,\Omega}$ has a genuine uncanceled pole at $s = s_\rho$.

But F_H^{can} and H_{tot} are holomorphic on Ω . Therefore the identity

$$F_H^{can} = H_{tot} - Z_{pole,\Omega}$$

cannot hold at $s = s_\rho$. This contradiction rules out every off-critical zero.

Hence every nontrivial zero satisfies

$$\Re \rho = 1/2$$

This proves the Riemann Hypothesis, assuming the stated Appendix D export, Appendix H pole package, and Appendix E local interface theorem.

Global zero-parameter convention.

For every nontrivial zero ρ of ζ , define

$$z_\rho := -i(\rho - \frac{1}{2})$$

Equivalently,

$$\rho = \frac{1}{2} + iz_\rho$$

Then

$$\rho(1 - \rho) = (\frac{1}{2} + iz_\rho)(\frac{1}{2} - iz_\rho) = \frac{1}{4} + z_\rho^2$$

Thus

$$K_t(z_\rho) = e^{-t(1/4+z_\rho^2)} = e^{-t\rho(1-\rho)}$$

No RH assumption is made here; z_ρ may be complex. Under RH, z_ρ is real and equals the usual ordinate γ .

All opposite-sign conventions for z_ρ are deprecated and must be replaced by

$$z_\rho = -i(\rho - \frac{1}{2})$$

Section 5

The RH closure uses the local transform-domain overlap identity proved in Appendix E, the Ω -holomorphic operator-side shield from Appendix D, the Ω -meromorphic zero-side package from Appendix H, and the meromorphic identity theorem applied in Appendix F.

Appendix E proves the identity only on the common overlap half-plane $U_{\sigma_0} \subset \Omega$; the global identity on Ω is obtained in Appendix F.

Conventions near §5.0

Fix

$$T(t) := \Theta_H(t) - A_{class}(t), \lambda_\rho := \rho(1 - \rho), \Omega := C \setminus (-\infty, 0].$$

Fix also once and for all the zero-sum convention $\sum_\rho^\dagger$ and the spike sign ε_{sp} , so that the target Gaussian identity is

$$T(t) = -\varepsilon_{sp} \sum_\rho^\dagger m(\rho) e^{-t\lambda_\rho} (Target)$$

Centered cutoff-Gaussian family.

Fix once and for all an even cutoff $\chi \in C_c^\infty(R)$ such that

$$0 \leq \chi \leq 1, \chi(\xi) = 1 \text{ for } |\xi| \leq 1, \chi(\xi) = 0 \text{ for } |\xi| \geq 2$$

For $R \geq 1$, define

$$\chi_R(\xi) := \chi(\xi/R), \widehat{G_{t,R}}(\xi) := e^{-t\xi^2} \chi_R(\xi), G_{t,R} := F^{-1}(\widehat{G_{t,R}}).$$

This is exactly the cutoff-Gaussian family used in Appendix H.3. In particular,

$$G_{t,R} \in W_{even}(t > 0, R \geq 1)$$

and

$G_{t,R} \rightarrow G_t$ in the Schwartz topology as $R \rightarrow \infty$.

Gaussian normalization and rigidity-kernel dictionary.

Throughout Section 5 and Appendix H we use

$$G_t(u) := (4\pi t)^{-1/2} e^{-u^2/(4t)}, \widehat{G_t}(z) = e^{-tz^2}$$

For the Laplace-rigidity closure we also use the renormalized kernel

$$K_t(u) := e^{-t/4} G_t(u)$$

so that

$$\widehat{K}_t(z) = e^{-t(1/4+z^2)}.$$

If $\rho = \frac{1}{2} + iz_\rho$ is a nontrivial zero and

$$\lambda_\rho := \rho(1 - \rho),$$

then

$$\lambda_\rho = \frac{1}{4} + z_\rho^2 \quad 2(z_\rho) = e^{-t\lambda_\rho} = e^{-t\rho(1-\rho)}.$$

Thus the Weil-Gaussian parameterization and the Laplace-rigidity parameterization differ by the fixed factor

$$K_t = e^{-t/4} G_t$$

No RH assumption enters this identity.

5.1.1 Kernel and pairing

For $t > 0$, define the Appendix-H Gaussian

$$G_t(u) := (4\pi t)^{-1/2} \exp\left(-\frac{u^2}{4t}\right), u \in R.$$

Given a locally finite Borel measure μ on R for which the integral converges, define the Gaussian pairing

$$\langle \mu, G_t \rangle := \int_R G_t(u) d\mu(u).$$

5.1.2 The spectral measure

(A) The observable and the weighted spectral measure

Let H be the PWQO Hamiltonian (as defined in Appendix A) and fix the localized observable

$$A_L := \Pi_L M_w \Pi_L,$$

where Π_L is the localization to the window $[-L, L]$ (or the corresponding projector in the chosen Hilbert model) and M_w is multiplication by the bump w .

Let $\{(\lambda_n, \psi_n)\}_{n \geq 1}$ be an orthonormal eigenbasis of H (discrete spectrum, Appendix A). Define the **A_L -weighted spectral measure**

$$\mu_H^{A_L} := \sum_{n \geq 1} \langle \psi_n, A_L \psi_n \rangle \delta_{\lambda_n}$$

Normalization convention for the interface. The weighted spectral measure and the trace $\text{Tr}(A_L e^{-tH})$ are written with the raw observable A_L . The prime-power package later identified with the canonical package B is the density-normalized D-side package $\widehat{P}_{L,R}$, obtained from $\widehat{g}_{t,L} = (2L)^{-1} g_{t,L}$. No change is made here to the operator-theoretic observable or its spectral measure.

(B) The operator-side functional used in the proof

Define the operator-side test functional (the one decomposed in Appendix B/D)

$$F_H^{A_L}(t) := \text{Tr}(A_L e^{-tH})(t > 0).$$

Lemma (spectral expansion). For all $t > 0$,

$$F_H^{A_L}(t) = \sum_{n \geq 1} \langle \psi_n, A_L \psi_n \rangle e^{-t\lambda_n} = \langle \mu_H^{A_L}, e^{-t\lambda} \rangle$$

(Proof: spectral theorem / trace in eigenbasis; absolute convergence holds by the trace-class bounds established in Appendix A and Appendix B/D where this functional is used.)

Important notational remark :

In the diagonal spike analysis (Appendix D), the symbol $G_t(\cdot)$ refers to the *Gaussian/heat kernel in the translation/displacement channel variable* u used inside $g_t(a) = \text{Tr}(G_t A_L T_a)$

Operator-side role in the RH spine (split form)/Canonical transform convention for the RH spine.

The transform consumed by Appendices E and F is the canonical displacement/commutator transform exported by Appendix D:

$$F_H(s) := F_H^{can}(s) = L[\Theta_H^{can} - A_{class}^{can}](s)$$

Here Θ_H^{can} is the canonical density-normalized displacement/commutator trace functional whose one-letter sector is

$$P_{L,R}^{can}(t) = \sum_{\log q \leq R} w(q) g_{t,L}(\log q), \quad g_{t,L}(a) = \text{Tr}(S_t A_L T_a).$$

This $FHcan$ is not the raw multiplication-bump heat trace. The raw multiplication-bump local-loop profile $J_{t,L,\alpha}$ is auxiliary and is not the object used in the RH rigidity closure.

In Section 5, the spectral pairing uses the Laplace kernel $e^{-t\lambda}$ acting on the eigenvalue parameter λ . These are different roles and should not be conflated.

5.1.3 The explicit-formula measure (unconditional)

Appendix H constructs the explicit-formula comparison measure in the **same Gaussian family** by starting on the Weil/Paley–Wiener class and passing to G_t via Fourier cutoffs. We denote that measure by μ_{Ξ} (in the Appendix-H symmetric “zero package” variable), and define

$$G_{\Xi}(t) := \langle \mu_{\Xi}, G_t \rangle.$$

Important (no hidden RH). At this stage, μ_{Ξ} is **not** taken to be the pure ordinate measure $\sum_k \delta_{\gamma k}$.

That reduction occurs only after the pole-rigidity forcing step in Appendix F (which enforces $\Re \rho = \frac{1}{2}$). Until then, Appendix H’s symmetric packaging is the correct unconditional object.

5.1.4 Kernel conversion line and package status.

The canonical shifted package used in the RH closure is denoted

$$B_{can}.$$

At finite cutoff R, the shifted package is

$$B_R^{can}(t) = e^{-t/4} B_R^{raw}(t) = \sum_{\log q \leq R} \frac{\Lambda(q)}{\sqrt{q}} K_t(\log q), K_t(u) = e^{-t/4} Gt(u).$$

The ordered Appendix-D construction identifies the limiting canonical package on a nonempty overlap half-plane $U_{\sigma_0} \subset \Omega$. On that half-plane,

$$F_H(s) = B_{can}(s) + R_H(s), R_H \in O(\Omega)$$

Appendix D does not assert that B_{can} is holomorphic on all of Ω . The global continuation of the canonical package is supplied independently by Appendix H, where on U_{σ_0}

$$B_{can}(s) = H_{arch}(s) - Z_{pole,\Omega}(s),$$

with

$$H_{arch} \in O(\Omega)$$

and

$$Z_{pole,\Omega}$$

meromorphic on Ω . Thus every later occurrence of B_{can} refers to the shared overlap package, not to a globally holomorphic D-side function.

D-side density bridge. For fixed (L,R), Appendix D assembles the raw explicit spike term

$$\sum_{\log q \leq R} w(q) g_{t,L}(\log q) \text{ For comparison with the canonical package defined here, we use the}$$

density-normalized package

$$\widehat{P}_{L,R}(t) := \frac{1}{2L} \sum_{\log q \leq R} w(q) g_{t,L}(\log q) = \sum_{\log q \leq R} w(q) \widehat{g}_{t,L}(\log q), \widehat{g}_{t,L}(a) := \frac{1}{2L} g_{t,L}(a).$$

By Lemma D.LNORM, $\widehat{g_{t,L}}(a) \rightarrow_{c_w} G_t(a)$ locally uniformly on compact a -intervals; hence $\widehat{P_{L,R}}(t) \rightarrow_{c_w} B_R(t)$, and after the ordered $R \rightarrow \infty$ passage to $c_w B(t)$. When the window family is normalized so that $c_w = 1$, this is exactly the canonical package B used below.

Kernel normalization (Appendix D $\leftrightarrow$ Appendix H).

Throughout the paper we use the 1D heat-kernel normalization

$$G_t(u) = (4\pi t)^{-1/2} e^{-u^2/(4t)}$$

Its Laplace transform in t is $G_s^{\sim}(u) = \frac{1}{2\sqrt{s}} e^{-u\sqrt{s}}$ (principal branch on $\Omega = \mathbb{C} \setminus (-\infty, 0]$).

Kernel conversion and package role

The one-dimensional heat kernel

$$G_t(u) = (4\pi t)^{-1/2} e^{-u^2/(4t)}$$

is the shared kernel used to describe the explicit prime-power package B on both the D-side spike analysis and the H-side Gaussian explicit-formula analysis.

This kernel agreement identifies the shared package B ; it does not by itself identify the full operator transform F_H , which is handled separately by Appendix D.

Definition (raw displacement package and canonical closure package).

For $t > 0$, define the raw displacement-side package

$$B_{raw,R}(t) := \sum_{\log q \leq R} w(q) G_t(\log q), \quad w(q) := \frac{\Lambda(q)}{\sqrt{q}},$$

and

$$B_{raw}(t) := \lim_{R \rightarrow \infty} B_{raw,R}(t)$$

whenever the limit is taken in the sense established in Appendix D.

Define the canonical closure package by

$$B_R(t) := e^{-t/4} B_{raw,R}(t), B(t) := e^{-t/4} B_{raw}(t).$$

Equivalently,

$$B_R(t) = \sum_{\log q \leq R} w(q) K_t(\log q), B(t) = \sum_q w(q) K_t(\log q)$$

in the corresponding cutoff and limiting senses.

For $\Re(s) > \sigma_0$

Define

$$B_{can}(s) := L[B_{can}](s)$$

This is the overlap representative used in the D/E/H interface. Its global expression on Ω is not supplied by Appendix D as a holomorphic prime-side series; it is supplied by Appendix H as the meromorphic function

$$H_{arch}(s) - Z_{pole,\Omega}(s)$$

Convention.

B_{raw} is the raw displacement-side package extracted in Appendix D.

B is the canonical shifted closure package used in Appendices E, F, and H.

Whenever the R_H closure refers to the canonical package or to B_{can} , it means B , not B_{raw} .

Clarifying note.

No equality between the full operator-side transform and the zero-side meromorphic package is asserted at this stage. Section 5 isolates the shared package only. The canonical whole-line operator transform is constructed independently in Appendix D, and the local overlap identity is proved later in Appendix E.

5.2.0 Explicit spike package (shared by D and H)

Define the prime-power spike series in the G_t kernel by

$$Ss_p(t) := \sum_{p \in P} \sum_{m \geq 1} (\log p) p^{-m/2} G_t(m \log p).$$

(Absolute/local uniform convergence on $t \in (0, t_0]$ is established in Appendix D/H under the stated truncation-then-limit procedure.)

Let $A_{class}(t)$ denote the explicit archimedean/trivial package in the same test family (the “bulk” term common to the explicit formula and the regularized trace expansion).

A) Theorem 5.2 (EGI) —

Theorem 5.2 (Short-time operator decomposition)

For $0 < t \leq t_0$, the operator-side Gaussian functional admits the decomposition

$$\Theta_{op}(t) = A_{class}(t) + S_{sp}(t) + R_{op}(t)$$

where:

- $A_{class}(t)$ is the common bulk/classical term;
- $S_{sp}(t)$ is the explicit prime-power spike package with global coefficient $w(q) = \frac{\Lambda(q)}{\sqrt{q}}$
- $R_{op}(t)$ is super-smoothing on $(0, t_0]$.

This theorem is an operator-side decomposition only. It makes **no claim** of equality with a zero-side object, no claim of transform-domain equality, and no R_H closure claim.

Remark 5.2.1:

Theorem 5.2 is preparatory and operator-side only.

The exact centered cutoff-Gaussian bridge is supplied later by Theorem H.CD-center in Appendix H.3.

Section 5 combines that exact cutoff identity with operator cutoff removal and the Gaussian explicit-formula limit.

Proposition 5.3.1 (centered Gaussian package identity).

For $0 < t \leq t_0$ and $R \geq 1$, define

$$CD_H(G_{t,R}) := \sum_{\log q \leq R} w(q) G_{t,R}(\log q), w(q) = \frac{\Lambda(q)}{\sqrt{q}}.$$

Then

$$CD_H(G_{t,R}) = \langle \mu_{pr}, G_{t,R} \rangle.$$

Lemma 5.3.2a (operator cutoff removal)

The exact cutoff identity of Proposition 5.3.1 becomes the Gaussian trace identity after passing $R \rightarrow \infty$. The operator-side limit is supplied by the following cutoff-removal lemma.

For every $0 < t \leq t_0$

$$\langle \mu_H, G_{t,R} \rangle \rightarrow \Theta_H(t)(R \rightarrow \infty). (STG. 2)$$

Corollary 5.3.2 (raw Gaussian package and shifted closure package).

Let

$$B_{raw}(t) := \langle \mu_{pr}, G_t \rangle.$$

Then the raw Gaussian package satisfies

$$B_{raw}(t) = A_{class,raw}(t) - Z_{\lambda}(G_t), t > 0$$

where $Z_{\lambda}(G_t)$ denotes the zero-side Gaussian package in the unshifted Gt-kernel.

Define the shifted closure package

$$B(t) := e^{-t/4} B_{raw}(t) = \langle \mu_{pr}, K_t \rangle$$

Then

$$B(t) = A_{class}(t) - Z_{\lambda}(K_t), t > 0,$$

where $A_{class}(t)$ is the corresponding archimedean term shifted by the same factor $e^{-t/4}$.

Only the shifted package B is used in the RH closure. The raw package B_{raw} remains the natural displacement-side package extracted in Appendix D and serves as the local Gaussian package from which B is obtained by the universal factor $e^{-t/4}$. This package-level shift does not alter the local *fixed* – (L, R) extraction in Appendix D; it specifies only the package used in the closure spine.

5.4 Bridge to Appendix F: Transform-Domain Closure

5.4 Bridge to Appendix F: transform-domain closure.

Appendix F works with the canonical Appendix-D transform

$$FH(s) := F_H^{can}(s) = L[\Theta Hcan - Aclasscan](s),$$

defined on

$$\Omega = C \setminus (-\infty, 0].$$

This transform is not the raw multiplication-bump heat trace and is not an unregularized whole-line trace. Its load-bearing property is

$F_H^{can} \in O(\Omega)$. It is not a raw finite-volume resolvent trace and it is not an unregularized whole-line trace.

More precisely, Appendix D constructs

$$F_H^{can} \in O(\Omega)$$

as the canonical bulk-subtracted analytic transform

$$F_H^{can}(s) = L[\Theta_H^{can} - A_{class}^{can}](s)$$

When the notation

$$RegTr^{can}(H^\sim + s)^{-1}$$

is used, it denotes this canonical transform, not a raw finite-volume or raw whole-line resolvent trace.

The finite-cutoff transforms

$$F_{L,R}(s)$$

are auxiliary objects used only in the construction of F_H^{can} , the canonical shifted prime-power package, and the holomorphic remainder. They are not the final object used in the RH rigidity step.

The load-bearing output of Appendix D is the canonical decomposition

$$F_H^{can}(s) = B_{can}(s) + R_H^{can}(s), \quad R_H^{can} \in O(\Omega),$$

on a nonempty overlap half-plane

$$U_{\sigma_0} \subset \Omega.$$

Here B_{can} is the shifted closure package

$$B_{can}(t) = \sum_q w(q) K_t(\log q), \quad K_t(u) = e^{-t/4} G_t(u),$$

constructed from the density-normalized one-letter spike package in Appendix D. The raw displacement package extracted from G_t is not substituted directly into the pole-rigidity argument. This is an overlap identity only. Appendix D exports $F_H \in O(\Omega)$ and $R_H \in O(\Omega)$; it does not export $B_{can} \in O(\Omega)$.

Appendix H independently constructs the zero-side meromorphic pole package on Ω . On the same overlap half-plane U_{σ_0} , Appendix H identifies the canonical shifted package with the explicit-formula package:

$$B_{can}(s) = H_{arch}(s) - Z_{pole,\Omega}(s),$$

where

$$H_{arch} \in O(\Omega)$$

and

$$Z_{pole,\Omega}(s) = \sum_{\rho}^{\dagger} \frac{m(\rho)}{s + \rho(1 - \rho)}$$

is meromorphic on Ω .

Substituting the Appendix-H identity into the Appendix-D decomposition gives, on U_{σ_0} ,

$$F_H(s) = -Z_{pole,\Omega}(s) + H_{tot}(s),$$

where

$$H_{tot}(s) := R_H(s) + H_{arch}(s) \in O(\Omega).$$

Appendix E supplies this overlap identity on U_{σ_0} . Appendix F then extends the identity to Ω by the meromorphic identity theorem.. Appendix F then applies rigidity: since H and H_{tot} are holomorphic on Ω , the meromorphic function $Z_{pole,\Omega}$ cannot have an isolated pole in Ω . Therefore an off-critical zero, which would place

$$s_\rho = -\rho(1 - \rho)$$

inside Ω , is impossible.

Thus the Section 5 role is only to identify and normalize the shared canonical package B_{can} . The R_H contradiction itself uses the Appendix-D operator export, the Appendix-H pole package, the Appendix-E overlap identity, and the Appendix-F rigidity argument.

Theorem 5.5 — Dependency spine for the RH closure.

Let

$$\Omega = \mathcal{C} \setminus (-\infty, 0]$$

The RH closure uses exactly the following four inputs.

(D) Canonical operator-side export. Appendix D constructs

$$F_H^{can} \in O(\Omega), R_H^{can} \in O(\Omega),$$

and proves that on a nonempty overlap half-plane $U_{\sigma_0} \subset \Omega$,

$$F_H^{can}(s) = B_{can}(s) + R_H^{can}(s)$$

(H) Zero-side meromorphic package. Appendix H constructs

$$Z_{pole,\Omega}(s) = \sum_{\rho}^{\dagger} \frac{m(\rho)}{s + \rho(1 - \rho)}$$

as a meromorphic function on Ω , together with

$$H_{arch} \in O(\Omega)$$

(E) Local interface. Appendix E proves, on U_{σ_0} ,

$$B_{can}(s) = H_{arch}(s) - Z_{pole,\Omega}(s)$$

(F) Rigidity. Combining the D and E identities gives

$$F_H^{can}(s) = H_{tot}(s) - Z_{pole,\Omega}(s), H_{tot} := R_H^{can} + H_{arch} \in O(\Omega),$$

first on U_{σ_0} , and then on all of Ω by the meromorphic identity theorem. Since F_H^{can} and H_{tot} are holomorphic on Ω , $Z_{pole,\Omega}$ can have no uncanceled pole in Ω . An off-critical zero would create such a pole at

$$s_\rho = -\rho(1 - \rho) \in \Omega$$

contradiction. Therefore every nontrivial zero lies on the critical line.

Lemma E.U (Overlap legality on U_σ)

Let

$$T(t) := \Theta_H(t) - A_{class}(t),$$

and let $\sigma > \max\{0, \sigma_{op}\}$, where σ_{op} is any half-plane threshold on which Appendix D identifies the canonical operator transform with the Laplace transform of T. Set

$$U_\sigma := \{s \in \mathbb{C} : \Re s > \sigma\} \subset \Omega.$$

Then the following hold.

(i) Operator-side Laplace legality on U_σ .

For every $s \in U_\sigma$, the integral

$$F_H(s) := \int_0^\infty e^{-st} T(t) dt$$

converges absolutely and defines a holomorphic function on U_σ . Moreover, all trace/integral interchanges used to identify this Laplace transform with the canonical operator-side transform are justified by the Appendix-D trace/localization estimates, Tonelli/Fubini, and dominated convergence.

(ii) Zero-side termwise Laplace legality on U_σ .

Define

$$T_{zero}(t) := -\varepsilon_{sp} \sum_{\rho}^{\dagger} m(\rho) e^{-t\rho(1-\rho)}$$

Then for every $s \in U_\sigma$,

$$\int_0^\infty e^{-st} T_{zero}(t) dt = -\varepsilon_{sp} \sum_{\rho}^{\dagger} \frac{m(\rho)}{s+\rho(1-\rho)}$$

and the series on the right converges absolutely and locally uniformly on U_σ .

(iii) Scope.

This lemma asserts only legality and convergence of the relevant Laplace transforms on $U\sigma U\sigma$. It does **not** assert the bridge identity

$$F_H(s) = Z_{pole}(s)$$

on U_σ ; that identity is proved separately in Theorem E.INTERFACE λ .

Proof.

Part (i) is Appendix D. Part (ii) is exactly the zero-package Laplace legality established in H.DOM-LAPLACE; see p. 178. Part (iii) is a scope statement.

Remark E.U.spikes

The absolute convergence of the full prime-power Gaussian series with global normalization

$$w(q) = \frac{\Lambda(q)}{\sqrt{q}}$$

is established earlier and is used in the derivation of the Section-5 / H.5 Gaussian identity; see p. 177. It is not part of the final Appendix-E bridge once the bulk subtraction $A_{class}(t)$ has been made.

Remark.

The logical roles are strictly separated. Appendix D supplies the operator-side holomorphic function on Ω ; Appendix H supplies the zero-side meromorphic package on Ω ; Appendix E / Proposition F.8 proves only the local overlap identity on a nonempty open set $U \subset \Omega$. The present theorem performs the global identification on Ω and the resulting rigidity contradiction. No global continuation is constructed inside the bridge step.

Reader note on scope of Section 5. Section 5 isolates the canonical shared spike package B in the Gaussian family and fixes the frozen normalization. The RH contradiction does not follow from Section 5 alone. The closure uses, in addition, the canonical whole-line operator transform F_H from Appendix D, the independent Ω -meromorphic pole package from Appendix H, and the local split identity from Appendix E employed in Appendix F. No identity between a finite-interval Laplace transform and the canonical whole-line transform F_H is asserted or used in the closure.

Package firewall for the RH closure.

The manuscript uses two package levels. The raw displacement-side package B_{raw} is extracted in Appendix D from the spike kernel G_t in the displacement variable $u = \log q$. The RH contradiction, however, uses the shifted closure package

$$B(t) = e^{-t/4} B_{raw}(t),$$

equivalently the package built from $K_t = e^{-t/4} G_t$. No later theorem should identify the raw G_t -package itself with poles at $s = -\rho(1 - \rho)$.

Section 6. Ξ -factorisation and its role in the proof

This section collects, in one place, the analytic properties of the Gaussian–Laplace (Mellin–Gaussian) transform used in the proof and clarifies exactly how the Ξ -factorisation is

invoked in Section 5. No new assertions are made here; proofs are in Sections 3–5 and Appendices C–F.

6.1 Transform and growth (recall)

Let (λ_n, ψ_n) be an L2-normalised eigenpair of the PWQO H. Define the Gaussian–Laplace transform

$$\Phi_n(s) := \int_{-\infty}^{\infty} \psi_n(t) e^{-t^2/2} e^{st} dt, \quad s \in \mathbb{C}. \quad (6.1)$$

this integral is well-defined for all s (Gaussian weight + uniform Agmon decay), and Φ_n extends to an entire function of order ≤ 1 with bounds uniform in n :

$$\log |\Phi_n(\sigma + iy)| \leq C_0 + C_1 (1 + |\sigma|) \log(2 + |\sigma|), \quad (\sigma, y \in \mathbb{R}), \quad (6.2)$$

and in particular $|\Phi_n(\sigma + iy)|$ is uniformly bounded in y along any fixed vertical line $\sigma = \text{const.}$

It is sometimes convenient to divide out the Gamma factor and work with the “de-Gamma’d” normalisation

$$G_n(s) := 2^{1-\frac{s}{2}} \Gamma\left(\frac{s}{2}\right)^{-1} \Phi_n(s), \quad (6.3)$$

for which G_n is entire of finite exponential type uniformly in n . Either normalisation may be used downstream.

6.2 Zero-annihilation and Ξ –factorisation (no RH assumed; no poles)

Let

$$\Xi(s) := \frac{1}{2} s^{(s-1)} \pi^{-s/2} \Gamma\left(\frac{s}{2}\right) \zeta(s) \quad (6.4)$$

be the completed zeta function (entire), so that we avoid the pole of ζ at $s = 1$.

We use only the following facts (proved in Section 4 / Appendix C):

1. Zero-annihilation. If $\Xi(\rho) = 0$, then for every $n \geq 1$,

$$\Phi_n(\rho) = 0. \quad (6.5)$$

2. **Ξ -factorisation.** For each n there exists an entire function $F_n^\sim$ such that

$$\Phi_n(s) = \Xi(s) F_n^\sim(s), \quad (6.6)$$

and $F_n^\sim$ is zero-free in a neighbourhood of every zero of Ξ (local non-vanishing).

3. **Uniform control off the zero set.** If $K \subset \mathcal{C}$ is compact with $K \cap \{\Xi = 0\} = \emptyset$, then there is $C_K > 0$ such that for all n ,

$$\sup_{s \in K} |F_n^\sim(s)| \leq C_K, \quad \inf_{s \in K} |F_n^\sim(s)| \geq C_K^{-1} - 1. \quad (6.7)$$

These are the only analytic properties of Φ_n and $F_n^\sim$ used later.

6.3 How the factorisation is used (and what it does not assert)

From (6.6) and local non-vanishing, we immediately obtain:

- Every nontrivial zero of Ξ is a zero of Φ_n for every n (zero-annihilation).
- Locally near a zero of Ξ , the zero multiplicities of Φ_n and Ξ agree, since the quotient $F_n^\sim$ stays nonzero.
- Crucially, at this stage we do *not* assert that

$$\Phi_n\left(\frac{1}{2} + i\lambda_n\right) = 0 \quad (6.8)$$

for a fixed n . That statement is a *consequence* of the later counting/measure identification, not an input here.

More precisely: Appendices D–F produce equality of the eigenvalue counting function $N_H(T)$ and the zero counting function $N_\Xi(T)$, and hence a matching of ordinates γ_n with eigenvalues λ_n (with multiplicities).

Once that bijection is established, (6.5) applied at $\rho_n = \frac{1}{2} + i\gamma_n$

yields

$$\Phi_n\left(\frac{1}{2} + i\lambda_n\right) = \Phi_n\left(\frac{1}{2} + i\gamma_n\right) = 0, F_n^{\sim}\left(\frac{1}{2} + i\lambda_n\right) \neq 0, (6.9)$$

exactly as summarised in the current draft.

The simplicity of eigenvalues of H (Appendix B) is used only to index $\{\lambda_n\}$ cleanly; no simplicity assumption is made about zeros of Ξ .

6.4 Trace, primes, and where the explicit-formula content lives

All trace identities and explicit prime-encoding statements live in Section 3 and Appendix D.

The structural items imported into Section 5 are:

1. **Trace-class regularisation.** One works with a regularised heat trace (finite-volume difference against the free operator, then limits), justified by a trace-norm estimate of the form

$$\|e^{-tH} - e^{-tH_0}\|_1 \leq C t^{1/2}, 0 < t \leq 1, (6.10)$$

which permits interchanging limits and traces in the regularisation procedure.

(Here H_0 is the free Laplacian; the “free contribution” is the reference term subtracted in the regularisation, not a constant.)

2. **Short-time expansion.** The regularised trace admits a controlled small- t expansion of the form

$$\Theta_{reg}(t) = \frac{1}{2\pi} \log \frac{1}{t} + B + O(t^{1/2}), (t \rightarrow 0^+), (6.11)$$

with explicit leading constant and a finite constant B .

3. **Prime-power spike structure (canonical variable).** The oscillatory part is decomposed into a superposition of Gaussian “spikes” supported at the *arithmetic displacements*

$$u_{p,m} = m \log p, (6.12)$$

with amplitudes proportional to $(\log p)p^{-m/2}$. This is the Selberg-type prime encoding: primes enter through the discrete set $\{m \log p\}$ and their fixed coefficients, with all Fubini/Tonelli interchanges dominated.

(Important consistency note: the spike centres are at $u = m \log p$ in the displacement variable, not “centred at $t = m \log p$ ”.)

. What to remember (one paragraph)

Downstream, we use only: (i) Φ_n is entire of order ≤ 1 with bounds uniform in n

(ii) Ξ -zeros annihilate Φ_n for every n

(iii) the factorisation $\Phi_n = \Xi F_n^\sim$ with $F_n^\sim$ locally nonvanishing near Ξ -zeros and uniformly controlled off the zero set

and (iv) the trace/prime content (regularised trace bounds, short-time expansion, prime-power spike encoding) feeds into the EGI $\langle \mu_H, \phi_t \rangle = \langle \mu_\Xi, \phi_t \rangle$ on an interval, which Appendix F upgrades to equality of measures and a bijection $\lambda_n \leftrightarrow \gamma_n$.

Once that bijection is in place, Section 4’s factorisation yields $\Phi_n(\frac{1}{2} + i\lambda_n) = 0$ without circularity.

Appendix A.1

Model Definition (Operator Framework Used Throughout)

We work on the Hilbert space $L^2(R)$.

The primary spectral object is the confining Schrödinger operator

$$H_\varepsilon = -\partial_t^2 + V_\varepsilon^\sim(t),$$

constructed via Friedrichs extension and having compact resolvent (Appendix A).

Finite-volume operators $H_{L,R}$ and localization operators Π_L are used only as trace regularizations and auxiliary tools in the heat-trace expansion.

All translation operators T_a act on $L^2(R)$.

Whenever Dirichlet restrictions are used, translations are applied before localization, and any boundary effects are handled via Lemma D.MISALIGN.

Thus all operator identities are ultimately statements on the whole-line confining operator H_ε , with finite-volume objects serving only intermediate analytic roles.

Standing convention for Appendix A. Standing convention for Appendix A. Throughout this appendix, H denotes the Section 2 prime-power weighted quantum operator with the same frozen global normalization used in the RH spine, namely $w(q) = \frac{\Lambda(q)}{\sqrt{q}}$. Nothing in Appendix A redefines the global operator used later in Section 5 and in Appendices D, E, F, and H. Any prime-only comparison family introduced below is auxiliary and is used only for the local confinement/comparison argument inside Appendix A.

Spectral normalization convention

The finite-window cutoff family used in Appendix D is normalized by the single global shift M supplied by Proposition A/B.UNIFORM-LOWER-BOUND. Thus

$$H_{L,R}^\sim = H_{L,R}^{orig} + M I \geq 0$$

for all finite cutoff stages (L,R) , with M independent of L and R .

For the whole-line confining operator, the same convention is understood through the ordered cutoff construction: all Stieltjes/resolvent transforms used in the R_H closure are formed from the globally normalized, regularized, bulk-subtracted operator-side transform.

No L-dependent, R-dependent, or separately chosen spectral shift is introduced in the D/E/F/H spine.

The fixed global shift is included from the beginning in the definition of the Stieltjes-compatible operator-side transform. It is therefore already part of the pole-location convention used in Appendices D–F.

Lemma A.GROWTH (Dirichlet eigenvalue growth)

Setup. Fix $L > 0$ Let

$$H_{L,R} = -\partial_u^2 + V_{L,R}(u) \text{ on } L^2([-L, L]),$$

with Dirichlet boundary conditions at $\pm L$. Given $V_{L,R} \in L^\infty([-L, L])$ is real-valued and set $M_{L,R} := \|V_{L,R}\|_\infty$. Let $\{\lambda_n(L, R)\}_{n \geq 1}$ be the **eigenvalues of** , listed nondecreasing with multiplicity.

Claim. For every $n \geq 1$,

$$\lambda_n(L, R) \geq \left(\frac{n\pi}{2L}\right)^2 - M_{L,R}$$

Proof. Let $A_0 = -\partial_u^2$ on $[-L, L]$ with Dirichlet boundary conditions. Then A_0 has eigenvalues $\mu_n(L) = \left(\frac{n\pi}{2L}\right)^2$. Writing W for multiplication by $V_{L,R}$ we have for all f ,

$$\langle f, Wf \rangle \geq -\|V_{L,R}\|_\infty \|f\|^2 = -M_{L,R} \|f\|^2,$$

hence as quadratic forms $H_{L,R} = A_0 + W \geq A_0 - M_{L,R}I$ By min-max,

$$\lambda_n(H_{L,R}) \geq \lambda_n(A_0 - M_{L,R}I) = \mu_n(L) - M_{L,R}$$

Corollary A.TRACE (Trace-class resolvent and absolute convergence)

Let $\Omega = \mathbb{C} \setminus (-\infty, 0]$ For any compact $K \subset \Omega$,

$$\sum_{n=1}^{\infty} \sup_{s \in K} \frac{1}{|s + \lambda_n(L,R)|} < \infty$$

In particular, for each $s \in \Omega$, $(H_{L,R} + s)^{-1}$ is trace-class and

$$\text{Tr}(H_{L,R} + s)^{-1} = \sum_{n=1}^{\infty} \frac{1}{s + \lambda_n(L,R)} \text{ (absolute convergence).}$$

Proof. Since $K \subset \Omega$ is compact,

$$d_K := \inf_{s \in K} \text{dist}(s, (-\infty, 0]) > 0. \text{ For } x \geq 0, \sup_{s \in K} |s + x|^{-1} \leq C_K (1 + x)^{-1}$$

. By Lemma A.GROWTH,

$$\lambda_n(L, R) \geq cn^2 - C \lambda_n(L, R) \geq cn^2 - C, \text{ hence } \sum (1 + \lambda_n)^{-1} \lesssim \sum (1 + n^2)^{-1} < \infty. \text{ Combine.}$$

Corollary A.HOLO (Holomorphy of the resolvent trace on Ω)

The function

$$F_{L,R}(s) := \text{Tr}(H_{L,R} + s)^{-1} = \sum_{n=1}^{\infty} \frac{1}{s + \lambda_n(L,R)}$$

is holomorphic on Ω , with uniform convergence on Ω -compacts and termwise differentiation valid.

Proof. Uniform absolute convergence on compacts plus Weierstrass theorem.

Stieltjes-normalized convention

In the Stieltjes-compatible statements below, $H_{L,R}$ denotes the globally shifted nonnegative operator

$$H_{L,R}^{\sim} = H_{L,R}^{orig} + MI$$

For the unshifted operator $H_{L,R}^{orig}$, the same finite-volume trace-class estimates hold after translating by the fixed global shift M .

The cutoff-dependent bound in Lemma A.GROWTH is used only for fixed finite-volume trace-class legality. The cutoff-uniform lower bound needed for the large-time density tail is supplied separately by Proposition A/B.UNIFORM-LOWER-BOUND.

Appendix A.1: Self-Adjointness and Discreteness of the PWQO Operator

A.1 Operator foundation for the confining PWQO: self-adjointness and discrete spectrum

A.1.1 The confining operator used in the proof

$$H_{\varepsilon,R} := -\partial_t^2 + V_{conf}(t) - V_{pp}^{(R)}(t), \quad V_{conf}(t) := V_{env}(t) + \varepsilon_{Uleft}(t),$$

where:

- $V_{conf}(t) \rightarrow +\infty$ as $|t| \rightarrow \infty$,
- $V_{pp}^{(R)}$ is the truncated prime-weighted well term,
- for each fixed R , $V_{pp}^{(R)}$ is bounded (or at least form-bounded relative to the confining baseline).

A.1.2 Quadratic form definition and self-adjointness

Define the quadratic form on $C_c^\infty(R)$:

$$q_\varepsilon[\psi] := \int_R |\psi'(t)|^2 dt + \int_R V_\varepsilon^\sim(t) |\psi(t)|^2 dt.$$

“Since $\tilde{V}_\varepsilon$ is bounded below and locally integrable, the form is lower semibounded and closable; by KLMN (Reed–Simon I, Thm X.17), the associated operator is self-adjoint.”

Suppose $\tilde{V}_\varepsilon \in L^1_{loc}(R)$ is real-valued and bounded below in the form sense (e.g. $\tilde{V}_\varepsilon \geq -C$ a. e.). Then q_ε is densely defined, lower bounded, and closable. Let $\overline{q}_\varepsilon$ denote its closure. By the representation theorem for closed lower-bounded forms, there exists a unique self-adjoint operator H_ε such that

$$\overline{q}_\varepsilon[\psi] = \langle H_\varepsilon^{1/2} \psi, H_\varepsilon^{1/2} \psi \rangle \quad (\psi \in D(\overline{q}_\varepsilon))$$

Equivalently, H_ε is the Friedrichs operator associated to $\overline{q}_\varepsilon$. In particular, H_ε is self-adjoint and lower semibounded, hence has real spectrum.

A.1.3 Confinement from the background envelope.

The compact-resolvent property used in the R_H closure is supplied by the fixed confining background

$$V_{conf}(u) := V_{env}(u) + \varepsilon U_{left}(u),$$

not by the arithmetic spike package. We suppose

$$V_{conf}(u) \rightarrow +\infty (|u| \rightarrow \infty).$$

The prime-power potential

$$V_{pp}^{(R)}(u) = \sum_{\log q \leq R} \frac{\Lambda(q)}{\sqrt{q}} W_\alpha(u - \log q)$$

is an attractive perturbation controlled by the envelope-domination estimate below. Thus the arithmetic wells do not supply the compactness mechanism; they are perturbative relative to the confining baseline.

A.1.4 Confinement at $-\infty$ via the penalty U

On the left end, $V(t)$ may fail to confine. We therefore choose $U \geq 0$ so that

$$V_{\varepsilon}^{\sim}(t) = V(t) + \varepsilon U(t) \rightarrow +\infty \text{ as } t \rightarrow -\infty.$$

Together with A.1.3, this implies

$$V_{\varepsilon}^{\sim}(t) \rightarrow +\infty \text{ as } |t| \rightarrow \infty. \text{ (A. 1. 4. 1)}$$

Proposition A.ENV-DOM

Envelope domination of the prime-power perturbation

For the R_H closure spine, fix a nonnegative admissible bump W_{α} satisfying a Gaussian-tail bound

$$0 \leq W_{\alpha}(x) \leq C_{\alpha} e^{-\kappa x^2} \quad (x \in R)$$

for some constants $C_{\alpha, \kappa} > 0$. In particular, the Gaussian profile

$$W_{\alpha}(x) = (2\pi\alpha)^{-1/2} e^{-x^2/(2\alpha)}$$

is admissible.

For each finite prime-power cutoff $R < \infty$, define the nonnegative majorant

$$V_R^{\#}(u) := \sum_{\log q \leq R} \frac{\Lambda(q)}{\sqrt{q}} W_{\alpha}(u - \log q), \quad q = p^m$$

Then there exists a constant $C < \infty$, independent of R , such that

$$V_R^{\#}(u) \leq C(1 + u^2 + eu) \quad (u \in R, \quad R < \infty).$$

Consequently, for every $0 < \theta < 1$, if

$$V_{env}(u) := A(1 + u^2 + eu)$$

with $A \geq C/\theta$ then

$$V_R^{\#}(u) \leq \theta V_{env}(u) \quad (u \in R, \quad R < \infty)$$

Thus A.ENV-DOM holds with $C_0 = 0$. In particular, the attractive prime-power perturbation is uniformly dominated by a fixed confining envelope. No zeta-zero information or zero-side input enters this estimate.

Proof

It is enough to estimate the full infinite majorant

$$V_{\infty}^{\#}(u) := \sum_{q \geq 2} \frac{\Lambda(q)}{\sqrt{q}} W_{\alpha}(u - \log q),$$

since

$$V_R^{\#}(u) \leq V_{\infty}^{\#}(u)$$

for every finite R.

Group the prime powers into logarithmic shells

$$k \leq \log q < k + 1, k = 0, 1, 2, \dots$$

For $q \in [e^k, e^{k+1})$, we have

$$\frac{\Lambda(q)}{\sqrt{q}} \leq \frac{\log q}{\sqrt{q}} \leq C(k + 1)e^{-k/2}.$$

The number of integers n in the shell $ek \leq n < ek + 1$ is $O(ek)$. Since prime powers form a subset of the integers,

$$\sum_{e^k \leq q < e^{k+1}} \frac{\Lambda(q)}{\sqrt{q}} \leq \sum_{e^k \leq n < e^{k+1}} \frac{\log n}{\sqrt{n}} \leq C(k + 1)e^{k/2}.$$

Next, for $k \leq \log q < k + 1$,

$$|u - \log q| \geq |u - k|^{-1}$$

Therefore, for any $0 < \kappa' < \kappa$ there exists a constant $C_{\kappa, \kappa'}$ such that

$$e^{-\kappa}(u - \log q)^2 \leq C_{\kappa, \kappa'} e^{-\kappa'(u-k)^2}$$

Using the Gaussian-tail bound for W_{α} , we obtain

$$W_{\alpha}(u - \log q) \leq C e^{-\kappa'(u-k)^2}$$

uniformly for all q in the k -th shell. Hence

$$V_{\infty}^{\#}(u) \leq C \sum_{k \geq 0} (k+1) e^{k/2} e^{-\kappa'(u-k)^2}.$$

We now estimate this shell sum.

If $u \leq 0$, then $k \geq 0$ implies

$(u - k)^2 \geq c(u^2 + k^2)$ for some $c > 0$. Thus

$$\sum_{k \geq 0} (k+1) e^{k/2} e^{-\kappa'(u-k)^2} \leq C e^{-cu^2} \sum_{k \geq 0} (k+1) e^{k/2} e^{-ck^2} \leq C. \quad k \geq 0$$

If $u > 0$, write $j = k - u$ Then

$$\frac{k}{2} - \kappa'(u - k)^2 = \frac{u}{2} + \frac{j}{2} - \kappa'j^2 \leq u^2 + C_{\kappa'}.$$

Therefore

$$\sum_{k \geq 0} (k+1) e^{k/2} e^{-\kappa'(u-k)^2} \leq C(1+u) e^{u/2}.$$

Combining the cases,

$$V_{\infty}^{\#}(u) \leq C(1 + (1+u)) e^{u/2}$$

Finally,

$$(1+u) e^{u/2} \leq C(1 + u^2 + u^{u/2}),$$

so

$$V_{\infty}^{\#}(u) \leq C(1 + u^2 + e^u).$$

Since $V_R^{\#} \leq V_{\infty}^{\#}$, the same estimate holds uniformly for every finite cutoff R :

$$V_R^{\#}(u) \leq C(1 + u^2 + e^u).$$

Now choose

$$V_{env}(u) = A(1 + u^2 + e^u)$$

with $A \geq C/\theta$. Then

$$V_R^\#(u) \leq \theta V_{env}(u)$$

for all $u \in \mathbb{R}$ and all finite R . Hence A.ENV-DOM holds with $C_0 = 0$.

Scope note for Proposition A.ENV-DOM

This proposition estimates the nonnegative majorant $V_R^\#$. The attractive sign in the Hamiltonian is handled only after this domination estimate, through the KLMN form bound in Lemma A.ENV-FORM.

For the R_H closure spine, the bump W_α is fixed in the Gaussian-tail admissible class above. Broader Schwartz profiles may be used in auxiliary trace-expansion variants only when they satisfy the same

Lemma A.ENV-FORM

Uniform KLMN form bound

Under Hypothesis A.ENV-DOM, the arithmetic perturbations $V_R^\#$ are uniformly form-bounded with respect to the confining baseline

$$H_{base} = -\partial_u^2 + V_{env} + \varepsilon U_{left}.$$

More precisely, for every $f \in D(H_{base})$

$$\int_R V_R^\#(u) |f(u)|^2 du \leq \theta H_{base}[f] + C_0 \|f\|_2^2,$$

with the same $\theta < 1$ and C_0 for every R .

Consequently,

$H_R := H_{base} - V_R^\#$ is self-adjoint and lower semibounded uniformly in R , and after the fixed spectral shift used elsewhere in the paper,

$$\sigma(H_R) \subset [0, \infty)$$

Proof

By A.ENV-DOM,

$$V_R^\#(u) \leq \theta V_{env}(u) + C_0$$

Therefore

$$\int V_R^\# |f|^2 \leq \theta \int V_{env} |f|^2 + C_0 \|f\|_2^2$$

Since

$$h_{base}[f] = \int |f'|^2 + \int V_{env} |f|^2 + \varepsilon \int U_{left} |f|^2,$$

we have

$$\int V_R^\# |f|^2 \leq \theta h_{base}[f] + C_0 \|f\|_2^2$$

Because $\theta < 1$ KLMN gives a closed lower-semibounded form

$$h_R = h_{base} - \int V_R^\# |f|^2$$

hence a self-adjoint operator H_R . The lower bound is uniform in R , and the standard spectral shift makes all H_R nonnegative.

(Appendix A.1.5 —)

Spectral normalization and nonnegativity

Since H_ε is self-adjoint and lower semibounded, let

$$\lambda_{\min} := \inf \sigma(H_\varepsilon) > -\infty.$$

Define the spectrally shifted operator

$$\tilde{H} := H_\varepsilon - \lambda_{\min} I.$$

Then $\tilde{H} \geq 0$ and $\tilde{H}$ is self-adjoint with compact resolvent and discrete spectrum

$$0 \leq \lambda_1^\sim \leq \lambda_2^\sim \leq \dots \rightarrow +\infty$$

Since spectral shifting by a constant only reparameterizes the Laplace variable,

$$\text{Tr}((H_\varepsilon + s)^{-1}) = \text{Tr}((\tilde{H} + (s + \lambda_{\min}))^{-1}),$$

all transform identities, spike decompositions, and Ω -holomorphy arguments are unchanged up to the harmless translation $s \mapsto s + \lambda_{\min}$

Accordingly, throughout the remainder of the manuscript we assume without further comment that the PWQO operator has been normalized so that

$$H \geq 0$$

Under this normalization, the poles of $(H + s)^{-1}$ occur only at

$$s = -\lambda_n \in (-\infty, 0],$$

and therefore the resolvent trace is holomorphic on

$$\Omega = \mathbb{C} \setminus (-\infty, 0].$$

A.1.6 Trace consequences used in Appendix F

Compact resolvent implies e^{-tH_ε} is trace class for every $t > 0$ and

$$\Theta_H(t) := \text{Tr}(e^{-tH_\varepsilon}) = \sum_{n \geq 1} e^{-t\lambda_n}$$

Moreover for $\Re s > 0$

$$\text{Tr}((s + H_\varepsilon)^{-1}) = \sum_{n \geq 1} \frac{1}{s + \lambda_n}$$

a Stieltjes-type meromorphic function with poles only at $-\lambda_n \in (-\infty, 0]$. These finite-stage trace identities supply operator-theoretic legality for the Appendix-D construction. They are not the final object used in the Appendix-F RH contradiction. The final Appendix-F input is the canonical exported transform

$$F_H^{can} \in O(\Omega)$$

A.1.7 Notational convention

For the remainder of the manuscript we fix $\varepsilon \in (0, 1]$ and write $H := H_\varepsilon$, $V^\sim := V_\varepsilon^\sim$

Smoothness and Growth of the Prime-Weighted Potential

Throughout we fix

$$H_{\varepsilon,R} = -\partial_u^2 + V_{env}(u) + \varepsilon U_{left}(u) - V_{pp}^{(R)}(u),$$

with

$$V_{pp}^{(R)}(u) = \sum_{\log q \leq R} \frac{\Lambda(q)}{\sqrt{q}} W_\alpha(u - \log q).$$

Lemma A 1.8 (Uniform C^∞ convergence on compact sets)

For every compact interval $K = [T_1, T_2] \subset \mathbb{R}$ and for every integer $k \geq 0$

$$\sum_{p \in P} V_p^{(k)} \text{ converges uniformly on } K.$$

Proof. Fix K . Because $\{\ln \sqrt{p}\}$ is discrete and tends to $+\infty$, choose $\delta_K > 0$ and a finite set $P_K \subset P$ so that $|t - \ln p| \geq \delta_K$ for all $t \in K$ and $p \notin P_K$

Write $V_p^{(k)} = P_k(t - \ln p) V_p(t)$ where P_k is a polynomial of degree $\leq 2k + 1$
Hence, for $p \notin P_K$ and $t \in K$, $C_{k,K}$

$$|V_p^{(k)}(t)| \leq C_{k,K} \exp(-\delta_K^2/(2\alpha)),$$

with $C_{k,K}$ independent of p . The number of primes in $[X, X + 1]$ is $O(e^X/X)$; summing that geometric tail yields uniform convergence.
The same bound holds for the finitely many derivatives $k \leq m$, so $V \in C^\infty(R)$.

Lemma A 1.9 (Super-linear confining bound as $t \rightarrow +\infty$)

There exist constants $C > 0$ and $T_0 > 0$ such that for all $t \geq T_0$,

$$V(t) \geq C \frac{e^t}{t}.$$

Proof. Let $\delta = 1$ The Prime Number Theorem implies $\#\{p: |\ln \sqrt{p} - t| < 1\} \sim e^t/t$. For each such p one has $V_p(t) \geq (\ln \sqrt{p})/\sqrt{2\pi\alpha} \cdot e^{-1/(2\alpha)} \geq c_0$. Summing over $\sim e^t/t$ primes gives the claimed lower bound.

Lemma A 10 (Boundedness as $t \rightarrow -\infty$)

There exists $V_{-\infty} < \infty$ such that $V(t) \leq V_{-\infty}$ for all $t \leq 0$.

Proof. For $t \leq 0$ one has $|t - \ln p| \geq \ln p$. Hence $V_p(t) \leq (\ln p)/\sqrt{2\pi\alpha} e^{-(\ln p)^2/(2\alpha)}$, and the series $\sum_p (\ln p) e^{-(\ln p)^2/(2\alpha)}$ converges.

Appendix A.2 — Domain, symmetry, and discreteness for the confining operator

A.2.1 Operator definition and standing convention

Fix $\varepsilon \in (0, 1]$ Let

$$V_\varepsilon^\sim(t) := V(t) + \varepsilon U(t), \quad H_\varepsilon := -\frac{d^2}{dt^2} + V_\varepsilon^\sim(t) \text{ on } L^2(R), \quad (\text{A. 2. 1. 1})$$

where H_ε is defined via the closed, lower-bounded quadratic form on $C_c^\infty(R)$

$$q_\varepsilon[\psi] = \int_R |\psi'(t)|^2 dt + \int_R \tilde{V}_\varepsilon(t) |\psi(t)|^2 dt, \psi \in C_c^\infty(R), (A. 2. 1. 2)$$

and its Friedrichs extension (Appendix A.1). We write $H := H_\varepsilon$ and $V := V_\varepsilon$ for the remainder of Appendix A unless stated otherwise.

Standing hypotheses. $\tilde{V} \in L_{loc}^1(R)$ is real-valued, $\tilde{V}$ is bounded below (in the form sense), and $\tilde{V}(t) \rightarrow +\infty$ as $|t| \rightarrow \infty$ (Appendix A.5).

A.2.2 Symmetry on $C_c^\infty(R)$

Lemma A.2.2 (Symmetry). The operator H is symmetric on $C_c^\infty(R)$: for all $f, g \in C_c^\infty(R)$,

$$\langle Hf, g \rangle = \langle f, Hg \rangle. (A. 2. 2. 1)$$

Proof. Since $\tilde{V}$ is real-valued and locally integrable, for $f, g \in C_c^\infty$ we have

$$\langle (-f''), g \rangle = \langle f, (-g'') \rangle$$

by integration by parts with vanishing boundary terms, and

$$\langle \tilde{V}f, g \rangle = \langle f, \tilde{V}g \rangle.$$

Adding yields (A.2.2.1).

A.2.3 Closedness and self-adjoint realization

Proposition A.2.3 (Closed form and self-adjoint operator). The quadratic form q_ε defined on $C_c^\infty(R)$ is densely defined, lower bounded, and closable. Its closure $\overline{q_\varepsilon}$ determines a unique self-adjoint, lower semibounded operator H (the Friedrichs operator associated to $\overline{q_\varepsilon}$).

Proof. This is standard: lower boundedness and local integrability of $\tilde{V}$ ensure q_ε is closable; the representation theorem for closed, lower-bounded forms gives a unique associated self-adjoint operator. (See Appendix A.1 for the cited theorem and setup.)

note: This section replaces any “limit-point” reliance. The Friedrichs construction is the cleanest route and aligns with Theorem F.RH

A.2.4 Compact resolvent (the only point needed for Theorem F.RH)

Theorem A.2.4 (Compact resolvent). Under the standing hypothesis $V^\sim(t) \rightarrow +\infty$ as $|t| \rightarrow \infty$, the resolvent $(H + 1)^{-1}$ is compact on $L^2(R)$.

Proof (citation). Confinement implies compactness of the form-domain embedding into $L^2(R)$, hence compact resolvent for the associated Schrödinger operator; see e.g. Reed–Simon IV, Theorem XIII.67 (or any equivalent compactness criterion). This is also established directly in Appendix A.5.

A.2.5 Discrete spectrum and eigenfunction basis

Corollary A.2.5 (Purely discrete spectrum). The spectrum of H is purely discrete: there exists a sequence of real eigenvalues

$$\lambda_1 \leq \lambda_2 \leq \dots, \lambda_n \rightarrow +\infty, \quad (A.2.5.1)$$

each of finite multiplicity, and an orthonormal basis $\{\psi_n\}_{n \geq 1}$ of $L^2(R)$ consisting of eigenfunctions $H\psi_n = \lambda_n \psi_n$.

Proof. Compact resolvent implies the spectrum consists only of eigenvalues of finite multiplicity accumulating at $+\infty$, and yields an orthonormal eigenbasis.

Scope of the trace identities

The following trace identities are used in two settings only:

1. fixed finite-volume Dirichlet operators, where the one-dimensional eigenvalue growth gives trace-class resolvent legality; and
2. the regularized/bulk-subtracted whole-line transform F_H constructed by the ordered Appendix-D limiting procedure.

They are not used as an unqualified raw whole-line identity

$$F_H(s) = \text{Tr}(H + s)^{-1}$$

Then replace any proof sentence saying “compact resolvent implies the resolvent trace is trace class” with:

In the one-dimensional finite-volume setting, the trace-class resolvent statement follows from the eigenvalue growth estimate

$$\lambda_n \gtrsim n^2$$

after the fixed global shift. Thus

$$\sum_n (1 + \lambda_n)^{-1} < \infty$$

For the whole-line R_H closure, the object used downstream is the regularized transform

$$F_H^{can}(s) = L[\Theta_H^{can} - A_{class}^{can}](s),$$

not an unqualified raw trace

A.2.6 A.GROWTH $\Rightarrow \sum (s + \lambda_n)^{\{-1\}}$ converges uniformly on Ω -compacts”

Proposition A.2.6 (Trace identities). For every $t > 0$, the heat semigroup e^{-tH} is trace class and

$$\Theta_H(t) := \text{Tr}(e^{-tH}) = \sum_{n \geq 1} e^{-t\lambda_n}. \quad (\text{A. 2. 6. 1})$$

For $\Re s > 0$, $(s + H)^{-1}$ is trace class and

$$F_H^{can}(s) = \int_0^\infty e^{-st} (\Theta_H^{can}(t) - A_{class}^{can}(t)) dt \quad (\text{A. 2. 6. 2})$$

on the initial half-plane, and then

$$F_H^{can} \in \mathcal{O}(\Omega).$$

Appendix A supplies finite-stage spectral and trace-class legality; Appendix F uses the holomorphic exported object, not a raw Stieltjes trace.

Proof. Both statements follow from (A.2.5.1) and functional calculus for self-adjoint operators with compact resolvent: eigenvalues are summable against $e^{-t\lambda_n}$ for each $t > 0$, and $\sum (s + \lambda_n)^{-1}$ converges for $\Re s > 0$.

Connection to Theorem F.RH: this supports the Appendix-D construction of F_H^{can} .

The Appendix-F rigidity step uses only $F_H^{can} \in O(\Omega)$.

Appendix A 3

Proof

Lemma A 3.1 (Infinitesimal form bound).

Let

$$q_0[\psi] = \int_R |\psi'(t)|^2 dt, \quad v[\psi] = \int_R V(t) |\psi(t)|^2 dt, \quad \psi \in C_c^\infty(R).$$

For every $\varepsilon > 0$ there exists $b(\varepsilon) > 0$ s.t.

$$v[\psi] \leq \varepsilon q_0[\psi] + b(\varepsilon) \|\psi\|_{L^2}^2 \quad \forall \psi \in C_c^\infty(R).$$

Proof.

Choose $R = R(\varepsilon)$ large so that $V(t) \geq 4\varepsilon$ for $|t| \geq R$ (Lemma A 2.3).

Split $R = (-R, R) \cup \{|t| > R\}$

Interior. V is bounded on $[-R, R]$; say $V \leq M_R$. Then

$$\int_{-R}^R V |\psi|^2 \leq M_R \|\psi\|_{L^2}^2.$$

Exterior. On $|t| > R$, we have $V \geq 4\varepsilon$. One-dimensional Hardy gives

$$\int_{|t| > R} \frac{|\psi|^2}{t^2} dt \leq 4q_0[\psi].$$

Because $|t| > R \Rightarrow 1/t^2 \leq 1/R^2$,

$$\int_{|t| > R} |\psi|^2 \leq 4R^2 q_0[\psi].$$

Multiplying by $V \geq 4\varepsilon$ gives the exterior bound $\leq \varepsilon q_0[\psi]$.

Adding the two parts proves (A.3.6.1).

Corollary A.3.1 (KLMN self-adjointness).

The quadratic form $q = q_0 + v$ is closed and lower bounded; hence

$H_\varepsilon = -d^2/dt^2 + V$ is the unique self-adjoint operator associated with q .

Agmon decay (needed in Appendix B)

Lemma A.4.2 (Uniform Agmon estimate on a bounded spectral window)

Let

$$H_\varepsilon := -d^2/dt^2 + \tilde{V}_\varepsilon(t) \text{ on } L^2(R), \tilde{V}_\varepsilon(t) := V(t) + \varepsilon U(t),$$

where $\varepsilon \in (0, 1]$, $\tilde{V}_\varepsilon \in L^1_{loc}(R)$ is real-valued, and

$$\tilde{V}_\varepsilon(t) \rightarrow +\infty \text{ as } |t| \rightarrow \infty \text{ (A.4.2.0)}$$

Let (λ_n, ψ_n) be an L^2 -normalized eigenpair of H_ε :

$$H_\varepsilon \psi_n = \lambda_n \psi_n, \quad \|\psi_n\|_{L^2} = 1$$

Claim. For every $\Lambda > 0$ there exist constants $\gamma = \gamma(\Lambda) > 0$ and $C = C(\Lambda) > 0$ such that for every eigenfunction ψ_n with $\lambda_n \leq \Lambda$,

$$|\psi_n(t)| \leq C e^{-\gamma|t|} \quad \forall t \in \mathbb{R}. \quad (A.4.2.1)$$

Equivalently: eigenfunctions with eigenvalue bounded by Λ decay exponentially at infinity with constants uniform in the eigen-index n (within that spectral window).

Proof

Fix $\Lambda > 0$. By (A.4.2.0), choose $R = R(\Lambda)$ so large that

$$V_\varepsilon^\sim(t) \geq \Lambda + 2\gamma^2 \text{ for all } |t| \geq R, \quad (A.4.2.2)$$

where $\gamma > 0$ will be chosen momentarily (any fixed $\gamma > 0$ works once R is taken large enough).

Let $\chi \in C^\infty(\mathbb{R})$ be a cutoff with $\chi(t) = 0$ for $|t| \leq R$ and $\chi(t) = 1$ for $|t| \geq R + 1$, and define the Agmon weight

$$\phi(t) := \gamma|t|.$$

Set $u := \chi e^\phi \psi_n$. Since $\psi_n \in D(H_\varepsilon) \subset H_{loc}^1$ we may use u as a test function in the standard Agmon identity (derived by multiplying the eigenvalue equation by $\chi^2 e^{2\phi} \overline{\psi_n}$ and integrating by parts):

$$\int_{\mathbb{R}} (|\chi \psi_n'|^2 + (V_\varepsilon^\sim - \lambda_n) \chi^2 |\psi_n|^2) e^{2\phi} dt = \int_{\mathbb{R}} \chi^2 |\psi_n|^2 |\phi'|^2 e^{2\phi} dt + \int_{\mathbb{R}} |\chi'|^2 |\psi_n|^2 e^{2\phi} dt. \quad (A.4.2.3)$$

On the region $|t| \geq R + 1$, we have $\chi \equiv 1$ and $|\phi'| = \gamma$, so using (A.4.2.2) and $\lambda_n \leq \Lambda$,

$$(V_\varepsilon^\sim - \lambda_n) - |\phi'|^2 \geq (\Lambda + 2\gamma^2 - \Lambda) - \gamma^2 = \gamma^2 \text{ for } |t| \geq R + 1. \quad (A.4.2.4)$$

Thus the left-hand side of (A.4.2.3) controls a positive multiple of

$$\int_{|t| \geq R+1} \gamma^2 |\psi_n(t)|^2 e^{2\gamma|t|} dt$$

up to terms supported where $\chi' \neq 0$, i.e. in the compact annulus $R \leq |t| \leq R + 1$. The χ' -term on the right-hand side of (A.4.2.3) is also supported in that compact

annulus, so it is bounded by a constant times $\|\psi_n\|_{L^2(R \leq |t| \leq R+1)}^2 \leq 1$ with a constant depending only on χ and R , hence only on Λ .

Consequently, there exists $C_1 = C_1(\Lambda)$ such that

$$\int_{|t| \geq R+1} |\psi_n(t)|^2 e^{2\gamma|t|} dt \leq C_1(\Lambda) \text{ for all } n \text{ with } \lambda_n \leq \Lambda. \quad (\text{A.4.2.5})$$

Finally, convert the weighted L^2 bound (A.4.2.5) into a pointwise bound using a 1D Sobolev/elliptic estimate on unit intervals together with the eigenvalue equation (standard; e.g. apply Caccioppoli-type bounds to ψ_n on $[t-1, t+1]$ and then Sobolev embedding in 1D). This yields constants $C = C(\Lambda)$ such that

$$\sup_{|t| \geq R+2} (e^{\gamma|t|} |\psi_n(t)|) \leq C(\Lambda),$$

which is equivalent to (A.4.2.1). Adjust C to cover the compact region $|t| \leq R+2$.

Appendix A.5 Left-end confinement and compact resolvent for the full-line operator

A.5.1 Motivation: why we add a left-end penalty

The background confining potential $V_{env}(u) + \varepsilon U_{left}(u)$ is confining at both ends. The prime-power well potential is not used to prove compact resolvent; it is controlled as a relatively form-bounded attractive perturbation.

$-\frac{d^2}{dt^2} + V(t)$ on $L^2(R)$ need not have compact resolvent. Since Theorem [F.RH](#)

Since the Appendix F rigidity closure requires a Stieltjes-compatible operator-side transform... we introduce a deterministic confining penalty on the left end.

A.5.2 Definition of the confining perturbation

Fix parameters $\beta > 0$, $T < 0$ and choose $\chi \in C^\infty(R)$ such that

$$\chi(t) = 1 \text{ for } t \leq T - 1, \chi(t) = 0 \text{ for } t \geq T.$$

Define

$$U(t) := \chi(t) e^{-\beta t}. \quad (\text{A.5.2.1})$$

Then $U \geq 0$, $U \in C^\infty(R)$, $U(t) = 0$ for all $t \geq T$, and

$$U(t) \rightarrow +\infty \text{ as } t \rightarrow -\infty. \quad (\text{A.5.2.2})$$

For $\varepsilon \in (0, 1]$, set

$$V_\varepsilon^\sim(t) := V(t) + \varepsilon U(t), H_\varepsilon := -\frac{d^2}{dt^2} + V_\varepsilon^\sim(t) \text{ on } L^2(R), \quad (\text{A.5.2.3})$$

where H_ε is defined via the closed, lower-bounded quadratic form

$$q_\varepsilon[\psi] = \int_R |\psi'(t)|^2 dt + \int_R V_\varepsilon^\sim(t) |\psi(t)|^2 dt, \psi \in C_c^\infty(R),$$

and Friedrichs extension (Appendix A.1).

A.5.3 Confinement at both ends

By Lemma A1.3, $V(t) \rightarrow +\infty$ as $t \rightarrow +\infty$. By (A.5.2.2), $\varepsilon U(t) \rightarrow +\infty$ as $t \rightarrow -\infty$. Hence

$$V_\varepsilon^\sim(t) \rightarrow +\infty \text{ as } |t| \rightarrow \infty. \quad (\text{A.5.3.1})$$

A.5.4 Compact resolvent and discrete spectrum

The confinement condition (A.5.3.1) implies that H_ε has **compact resolvent**, i.e.

$$(H_\varepsilon + 1)^{-1} \text{ is compact on } L^2(R), \quad (\text{A.5.4.1})$$

by standard Schrödinger operator theory (e.g. Reed–Simon IV, Theorem XIII.67, or equivalent compactness criteria for the embedding of the form domain into L^2).

Consequently, the spectrum of H_ε is purely discrete:

$$0 \leq \lambda_1 \leq \lambda_2 \leq \dots, \lambda_n \rightarrow +\infty,$$

each eigenvalue has **finite multiplicity**, and there exists an orthonormal basis of $L^2(R)$ consisting of eigenfunctions of H_ε . In particular, all eigenvalues are real.

A.5.5 Convention for the remainder of the proof

Fix once and for all $\varepsilon \in (0, 1]$ and set

$H := H_\varepsilon, \tilde{V} := V_\varepsilon$. All trace and resolvent identities used in Appendix F rigidity closure are applied to this confining operator H .

A.6. Left-end penalization: self-adjointness and compact resolvent

Proposition A.6 (Confining left-end penalization).

Let $V: \mathbb{R} \rightarrow \mathbb{R}$ be measurable with $V \in L^1_{loc}(\mathbb{R})$ and bounded below, and assume $V(t) \rightarrow +\infty$ as $t \rightarrow +\infty$ (as established earlier in Appendix A). Fix $T < 0$, choose $\chi \in C^\infty(\mathbb{R})$ with $0 \leq \chi \leq 1$, $\chi(t) = 1$ for $t \leq T - 1$, and $\chi(t) = 0$ for $t \geq T$. For $\beta > 0$ define

$$U(t) := e^{-\beta t} \chi(t),$$

so that $U(t) \geq 0$, $U \in C^\infty(\mathbb{R})$, and $U(t) \rightarrow +\infty$ as $t \rightarrow -\infty$. For $\varepsilon \in (0, 1]$ define

$$H_\varepsilon := -\frac{d^2}{dt^2} + V(t) + \varepsilon U(t) \text{ on } L^2(\mathbb{R}).$$

1. (*Self-adjointness and lower semiboundedness.*) The quadratic form

$$q_\varepsilon[\varphi] := \int_{\mathbb{R}} (|\varphi'(t)|^2 + (V(t) + \varepsilon U(t))|\varphi(t)|^2) dt, \quad \varphi \in H^1(\mathbb{R}) \cap L^2((V + \varepsilon U) + dt),$$

is densely defined, closed, and bounded below. Hence q_ε defines a unique self-adjoint operator H_ε (KLMN theorem).

2. (*Compact resolvent / purely discrete spectrum.*) The confining potential

$$\tilde{V}_\varepsilon(t) := V(t) + \varepsilon U(t) \text{ satisfies } \tilde{V}_\varepsilon(t) \rightarrow +\infty \text{ as } |t| \rightarrow \infty.$$

Therefore H_ε has **empty essential spectrum** and **compact resolvent**, hence its spectrum consists only of isolated eigenvalues of finite multiplicity accumulating at $+\infty$. (For example, by Reed–Simon IV, Thm. XIII.67 / Persson-type criteria.)

Remark (notation used later). In the remainder of the manuscript we work with a fixed $\varepsilon \in (0, 1]$ and write $H := H_\varepsilon$, $\tilde{V} := \tilde{V}_\varepsilon$.

Proposition A/B.UNIFORM-LOWER-BOUND

Cutoff-uniform lower form bound for the finite-window operators

Let $H_{L,R}^{orig}$ denote the unshifted finite-window Dirichlet operator used in Appendix D, obtained by restricting the truncated PWQO operator to $[-L, L]$:

$$H_{L,R}^{orig} = -\partial_u^2 + V_{env}(u) + \varepsilon U_{left}(u) - V_{(R)}^{pp}(u),$$

with Dirichlet boundary conditions at $\pm L$, where

$$V_{(R)}^{pp}(u) = \sum_{\log q \leq R} \frac{\Lambda(q)}{\sqrt{q}} W_\alpha(u - \log q).$$

Suppose the envelope decomposition of Appendix B:

$$V_{env}(u) + \varepsilon U_{left}(u) - V_{(R)}^{pp}(u) = V_{env}(u) + V_{osc,R}(u),$$

where V_{env} is coercive and where the oscillatory/truncated arithmetic part satisfies a cutoff-uniform lower bound

$$\inf_{R,u} V_{osc,R}(u) > -\infty.$$

Then there exists a constant $M < \infty$, independent of L and R , such that

$$H_{L,R}^{orig} \geq -M$$

as quadratic forms on $H_0^1([-L, L])$

Consequently, after the single global shift

$$\tilde{H}_{L,R} = H_{L,R}^{orig} + MI,$$

one has

$$\tilde{H}_{L,R} \geq 0$$

for all finite windows L and all prime-power cutoffs R .

Proof

By Appendix B.Coercive, the background/envelope decomposition gives constants $\alpha > 0$, $\beta \in \mathbb{R}$, and $M_{osc} \geq 0$, independent of L and R , such that

$$V_{env}(u) + V_{osc,R}(u) \geq \alpha u^2 - \beta - M_{osc}$$

outside a fixed compact region, and is bounded below on that compact region uniformly in R . Hence there exists $M < \infty$, independent of L and R , such that

$$V_{env}(u) + \varepsilon U_{left}(u) - V_{(R)}^{pp}(u) \geq -M$$

for all $u \in [-L, L]$. Therefore, for every $\psi \in H_0^1([-L, L])$,

$$\langle \psi, H_{L,R}^{orig} \psi \rangle = \int_{-L}^L |\psi'(u)|^2 du + \int_{-L}^L (V_{env} + \varepsilon U_{left} - V_{(R)}^{pp}) |\psi(u)|^2 du \geq -M \|\psi\|_2^2$$

Thus $H_{L,R}^{orig} \geq -M$ uniformly in L, R . The shifted operator $\tilde{H}_{L,R} = H_{L,R}^{orig} + MI$ is therefore nonnegative for all L, R .

Referee note

This proposition replaces any cutoff-dependent spectral normalization $c_0(R)$. From this point onward, the D-side Stieltjes normalization uses one global shift M , not a separately chosen shift depending on L or R .

Theorem A/B.UNIFORM-LOWER-BOUND, strengthened form.

Let

$$H_{L,R}^{orig} = -\partial_u^2 + V_{env}(u) + \varepsilon U_{left}(u) - V_{pp}^{(R)}(u)$$

be the finite-window Dirichlet operator used in Appendix D, where

$$V_{pp}^{(R)} = \sum_{\log q \leq R} \frac{\Lambda(q)}{\sqrt{q}} W_\alpha(u - \log q)$$

Suppose A.ENV-DOM and A.ENV-FORM, namely

$$0 \leq V_R^\# \leq \theta V_{env}, 0 < \theta < 1,$$

as quadratic forms, uniformly in R. Then there exists $M < \infty$, independent of L and R, such that

$$H_{L,R}^{orig} \geq -M$$

as quadratic forms on $H_0^1([-L, L])$ Consequently, the shifted operators

$$\tilde{H}_{L,R} = H_{L,R}^{orig} + MI$$

satisfy

$$\tilde{H}_{L,R} \geq 0$$

for every finite cutoff stage.

Proof.

By A.ENV-DOM and A.ENV-FORM,

$$\int V_R^\# |\psi|^2 \leq \theta \int V_{env} |\psi|^2 + C_0 \|\psi\|_2^2$$

with constants independent of R. Hence

$$\langle \psi, H_{L,R}^{orig} \psi \rangle = \int |\psi'|^2 + \int (V_{env} + \varepsilon U_{left}) |\psi|^2 - \int V_R^\# |\psi|^2 \geq \int |\psi'|^2 + (1 - \theta) \int V_{env} |\psi|^2 + \varepsilon \int U_{left} |\psi|^2 - C_0 \|\psi\|_2^2$$

The right side is bounded below by $-M \|\psi\|_2^2$, with M depending only on the fixed envelope data and not on L or R. Therefore

$$H_{L,R}^{orig} \geq -M$$

The shifted nonnegativity follows immediately.

Appendix B Heat-Trace Expansion and Prime Encoding

(Global coercivity from an envelope)

(envelope decomposition).

Suppose $V = V_{env} + V_{osc}$ with

$$V_{env}(t) \geq \alpha t^2 - \beta \text{ for } |t| \geq R, \inf_{t \in R} V_{osc}(t) \geq -M,$$

for some explicit $\alpha > 0$, $\beta \in \mathbb{R}$, $M \geq 0$, $R \geq 0$. (In your Gaussian-well model, V_{env} is the Gaussian envelope; V_{osc} is the sum of localized prime spikes minus the envelope; α, β follow from the envelope's tail, M from the spikes' minimum.)

Lemma B.Coercive. There exist explicit $\tilde{\alpha} > 0$, $\tilde{\beta} \in \mathbb{R}$ such that

$$V(t) \geq \tilde{\alpha} t^2 - \tilde{\beta} \text{ for all } |t| \geq R, \|f\|_{L^2}^2$$

namely $\tilde{\alpha} = \alpha$ and $\tilde{\beta} = \beta + M$. Consequently,

$$\langle f, Hf \rangle \geq \int_{\mathbb{R}} (|f'|^2 + \alpha t^2 |f|^2) dt - \tilde{\beta} \|f\|_{L^2}^2,$$

and by Persson's lemma the essential spectrum is empty; thus H has **purely discrete spectrum**.

Proof. For $|t| \geq R$

$$V(t) = V_{env}(t) + V_{osc}(t) \geq (\alpha t^2 \tilde{\beta}) + (-M) = \alpha t^2 - (\beta + M).$$

Set $\tilde{\alpha} = \alpha$, $\tilde{\beta} = \beta + M$. The quadratic confinement implies H is a compact perturbation of a harmonic-oscillator-type operator; discreteness follows (e.g. by Rellich compactness or Persson's lemma).

Implementation note (parameters). If your envelope is $V_{env}(t) = \frac{1}{2\sigma^2} t^2$ for $|t|$ larger than a fixed threshold (or dominates that), take $\alpha = \frac{1}{2\sigma^2}$; β is the maximal downward deviation of the envelope on the matching region; M is the uniform lower bound of the oscillatory part (finite because spikes are localized and bounded below). Put the resulting $\tilde{\alpha}, \tilde{\beta}$ into the statement.

B 1 Trace class and analyticity

Lemma B 1.1 (Trace class).

For every $t > 0$ the operator e^{-tH} is trace-class and $Tr(e^{-tH})$ is real-analytic on $(0, \infty)$.

Proof. By Proposition A 3.4, the spectrum of H is purely discrete $0 < \lambda_1 < \lambda_2 < \dots \rightarrow \infty$ Then

$$Tr(e^{-tH}) = \sum_{n \geq 1} e^{-t\lambda_n} < \infty, \quad t > 0,$$

and term-wise differentiation is justified.

B 1.2 Feynman–Kac representation

The integral kernel of e^{-tH} satisfies

$$K(t; x, y) = E_{x \rightarrow y}^t \left[\exp \left(- \int_0^t V(B_s) ds \right) \right] \quad (\text{B 2.1})$$

where B_s is Brownian bridge from x to y in time t .

Lemma A 4.1 (uniform Agmon decay) implies $K(t; x, x)$ is continuous and that the integral

$$Tr(e^{-tH}) = \int_R K(t; x, x) dx. \quad (\text{B 2.2})$$

converges absolutely.

B 1. 3 Single-well contribution

Write $V = \sum_p V_p$ with

$$V_p(t) = \frac{\ln \sqrt{p}}{\sqrt{2\pi\alpha}} \exp \left[-\frac{(t - \ln p)^2}{2\alpha} \right].$$

For each prime p let $H_p := -d^2/dt^2 + V_p$ and denote by

$$f(u) := \text{Tr}(e^{-uH_p}) - \text{Tr}(e^{-uH_0}), \quad H_0 = -d^2/dt^2. \quad (\text{B 3.1})$$

Lemma B 3.1 (Localised kernel).

$f(u)$ extends to a smooth, rapidly decaying function:

$$|f(u)| \leq C_1 e^{-C_2 u} \text{ for } u \geq 1.$$

Proof. Since V_p is compactly supported in Fourier space and analytic, the usual

Gel'fand–Yaglom comparison plus Agmon bounds yields exponential decay of the heat trace difference. (Full details mirror Helffer–Robert, Ann. Éc. Norm. Sup. (1983).)

Lemma B.K0 (Free kernel; trace formula and Gaussian decay for g_t).

Let $H_0 = -\partial_u^2$ on $L^2(\mathbb{R})$ and $G_t = e^{-tH_0}$ for $t > 0$. Then G_t has integral kernel

$$K_0(t; u, v) = \frac{1}{\sqrt{4\pi t}} \exp \left(-\frac{(u-v)^2}{4t} \right).$$

Fix $L > 0$ and let Π_L denote multiplication by $1_{[-L, L]}$. Let $w \in L^\infty(\mathbb{R})$ with $\text{supp}(w) \subset [-L, L]$, and set

$$A_L = \Pi_L M_w \Pi_L.$$

For each $a \in \mathbb{R}$ define

$$g_t(a) := \text{Tr}(G_t A_L T_a), \quad (T_a f)(u) := f(u - a).$$

Then

$$g_t(a) = \int_R K_0(t; u, u - a) w(u - a) 1_{[-L, L]}(u) 1_{[-L, L]}(u - a) du,$$

and in particular

$$|g_t(a)| \leq \|w\|_\infty \frac{2L}{\sqrt{4\pi t}} \exp\left(-\frac{a^2}{4t}\right).$$

Proof.

The kernel of T_a is $K_{T_a}(u, v) = \delta(v - (u - a))$. The product $G_t A_L T_a$ has kernel

$$(u, v) \mapsto \int_R K_0(t; u, x) w(x) 1_{[-L, L]}(x) \delta(v - (x + a)) dx,$$

so the diagonal $u = v$ gives $x = u - a$ and yields the stated trace integral. The bound follows

from $|w| \leq \|w\|_\infty$ and $K_0(t; u, u - a) = \frac{1}{\sqrt{4\pi t}} e^{-a^2/(4t)}$

B.TC Trace-class legality + Duhamel expansion in trace norm

We'll write the truncated potential in the standard finite-sum form already being used in our "finite I_R " truncation.

Standing truncation model.

Fix $R < \infty$ and an index set I_R (finite). Let

$$H_R := H_0 + V_R, \quad V_R(u) := \sum_{\alpha \in I_R} v_\alpha W(u - u_\alpha),$$

where:

- $W \in L^\infty(R) \cap L^1(R)$ has compact support,
- $v_\alpha \in R$
- $\{u_\alpha\}_{\alpha \in I_R} \subset R$ are the "centers" (later specialized to $u_{p,m} = m \log p$).

Lemma B.TC (Trace-class localization; Dyson–Phillips/Duhamel legality).

Fix L, R and define $A_L = \Pi_L M_w \Pi_L$ as above. Then for each $t > 0$,

$$G_t A_L e^{-tH_R} \text{ and } A_L e^{-tH_R}$$

are trace class. Moreover, the Dyson–Phillips expansion

$$e^{-tH_R} = e^{-tH_0} + \sum_{k \geq 1} (-1)^k \int_{0 < s_1 < \dots < s_k < t} e^{-(t-s_k)H_0} V_R e^{-(s_k-s_{k-1})H_0} \dots V_R e^{-s_1 H_0} ds$$

converges **absolutely in trace norm** after left/right multiplication by A_L (or Π_L), uniformly for $0 < t \leq t_0(L, R)$ for some $t_0(L, R) > 0$. In particular, termwise trace and Fubini/Tonelli manipulations used in the word expansion are justified on $(0, t_0]$.

Proof.

Because I_R is finite and $W \in L^\infty$, we have $V_R \in L^\infty$ and $\|V_R\|_\infty \leq C(R)$

1. Localized free semigroup is Hilbert–Schmidt.

The kernel of $\Pi_L e^{-tH_0}$ is $1_{[-L, L]}(u) K_0(t; u, v)$. By the explicit formula, its square is integrable on $\mathbb{R}^2$, hence $\Pi_L e^{-tH_0}$ is Hilbert–Schmidt and

$$\|\Pi_L e^{-tH_0}\|_2 \leq C(L) t^{-1/4}.$$

Similarly $e^{-tH_0} \Pi_L$ is Hilbert–Schmidt. Therefore $\Pi_L e^{-tH_0} \Pi_L$ is trace class with

$$\|\Pi_L e^{-tH_0}\|_1 \leq \|\Pi_L e^{-tH_0}\|_2 \|e^{-tH_0} \Pi_L\|_2 \leq C(L) t^{-1/2}.$$

One Duhamel insertion is trace class after localization.

For $s \in (0, t)$

$$\Pi_L e^{-(t-s)H_0} V_R e^{-sH_0} \Pi_L = (\Pi_L e^{-(t-s)H_0}) (V_R) (e^{-sH_0} \Pi_L).$$

The first and third factors are Hilbert–Schmidt with bounds $C(L)(t-s)^{-1/4}$ and $C(L)s^{-1/4}$; the middle factor is bounded with norm $\|V_R\|_\infty$. Hence the product is trace class and

$$\|\Pi_L e^{-(t-s)H_0} V_R e^{-sH_0} \Pi_L\|_1 \leq C(L, R) (t-s)^{-1/4} s^{-1/4}.$$

Iterated insertions and simplex integrability.

At order k , the localized integrand is a product of $k+1$ free semigroups and k bounded multipliers V_R . Insert Π_L at the left and right ends. Group as successive Hilbert–Schmidt factors so that each free piece contributes a $(\Delta s)^{-1/4}$ factor. One obtains

$$\|\Pi_L e^{-(t-s_k)H_0} V_R e^{-(s_k-s_{k-1})H_0} \dots V_R e^{-s_1 H_0} \Pi_L\|_1 \leq C(L, R)^k \prod_{j=0}^k (s_{j+1} - s_j)^{-1/4},$$

with $s_0 = 0, s_k + 1 = t$

The simplex integral of $\prod (s_j + 1 - s_j)^{-1/4}$ is finite and scales like $t^{(3/4)k}$ times a Beta-function constant. Hence the k -th Duhamel term is bounded by $(C(L, R)t^{3/4})^k$. For $t \leq t_0(L, R)$ sufficiently small, the series $\sum k(Ct^{3/4})^k$ converges, giving absolute convergence in trace norm uniformly on $(0, t_0]$.

4. Inclusion of A_L .

Since A_L is bounded (norm $\leq \|w\|_\infty$) and supported in $[-L, L]$, multiplying by A_L only improves trace-ideal bounds.

This proves trace-class legality, convergence of the word expansion after localization, and justifies termwise trace/Fubini used later.

B.MIX Mixed words are super-smoothing

Definition (cluster merging).

Let $D := \text{diam}(\text{supp } W)$. Declare $\alpha \sim \beta$ if $|u_\alpha - u_\beta| \leq D$. Replace each equivalence class (“cluster”) by a single effective bump (sum of the bumps in the cluster). After this replacement, distinct clusters have separation

$$\delta_R := \min_{\text{clusters } C \neq C'} \text{dist}(\text{supp } V_C, \text{supp } V_{C'}) > 0.$$

This does not change the operator H_R ; it only changes notation.

Lemma B.MIX (Mixed-word super-smoothing at fixed R).

Fix L, R . Let “mixed words” denote Duhamel terms in which at least two distinct clusters appear. Then for each $N \geq 1$ there exist constants $C_N(L, R)$, $t_0(L, R) > 0$ such that the total mixed contribution $R_{L,R}^{\text{mix}}(t)$ satisfies

$$|R_{L,R}^{\text{mix}}(t)| \leq C_N(L, R) t^N, \quad 0 < t \leq t_0(L, R).$$

Proof.

In any mixed word, there is at least one adjacent transition from an insertion supported in cluster

C to an insertion supported in cluster $C' \neq C$. Between those insertions sits a free heat factor $e^{-\tau H_0}$ (some increment $\tau = s_j + 1 - s_j$). Because cluster supports are separated by $\delta_R > 0$, the corresponding kernel contribution includes an off-diagonal Gaussian penalty:

$$|K_0(\tau; u, v)| \leq \frac{1}{\sqrt{4\pi\tau}} \exp\left(-\frac{\delta_R^2}{4\tau}\right) \text{ whenever } u \in \text{supp}(V_C), v \in \text{supp}(V_{C'}).$$

Squaring and integrating shows the corresponding localized operator is Hilbert–Schmidt with bound

$$\|1_{\text{supp}(V_C)} e^{-\tau H_0} 1_{\text{supp}(V_{C'})}\|_2 \leq C \tau^{-1/4} \exp\left(-\frac{\delta_R^2}{c\tau}\right).$$

All other links are bounded by $\tau^{-1/4}$ as in Lemma B.TC. Therefore each mixed-word integrand gains a factor $\exp(-\delta_R^2/(c\tau))$. Since for any fixed $\delta_R > 0$ and any N , one has $\exp(-\delta_R^2/(c\tau)) \leq C_{N,R} \tau^N$, for τ small, the simplex integral yields a factor t^N uniformly for $0 < t \leq t_0(L, R)$ after choosing t_0 small. Summing over word length k converges by the same geometric-series argument used in Lemma B.TC.

Corollary B.4.1 Proposition B.4.1 (Heat-trace/word decomposition; unconditional).

Under the standing truncation model and the localized observable A_L , the localized trace functional admits

$\Theta_{L,R}(t) = A_{\text{class},L}(t) + P_{L,R}(t) + R_{L,R}(t)$ where $P_{L,R}(t)$ is the explicit diagonal spike sum (built from $g_t(u_\alpha)$ and later specialized to prime-power centers), and for each $N \geq 1$,

$$|R_{L,R}(t)| \leq C_N(L, R) t^N (0 < t \leq t_0(L, R)).$$

Proof.

Lemma B.TC gives trace-class legality and termwise Duhamel expansion. The diagonal terms reduce to the spike kernel $g_t(\cdot)$ by Lemma B.K0. The mixed terms are $O(t^N)$ for all N by Lemma B.MIX. Collect the bulk/free subtraction into $A_{\text{class},L}$

Corollary B.4.2 (Ω -holomorphy of the mixed remainder transform)

Let

$$R_{mix,L,R}^{\sim}(s) := \int_0^{\infty} e^{-st} R_{mix,L,R}(t) dt$$

be defined on a right half-plane. Then $R_{mix,L,R}^{\sim}$ extends holomorphically to $\Omega = \mathcal{C} \setminus (-\infty, 0]$

B1. 5 Fourier–Laplace spectrum of the oscillatory part

Define

$$T(t) := \sum_{p,m} \frac{\ln p}{p^{m/2}} f(t \cdot m \ln p).$$

Because $f(u)$ decays exponentially, the Mellin–Laplace transform

$$\widehat{T}(\xi) = \int_0^{\infty} e^{-i\xi t} T(t) dt \quad (\text{B 5.1})$$

converges absolutely and equals

$$\widehat{T}(\xi) = \sum_{p,m} \frac{1}{m p^{m/2}} \widehat{f}(\xi/(m \ln p)), \quad \widehat{f}(\omega) = \int_0^{\infty} e^{-i\omega u} f(u) du. \quad (\text{B 5.2})$$

Because $\widehat{f}$ is peaked at $\omega = 0$, $\widehat{T}$ exhibits resonances at frequencies

$$\xi = m \ln p (p \in P, m \geq 1),$$

mirroring the oscillatory sum in the classical explicit formula. Thus the spectral side of H “knows” the prime powers.

Appendix B 2 Uniform Localization and Decay of Eigenfunctions

Appendix B.2 Uniform localization and decay of eigenfunctions (fixed R).

Fix $R < \infty$. Let $\{(\lambda_n, R, \psi_n, R)\}_{n \geq 1}$ be the eigenpairs of the ε -penalized truncated PWQO Hamiltonian on $L^2(R)$

$$H_{\varepsilon, R} := -\partial_u^2 + V_{env}(u) + \varepsilon_{Uleft}(u) - V_{pp}^{(R)}(u),$$

where

$$V_{pp}^{(R)}(u) := \sum_{\log q \leq R} \frac{\Lambda(q)}{\sqrt{q}} W_\alpha(u - \log q) (q = p^m).$$

Normalize $\|\psi_n\|_{L^2} = 1$

Agmon metric + exponential decay

B.2.1 Agmon metric and classically allowed set

Fix $\varepsilon \in (0, 1]$ and let

$$H_\varepsilon := -\frac{d^2}{dt^2} + \tilde{V}_\varepsilon(t), \quad \tilde{V}_\varepsilon(t) := V(t) + \varepsilon U(t),$$

as in Appendix A. (Recall $U(t) = 0$ for $t \geq T$, so the right-end prime-well geometry is unchanged by the left-end penalty.)

Define the “prime-trench distance” function

$$d(t) := \min(p, m) \in \mathbb{R} | t - m \log p |, \quad \mathbb{R} = \{(p, m) : m \log p \leq R\}.$$

and for parameters $\gamma > 0$ set the Agmon weight

$$\Phi(t) := \gamma d(t). \quad (B.2.1.0)$$

For an eigenpair (λ_n, ψ_n) of H_ε , define the classically allowed region (for a chosen threshold Λ)

$$\Omega_\Lambda := \{t \in \mathbb{R} : V(t) \leq \Lambda + 1\}. \quad (B.2.1.1)$$

"Since V_R is a sum of narrow Gaussian wells centered at the **prime-power lattice** $m \log p \dots$ ", Ω_Λ is contained in a union of small neighborhoods of the prime centers:

$$\Omega_\Lambda \subset \bigcup_{(p,m) \in IR} I_{p,m}(\delta), \quad I_{p,m}(\delta) := (m \log p - \delta, m \log p + \delta).$$

for some $\delta = \delta(\Lambda) > 0$.

Lemma B.2.1 (Agmon decay outside the prime trenches; uniform on a bounded spectral window)

For every $\Lambda > 0$ there exist constants $\gamma = \gamma(\Lambda) > 0$ and $C = C(\Lambda) > 0$ such that for every L^2 -normalized eigenfunction ψ_n of H_ε with eigenvalue $\lambda_n \leq \Lambda$,

$$|\psi_n(t)| \leq C e^{-\gamma d(t)} \quad \forall t \in R. \quad (B.2.1.3)$$

Proof

Fix $\Lambda > 0$. By confinement of $V_\varepsilon^\sim$ (Appendix A.5) and by the local structure of V , choose $\delta = \delta(\Lambda)$ so that on the complement $\{d(t) \geq \delta\}$ we have the uniform lower bound

$$V_\varepsilon^\sim(t) \geq \Lambda + 2\gamma^2 \text{ whenever } d(t) \geq \delta, (B.2.1.4)$$

for some $\gamma > 0$ (choose γ first, then enlarge δ if needed; existence follows since $V_\varepsilon^\sim(t) \rightarrow \infty$ as $|t| \rightarrow \infty$ and V is away from its wells whenever $d(t) \geq \delta$).

Let $\chi \in C^\infty(R)$ be a cutoff supported in $\{d(t) \geq \delta/2\}$ with $\chi \equiv 1$ on $\{d(t) \geq \delta\}$. Multiply the eigenvalue equation

$(H_\varepsilon - \lambda_n)\psi_n = 0$ by $\chi^2 e^{2\Phi} \overline{\psi_n}$ and integrate by parts to obtain the standard Agmon identity

$$\int_R (|\chi \psi_n'|)^2 + (V_\varepsilon^\sim - \lambda_n) \chi^2 |\psi_n|^2 e^{2\Phi} dt = \int_R \chi^2 |\psi_n|^2 |\Phi'|^2 e^{2\Phi} dt + \int_R |\chi'|^2 |\psi_n|^2 e^{2\Phi} dt. \quad (B.2.1.5)$$

On $\{d(t) \geq \delta\}$ we have $\chi \equiv 1$, $|\Phi'| \leq \gamma$ a. e., and by (B.2.1.4) and $\lambda_n \leq \Lambda$

$$(V_\varepsilon^\sim - \lambda_n) - |\Phi'|^2 \geq (\Lambda + 2\gamma^2 - \Lambda) - \gamma^2 = \gamma^2. \quad (B.2.1.6)$$

Therefore (B.2.1.5) yields a uniform weighted L^2 estimate

$$\int_{\{d \geq \delta\}} |\psi_n(t)|^2 e^{2\gamma d(t)} dt \leq C_1(\Lambda), \quad (B.2.1.7)$$

since the $\chi' -$ term is supported where $\delta/2 \leq d(t) \leq \delta$ (a region controlled uniformly once Λ is fixed) and $\|\psi_n\|_{L^2} = 1$.

Lemma B.i.X (Local $L^2 \rightarrow L^\infty$ upgrade on a bounded spectral window).

Fix $\Lambda > 0$. There exists $C = C(\Lambda)$ such that for every eigenpair (λ_n, ψ_n) of H_ε with $\lambda_n \leq \Lambda$, and every unit interval $I = [x-1, x+1]$ we have

$$\|\psi_n\|_{L^\infty(I)} \leq C(\Lambda) (\|\psi_n\|_{L^2(I)} + \|\psi_n\|_{L^2(I)}) \leq C(\Lambda) \|\psi_n\|_{L^2([x-2, x+2])}$$

Consequently, the weighted L^2 estimate (B.2.1.7) implies the pointwise bound (B.2.1.3), with constants depending only on Λ .

Sketch of proof. On $\mathbb{R}$, use the eigenvalue equation $-\psi_n'' = (\lambda_n - V_\varepsilon^\sim)\psi_n$ and local boundedness of $V_\varepsilon^\sim$ to obtain a Caccioppoli/Sobolev estimate $\|\psi_n'\|_{L^2(I)} \leq C(\Lambda) \|\psi_n\|_{L^2(I^*)}$ // for a slightly larger interval $I^* \supset I$. Then apply the 1D Sobolev embedding

$$\|f\|_{L^\infty(I)} \leq C(\|f\|_{L^2(I)} + \|f'\|_{L^2(I)})$$

Comment (scope of uniformity). The constants are uniform only for $\lambda_n \leq \Lambda$. No global “uniform in n ” statement is claimed or needed: when $\lambda_n \rightarrow \infty$, such global uniformity is generally false.

Lemma B 2.2 (Uniform derivative bounds)

For each k and each Λ , $\sup_{\lambda_n, R \leq \Lambda} \|\psi_n, R(k)\|_\infty \leq C_{k, \Lambda, R}$,

$$\|\psi_n^k\|_{L^\infty(R)} \leq C_k \quad (\text{B 2.2.1})$$

Proof.

Fix k . On every finite interval $[-R, R]$ the potential V is smooth and bounded, so classical elliptic estimates yield

$$\|\psi_n^k\|_{H^1([-R, R])} \leq C(R, k)(1 + \lambda_n).$$

Outside $[-R, R]$ we use the decay (B 2.1.1) plus induction on the eigen-equation to bound derivatives; the exponential factor kills any λ_n -growth. As λ_n enters only quadratically and $C(R, k)$ can be absorbed into a global constant, (B 2.2.1) follows.

Appendix B 3 Absolute Convergence and Sum–Integral Interchange

Lemma B.3.0 (Tonelli–Fubini / majorant template)

Let (A, μ) be a measure space and let $\{F_\alpha\}_{\alpha \in I}$ be a family of measurable functions on A , indexed by a countable set I . Given

$$\sum_{\alpha \in I} \int_A |F_\alpha(x)| d\mu(x) < \infty. \quad (\text{B. 3. 0. 1})$$

Then:

1. The series $\sum_{\alpha \in I} F_\alpha(x)$ converges absolutely for μ – a. e. $x \in A$, and its sum $F(x) := \sum_{\alpha \in I} F_\alpha(x)$ lies in $L^1(A, \mu)$.
2. One may interchange summation and integration:

$$\int_A \sum_{\alpha \in I} F_\alpha(x) d\mu(x) = \sum_{\alpha \in I} \int_A F_\alpha(x) d\mu(x), \quad (\text{B. 3. 0. 2})$$

and both sides converge absolutely.

Remark (what we use). For each interchange in the proof we supply either

- (i) a nonnegative integrable majorant (Tonelli), or
 - (ii) an absolute/uniform majorant on compact parameter sets (Weierstrass M-test / dominated convergence), depending on context.
-

B.3.1 Heat-trace series (prime-power normalization)

Recall from Proposition B.4.1 that the diagonal spike contribution can be written in prime-power form with coefficients $\frac{\Lambda(q)}{\sqrt{q}}$ and that the scalar test factor f satisfies exponential decay:

$$f(u) = O(e^{-cu})(c > 0). \quad (B.3.1.0)$$

Lemma B.3.1 (Absolute convergence of the prime-power heat term)

Assume $|f(u)| \leq C e^{-cu}$ for some $C, c > 0$. For $t > 0$ define

$$T(t) := \sum_{q \text{ prime power}} \frac{\Lambda(q)}{\sqrt{q}} |f(t \log q)|. \quad (B.3.1.1)$$

Then for every $t \geq t_0 > 0$ the series in (B.3.1.1) converges absolutely, and the convergence is uniform for $t \in [t_0, \infty)$. In particular $T(t) < \infty$ for all $t \geq t_0$.

Proof. Write $q = p^m$ with $m \geq 1$. Then $\frac{\Lambda(q)}{\sqrt{q}} = \log p$, $\sqrt{q} = p^{m/2}$, and $\log q = m \log p$.

Using $|f(u)| \leq C e^{-cu}$

$$T(t) \leq C \sum_p \sum_{m \geq 1} \frac{\log p}{p^{m/2}} e^{-ct} m \log p = C \sum_p (\log p) \sum_{m \geq 1} p^{-m(1/2+ct)}$$

The inner sum is geometric, hence

$$\sum_{m \geq 1} p^{-m(1/2+ct)} = \frac{p^{-(1/2+ct)}}{1 - p^{-(1/2+ct)}} = \frac{1}{p^{1/2+ct} - 1}$$

Therefore

$$T(t) \leq C \sum_p \frac{\log p}{p^{1/2+ct} - 1} \leq C \sum_p \frac{\log p}{p^{1/2+ct_0} - 1} \quad (B.3.1.2)$$

Since $1/2 + ct_0 > 1$ the denominator grows like $p^{1/2+ct_0}$ and the prime sum in (B.3.1.2) converges (e.g., by comparison with the PNT integral test). Uniformity for $t \geq t_0$ is immediate from the monotonicity of $p^{1/2+ct} - 1$ in t .

Corollary (termwise Laplace legality). For any s with $\Re(s) \geq \sigma_0 > 0$,

$$\int_0^\infty e^{-st} T(t) dt = \sum_q \frac{\Lambda(q)}{\sqrt{q}} \int_0^\infty e^{-st} |f(t \log q)| dt$$

Lemma B.3.2 (Absolute convergence of Mellin prime expansion).

Fix $\sigma > 0$. For each eigenfunction ψ_λ with eigenvalue λ_λ in a bounded spectral window, the Mellin–Gaussian transform

$$\Phi_n(s) = \int_0^\infty \psi_n(\log x) x^{s-1} e^{-x^2/2} dx$$

admits a prime-power expansion that converges absolutely for $\Re(s) = \sigma$.

Proof.

From the Agmon decay (Appendix B.2), there exist constants $C, \gamma > 0$ such that

$$|\psi_n(u)| \leq C e^{-\gamma|u|}$$

uniformly for λ_λ in a bounded window.

Substituting $u = \log x$, we obtain

$$|\psi_n(\log x)| \leq C x^{-\gamma} + C x^\gamma$$

Multiplying by $x^{\sigma-1} e^{-x^2/2}$, the integrand is bounded by

$$C x^{\sigma-1-\gamma} e^{-x^2/2} + C x^{\sigma-1+\gamma} e^{-x^2/2}.$$

Both are integrable on $(0, \infty)$ for any fixed σ .

For the prime expansion terms, one obtains a bound of the form

$$Cp^{-m(1/2+\epsilon)}$$

for some $\epsilon > 0$ depending on σ and γ .

Thus the sum over p, m converges absolutely by comparison with

$$\sum_p \sum_{m \geq 1} p^{-m(1/2+\epsilon)}.$$

This bound is independent of n in the fixed spectral window, hence the convergence is uniform in n .

Appendix C

(Non-load-bearing note. Appendix C is not used in the proof chain establishing Theorem F.RH; the RH closure proceeds via Appendices A, B, D, H, and F. In particular, Lemma C.3.1 is auxiliary and may be omitted without affecting the RH implication.)

Appendix C — Mellin–Gaussian transforms of PWQO eigenfunctions, basic complex-analytic control, and the (only) local factorization we actually use

C.0. Purpose and scope (what this appendix *does* and *does not* claim)

For each eigenpair (λ_n, ψ_n) of the prime-weighted operator H , we define a Mellin–Gaussian transform $\Phi_n(s)$ and prove three structural facts :

1. Φ_n is entire (holomorphic on $\mathbb{C}$).

2. Φ_n has finite exponential type (hence order ≤ 1), which is what later justifies genus-1 Hadamard factorisations used elsewhere.

We also record a purely local factorization statement at *common* zeros: if Ξ and Φ_n vanish at the same point, then Φ_n/Ξ is holomorphic there. This appendix **does not** claim global divisibility of Φ_n by Ξ , does not claim that Φ_n shares *all* zeros of Ξ , and does not claim any global constancy or zero-freeness of a quotient—those would require additional arguments not contained here.

C.1. Definition of $\Phi_n(s)$ and two equivalent forms

Let (λ_n, ψ_n) be an eigenpair of H . Define, for complex s ,

$$\Phi_n(s) := \int_0^\infty \psi_n(\log x) x^{s-1} e^{-x^2/2} dx.$$

With the substitution $x = e^t$ so $dx = e^t dt$, this becomes the equivalent “t-line” form

(These are exactly the definitions already in your draft.

Standing decay input. From Appendix A (Agmon-type bounds), assume: there exist constants $C_0, \kappa > 0$ (depending on n) such that

$$|\psi_n(t)| \leq C_0 e^{-\kappa|t|} \quad (t \in \mathbb{R}).$$

C.2 Entirety of Φ_n

Lemma C.2.1 (Holomorphy in a strip; controlled analytic continuation)

Fix n . Let ψ_n be an L^2 –normalized eigenfunction of H with eigenvalue λ_n , and assume the Agmon bound

$$|\psi_n(t)| \leq C_0 e^{-\kappa|t|}, t \in \mathbb{R},$$

as recorded in C.1. Define

$$f_s(t) := e^{st} e^{-e^{2t}/2}, \Phi_n(s) := \langle \psi_n, f_s \rangle = \int_{-\infty}^{\infty} \psi_n(t) e^{st} e^{-e^{2t}/2} dt.$$

Then:

1. Φ_n is holomorphic on the open strip

$$S_\kappa := \{s \in \mathbb{C} : |\Re s| < \kappa\}.$$

Φ_n admits analytic continuation from S_κ along any path in $\mathbb{C}$ that avoids a discrete set of possible singularities arising from the analytic continuation mechanism described below. In particular, if the continuation lemma C.4.1 (or its replacement) yields an identity of the form $(L\Phi_n)(s) = 0$ with L a holomorphic-coefficient linear differential operator on $\mathbb{C}$, then Φ_n extends uniquely to an entire function.

Referee note. The proof of RH does **not** require global entirety of each individual Φ_n ; only strip holomorphy and local factorization near zeta zeros are used downstream. Entirety is included here as an optional strengthening contingent on the continuation mechanism (Lemma C.4.1).

Proof

(1) Let $K \Subset S_\kappa$ be compact. Write $s = \sigma + i\tau$. Using the Agmon bound and $\sup_{s \in K} |\Re s| < \kappa$, we have for all $t \in \mathbb{R}$

$$|\psi_n(t) f_s(t)| \leq C_0 e^{-\kappa|t|} e^{(\sup_{s \in K} |\Re s|)|t|} e^{-e^{2t}/2} = C_0 e^{-(\kappa - \sup_{s \in K} |\Re s|)|t|} e^{-e^{2t}/2}.$$

The right-hand side is integrable on $\mathbb{R}$ (exponential decay as $t \rightarrow -\infty$ and super-exponential decay as $t \rightarrow +\infty$), and is independent of $s \in K$. Since $t \mapsto f_s(t)$ is entire in s pointwise and the integrand is dominated by an L^1 majorant uniform in $s \in K$, Morera's theorem (or dominated convergence applied to ∂_s -difference quotients) gives holomorphy on K . As $K \Subset S_\kappa$ was arbitrary, Φ_n is holomorphic on S_κ .

Lemma C.2.1 (Strip holomorphy of the test pairing).

Fix n . Let ψ_n be an L^2 -normalized eigenfunction of H with eigenvalue λ_n , and assume the Agmon bound

$$|\psi_n(t)| \leq C_0 e^{-\kappa|t|}, t \in R,$$

as in C.1. Define for $s \in C$

$$f_s(t) := e^{st} e^{-e^2 t/2}, \Phi_n(s) := \langle \psi_n, f_s \rangle = \int_{-\infty}^{\infty} \psi_n(t) e^{st} e^{-e^2 t/2} dt.$$

Then Φ_n is holomorphic on the open strip

$$S_\kappa := \{s \in C: |\Re(s)| < \kappa\}$$

Remark (scope). The RH closure does not require global analyticity properties of each individual Φ_n beyond strip holomorphy. Any stronger continuation of Φ_n (e.g., via auxiliary identities in s) is optional and not used in the A–B–D–H–F proof chain.

C.3. Finite exponential type (order ≤ 1)

Non-load-bearing note. Appendix C is not used in the proof chain establishing Theorem F.RH; the RH closure proceeds via Appendices A, B, D, H, and F. In particular, Lemma C.3.1 is auxiliary and may be omitted without affecting the RH implication.

Lemma C.3.1 (Strip-bound only).

Assume $|\psi_n(t)| \leq C_0 e^{-\kappa|t|}$. For each compact $K \subseteq S_\kappa := \{s \in C: |\Re(s)| < \kappa\}$, there exists an integrable majorant $M_K(t) \in L^1(R)$ such that for all $s \in K$,

$$|\psi_n(t) e^{st} e^{-e^2 t/2}| \leq M_K(t) (t \in R)$$

Consequently $\Phi_n(s) = \int R \psi_n(t) e^{st} e^{-e^2 t/2} dt$ is holomorphic on S_κ

Proof. Let $\eta := \kappa - \sup_{s \in K} |\Re(s)| > 0$. Then

$$|\psi_n(t) e^{st} e^{-e^2 t/2}| \leq C_0 e^{-(\kappa - |\Re(s)|)|t|} e^{-e^2 t/2} \leq C_0 e^{-\eta|t|} e^{-e^2 t/2},$$

and the right-hand side is integrable on R . Holomorphy follows by dominated convergence (or Morera).

Remark. No bound is asserted outside S_κ . Any extension beyond the strip would require an additional continuation mechanism and is not used in the RH chain.

C.4 Remark (No premature evaluation; non-circular use)

Remark C.4.1 (No premature evaluation at spectral points; Appendix C is non-load-bearing)

In Sections 4–6 and in the closure Appendix F, Appendix C is used only for local analytic control near nontrivial zeros of Ξ (e.g., local factorization and multiplicity bookkeeping). In particular:

1. At this stage we use only the one-way implication furnished by the trace/test-function interface:

$$\Xi(\rho) = 0 \Rightarrow \Phi_n(\rho) = 0 \text{ for each } n,,$$

and the corresponding local factorization near such ρ :

$$\Phi_n(s) = \Xi(s) \tilde{F}_n(s) \text{ in a neighborhood of } \rho, \Phi_n(s) = \Xi(s) F_n(s) \text{ in a neighborhood of } \rho,$$

with $\tilde{F}_n$ holomorphic and nonvanishing near ρ . (This is a local statement: it does not assert any global quotient Φ_n/Ξ .)

2. We do not assert any “spectral vanishing” statement at

$$s_n := \frac{1}{2} + i\lambda_n$$

in Appendix C. In particular, we do not evaluate Φ_n at $s = s_n$ here and do not use any claim of the form $\Phi_n(s_n) = 0$ at this stage.

3. The evaluation at points $s = \frac{1}{2} + i\lambda_n$ is made only after Appendix F establishes the eigenvalue/zero-ordinate matching (including multiplicities). This ordering prevents circularity and ensures Appendix C imports no RH content into the proof engine.

C.5. Local Ξ -factorization at common zeros (the only correct “local factorization” statement)

Let $\Xi(s)$ denote the completed zeta function.

Lemma C.5.1’ (Local factorization at a common zero, general multiplicity).

Let $s^* \in C$. Suppose Ξ has a zero of order $m \geq 1$ at s^* , i.e.

$$\Xi(s) = (s - s^*)^m g(s) \text{ with } g \text{ holomorphic and } g(s^*) \neq 0.$$

Assume Φ_n vanishes at s^* to order at least m , i.e.

$$\Phi_n(s) = (s - s^*)^m h(s) \text{ with } h \text{ holomorphic.}$$

Then there exists a neighborhood U of s^* and a holomorphic function $F_{n,*}$ on U such that

$$\Phi_n(s) = \Xi(s) F_{n,*}(s) \quad (s \in U).$$

Moreover $F_{n,*}(s^*) \neq 0$ iff and only if $\text{ord}_{s^*} \Phi_n = \text{ord}_{s^*} \Xi$.

Proof. By hypothesis, $\Xi(s) = (s - s^*)^m g(s)$ with $g(s^*) \neq 0$, and $\Phi_n(s) = (s - s^*)^m h(s)$. Hence on a small disk U where g has no zeros,

$$\frac{\Phi_n(s)}{\Xi(s)} = \frac{h(s)}{g(s)}$$

is holomorphic, and the factorization follows. The nonvanishing criterion is immediate. **Remark (what we actually use).** In the main proof chain, we never use any claim of global divisibility $\Phi_n = \Xi \cdot F_n$ or global nonvanishing of a quotient. The only use of local factorization is: *once* a bijection between spectral parameters and Ξ -zeros is established elsewhere (Appendices D/F), local nonvanishing can be invoked to control local multiplicity behavior.

A/B infrastructure firewall.

Appendices A and B are operator-side infrastructure only. Appendix A supplies the self-adjointness, lower semiboundedness, compactness/finite-volume trace legality, and the single global Stieltjes-compatible spectral shift for the PWQO family. Appendix B supplies fixed-(L,R) localized trace-class/Duhamel legality and kernel extraction. Neither Appendix A nor Appendix B constructs a zero-side object, uses the explicit formula, or contributes directly to the final pole-rigidity contradiction. All Duhamel word expansions are performed at fixed (L,R); cutoff removal and the Ω -holomorphic remainder are owned by Appendix D.

Appendix D

Limit Protocol (strict fixed-(L,R) word analysis; limits taken only after Laplace transform)

Throughout Appendix D we fix truncation parameters

$$L < \infty(\text{window size}), \quad R < \infty(\text{prime} - \text{power cutoff } \log q \leq R).$$

All Duhamel expansions, word decompositions, diagonal/mixed classifications, and trace-class estimates are carried out **at fixed (L,R)**.

Throughout the proof spine, the perturbation is decomposed by prime powers $q = p^m$, with frozen weight $w(q) = \frac{\Lambda(q)}{\sqrt{q}}$. The explicit spike package is

$$P_{L,R}(t) = \sum_{\log q \leq R} w(q) g_{t,L}(\log q)$$

The equivalent two-level notation

$\sum_{p^m \leq e^R} (\log p) p^{-m/2} g_{t,L}(m \log p)$ is purely a rewriting of the same sum.

Window convention. The window operator A_L is an auxiliary regularization device. The R_H closure theorem is stated for the canonical whole-line transform F_H . No separate $L \rightarrow \infty$ pole-passing argument is invoked in Theorem F.RH unless explicitly stated.

Notation firewall (operator semigroup vs. scalar Gaussian).

In this appendix we distinguish strictly between two different objects that are denoted by similar Gaussian symbols elsewhere in the manuscript:

1. Operator semigroup.

We write

$$S_t := \exp(-t H_0)$$

for the heat semigroup generated by the base operator H_0 (or by $H_{\{L,R\}}$ in finite volume).

When appearing inside traces — for example in expressions of the form

$$\text{Tr}(S_t A_L T_\alpha) —$$

this denotes an operator acting on $L^2(\mathbb{R})$.

Its integral kernel will be denoted by $K_t(u, v)$.

2. Scalar Gaussian kernel.

We reserve

$$G_t(u) = (4\pi t)^{(-1/2)} \exp(-u^2/(4t))$$

exclusively for the one-dimensional scalar heat kernel in the displacement variable.

This is the same Gaussian family used in Appendix H (explicit formula side), with identical normalization.

Thus:

- S_t is an operator (a semigroup).
- $K_t(u, v)$ is its integral kernel.
- $G_t(u)$ is the scalar Gaussian in the displacement variable.

This separation prevents any ambiguity between:

- the semigroup generator sign ($\exp(-tH)$ convention), and
- the scalar Gaussian used in the spike package and explicit formula.

In particular, no identification is made between operator-valued expressions and scalar Gaussian test functions unless explicitly stated via the kernel $K_t(u, v)$.

(1) Fixed-(L,R) Heat-Trace Decomposition

For each fixed (L,R), Theorem D.ONE-STEP' yields

$$\Theta_{L,R}(t) = A_{class,L}(t) + \sum_{\log q \leq R} \frac{\Lambda(q)}{\sqrt{q}} g_{t,L}(\log q) + R_{L,R}(t),$$

where:

- the spike kernel $g_{t,L}(a) = \text{Tr}(S_t A_L T_a)$,
- the diagonal extraction uses Lemma D.K0' and Lemma D.DIAG-SUM,
- the mixed-word contribution is estimated **at fixed (L,R)** via Lemma B.MIX (fixed-R version),
- and the remainder satisfies, for every N,

$$|R_{L,R}(t)| \leq C_N(L, R) t^N (0 < t \leq t_0(L, R))$$

No limit in R is taken at this stage.

All constants are allowed to depend on (L, R).

(2) Passage R→ Not at Word Level

We do **not** pass to the limit R→inside the Duhamel/word expansion.

Instead, we proceed as follows:

1. Take the Laplace transform (justified for fixed R):

$$F_{L,R}(s) = \int_0^\infty e^{-st} \Theta_{L,R}(t) dt = \text{Tr}((H_{L,R} + s) - 1), \Re s > \sigma_0$$

Use the Gaussian bound on $g_{t,L}(a)$,

$$|g_{t,L}(a)| \leq C(L) t^{-1/2} e^{-a^2/(4t)},$$

to obtain a summable majorant for

$$\frac{\Lambda(q)}{\sqrt{q}} g_{t,L}(\log q), q$$

uniform on compact t-intervals.

3. Apply Lemma D.LIMIT / D.NORM Ω to obtain:
 - normal convergence of the spike package on Ω -compacts,
 - convergence of $F_{L,R} \rightarrow F_L$ locally uniformly on Ω ,
 - and definition of the Ω -holomorphic remainder by subtraction:

$$R_L^\Omega(s) := F_L(s) - B_L(s).$$

At no point is uniform-in-R cluster separation required.

(3) Passage $L \rightarrow \infty$ (Window Normalization)

Write

$$\widehat{g_{t,L}}(a) := \frac{1}{2L} g_{t,L}(a), \widehat{P_{L,R}}(t) := \frac{1}{2L} \sum_{\log q \leq R} w(q) g_{t,L}(\log q).$$

Then $\widehat{g_{t,L}}(a) \rightarrow_c G_t(a)$ on compact aa-intervals, so the fixed-cutoff explicit package passes to the canonical intensive package of Section 5.1.4.

.

Notation (prime-power letters).

Fix $R > 0$. For each prime power q with $\log q \leq R$, define the bump

$$V_q(u) := \frac{\Lambda(q)}{\sqrt{q}} W_\alpha(u - \log q), M_q := MV_q$$

Convention (Prime-power weights). We will write the diagonal spike weights in either of the equivalent forms

$$\sum_q p^m \frac{\Lambda(q)}{\sqrt{q}} \Psi(\log q) \text{ or } \sum_p \sum_{m \geq 1} (\log p) p^{-m/2} \Psi(m \log p),$$

using $\Lambda(p^m) = \log p$ and $p^m = p^{m/2}$ These are identical notations.

Remark : “All spike/package formulas in this paper use the scale $\frac{\Lambda(q)}{\sqrt{q}}$ (equivalently $(\log p)p^{-m/2}$;

Lemma D.TC (Trace-class difference + Duhamel validity)

Lemma D.TC (Trace-class difference and trace-norm Duhamel expansion).

Work on $L^2(R, dt)$. Let H_0 be the base self-adjoint operator (e.g. $H_0 = -\partial_t^2$ or our PWQO base part) and let $G_t := e^{-tH_0}$. Fix a cutoff $\chi \in C_c^\infty(R)$ with $\chi(t) = 1$ for $|t| \leq 1$ and $\chi(t) = 0$ for $|t| \geq 2$. For $L > 0$, define the localizer

$$(\Pi_L f)(t) := \chi(t/L) f(t),$$

and optionally the smoothing upgrade

$$\Pi_L := M_{\chi(\cdot/L)} (1 + H_0)^{-\sigma/2} \text{ for some } \sigma > 1,$$

(where $M_{\chi(\cdot/L)}$ denotes multiplication by $\chi(t/L)$).

Let V_R be the truncated prime-channel / prime-power-channel perturbation (finite truncation at level R), and set

$$B_{L,R} := \Pi_L V_R \Pi_L, H_{L,R} := H_0 + B_{L,R}.$$

Consider either of the following packages:

(TC-A) Smoothing Π -package. Use the smoothing choice $\Pi_L = M_{\chi(\cdot/L)} (1 + H_0)^{-\sigma/2}$ with $\sigma > 1$, and assume $R < \infty$ so V_R (hence $B_{L,R}$) is bounded for each fixed R . Then for every $t > 0$,

1. $\Pi_L S_t \in S_2$ and $S_t \Pi_L \in S_2$;
2. $B_{L,R} S_t \in S_1$ and $S_t B_{L,R} \in S_1$;
3. $e^{-tH_{L,R}} - e^{-tH_0} \in S_1$.

Moreover, the Dyson–Duhamel expansion

$$e^{-t(H_0+B_{L,R})} - e^{-tH_0} = \sum_{k \geq 1} (-1)^k \int_{\Delta_k} (t) S_{t-t_k} B_{L,R} S_{t_k-t_k-1} \cdots B_{L,R} S_{t_1} d_{t_1} \cdots d_{t_k}$$

converges absolutely in S_1 on each interval $0 < t \leq t_0$, and the trace may be taken termwise:

$$\text{Tr}(e^{-tH_{L,R}} - e^{-tH_0}) = \sum_{k \geq 1} (-1)^k \int_{\Delta_k} (t) \text{Tr}(S_{t-t_k} B_{L,R} S_{t_k-t_k-1} \cdots B_{L,R} S_{t_1}) d_{t_1} \cdots d_{t_k}$$

$$d_{t_1} \cdots d_{t_k}$$

(TC-B) Word-estimate package. Keep $\Pi_L = \chi(\cdot/L)$ (pure cutoff). Given that for each word length k , the k -fold Duhamel term is trace class and satisfies a uniform bound

$$\|W_k(t)\|_{S_1} \leq C_k t^{k/2} (0 < t \leq t_0),$$

uniformly in L and for fixed finite truncation R . Then $e^{-tH_{L,R}} - e^{-tH_0} \in S_1$, the Dyson–Duhamel expansion above converges absolutely in S_1 on $0 < t \leq t_0$, and termwise tracing/regrouping is justified by absolute summability.

Conclusion (common to TC-A and TC-B).

Under either (TC-A) or (TC-B), for every $t > 0$ and finite L, R ,

$$e^{-tH_{L,R}} - e^{-tH_0} \in S_1,$$

the Duhamel/Dyson series converges in trace norm, and the trace may be taken termwise and regrouped by “word type” (diagonal vs mixed) in subsequent arguments.

Lemma D.DIAG→SHIFT (Diagonal channel contribution reduces to shift-trace spikes up to super-smoothing).

Auxiliary bookkeeping lemma only

This shift-channel model is not the genuine shifted-bump perturbation used in the final one-letter spike package. It is retained only to motivate the one-letter displacement mechanism. The load-bearing fixed-cutoff extraction for the genuine shifted-bump model is Theorem D.ONE-STEP” together with Lemma D.DIAG-SUM.

In particular, repeated insertions of the genuine shifted-bump potential are not exported as new spike locations

$$ka_q.$$

They remain in the fixed-cutoff operator-side remainder.

Context.

Given Lemma **D.TC** (trace-norm Duhamel validity) is available, so the Duhamel/Dyson series for $e^{-tH_{L,R}} - e^{-tH_0}$ converges in S_1 and may be traced termwise. (We treat this as done.)

Fix a single prime p and consider the “p-channel” perturbation acting by translation in log-coordinates. Let T_u denote translations on $L^2(R, dt)$: $(T_u f)(t) = f(t - u)$. Let A be the generator, so $T_u = e^{-uA}$ (in the weak/strong sense appropriate to your setup). Let $S_t := e^{-tH_0}$ be the background heat operator.

Let the p – channel well insertion be represented (after localization) by an operator of the form

$$B_p := \Pi_L M_w T_{\log p} \Pi_L,$$

or more generally as a finite sum of such terms (same argument applies termwise).

Define the purely diagonal p-word contribution to the traced Duhamel expansion by restricting to words that use only B_p repeatedly (no other channels):

$$\Theta_p^{diag}(t) := \sum_{k \geq 1} (-1)^k \int_{\Delta_k(t)} \text{Tr}(S_{t-t_k} B_p S_{t_k-t_{k-1}} B_p \cdots B_p S_{t_1}) d_{t_1} \cdots d_{t_k}.$$

Proof

Step 1 — Move all translations to the right (translation covariance)

Use the algebra:

$$T_u M_w T_{-u} = M_{w(\cdot - u)}, \quad T_u \Pi_L T_{-u} = \Pi_L(u),$$

and the key covariance hypothesis for the background heat operator:

There is a controlled relation between $T_u S_t T_{-u}$ and S_t .

- If H_0 is translation-invariant in tt, then $T_u S_t T_{-u} = S_t$ exactly.
- If not, we need a localization statement: for the diagonal channel terms, the conjugation defect contributes only to $O(t^N)$ after tracing with Π_L (this becomes part of the super-smoothing remainder).

This step rewrites each word trace as a trace of the form

$$\text{Tr} (S_{s_0} B_p^{\sim(0)} G_{s_1} B_p^{\sim(1)} \dots B_p^{\sim(k-1)} S_{s_k} T_{klogp}),$$

i.e. the net translation after k repetitions is exactly T_{klogp} .

This is the key combinatorial fact: repeating the same channel k times yields a net shift by $klogp$.

Step 2 — Collapse the simplex integrals (replace by a single operator power)

By the standard Duhamel identity (in trace norm, justified by D.TC),

$$\int_{\Delta_k(t)} S_{s_0} B_p S_{s_1} \dots B_p S_{s_k} ds = \frac{t^k}{k!} W_p(t)^k,$$

for a suitable “time-averaged insertion” operator $W_p(t)$ that is trace-class after localization.

After Step 1, this becomes

$$\Theta_p^{diag}(t) = \sum_{k \geq 1} (-1)^k \text{Tr} (W_p(t)^k T_{klogp}) + (\text{terms absorbed into } r_p(t)).$$

Step 3 — Replace $W_p(t)$ by its leading small-t model A

(Lead) There exists a fixed operator A such that

$$\|W_p(t) - C_p S_t A\|_{S_1} \leq C_{p,N} t^N \quad \forall N.$$

Interpretation: the localized time-averaged insertion is asymptotic to a fixed “one-step insertion” sandwiched with S_t , with error super-smoothing.

This is exactly the same kind of estimate as your mixed-word remainder bound; it’s a “small-t parametrix / smoothing” statement.

Given, we get

$$\text{Tr}(W_p(t)^k T_{klogp}) = C_p \text{Tr}(S_t A T_{klogp}) + O(t^N)$$

uniformly in k over the truncated regime you need (or absolutely summably in k due to decay).

Step 4 — Sum over k and rename $m = k$

Putting Steps 2–3 together:

$$\Theta_p^{diag}(t) = C_p \sum_{m \geq 1} \text{Tr}(S_t A T_{mlogp}) + r_p(t),$$

with $r_p(t) = O(t^N) \forall N$, as claimed.

Remark (order of limits, interchanges, and the holomorphic remainder shield).

All Duhamel/word expansions, freezing replacements, and simplex-integral interchanges are performed at fixed truncation parameters (L, R) . No step takes the limit $R \rightarrow \infty$ inside the Duhamel/word expansion, and no uniformity in R is required for the fixed- (L, R) constants appearing there.

The explicit spike package is first separated at fixed (L, R) , and the remaining mixed-word and diagonal-tail contribution is placed into the remainder term. Only after this separation do we pass to the transform/resolvent level and take limits. The passage $R \rightarrow \infty$ is then carried out by the finite-volume norm-resolvent bridge, locally uniformly on compact subsets $K \subset \Omega$. After that transform-level $R \rightarrow \infty$ passage is established, the window normalization $L \rightarrow \infty$ is taken only at the package/transform level, not inside the Duhamel expansion.

Accordingly, all interchanges of sums, integrals, and limits are carried out either on a fixed overlap half-plane U_σ or on compact subsets of Ω away from the pole set, and only after the explicit package has been isolated. The remainder terms converge uniformly on such compacta by the estimates established above. Consequently, the limiting remainder remains holomorphic, and the passages $R \rightarrow \infty$ and $L \rightarrow \infty$ do not create new poles beyond those already carried by the explicit package.

In particular, any R -dependence in mixed-word or Duhamel constants is confined to the fixed- (L, R) remainder estimates before package separation, and the final Ω -holomorphic remainder is obtained only after this separation.

Reader note on scope of the channel algebra.

The auxiliary channel operator

$$B_p := \Pi_L M_w T_{\log p} \Pi_L$$

is used only as a bookkeeping model for one-letter displacement extraction.

For the genuine prime-power perturbation,

$$V_q(u) = w(q) W_\alpha(u - \log q) = w(q) T_{a_q} M_{W_\alpha} T_{-a_q}, \quad a_q := \log q,$$

higher same-channel words are treated separately below. In particular, for the genuine shifted-bump model, repeated insertions are **not** identified with a net displacement ka_q . The explicit spike package records only the one-letter diagonal sector; higher same-channel sectors are retained inside the fixed-cutoff operator-side remainder.

Lemma D.FREEZE (free baseline version)

Remark:

All Duhamel/word expansions (including freezing replacements) are performed at fixed truncation cutoff $R < \infty$. No step requires constants uniform in R ; indeed we allow $C = C(L, R)$ and $t_0 = t_0(L, R)$ throughout the fixed-(L, R) trace-class analysis. We do not take $R \rightarrow \infty$ inside the word expansion. Instead, we first pass to the Stieltjes/Laplace transform

$F_{L,R}(s) = \text{Tr}(H_{L,R} + s)^{-1}$, and only then take $R \rightarrow \infty$ using the finite-volume norm-resolvent bridge $\|V_\infty - V_R\|_\infty \rightarrow 0$ on $[-L, L]$ together with Hilbert–Schmidt factorization of localized resolvents, yielding $\|(H_{L,R} + s)^{-1} - (H_{L,\infty} + s)^{-1}\|_1 \rightarrow 0$ uniformly on compact $K \Subset \Omega$. The short-time spike transform $P_L^\sim < P_L$ converges normally on Ω -compacts. This statement does **not** refer to the full prime-side kernel series B_{pr} , whose normal convergence near the cut is not asserted and is not used for the global Ω -interface.

This establishes local uniform convergence $F_{L,R} \rightarrow F_L$ on Ω , and the Ω -holomorphic remainder is then defined by subtraction (the Ω -shield), not by analytic continuation.

Lemma D.FREEZE (Exact freezing for free baseline).

Let $H_0 = -\partial_u^2$ on $L^2(R, du)$, and $S_t = e^{-tH_0}$. Let T_a be translations $(T_a f)(u) = f(u - a)$. Then for all $a \in R$ and $t > 0$,

$$T_a S_t T_{-a} = S G_t$$

Consequently, for every trace-class expression of the form $G_t B T_a$,

$$\text{Tr}(T_a S_t T_{-a} B T_a) = \text{Tr}(S_t B T_a)$$

and there is no freezing error term.

Proof. Since H_0 is translation invariant, $T_a H_0 T_{-a} = H_0$. Functional calculus gives

$$T_a e^{-tH_0} T_{-a} = e^{-tH_0}.$$

Lemma D.MISALIGN (cutoff misalignment $\rightarrow$ boundary insertion). For $a \geq 0$

$$\Pi_L - T_a \Pi_L T_{-a} = \int_0^a \text{Tr}(\partial_L \Pi_L) T_{-r} dr,$$

where $\partial_L \Pi_L$ is the boundary operator. Consequently, any localized trace containing $T_a(\cdot)T_{-a}$ can be rewritten as a boundary-trace containing a *single* T_a (up to an explicitly controlled remainder).

Lemma D.SPIKE-TRANSFER (free kernel substitution absorbed into the remainder).

Fix $L > 0$ and $R < \infty$. Let $H_{L,R} = H_0 + V_{L,R}$ with $V_{L,R} \in L^\infty([-L, L])$. Define the “true” localized spike response

$$g_{t,L,R}^{\text{true}}(a) := \text{Tr}(\Pi_L e^{-tH_{L,R}} \Pi_L A_L T_a)$$

and the free spike kernel

$$g_{t,L}^0(a) := \text{Tr}(\Pi_L e^{-tH_0} \Pi_L A_L T_a)$$

Then for every $N \geq 1$ there is $C_N(L, R)$ such that for all $0 < t \leq t_0(L, R)$ and all shifts a in the truncated regime $|a| \leq R$,

$$|g_{t,L,R}^{\text{true}}(a) - g_{t,L}^0(a)| \leq C_N(L, R) t^N$$

Consequently, every occurrence of $g_{t,L}$ in the diagonal spike package may be taken to be the *free* spike kernel, and the discrepancy is part of the super-smoothing remainder $R_{L,R}(t)$.

Proof : Apply Duhamel once,

$$e^{-tH_{L,R}} - e^{-tH_0} = - \int_0^t e^{-(t-u)H_{L,R}} V_{L,R} e^{-uH_0} du,$$

localize by Π_L , take trace norms, use HS factorization and heat smoothing on the finite window; each Duhamel insertion produces an extra tt factor, and iterating gives t^N .

Let $K(u, v) = (4\pi t)^{-1/2} \exp^{-(u-v)^2/(4t)}$ be the free heat kernel.

Lemma D.BRIDGE I (Diagonal word $\Rightarrow$ shift-trace form).

Setup.

Work on $L^2(R, dt)$. Let H_0 be the base self-adjoint operator and $S_t := e^{-tH_0}$. Fix a cutoff/localizer Π_L as in Lemma D.TC, and define the localized trace functional using Π_L

(e.g. $\Theta_{L,R}(t) = \text{Tr}(\Pi_L e^{-tH_{L,R}} \Pi_L)$) We assume TC-B so all word terms are trace class and termwise tracing is legal.

Let T_u denote translations: $(T_u f)(t) = f(t - u)$ Let p be a fixed prime and set $u_p := \log p$.

Define the localized p-channel insertion operator B_p in one of the following equivalent forms.

Variant A (single multiply-shift form)

Given

$$B_p = \Pi_L M_w T_{u_p} \Pi_L$$

for some bounded, localized multiplier M_w (multiplication by a fixed profile $w(t)$).

For $k \geq 1$ define the diagonal p-word operator appearing in the Duhamel expansion:

$$W_{p,k}(t) := \int_{\Delta_k} (t) S_{s_0} B_p G_{s_1} B_p \cdots B_p S_{s_k} ds, s_0 + \cdots + s_k = t.$$

Claim (exact shift extraction).

For each $k \geq 1$,

$$W_{p,k}(t) = Q_{p,k}(t) T_{ku_p},$$

where $Q_{p,k}(t)$ is an operator that depends on H_0, w, Π_L and k, t but contains no remaining translations.

Consequently,

$$Tr(W_{p,k}(t)) = Tr(Q_{p,k}(t) T_{klogp}).$$

Variant B (finite sum of multiply-shift pieces)

Assume

$$B_p = \sum_{j=1}^J \Pi_L M_{w_j} T_{u_p} \Pi_L$$

with a fixed finite J and bounded localized multipliers M_{w_j} .

Then for each $k \geq 1$,

$$W_{p,k}(t) = Q_{p,k}(t) T_{ku_p},$$

where $Q_{p,k}(t)$ is now a finite sum (over J^k terms) of translation-free operators built from G_s, Π_L , and multipliers $M_{w_j(\cdot - \ell u_p)}$ with $\ell \in \{0, \dots, k-1\}$. In particular,

$$Tr(W_{p,k}(t)) = Tr(Q_{p,k}(t) T_{klogp}).$$

Proof

Prove Variant A; Variant B is identical after expanding the finite sum.

Start with the operator product inside the simplex integral:

$$S_{s_0} B_p S_{s_1} B_p \cdots B_p S_{s_k} = S_{s_0} \Pi_L M_{w_p} T_{u_p} \Pi_L S_{s_1} \Pi_L M_{w_p} T_{u_p} \Pi_L \cdots \Pi_L M_{w_p} T_{u_p} \Pi_L S_{s_k}$$

Now use the basic commutation identity for translations and multipliers:

$$T_{u_p} M_w = M_{w(\cdot - u_p)} T_{u_p}, \text{ equivalently } M_w T_{u_p} = T_{u_p} M_{w(\cdot - u_p)}.$$

Push the **leftmost** T_{u_p} to the far right through everything to its right, one factor at a time:

- Through multipliers: $T_{u_p} M_w = M_{w(\cdot - u_p)} T_{u_p}$
- Through the localizer: $T_{u_p} \Pi_L = \Pi_L^{(u_p)} T_{u_p}$, where $\Pi_L^{(u)} := T_u \Pi_L T_{-u}$ (still a bounded localization operator).
- Through heat factors: we do **not** need $T_{u_p} S_s = S_s T_{u_p}$ for this lemma; you simply define the

conjugated heat factor $G_s^{(u)} := T_u S_s T_{-u}$. Then

$$T_{u_p} S_s = (T_{u_p} S_s T_{-u_p}) T_{u_p} = S_s^{(u_p)} T_{u_p}$$

- After commuting across the next block $\Pi_L S_{s_1} \Pi_L M_w \Pi_L S_{s_1} \Pi_L M_w$, the first T_{u_p} becomes “captured” on the right, and the operators left behind are the conjugated versions:
- multipliers shifted by u_p ,
- localizers conjugated by u_p ,
- heat factors conjugated by u_p .

Repeat this process for the second T_{u_p} , then the third, etc. After k such shifts, all translations have been pushed to the far right and multiply to T_{ku_p} . Everything else is translation-free by construction.

Therefore, for each simplex point $(s_0, \dots, s_k)(s_0, \dots, s_k)$,

$$S_{s_0} B_p S_{s_1} B_p \cdots B_p S_{s_k} = Q_{p,k}(s_0, \dots, s_k) T_{ku_p}$$

for an explicitly defined translation-free operator $Q_{p,k}$ built from conjugates $S_{s_j}(\ell u_p)$, $\Pi_L(\ell u_p)$, and shifted multipliers $M_{w(\cdot - \ell u_p)}$, with $\ell \in \{0, \dots, k-1\}$.

Integrating over the simplex preserves the factorization:

$$W_{p,k}(t) = \int_{\Delta_k(t)} Q_{p,k}(s) ds \cdot T_{ku_p} =: Q_{p,k}(t) T_{ku_p}.$$

Finally, since $W_{p,k}(t) \in S_1$ by TC-B, we can take traces and obtain

$$Tr(W_{p,k}(t)) = Tr(Q_{p,k}(t) T_{klogp})$$

Lemma D.g DIFF (shift differentiation and generator insertion)

Let T_a be translations on $L^2(R)$: $(T_a f)(u) = f(u - a)$. Let A denote the translation generator on a dense domain, so that

$$\frac{d}{da} T_a = -AT_a, \frac{d}{da} T_{-a} = T_{-a}A$$

Let M_w be multiplication by a fixed bounded w . Then the conjugated bump

$$M_w(a) := T_a M_w T_{-a}$$

is differentiable in a (in the strong operator topology on a suitable dense domain), and

$$\frac{d}{da} M_w(a) = T_a [A, M_w] T_{-a}.$$

Moreover, for any trace-class operator K , the map $a \mapsto Tr(K M_w(a))$ is differentiable and

$$\frac{d}{da} Tr(K M_w(a)) = Tr(K T_a [A, M_w] T_{-a})$$

Proof. Differentiate $T_a M_w T_{-a}$ using the product rule and the identities for dT_a/da and dT_{-a}/da

$$\frac{d}{da} (T_a M_w T_{-a}) = (-AT_a) M_w T_{-a} + T_a M_w (T_{-a} A) = T_a (-AM_w + M_w A) T_{-a} = T_a [A, M_w] T_{-a}$$

The trace differentiation follows from standard trace-class continuity and cyclicity bounds (dominated by $\|K\|_{S_1} \| [A, M_w] \|$).

Lemma D.SHIFT-CHANNEL (derivative channel is the extracted diagonal coefficient)

Let $M_w(a) = T_a M_w T_{-a}$. Let K_t be trace-class for $t \in (0, t_0]$. Define

$$J(t, a) := \text{Tr}(K_t M_w(a)), J_1(t, a) := \text{Tr}(K_t T_a [A, M_w] T_{-a}).$$

Then $a \mapsto J(t, a)$ is C^1 and

$$\partial_a J(t, a) = J_1(t, a).$$

Moreover, if the diagonal extraction step in D.ONE-STEP' is performed on the *commutator-letter perturbation*

$$V_q^{(1)} := \frac{\Lambda(q)}{\sqrt{q}} T_{a_q} [A, M_w] T_{-a_q},$$

then the diagonal q-letter contribution equals $\frac{\Lambda(q)}{\sqrt{q}} J_1(t, a_q)$, and this term is displacement-sensitive even when $J(t, a)$ is flat.

Proof. Differentiation under the trace is justified by trace-class bounds and the identity $\partial_a M_w(a) = T_a [A, M_w] T_{-a}$. The last statement is immediate because the diagonal word with letter produces

$$\text{Tr}(K_t V_q^{(1)}) = (\Lambda(q)/\sqrt{q}) J_1(t, a_q)$$

Lemma D.BRIDGE II' (prime-power channel produces a single-sided spike observable)

Fix a bump w and define $M_w(a) = T_a M_w T_{-a}$ as above. For prime powers $q = p^m$, set $a_q = \log q$ and define the q-letter perturbation operator

$$V_q := \frac{\Lambda(q)}{\sqrt{q}} M_w(a_q)$$

Let $H_R = H_0 + \sum_{\log q \leq R} V_q$ (finite prime-power truncation).

Define the spike functional (the displacement-sensitive observable)

$$g_t(a) := \text{Tr}(S_t A_L T_a)$$

where $G_t = e^{-tH_0}$ is the free heat semigroup and A_L is the one-sided/boundary-sensitive operator used throughout Appendix B–D (fixed once and for all), chosen so that A_L does **not** commute with translations and $S_t A_L$ is trace-class for $t \in (0, t_0)$

Then the a-derivative of the diagonal q-channel contribution is proportional to the spike observable:

$$\frac{d}{da} \Big|_{a=a_q} \text{Tr}(S_{t-s} A_L M_w(a) G_s) = \text{Tr}(S_{t-s} A_L T_{a_q} [A, M_w] T_{-a_q} S_s),$$

and after commuting translations through G_{t-s}, G_s (freezing exact for H_0) and using trace cyclicity in the one-sided A_L channel, this reduces to a fixed multiple of $g_t(a_q)$.

Lemma D. $R \rightarrow \infty$ (Uniform potential convergence on $[-L, L]$)

Statement.

Fix $L > 0$ and $\alpha > 0$. Let W_α be the Gaussian bump profile used for the arithmetic package and let

$$V_R(u) := \sum_{\log q \leq R} w(q) W_\alpha(u - \log q), \quad V_\infty(u) := \sum_q w(q) W_\alpha(u - \log q),$$

with the prime-power weights $w(q)$ in our global convention (equivalently $(\log p)p^{-m/2}$ for $q = p^m$).

Then:

1. The series defining V_∞ converges absolutely and uniformly on $[-L, L]$.
2. Hence $V_R \rightarrow V_\infty$ in $L^\infty([-L, L])$ as $R \rightarrow \infty$ in particular

$$\|V_\infty - V_R\|_{L^\infty([-L, L])} \xrightarrow{R \rightarrow \infty} 0, \text{ and } \sup_{R \geq 1} \|V_R\|_{L^\infty([-L, L])} < \infty.$$

Proof.

For $u \in [-L, L]$ and any $q \geq 2$,

$$|W_\alpha(u - \log q)| \leq \sup_{|u| \leq L} |W_\alpha(u - \log q)| \leq C_\alpha \exp\left(-\frac{(\log q - L)^2}{2\alpha}\right),$$

using the explicit Gaussian form of W_α .

Also $|w(q)| \leq C \log q \cdot q^{-1/2}$ under the intended normalization (equivalently $(\log p)p^{-m/2}$).

For q sufficiently large (depending on L, α), the Gaussian factor dominates any power of q ; concretely one can choose $q \geq q_0(L, \alpha)$ so that

$$\exp\left(-\frac{(\log q - L)^2}{2\alpha}\right) \leq q^{-2},$$

hence for $q \geq q_0$,

$$|w(q)| \sup_{|u| \leq L} |W_\alpha(u - \log q)| \leq C (\log q) q^{-1/2} q^{-2} = C (\log q) q^{-5/2}.$$

The tail $\sum_{q \geq q_0} (\log q) q^{-5/2}$ converges absolutely, so by Weierstrass M-test the series for V_∞ converges uniformly on $[-L, L]$. The uniform tail estimate also gives $\|V_\infty - V_R\|_{L^\infty([-L, L])} \rightarrow 0$.

Proposition D. $R \rightarrow \infty$ (Trace-norm resolvent convergence on Ω , uniform on compacts)

Statement.

Fix $L > 0$. Let $H_{L,R}$ denote the finite-volume (Dirichlet) Schrödinger operator on $[-L, L]$ with potential V_R and define H_L^∞ analogously with V_∞ (This is the finite-volume model used in your shield setup.)

Let $\Omega = \mathbb{C} \setminus (-\infty, 0]$ be the slit domain. For $s \in \Omega$ define

$$R_{L,R}(s) := (H_{L,R} + s)^{-1}, R_{L,\infty}(s) := (H_L^\infty + s)^{-1}$$

Then for every compact $K \Subset \Omega$,

$$\sup_{s \in K} \|R_{L,R}(s) - R_{L,\infty}(s)\|_{S^1} \xrightarrow{R \rightarrow \infty} 0$$

In particular,

$F_{L,R}(s) := \text{Tr}(R_{L,R}(s))$ converges locally uniformly on Ω to $F_{L,\infty}(s) := \text{Tr}(R_{L,\infty}(s))$,

and $F_{L,\infty}$ is holomorphic on Ω (being a locally uniform limit of holomorphic functions).

Proof.

Fix $K \Subset \Omega$ and set

$$\delta_{K^c} = \inf_{s \in K^c} \text{dist}(-s, [0, \infty)) > 0$$

This is strictly positive because K stays a positive distance from the cut.

Step 1: uniform operator-norm resolvent bounds.

Since each $H_{L,R}$ is self-adjoint with $\sigma(H_{L,R}) \subset [0, \infty)$ in your finite-volume setup, we have for all $s \in K$,

$$\|R_{L,R}(s)\| \leq \frac{1}{\delta_{K^c}}, \quad \|R_{L,\infty}(s)\| \leq \frac{1}{\delta_{K^c}}$$

Step 2: uniform Hilbert–Schmidt bounds.

By Lemma D.R $\rightarrow$ above, $M_L := \sup_{R \geq 1} \|V_R\|_{L^\infty([-L, L])} < \infty$

Let μ_n be Dirichlet eigenvalues of $-\partial_u^2$ on $[-L, L]$, so $\mu_n \asymp n^2$. **By the min–max principle applied** to the bounded perturbation V_R , there exist constants $c_L > 0$ and $C_L \geq 0$ (depending on L and M_L , but not on R) such that the eigenvalues $\lambda_n(R)$ of $H_{L,R}$ satisfy

$$\lambda_n(R) \geq c_L n^2 - C_L \quad (\forall n \geq 1, \forall R \geq 1)$$

Hence for $s \in K$,

$$\|R_{L,R}(s)\|_{S_2}^2 = \sum_{n \geq 1} \frac{1}{|\lambda_n^{(R)} + s|^2} \leq \sum_{n \geq 1} \frac{1}{(c_L n^2 - C_L')^2} =: C_{L,K}^2 < \infty,$$

where C_L' absorbs $\sup_{s \in K} |s|$. This yields

$\sup_{R \geq 1} \sup_{s \in K} \|R_{L,R}(s)\|_{S_2} \leq C_{L,K}$, and similarly $\sup_{s \in K} \|R_{L,\infty}(s)\|_{S_2} \leq C_{L,K}$. $R \geq 1$

Step 3: resolvent identity + trace norm estimate.

The resolvent identity gives (as you already write in outline form)

$$R_{L,R}(s) - R_{L,\infty}(s) = R_{L,R}(s) (V_\infty - V_R) R_{L,\infty}(s)$$

Taking S_1 norms and using $\|ABC\|_1 \leq \|A\|_2 \|B\|_\infty \|C\|_2$

$$\|R_{L,R}(s) - R_{L,\infty}(s)\|_{S_1} \leq \|R_{L,R}(s)\|_{S_2} \|V_\infty - V_R\|_{L^\infty([-L,L])} \|R_{L,\infty}(s)\|_{S_2}$$

Taking $\sup_{s \in K}$ and using the uniform S_2 bounds from Step 2,

$$\sup_{s \in K} \|R_{L,R}(s) - R_{L,\infty}(s)\|_{S_1} \leq C_{L,K}^2 \|V_\infty - V_R\|_{L^\infty([-L,L])} \rightarrow R \rightarrow \infty 0$$

by Lemma D. $R \rightarrow \infty$ (uniform potential convergence).

D.CANONICAL-FUNCTIONAL.

The load-bearing object exported by Appendix D is the canonical displacement/commutator trace transform

$$F_H^{can}(s) = L [\Theta_H^{can} - A_{class}^{can}](s).$$

Its one-letter sector is

$$P_{L,R}^{can}(t) = \sum_{\log q \leq R} w(q) g_{t,L}(\log q), \quad g_{t,L}(a) = \text{Tr}(S_t A_L T_a)$$

The raw local-loop kernel $J_{t,L,\alpha}$ computed in D.KERNEL-LOCAL belongs to the auxiliary multiplication-bump model and is not identified with $g_{t,L}$. No equality

$$J_{t,L,\alpha} = g_{t,L}$$

is asserted or used in the RH closure.

The object exported to Appendices E and F is F_H^{can} , not the raw multiplication-bump heat trace.

Proposition D.SPIKE-TRANSFER

Fixed-cutoff spike transfer; only the free leading kernel is exported

Fix $L > 0$ and $R < \infty$ Let

$$H_{L,R} = H_0 + V_{L,R}$$

be the fixed-cutoff localized operator on the L-window, where

$$H_0 = -\partial_u^2$$

is the free baseline and $V_{L,R}$ is the bounded fixed-cutoff perturbation. Let A_L be the one-sided localized observable fixed in Appendix D, and for $a \in \mathbb{R}$ define

$$g_{t,L,R}^{true}(a) := \text{Tr}(\Pi_L e^{-tH_{L,R}} \Pi_L A_L T_a)$$

and

$$g_{t,L}^0(a) := \text{Tr}(\Pi_L e^{-tH_0} \Pi_L A_L T_a)$$

Then, for every fixed integer $M \geq 1$, there exist fixed-cutoff local correction profiles

$$c_{1,L,R}(a), \dots, c_{M-1,L,R}(a)$$

and a remainder $E_{M,L,R}(t, a)$ such that, uniformly for $|a| \leq R$,

$$g_{t,L,R}^{true}(a) = g_{t,L}^0(a) + \sum_{j=1}^{M-1} t^j c_{j,L,R}(a) + E_{M,L,R}(t, a),$$

with

$$\sup_{|a| \leq R} |E_{M,L,R}(t, a)| \leq C_M(L, R) t^M, \quad 0 < t \leq t_0(L, R).$$

The leading free kernel $g_{t,L}^0(a)$ is the only part exported into the explicit one-letter prime-power package. The finite correction terms

$$\sum_{j=1}^{M-1} t^j c_{j,L,R}(a)$$

are fixed-cutoff local heat corrections. They are absorbed into $A_{class,L,R}$ or into the fixed-cutoff operator-side remainder. In particular, no finite correction term is part of the canonical prime-power package B_{can}

Proof

Since $V_{L,R}$ is bounded on the fixed window, the Duhamel expansion is valid in trace norm after localization by Lemma D.TC. Hence, for fixed L,R ,

$$e^{-tH_{L,R}} = e^{-tH_0} + \sum_{k=1}^{M-1} (-1)^k \int_{\Delta_k} (t) e^{-(t-s_k)H_0} V_{L,R} e^{-(s_k-s_{k-1})H_0} \dots V_{L,R} e^{-s_1 H_0} ds + E_M(t),$$

where the localized remainder satisfies

$$\|\Pi_L E_M(t) \Pi_L A_L T_a\|_{S_1} \leq C_M(L, R) t^M$$

uniformly for $|a| \leq R$ and $0 < t \leq t_0(L, R)$

Taking traces after composition with $\Pi_L A_L T_a$ the zeroth-order term gives

$$g_{t,L}^0(a) := \text{Tr}(\Pi_L e^{-tH_0} \Pi_L A_L T_a)$$

The finitely many terms with $1 \leq k \leq M-1$ define the local correction profiles $c_{j,L,R}(a)$ after collecting equal powers of t in the fixed-cutoff short-time expansion. These profiles depend on the fixed data $L, R, V_{L,R}, A_L$ but they do not alter the prime-power coefficient normalization and do not create a new canonical arithmetic package.

The remainder estimate follows from the trace-norm Duhamel remainder bound above. Thus

$$g_{t,L,R}^{true}(a) = g_{t,L}^0(a) + \sum_{j=1}^{M-1} t^j c_{j,L,R}(a) + O_{M,L,R}(t^M)$$

uniformly for $|a| \leq R$

By the bookkeeping convention of Appendix D, the leading free term $g_{t,L}^0$ supplies the explicit one-letter spike package, while the finite local correction terms are absorbed into $A_{class,L,R}$ or into the holomorphic operator-side remainder. Therefore D.SPIKE-TRANSFER contributes no additional pole-carrying package beyond the canonical one-letter prime-power package.

No $R \rightarrow \infty$ passage is taken here.

Reader note

This proposition is a fixed-(L,R) local transfer statement only. It does not assert

$$g_{t,L,R}^{true}(a) - g_{t,L}^0(a) = O(t^N) \text{ for every } N$$

without lower heat-correction terms. The R_H closure uses only the free leading spike kernel in the canonical package. All finite heat-kernel corrections are assigned to A_{class} or to the Ω -holomorphic operator-side remainder.

Theorem D.ONE-STEP" (fixed-(L,R) one-letter explicit package and residual sector decomposition).

Let

$$H_0 = -\partial_u^2, V_R(u) = \sum_{\log q \leq R} w(q) W_\alpha(u - \log q),$$

where $q = p^m$ runs over prime powers with $\log q \leq R$ and let $H_{L,R}$ and $\Theta_{L,R}(t)$ be the localized/regularized fixed-cutoff objects already fixed in Appendix D. Let

$$S_t := e^{-tH_0}, \quad g_{t,L}(a) := \text{Tr}(S_t A_L T_a),$$

where A_L is the one-letter boundary/localization insertion supplied by the earlier kernel lemmas.

Then there exists $t_0 = t_0(L, R) > 0$ such that for all $0 < t \leq t_0$,

$$\Theta_{L,R}(t) = A_{class,L}(t) + P_{L,R}(t) + E_{L,R}^{rep}(t) + R_{L,R}^{mix}(t),$$

where:

1. the explicit one-letter spike package is

$$P_{L,R}(t) = \sum_{\log q \leq R} w(q) g_{t,L}(\log q);$$

$E_{L,R}^{rep}(t)$ is the total repeated same-channel sector from Lemma D.DIAG-SUM, and contributes no new explicit spike locations beyond the one-letter package;

2. $R_{L,R}^{mix}(t)$ is the mixed-channel sector, and for every $N \geq 1$ there exists $C_N(L, R) < \infty$ such that

$$|R_{L,R}^{mix}(t)| \leq C_N(L, R) t^N (0 < t \leq t_0)$$

Proof.

Write the perturbation as the finite prime-power sum

$$V_R = \sum_{\log q \leq R} V_q, V_q(u) = w(q) W_\alpha(u - \log q).$$

By the fixed-cutoff trace-class legality package, the Duhamel expansion for the localized trace is valid in trace norm for fixed L and R, and the resulting sum over ordered words in the letters q is absolutely convergent for $0 < t \leq t_0(L, R)$

The load-bearing one-letter sector is the canonical displacement/commutator channel of D.CANONICAL-FUNCTIONAL. By D.SHIFT-CHANNEL / D.BRIDGE-II' and D.K0', this channel yields

$$P_{L,R}^{can}(t) = \sum_{\log q \leq R} w(q) g_{t,L}(\log q), \quad g_{t,L}(a) = \text{Tr}(S_t A_L T_a).$$

D.KERNEL-LOCAL computes only the auxiliary raw multiplication-bump loop profile $J_{t,L,\alpha}$. That profile is not the exported Bcan-package.

For $k \geq 2$, split the higher words into two classes.

If all letters agree, the word is a repeated same-channel word. By Lemma D.DIAG-SUM, these terms remain same-center local terms for the genuine shifted-bump perturbation, create no new explicit spike locations, and are collected into

$$E_{L,R}^{rep}(t)$$

If at least two channel labels differ, the word is mixed. Since $R < \infty$, the set of active centers is finite, so distinct centers have a positive minimum separation. The off-diagonal Gaussian bound for the free heat kernel, together with the fixed-cutoff localization and trace-class bounds already proved in Appendix D, implies that the total mixed sector is super-smoothing to all orders. This yields

$$R_{L,R}^{mix}(t) = O_N(t^N)$$

for every N .

Collecting the one-letter, repeated, and mixed sectors yields the stated decomposition.

Audit note for Theorem D.ONE-STEP" (source map for the fixed-(L,R) decomposition).

The proof of Theorem D.ONE-STEP" uses only previously established fixed-cutoff ingredients, with no cutoff removal and no appeal to the global Ω -shield.

1. Trace legality and regrouping.
The fixed-cutoff trace-class/Duhamel package justifies termwise tracing and regrouping of word contributions.
2. One-letter diagonal term.
The free spike kernel $g_{t,L}$ is identified by the earlier kernel computation.
3. True-kernel to free-leading-kernel extraction.

At fixed (L,R) , D.SPIKE-TRANSFER decomposes the true localized one-letter spike response into the free leading spike kernel, finitely many local heat-correction terms, and an arbitrarily high-order fixed-cutoff remainder. Only the free leading spike kernel is exported into the one-letter arithmetic package. The finite correction terms are absorbed into

$$A_{class,L,R}$$

or into the fixed-cutoff operator-side remainder.

4. Higher same-channel repeat words. Lemma D.DIAG-SUM shows that repeated single-channel words of length $k \geq 2$ remain same-center local terms for the genuine shifted-bump perturbation. They do not create new explicit spike locations and are retained in the fixed-cutoff operator-side remainder.

5. Mixed words.

The mixed-channel sector is super-smoothing to all orders.

Accordingly, Theorem D.ONE-STEP" is a purely fixed-(L,R) decomposition theorem whose only explicit arithmetic package is the one-letter spike package $P_{L,R}$.

Lemma D.K0' : Free kernel evaluation of the spike trace

Lemma D.K0' (Free kernel; shift–trace formula and Gaussian decay).

Let $H_0 = -\partial_u^2$ on $L^2(R)$, and let $S_t = e^{-tH_0} = e^{t\partial_u^2}$ for $t > 0$. Its integral kernel is

$$K_0(t; u, v) = (4\pi t)^{-1/2} \exp\left(-\frac{(u-v)^2}{4t}\right).$$

Fix $L > 0$ and a cutoff $\chi \in C_c^\infty(R)$ with $\chi \equiv 1$ on $[-1, 1]$, $\chi \equiv 0$ on $|x| \geq 2$. Define the localization operator

$$(\Pi_L f)(u) = \chi(u/L) f(u).$$

Let $w \in L^\infty(R)$ (typically supported where Π_L is nonzero) and define

$$A_L := \Pi_L M_w \Pi_L, (M_w f)(u) = w(u) f(u).$$

Let T_a denote translation:

$$(T_a f)(u) = f(u - a).$$

For $t > 0$ and $a \in R$ define the spike kernel trace

$$g_{t,L}(a) := \text{Tr}(S_t A_L T_a).$$

Then $G_t A_L T_a$ is trace class and

$$g_{t,L}(a) = \int_R K_0(t; u, u - a) w(u - a) \chi(u/L) \chi((u - a)/L) du. \quad (D.K0'.1)$$

In particular,

$$|g_{t,L}(a)| \leq (4\pi t)^{-1/2} \|w\|_{\infty} e^{-a^2/(4t)} \int R \chi(u/L) \chi((u-a)/L) du \lesssim L t^{-1/2} e^{-a^2/(4t)} \quad (\text{D.K0'.2})$$

If instead $\Pi_L = 1_{[-L,L]}$ and $w \equiv 1$ on $[-L, L]$ then

$$g_{t,L}(a) = (4\pi t)^{-1/2} e^{-a^2/(4t)} (2L - |a|)_+ \quad (\text{D.K0'.3})$$

Proof (clean kernel proof)

1. Trace class.

$\Pi_L S_t$ and $S_t \Pi_L$ are Hilbert–Schmidt for each $t > 0$ (**standard heat kernel estimate** under compact localization), A_L **is bounded**, and T_a **is unitary**; hence $S_t A_L T_a \in S_1$

2. Kernel computation.

The kernel of T_a **is** $T_a(u, v) = \delta(u - v - a)$. The kernel **of** $A_L T_a$ is therefore

$$(A_L T_a)(u, v) = \chi(u/L) w(u) \chi(v/L) \delta(u - v - a)$$

Thus the diagonal kernel of $S_t A_L T_a$ **is**

$$(G_t A_L T_a)(u, u) = \int_R K_0(t; u, x) (A_L T_a)(x, u) dx = K_0(t; u, u - a) w(u - a) \chi(u/L) \chi((u - a)/L)$$

Integrate over u to get (D.K0'.1). Bounds (D.K0'.2)–(D.K0'.3) follow immediately using the explicit Gaussian form of K_0 and the overlap length of the cutoff region.

Remark: “Push translations through the free kernels; the remaining aa -dependence occurs only through the shifted cutoff $T_{-a} \Pi_L T_a$. Apply Lemma D.MISALIGN to replace cutoff misalignment by a boundary insertion A_L , yielding the spike kernel $g_{t,L}(a)$.”

Remark D.BULK (explicit inoculation against the “flat trace” test)

A symmetric bulk trace such as $Tr(\Pi_L G M_w(a) G \Pi_L)$ is essentially independent of a when the support is away from the boundary (translation symmetry + cyclicity). Our spike extraction therefore uses the one-sided operator A_L and the generator insertion from Lemma D.DIFF, which makes the diagonal channel displacement-sensitive and yields the spike kernel

$$G_t(a) = Tr(G_t \Pi_L T_a)$$

Lemma D.L NORM (Window normalization of the spike kernel)

Fix $t \in (0, t_0] 1_{[-L, L]}$, let $w \in L^\infty(R)$ be supported in $[-L, L]$, and define $A_L := \Pi_L M_w \Pi_L$

Let $G_t = e^{-tH_0}$ with $H_0 = -\partial_u^2$ and define

$$g_{t,L}(a) := Tr(G_t A_L T_a).$$

Given that $w = w_L$ is chosen so that $\frac{1}{2L} \int_{-L}^L w(u) du \rightarrow c_w$

as $L \rightarrow \infty$. Then for each fixed $R > 0$,

$$\sup_{|a| \leq R} \left| \frac{1}{2L} \widehat{g_{t,L}}(a) - c_w G_t(a) \right| \rightarrow 0 (L \rightarrow \infty),$$

where $G_t(a) = (4\pi t)^{-1/2} e^{-a^2/(4t)}$ is the free 1D heat kernel evaluated at displacement a .

Proof. Using the free kernel $K_t(u, v) = (4\pi t)^{-1/2} e^{-(u-v)^2/(4t)}$

$$g_{t,L}(a) = \int_R K_t(u, u-a) (\Pi_L M_w \Pi_L)(u-a) du = \int_{-L}^L K_t(u, u-a) w(u-a) 1_{[-L, L]}(u-a) du$$

Thus for $|a| \leq R$ and $L \geq R$, one has

$$g_{t,L}(a) = S_t(a) \int_{-L}^L w(u-a) 1_{[-L, L]}(u-a) du = S_t(a) \int_{-L-a}^{L-a} w(v) 1_{[-L, L]}(v) dv$$

▪

Hence

$$\frac{1}{2L} \widehat{g_{t,L}}(a) = G_t(a) \cdot \frac{1}{2L} \int_{-L}^L w(v) dv + G_t(a) \cdot \frac{1}{2L} \int_{[-L,L] \Delta [-L-a, L-a]} w(v) dv$$

The symmetric difference has measure $\leq 2|a| \leq 2R$, so the second term is bounded by

$$|G_t(a)| \cdot \frac{1}{2L} \cdot (2R) \|w\|_{\infty} \leq \frac{R}{L} \|w\|_{\infty} \sup_{|a| \leq R}$$

This tends to 0 uniformly in $|a| \leq R$. The first term tends to $c_w S_t(a)$ uniformly on $|a| \leq R$ because $G_t(a)$ is bounded there and $\frac{1}{2L} \int_{-L}^L w \rightarrow c_w$ by hypothesis.

Remark (order of limits). In the proof engine, keep L fixed while extracting spikes and passing $R \rightarrow \infty$ the level of trace identities; then apply Lemma D.LNORM to pass $L \rightarrow \infty$ on compact $|a| \leq R$, yielding the EF Gaussian kernel $G_t(a)$.

Finite-volume exhaustion at fixed R.

In the fixed-R analysis, the finite-volume Dirichlet operators $H_{L,R}$ serve only as auxiliary regularizations. The following lemma records the $L \rightarrow \infty$ exhaustion step for the heat-trace functionals used below.

Lemma D.FV-LIMIT (finite-volume Dirichlet exhaustion at fixed R).

Fix $R < \infty$. Let

$$H_R$$

denote the whole-line truncated PWQO operator, and let

$$H_{L,R}$$

be its Dirichlet restriction to $[-L, L]$ Then for each fixed $t > 0$,

$$\text{Tr}(e^{-tH_{L,R}}) \rightarrow \text{Tr}(e^{-tH_R})(L \rightarrow \infty)$$

The same $L \rightarrow \infty$ passage holds for the finite-volume localized / regularized heat-trace functionals used in Appendix D.

Proof.

Since H_R is confining, it has compact resolvent and purely discrete spectrum. Let

$$\{\lambda_n(R)\}_{n \geq 1}, \{\lambda_n(L, R)\}_{n \geq 1}$$

be the eigenvalues of H_R and $H_{L,R}$ respectively, listed in nondecreasing order with multiplicity.

By Dirichlet bracketing and the min-max principle,

$$\lambda_n(L, R) \geq \lambda_n(R) \quad (n \geq 1)$$

Hence, for every fixed $t > 0$,

$$e^{-t\lambda_n(L,R)} \leq e^{-t\lambda_n(R)}.$$

Because H_R has compact resolvent,

$$\sum_{n \geq 1} e^{-t\lambda_n(R)} = \text{Tr}(e^{-tH_R}) < \infty.$$

Thus the whole-line truncated heat trace furnishes a summable dominating sequence.

As $L \rightarrow \infty$, the Dirichlet restrictions exhaust the whole line, and standard domain exhaustion for confining Schrödinger operators gives

$$\lambda_n(L, R) \rightarrow \lambda_n(R) \text{ for each fixed } n.$$

Therefore, by dominated convergence,

$$\text{Tr}(e^{-tH_{L,R}}) = \sum_{n \geq 1} e^{-t\lambda_n(L,R)} \rightarrow \sum_{n \geq 1} e^{-t\lambda_n(R)} = \text{Tr}(e^{-tH_R}).$$

The localized / regularized finite-volume functionals appearing in Appendix D are obtained from these heat traces by the fixed bounded localization and subtraction operations already introduced there. Hence the same cv passage holds for those D-side objects as well. By Lemma D.FV-LIMIT, the $L \rightarrow \infty$ passage is valid for the finite-volume heat-trace functionals at fixed R

Lemma D.DIAG-SUM (same-channel repeated words remain same-center local terms).

Fix $L > 0$ $R > 0$, and $0 < t \leq t_0(L, R)$.

For a prime power q with

$$a_q := \log q, V_q = w(q) T_{a_q} M_{W_\alpha} T_{a_q},$$

let $D_q^{(k)}$ denote the k -fold same-channel Duhamel word ($k \geq 2$), i.e. the contribution in which the letter V_q appears k times and no other channel is used.

Then:

1. there exist fixed-cutoff local trace profiles $J_{q,k,L}(t)$ such that

$$D_q^{(k)}(t) = w(q)^k J_{q,k,L}(t),$$

with the dependence on the channel entering through the same center a_q and **no new displacement** ka_q being created;

2. consequently, the repeated same-channel sector contributes **no new explicit spike locations** beyond the one-letter package indexed by $a_q = \log q$
3. for each fixed (L, R) , the total repeated same-channel contribution

$$E_{L,R}^{rep}(t) := \sum_{\log q \leq R} \sum_{k \geq 2} D_q^{(k)}(t)$$

is well-defined on $(0, t_0(L, R)]$ by the fixed-cutoff trace-class/Duhamel convergence, and is included in the fixed-cutoff operator-side remainder rather than in the explicit spike package.

Proof

Write

$$V_q = w(q) T_{a_q} M_{W_\alpha} T_{a_q}$$

Since the free/background heat factors commute with translations, every same-channel word with repeated letter V_q is conjugate to a centered local word at the same center a_q . Thus repeated insertions of the same shifted bump remain tied to that same center and do not generate a new displacement ka_q . This proves the first two assertions.

For fixed (L, R) , there are only finitely many prime powers with $\log q \leq R$, and the fixed-cutoff Duhamel expansion is absolutely convergent in trace norm by the fixed-cutoff legality package.

Hence the double sum defining $E_{L,R}^{rep}$ is well-defined. By construction these terms are not promoted to the explicit one-letter spike package and are retained inside the operator-side remainder.

Lemma D.RCONV (Uniform trace-norm resolvent convergence on Ω -compacts).

Fix $L > 0$.

Let

$$H_{L,R} = -\partial_u^2 + V_R(u) \text{ on } L^2([-L, L])$$

with Dirichlet boundary conditions at $\pm L$, where each $V_R \in L^\infty([-L, L])$ is real-valued and

$$V_R \rightarrow V_\infty \text{ uniformly on } [-L, L].$$

Let

$$H_{L,\infty} = -\partial_u^2 + V_\infty(u)$$

with the same boundary conditions.

Let

$$\Omega = \mathbb{C} \setminus (-\infty, 0]$$

and for $s \in \Omega$ define the resolvents

$$R_{L,R}(s) := (H_{L,R} + s)^{-1}, R_{L,\infty}(s) := (H_{L,\infty} + s)^{-1}$$

Then for every compact set $K \subset \Omega$,

$$\sup_{s \in K} \|R_{L,R}(s) - R_{L,\infty}(s)\|_{S_1} \rightarrow 0 \text{ as } R \rightarrow \infty,$$

where $\|\cdot\|_{S_1}$ denotes the trace-class norm.

Proof

We proceed in four steps.

Step 1. Resolvent identity

For every $s \in \Omega$ the resolvent identity gives

$$R_{L,R}(s) - R_{L,\infty}(s) = R_{L,R}(s) (V_\infty - V_R) R_{L,\infty}(s) \quad (1)$$

This identity holds in operator norm on $L^2([-L, L])$.

Step 2. Schatten factorization estimate

We use the Schatten inequality

$$\|ABC\|_{S_1} \leq \|A\|_{S_2} \|B\|_{L^\infty} \|C\|_{S_2} \quad (2)$$

Applying (2) to (1),

$$\|R_{L,R}(s) - R_{L,\infty}(s)\|_{S_1} \leq \|R_{L,R}(s)\|_{S_2} \|V_\infty - V_R\|_{L^\infty([-L, L])} \|R_{L,\infty}(s)\|_{S_2} \quad (3)$$

Thus everything reduces to:

- uniform Hilbert–Schmidt bounds for resolvents on K ,
 - uniform L^∞ convergence of V_R
-

Step 3. Uniform Hilbert–Schmidt bounds on Ω -compacts

Since $H_{L,R}$ is a Dirichlet Schrödinger operator on a finite interval, it has compact resolvent and discrete spectrum

$$0 \leq \lambda_1, R \leq \lambda_2, R \leq \dots \rightarrow \infty$$

By the min–max principle and boundedness of V_R , there exist constants $L > 0, C_L$ independent of R such that

$$\lambda_{n,R} \geq c_L n^2 - C_L \quad (4)$$

Fix a compact $K \subset \Omega$.

Since K stays a positive distance from the cut,

$$\delta_K := \inf_{s \in K} \text{dist}(-s, [0, \infty)) > 0 \quad (5)$$

Also set $S_K := \sup_{s \in K} |s|$

We compute

$$\|R_{L,R}(s)\|_{S_2}^2 = \sum_{n \geq 1} \frac{1}{|\lambda_{n,R} + s|^2} \quad (6)$$

Split into small and large n.

For large n, by (4),

$$|\lambda_{n,R} + s| \geq \lambda_{n,R} - |s| \geq \frac{1}{2} c_L n^2 \quad (n \geq n_0(K))$$

Thus

$$\sum_{n \geq n_0} \frac{1}{|\lambda_{n,R} + s|^2} \leq C \sum_{n \geq n_0} \frac{1}{n^4} < \infty$$

For finitely many small n, we use (5):

$$|\lambda_{n,R} + s| \geq \delta_K.$$

Hence

$$\sum_{n < n_0} \frac{1}{|\lambda_{n,R} + s|^2} \leq \frac{n_0}{\delta_K^2}$$

Combining both parts gives

$$\sup_R \sup_{s \in K} \|R_{L,R}(s)\|_{S_2} \leq C_{L,K} < \infty \quad (7)$$

The same estimate holds for $R_{L,\infty}(s)$

Step 4. Convergence

Insert (7) into (3):

$$\sup_{s \in K} \|R_{L,R}(s) - R_{L,\infty}(s)\|_{S_1} \leq C_{L,K}^2 \|V_\infty - V_R\|_{L^\infty([-L,L])}$$

By uniform convergence of potentials,

$$\|V_\infty - V_R\|_{L^\infty([-L,L])} \rightarrow 0$$

Therefore

$$\sup_{s \in K} \|R_{L,R}(s) - R_{L,\infty}(s)\|_{S_1} \rightarrow 0$$

This completes the proof.

Proposition D.LEGAL-S (Half-plane legality and prime-power expansion for $F'(s)$) (Non-Loadbearing).

Fix $\sigma_1 > 1/2$ set $U = \{s \in \Omega: \Re(s) \geq \sigma_1\}$ and define

$$F(s) = \text{Tr}((H + s)^{-1} - (H_0 + s)^{-1}).$$

Then:

1. *(Holomorphy and differentiation under trace.)*

$F \in O(\Omega)$ and for every $s \in U$,

with $(H + s)^{-2} - (H_0 + s)^{-2} \in S_1$ on U .

2. *(Prime-power expansion on U .)*

Proposition D.SPIKE-EXTRACTION (diagonal spike package) together with Lemma B.MIX (mixed terms are super-smoothing) and Lemma D.DIAG-SUM (higher-order diagonal repeats are absorbed into a holomorphic remainder), the only non-bulk contribution to $F'(s)$ on U is the

explicit prime-power package with coefficient $w(q) = \frac{\Lambda(q)}{\sqrt{q}}$

For every $s \in U$,

$$F'(s) = - \sum_{q \geq 2} \frac{\Lambda(q)}{q^{1/2+s}} + A'(s),$$

where $A \in O(\Omega)$ is the explicit archimedean/trivial term. The series converges absolutely and uniformly on U . (see (D.A1 (+ D.A2))

By lemma D.SPIKE-EXTRACTION,

Normalization note: $w(q) = \frac{\Lambda(q)}{\sqrt{q}}$ and $w(q) q^{-s} = \frac{\Lambda(q)}{q^{-(1/2+s)}}$

Proposition D.5.2 (Norm-resolvent / trace convergence bridge on Ω)

Let $L > 0$ be fixed. Let $\Omega := \mathbb{C} \setminus (-\infty, 0]$. For each $R \in (0, \infty]$, let $H_{L,R}$ be a self-adjoint operator on a finite-volume domain (e.g. Dirichlet Schrödinger operator on $[-L, L]$) with

$$\sigma(H_{L,R}) \subset [0, \infty), (H_{L,R} + 1)^{-1} \text{ compact}$$

Given the R -dependence is through a bounded potential perturbation $V_R \in L^2([-L, L])$ with a limit $V_\infty \in L^2([-L, L])$ such that

$$\|V_\infty - V_R\|_{L^\infty([-L, L])} \xrightarrow{R \rightarrow \infty} 0$$

Define the resolvent-trace observable

$$F_{L,R}(s) := \text{Tr}((H_{L,R} + s)^{-1}), s \in \Omega$$

.

Then:

1. For each fixed $R \leq \infty$, the map $s \mapsto F_{L,R}(s)$ is holomorphic on Ω . For every compact $K \Subset \Omega$,

$$\sup_{s \in K} \| (H_{L,R} + s)^{-1} - (H_{L,\infty} + s)^{-1} \|_1 \xrightarrow{R \rightarrow \infty} 0,$$

and hence

$$\sup_{s \in K} |F_{L,R}(s) - F_{L,\infty}(s)| \xrightarrow{R \rightarrow \infty} 0.$$

In particular, $F_{L,R} \rightarrow F_{L,\infty}$ locally uniformly on Ω , and $F_{L,\infty} \in O(\Omega)$

Proof

Fix a compact $K \Subset \Omega$

Since $\sigma(H_{L,R}) \subset [0, \infty)$ for all R , we have $-s \notin [0, \infty)$ for all $s \in \Omega$, so the resolvents $(H_{L,R} + s)^{-1}$ exist for all $s \in \Omega$.

Step 1: Resolvent identity.

For each $s \in \Omega$,

$$(H_{L,R} + s)^{-1} - (H_{L,\infty} + s)^{-1} = (H_{L,R} + s)^{-1} (V_\infty - V_R) (H_{L,\infty} + s)^{-1}$$

This is exactly the identity you cite.

Step 2: Trace-ideal factorization.

Using the Schatten inequality $\|ABC\|_1 \leq \|A\|_2 \|B\|_\infty \|C\|_2$, we obtain

$$\|(H_{L,R} + s)^{-1} - (H_{L,\infty} + s)^{-1}\|_1 \leq \|(H_{L,R} + s)^{-1}\|_2 \|V_\infty - V_R\|_\infty \|(H_{L,\infty} + s)^{-1}\|_2$$

Step 3: Uniform Hilbert–Schmidt bounds on K .

Because the domain is finite-volume and the resolvent is compact, $H_{L,R}$ has discrete eigenvalues $0 \leq \lambda_{1,R} \leq \lambda_{2,R} \leq \dots \rightarrow \infty$. For each $s \in K$,

$$\|(H_{L,R} + s)^{-1}\|_2^2 = \sum_{n \geq 1} \frac{1}{|\lambda_{n,R} + s|^2}$$

.

Concretely, in the Dirichlet model with bounded nonnegative potentials, one may use

$\lambda_{n,R} \geq \lambda_{n,0} \sim c(L)n^2$ to dominate the tail uniformly in R , and since K is compact away from the cut, $|\lambda_{n,R} + s| \gtrsim \lambda_{n,R} + 1$ uniformly for $s \in K$. This yields

$$\sup_{R \geq R_0} \sup_{s \in K} \|(H_{L,R} + s)^{-1}\|_2 \leq C(L, K) < \infty,$$

and similarly for $(H_{L,\infty} + s)^{-1}$.

Step 4: Conclude trace-norm convergence.

Combining Step 2 with Step 3 gives

$$\sup_{s \in K} \| (H_{L,R} + s)^{-1} - (H_{L,\infty} + s)^{-1} \|_1 \leq C(L, K)^2 \|V_\infty - V_R\|_\infty,$$

and the $RHS \rightarrow 0$ as $R \rightarrow \infty$ by hypothesis.

Step 5: Conclude local uniform convergence of traces and holomorphy of the limit.

Since $|Tr(T)| \leq \|T\|_1$, we get $\sup_{s \in K} |F_{L,R}(s) - F_{L,\infty}(s)| \rightarrow 0$

Because each $F_{L,R}$ is holomorphic on Ω (Lemma D-A.1 / D.5.2), local uniform convergence implies $F_{L,\infty} \in O(\Omega)$

.

Remark (localized/window model).

The same argument works for the window-localized model by inserting localization operators Π_L and using HS factorization of localized resolvents.

Lemma D.LIMIT (normal convergence of the spike package as $R \rightarrow \infty$, time-domain).

Fix $t_0 > 0$. For each $t \in (0, t_0]$, define the truncated spike sum

$$S_R(t) := \sum_{\log q \leq R} \frac{\Lambda(q)}{\sqrt{q}} g_t(\log q),$$

where $g_t(a)$ satisfies the Gaussian bound

$$|g_t(a)| \leq C t^{-1/2} \exp\left(-\frac{a^2}{4t}\right) (t \in (0, t_0], a \in \mathbb{R}). \quad (D.LIM1)$$

Then $S_R(t)$ converges as $R \rightarrow \infty$ for each $t \in (0, t_0]$, and the convergence is locally uniform in t on $(0, t_0]$. Moreover, the limit $S(t) = \lim_{R \rightarrow \infty} S_R(t)$ is C^∞ on $(0, t_0]$ and derivatives may be passed termwise:

$$\partial_t^k S(t) = \sum_q \frac{\Lambda(q)}{\sqrt{q}} \partial_t^k g_t(\log q), k \geq 0.$$

Justification. The Gaussian bound (D.LIM1) implies super-polynomial decay in q at fixed t : for each $t \in (0, t_0]$

$$|\frac{\Lambda(q)}{\sqrt{q}} g_t(\log q)| \ll \frac{\Lambda(q)}{\sqrt{q}} t^{-1/2} \exp(-\frac{(\log q)^2}{4t_0}),$$

and the right-hand side is absolutely summable over prime powers $q = p^m$ because $p^{-m/2}$ is geometric in m and the factor $\exp(-(\log p^m)^2/(4t_0))$ provides super-exponential decay in p . Uniformity in t on compact subintervals of $(0, t_0]$ follows similarly. Smoothness and termwise differentiation follow by dominated convergence using the corresponding derivative bounds on $\partial_t^k g_t(a)$, which have the same Gaussian-in-a structure (proved earlier in Appendix D/G).

Lemma D.MAJORANTS $\Omega(K)$ (Ω -compact majorants for Laplace transforms of the spike package).

Fix a compact set $K \subseteq \Omega$ and set $\sigma_K := \sup_{s \in K} \Re(s)$. Define the truncated Laplace transform of the spike package by

$$B_R(s) := \int_0^{t_0} e^{-st} S_R(t) dt = \sum_{\log q \leq R} \frac{\Lambda(q)}{\sqrt{q}} \int_0^{t_0} e^{-st} g_t(\log q) dt, \quad s \in K.$$

Then $B_R(s)$ converges uniformly on K as $R \rightarrow \infty$ to a holomorphic limit $B(s)$ on Ω . Moreover, the interchange of sum and integral is justified:

$$B(s) = \sum_q \frac{\Lambda(q)}{\sqrt{q}} \int_0^{t_0} e^{-st} g_t(\log q) dt = \int_0^{t_0} e^{-st} \left(\sum_q \frac{\Lambda(q)}{\sqrt{q}} g_t(\log q) \right) dt, \quad s \in K.$$

Justification (explicit majorant). For $s \in K$ and $t \in (0, t_0]$,

$$|e^{-st}| \leq e^{\sigma_K t} \leq e^{\sigma_K t_0}. \quad (D.M\Omega 1)$$

By the Gaussian bound (D.LIM1),

$$|g_t(\log q)| \leq C t^{-1/2} \exp(-\frac{(\log q)^2}{4t}) \leq C t^{-1/2} \exp(-\frac{(\log q)^2}{4t_0}). \quad (D.M\Omega 2)$$

Hence for all $s \in K$,

$$|\frac{\Lambda(q)}{\sqrt{q}} \int_0^{t_0} e^{-st} g_t(\log q) dt| \leq \frac{\Lambda(q)}{\sqrt{q}} \int_0^{t_0} e^{\sigma_K t} C t^{-1/2} \exp(-\frac{(\log q)^2}{4t_0}) dt = C_K \frac{\Lambda(q)}{\sqrt{q}} \exp(-\frac{(\log q)^2}{4t_0}) \quad (\text{D.M}\Omega 3)$$

where $C_K := e^{\sigma_K t_0} C \int_0^{t_0} t^{-1/2} dt < \infty$. The majorant on the right is summable over prime powers $q = p^m$: the factor $p^{-m/2}$ is geometric in m and $\exp(-\frac{(\log q)^2}{4t_0})$ gives super-exponential decay in p . Therefore the Weierstrass M-test implies uniform convergence of the q -series on K , justifying the sum–integral interchange and giving normal convergence on Ω -compacta. Holomorphy of the limit follows because each truncated $B_R(s)$ is holomorphic in s and convergence is uniform on compacta.

Scope of D.MAJORANTS Ω

The normal convergence proved in D.MAJORANTS Ω concerns the short-time Gaussian transform over the finite interval

$$(0, t_0]$$

It uses the super-Gaussian decay in $\log q \log q$ coming from the heat kernel.

It is not the unrestricted full prime-side Laplace series on all of Ω .

The full Laplace-side prime series is governed by D.NORM Ω -C and has only the restricted normal-convergence region where

$$c_K > \frac{1}{2}$$

The global Ω -closure does not rely on unrestricted prime-side normal convergence. It proceeds through the zero-side meromorphic package and the meromorphic identity theorem.

Remark (*tail* $t \geq t_0$). If the full Laplace transform integrates over $(0, \infty)$ the tail $\int_{t_0}^{\infty}$ is handled separately using the eigenvalue growth / trace-class decay package (Appendix A/D), yielding a holomorphic tail on Ω ; the majorant argument above treats only the spike package on $(0, t_0]$

Appendix D.NORM — Global Normalization and Conventions

This appendix fixes, once and for all, the normalization conventions used throughout the manuscript. No other scaling is used anywhere in Appendices D, E, F, or H.

Global normalization firewall

Throughout Sections 5 and Appendices D, E, F, and H, every arithmetic spike package is indexed by prime powers

$$q = p^m, m \geq 1,$$

with center

$$u_q = \log q = m \log p$$

and frozen spike weight

$$w(q) = \frac{\Lambda(q)}{\sqrt{q}} = (\log p) p^{-m/2}.$$

This is the only arithmetic normalization used in the RH closure spine. In particular, no step in the D/E/F/H argument uses $\frac{\Lambda(q)}{\sqrt{q}}$ as the spike coefficient. Any occurrence of $\frac{\Lambda(q)}{\sqrt{q}}$ in the RH spine is to be read as a typographical or extraction error unless it is immediately rewritten in the prime-power form $(\log p) p^{-m/2}$.

The canonical packages are therefore

$$B_{raw,R}(t) = \sum_{\log q \leq R} \frac{\Lambda(q)}{\sqrt{q}} G_t(\log q),$$

and

$$B_{can,R}(t) = e^{-t/4} B_{raw,R}(t) = \sum_{\log q \leq R} \frac{\Lambda(q)}{\sqrt{q}} K_t(\log q), \quad K_t(u) = e^{-t/4} G_t(u).$$

The raw G_t -package and shifted K_t -package have different Laplace-side pole geometry.
The RH closure uses only the shifted package B_{can}

1. Prime-power indexing and weight

All arithmetic spike sums are indexed by **prime powers**

$$q = p^m (m \geq 1),$$

and we write

$$u_q = \log q = m \log p.$$

The global spike weight is

$$w(q) = \frac{\Lambda(q)}{\sqrt{q}}$$

Equivalently, in prime-power coordinates,

$$w(p^m) = (\log p) \cdot \{p^{\{-m/2\}}\}$$

These two forms are mathematically identical, and either one may be used as a consistency check on the other.

There is **no alternative scaling anywhere in the proof**. In particular, the manuscript does **not** use $\Lambda(q)/q$ as the spike coefficient. Any appearance of $\Lambda(q)/q$ in older drafts or mis-rendered extraction text must be read as a typographical artifact. The authoritative normalization is $\frac{\Lambda(q)}{\sqrt{q}}$ confirmed by the prime-power form $(\log p) \cdot p^{\{-m/2\}}$.

2. Gaussian test kernel

The scalar test kernel used in spike extraction and in the explicit formula comparison is the standard one-dimensional heat kernel in the displacement variable:

$$G_t(u) = (4\pi t)^{\{-1/2\}} \exp^{(-u^2 / (4t))}, \text{ for } t > 0.$$

This Gaussian family is used consistently on:

- the operator-side diagonal spike extraction (Appendix D), and

- the explicit-formula side (Appendix H).

No alternate normalization of the Gaussian is used in the R_H closure argument.

3. Laplace transform convention and the slit plane

All Laplace transforms are taken in the complex parameter s on the slit plane

$$\Omega = \mathbb{C} \setminus (-\infty, 0],$$

with the principal branch of $\sqrt{s}$.

The Laplace transform of the Gaussian kernel satisfies

$$\int_0^\infty e^{\{-st\}} G_t(u) dt = (1 / (2\sqrt{s})) e^{\{-|u|\sqrt{s}\}},$$

valid for s in Ω .

This is the transform identity used when passing from the heat-trace spike representation to the resolvent-type expression in the s -domain.

4. Separation of operator objects and scalar kernels

We distinguish strictly between:

- The operator semigroup $e^{\{-tH_0\}}$, acting on $L^2(\mathbb{R})$,
- The scalar Gaussian $G_t(u)$, used as a test kernel in displacement space.

No identification between operator objects and scalar Gaussian expressions is made unless explicitly justified via the integral kernel representation.

Lemma D.NORM Ω -B (Uniform majorant for the spike kernel on Ω -compacts)

Statement. Let $K \in \Omega$. Let $g^\wedge(s; a)$ denote the Laplace-side spike kernel used in the explicit package (the transform of the displacement Gaussian), so that on $\Re(s) > 0$ it has the form

$$\widehat{g}(s; a) = \frac{1}{2\sqrt{s}} e^{-|a|\sqrt{s}} (\text{principal branch})$$

Then there exists a constant $c_K > 0$ and $c_K > 0$ such that for all $s \in K$ and all $a \geq 0$,

$$|\widehat{g}(s; a)| \leq c_K e^{-c_K a}$$

Proof. By Lemma D.NORMΩ-A, $\Re \sqrt{s} \geq c_K$ on K . Also s is continuous and nonvanishing on K , so $|1/s| \leq c'_K$. Hence

$$|\widehat{g}(s; a)| \leq \frac{c'_K}{2} 2e^{-a\Re \sqrt{s}} \leq c_K e^{-c_K a}$$

.

Corollary D.NORMΩ-C

Let $\Omega = \mathbb{C} \setminus (-\infty, 0]$ and let $\sqrt{s}$ denote the principal square root on Ω . For a compact set $K \subset \Omega$ define

$$c_K := \inf_{s \in K} \Re(\sqrt{s})$$

Then $c_K > 0$ because K stays a positive distance away from the cut.

Lemma (kernel decay on compact sets).

For each compact $K \subset \Omega$ there is a constant c_K such that for all $s \in K$ and all $a \geq 0$,

$$|\widetilde{g}(s, a)| \leq c_K e^{-c_K a}.$$

In particular, for $a = \log q$,

$$|\widetilde{g}(s, \log q)| \leq c_K q^{-c_K}.$$

The restricted convergence result above reflects the classical Dirichlet threshold $1/2 + c_K > 1$. In the RH closure, we do not rely on unrestricted prime-side convergence on Ω , but instead on meromorphic continuation through the zero–pole representation.”

Corollary D.NORM Ω -C (restricted normal convergence of the prime-side spike Laplace series).

Define the truncated spike Laplace sum by

$$B_R^{spike}((s)) := \sum_{\log q \leq R} \frac{\Lambda(q)}{\sqrt{q}} g^\sim(s, \log q).$$

Let $K \subset \Omega$ be compact and assume that $c_K > 1/2$. Then $B_R^{\{spike\}}(s)$ converges uniformly on K as $R \rightarrow \infty$, and the limit defines a holomorphic function on the open region

$$\Omega_{>1/2} := \{s \in \Omega: \Re(\sqrt{s}) > 1/2\}$$

Proof.

For $s \in K$ we have

$$|\frac{\Lambda(q)}{\sqrt{q}} g^\sim(s, \log q)| \leq \frac{\Lambda(q)}{q^{1/2+c_K}}$$

If $c_K > 1/2$ then $1/2 + c_K > 1$, and the comparison series

$$\sum_{q \geq 2} \frac{\Lambda(q)}{q^{1/2+c_K}}$$

converges. By the Weierstrass M-test, the series defining $B_R^{\{spike\}}(s)$ converges uniformly on K . Uniform convergence of holomorphic terms implies holomorphy of the limit on $\Omega_{\{>1/2\}}$

Remark (why this is not used to continue to all of Ω).

On a general compact $K \subset \Omega$ one only knows $c_K > 0$, not $c_K > 1/2$. Thus the infinite prime-side spike Laplace series need not converge normally on compact sets near the cut. For the global Ω -interface we therefore do not use prime-side Ω -compact normal convergence; we continue to Ω through the zero-side meromorphic package and the meromorphic identity theorem (see the interface continuation step below).

Step 2 — Theorem D.FREEZE (One-step reduction)

Theorem D.FREEZE (One-step short-time replacement / freezing lemma).

Fix $M \in \mathbb{N}$. Let G_t denote the Gaussian cutoff (as normalized in Appendix H), and let the localized operator factors A_L, T_u, U_p be as defined in §D.* (localizer, translation, and prime-dilation channel). For each fixed word length k (with) $k \leq k(M)$ as specified in the diagonal-word expansion), the diagonal contribution to the k -fold Duhamel/word expansion admits the short-time replacement

$$\text{Tr}(S_t W_{k,L}(t)) = \text{Tr}(S_t W_{k,L}^{fr}(t)) + O(t^M)(t \downarrow 0),$$

uniformly in the localization parameters in the regime $(L, R) \in RM$, where $W_{k,L}^{fr}$ is the “frozen” diagonal model word obtained by replacing each conjugated/local factor by its diagonal frozen representative (as defined immediately below). The implied constant depends on M and the admissible regime, but not on t .

Proposition D.SPIKE-EXTRACTION

Fix $L < \infty$ and a truncation $R < \infty$. Let $H_0 = -\partial_u^2$ on $L^2(R)$, $S_t = e^{-tH_0}$, and $T_a f(u) = f(u - a)$. Let Π_L be a cutoff/localizer and $A_L = \Pi_L M_w \Pi_L$ with bounded w . Define the spike kernel

$$g_{t,L}(a) := \text{Tr}(S_t A_L T_a).$$

(Explicit kernel formula and Gaussian decay are available.)

Let the truncated prime-power package be indexed by prime powers q with $a_q = \log q \leq R$, and weights $w(q) = \Lambda(q)/\sqrt{q}$

Then the localized/regularized trace functional $\Theta_{L,R}(t)$ admits, for $t \in (0, t_0)$

$$\Theta_{L,R}(t) = A_{\text{class},L}(t) + \sum_{\log q \leq R} \frac{\Lambda(q)}{\sqrt{q}} g_{t,L}(\log q) + R_{L,R}(t)$$

and the remainder is super-smoothing:

$$\forall N \geq 1, \quad |R_{L,R}(t)| \leq C_N(L, R) t^N \quad (t \downarrow 0).$$

Step 3:

For the head, reduce each k-term to a spike kernel $g_t(ku_p)$ plus $O(t^M)$

(OS_M) One-step short-time reduction.

For each fixed $k \in \{1, \dots, K - 1\}$ there exists a scalar $\beta_{p,k}$ such that

$$Tr(Q_{p,k}(t) T_{ku_p}) = \beta_{p,k} Tr(S_t A_{p,L} T_{ku_p}) + \varepsilon_{p,k}(t),$$

with

$$|\varepsilon_{p,k}(t)| \leq C_{p,k} M t^M.$$

Given (OS_M) , we can rewrite the head:

$$\sum_{k=1}^{K-1} (-1)^k Tr(Q_{p,k}(t) T_{ku_p}) = \sum_{k=1}^{K-1} (-1)^k \beta_{p,k} g_t(ku_p) + O(t^M),$$

and then combine with the tail bound from Step 1. This yields the stated decomposition with

$$\beta_{p,m} := (-1)^m \beta_{p,m}$$

all the real content collapses to proving (OS_M) for finitely many k.

Step 4 — Unpack $Q_{p,k}(t)$: it is an integral over conjugated pieces

$$G_{s_j}^{(\ell u_p)}, \Pi_L^{(\ell u_p)}, M_{w(-\ell u_p)}, \ell \in \{0, \dots, k - 1\}.$$

(OS_M) is exactly the statement that, to order t^M in trace,

We may replace those conjugated factors by their “frozen” counterparts at $\ell = 0$ (i.e., drop the ℓu_p -dependence inside $Q_{p,k}$), leaving only the net shift T_{ku_p} outside.

This is a local parametrix / localization claim: the heat kernel is concentrated on scales $\sim \sqrt{t}$, so changes at macroscopic scale ℓu_p don't affect the trace beyond $O(t^M)$ once we have Π_L -localization and TC-B bounds.

Proposition D.OSMM-FREEZE (fixed-cutoff freezing estimate for diagonal words)

Fix finite cutoff parameters L and R , a bounded shift range $|a| \leq A$, and an integer $M \geq 1$. Let

$$H_R = H_0 + V_R$$

be the truncated PWQO Hamiltonian, with V_R the bounded prime-power cutoff potential at level R . For each shift aa , write

$$H_R^{(a)} = T_a H_R T_{-a}, S_s^{(a)} = e^{-sH_R^{(a)}}, S_s = e^{-sH_R}.$$

Consider a diagonal word of length kk arising in the fixed-cutoff Duhamel expansion of Step 3, with $k \leq K(M) - 1$, written in the extracted form

$$W_{p,k}(t) = Q_{p,k}(t) T_{ku_p},$$

where $u_p = \log p$, and where the operator $Q_{p,k}(t)$ is built from bounded local factors and conjugated heat factors of the form $S_{s_j}^{(\ell j u_p)}$ on the simplex $\Delta_k(t)$

Define the frozen word $W_{p,k}^{fr}(t)$ by replacing every conjugated heat factor $S_{s_j}^{(\ell j u_p)}$ appearing inside $Q_{p,k}(t)$ by the corresponding frozen factor S_{s_j} leaving the external translation T_{ku_p} unchanged.

Then, after insertion of the fixed localizer Π_L and the same traced channel factor used throughout Appendix D, the freezing replacement is quantitative:

$$Tr(S_t \Pi_L W_{p,k}(t) \Pi_L) = Tr(S_t \Pi_L W_{p,k}^{fr}(t) \Pi_L) + O_{p,k,M,L,R,A}(t^{k/2+1}) \quad (t \downarrow 0)$$

In particular, the difference between the exact diagonal word and its frozen representative is one order flatter in t than the corresponding diagonal head term.

Proof

For each fixed cutoff R , the potential V_R is a bounded multiplication operator. For $|a| \leq A$, the translated defect

$$V_R^{(a)} - V_R$$

is also a bounded multiplication operator, with bound uniform in a on every fixed bounded shift interval.

For a single conjugated factor, Duhamel's formula gives

$$S_s^{(a)} - S_s = \int_0^s S_{s-\tau}^{(a)} (V_R^{(a)} - V) S_\tau d\tau.$$

Insert the localizer Π_L on both sides. By proved fixed-cutoff legality source in Appendix B, namely B.TC / Proposition B.4.1. , this localized defect satisfies

$$\|\Pi_L (S_s^{(a)} - S_s) \Pi_L\|_{S_1} \leq C_{L,R,A} s (0 < s \leq t_0, |a| \leq A).$$

Now apply this replacement inside the simplex integral defining a diagonal word of length k . Replacing one conjugated factor by its frozen version inserts exactly one Duhamel defect, and therefore contributes one additional simplex variable. After simplex integration, this yields one additional factor of t . The remaining factors are estimated by the same fixed-cutoff head bounds already used in Step 3. Since the unfrozen head term is of size $O(t^{k/2})$, the freezing defect is of size $O(t^{k/2+1})$.

There are only finitely many factors in any word of fixed length k , so summing over the possible replacement positions changes only the constant. This proves the stated estimate.

Definition (density-normalized fixed-cutoff spike package). For fixed (L, R) , define

$$\widehat{P}_{L,R}(t) := \frac{1}{2L} \sum_{\log q \leq R} w(q) g_{t,L}(\log q) = \sum_{\log q \leq R} w(q) \widehat{g}_{t,L}(\log q).$$

The decomposition proved above remains a raw fixed-cutoff trace identity. The factor $(2L)^{-1}$ is introduced only at the level of the assembled spike package, which is the intensive object compared with the canonical package of Section 5.1.4.

D.KERNEL-LOCAL scope note

Auxiliary raw-bump calculation.

This subsection computes the one-letter local-loop profile for the raw multiplication-bump model:

$$\Theta_{q,L}^{(1)}(t) = w(q)J_{t,L,\alpha}(\log q).$$

It is not the package exported to Appendices E and F. The RH closure uses the canonical displacement/commutator package defined in D.CANONICAL-FUNCTIONAL. In particular, D.KERNEL-LOCAL is not used to prove

$$J_{t,L,\alpha} = g_{t,L}$$

Lemma D.KERNEL-LOCAL.

For a single prime power q , write $u := \log_q q$, and set

$$\tau_{\sim}(s) := \frac{\alpha t}{2} + s(t - s), 0 < s < t$$

Let

$$K_0(r; x, z) = (4\pi r)^{-1/2} \exp\left(-\frac{(x-z)^2}{4r}\right)$$

denote the free heat kernel, and let

$$W_{\alpha}(u) = (2\pi\alpha)^{-1/2} \exp\left(-\frac{u^2}{2\alpha}\right)$$

Then for every $0 < s < t$

$$\int_R K_0(t - s; x, z) W_{\alpha}(z - u_q) K_0(s; z, x) dz = \frac{1}{4\pi} \tau_{\sim}(s)^{-1/2} \exp\left(-\frac{t(x-u_q)^2}{4\tau_{\sim}(s)}\right)$$

Proof

Write

$$G_r(u) := (4\pi r)^{-1/2} e^{-u^2/(4r)}$$

Then

$$K_0(r; x, z) = G_r(x - z) \text{ and } W_\alpha = G_{\alpha/2}$$

Also,

$$K_0(t - s; x, z) K_0(s; z, x) = \frac{1}{\sqrt{2\pi t}} G_{\beta(s)}(z - x), \beta(s) := \frac{s(t-s)}{t}.$$

Therefore

$$\int_R K_0(t - s; x, z) W_\alpha(z - u_q) K_0(s; z, x) dz = \frac{1}{2\pi t} \int_R G_{\beta(s)}(z - x) G_{\alpha/2}(z - u_q) dz$$

By Gaussian convolution, the inner integral is $G_{\beta(s) + \alpha/2}(x - u_q)$

Hence

$$\int_R K_0(t - s; x, z) W_\alpha(z - u_q) K_0(s; z, x) dz = \frac{1}{2\pi t} G_{\beta(s) + \alpha/2}(x - u_q)$$

Since

$$t(\beta(s) + \alpha/2) = s(t - s) + \frac{\alpha t}{2} = \tilde{\tau}(s),$$

this is exactly

$$\frac{1}{4\pi} \tilde{\tau}(s)^{-1/2} \exp\left(-\frac{t(x-u_q)^2}{4\tilde{\tau}(s)}\right)$$

This proves the claim.

Corollary D.KERNEL-LOCAL.2.

Let $q = p^m$ be a prime power, set $u_q := \log q$, and write

$$w(q) := \frac{\Lambda(q)}{\sqrt{q}}$$

Let ϕ_L denote the centered localizing profile used in the one-letter diagonal channel at fixed cutoff L . Define

$$\Theta_{q,L}^{(1)}(t) := w(q) \int_0^t \int_R \phi_L(x) \left[\int_R K_0(t-s; x, z) W_\alpha(z - u_q) K_0(s; z, x) dz \right] dx ds.$$

Then

$$\Theta_{q,L}^{(1)}(t) = w(q) J_{t,L,\alpha}(u_q),$$

where

$$J_{t,L,\alpha}(\mu) := \frac{1}{4\pi} \int_0^t \tau^\sim(s)^{-1/2} \left(\int_R \phi_L(x) \exp\left(-\frac{t(x-\mu)^2}{4\tau^\sim(s)}\right) dx \right) ds, \quad \tau^\sim(s) = \frac{\alpha t}{2} + s(t-s).$$

In particular, the one-letter localized diagonal term depends on q only through $u_q = \log q$, and its coefficient is exactly the frozen global spike weight $w(q) = \frac{\Lambda(q)}{\sqrt{q}}$.

Proof. Insert Lemma D.KERNEL-LOCAL into the definition of $\Theta_{q,L}^{(1)}(t)$ and use Tonelli-Fubini:

$$\Theta_{q,L}^{(1)}(t) = w(q) \frac{1}{4\pi} \int_0^t \tau^\sim(s)^{-1/2} \left(\int_R \phi_L(x) \exp\left(-\frac{t(x-u_q)^2}{4\tau^\sim(s)}\right) dx \right) ds.$$

This is exactly $w(q) J_{t,L,\alpha}(u_q)$

Normalization remark.

Lemma D.KERNEL-LOCAL and Corollary D.KERNEL-LOCAL.2 are raw fixed-cutoff local identities. The density factor $(2L)^{-1}$ is not inserted at the one-letter kernel level. It enters only when the full fixed-cutoff spike package is assembled as

$$\widehat{P}_{L,R}(t) := \frac{1}{2L} \sum_{\log q \leq R} w(q) g_{t,L}(\log q)$$

Thus the local Gaussian computation and the later density normalization are logically separate.

Auxiliary Gaussian subcase.

If, in this subsection only, the localizing profile is taken to be the centered Gaussian

$$\phi_L(x) = G_L(x) := (4\pi L)^{-1/2} e^{-x^2/(4L)},$$

then the x-integral can be evaluated explicitly. To avoid collision with the canonical large-L kernel used later in Lemma D.LNORM, we denote the resulting auxiliary model kernel by

$$J_{t,L,\alpha}^{Gauss}(\mu).$$

One has

$$J_{t,L,\alpha}^{Gauss}(\mu) = \frac{\sqrt{t}}{4\pi} \int_0^1 (L + \frac{\alpha}{2} + t v(1 - v))^{-1/2} \exp\left(-\frac{\mu^2}{4(L + \frac{\alpha}{2} + t v(1 - v))}\right) dv.$$

Proof.

For $\tilde{\tau}(s) = \alpha t/2 + s(t - s)$

$$\int_R G_L(x) \exp\left(-\frac{t(x-\mu)^2}{4\tilde{\tau}(s)}\right) dx = \left(\frac{\tilde{\tau}(s)}{Lt + \tilde{\tau}(s)}\right)^{1/2} \exp\left(-\frac{t\mu^2}{4(Lt + \tilde{\tau}(s))}\right)$$

Substituting into the definition of $J_{t,L,\alpha}$ gives

$$J_{t,L,\alpha}^{Gauss}(\mu) = \frac{1}{4\pi} \int_0^t (Lt + \tilde{\tau}(s))^{-1/2} \exp\left(-\frac{t\mu^2}{4(Lt + \tilde{\tau}(s))}\right) ds.$$

Now set $s = tv$ Since

$$Lt + \tilde{\tau}(tv) = t(L + \frac{\alpha}{2} + t v(1 - v)),$$

the stated formula follows.

Reader note.

The explicit Gaussian subcase above is an auxiliary local model and is not the window family used in Lemma D.LNORM. The large-L density bridge is applied only to the density-normalized package

$$\widehat{P}_{L,R}(t) := \frac{1}{2L} \sum_{\log q \leq R} w(q) \widehat{g}_{t,L}(\log q), \widehat{g}_{t,L}(a) := \frac{1}{2L} g_{t,L}(a),$$

built from an admissible averaging window family satisfying the D.LNORM hypothesis. Under that hypothesis,

$$\widehat{g}_{t,L}(a) \rightarrow_{c_w} G_t(a)$$

locally uniformly on compact a-intervals, and when $c_w = 1$ this yields the canonical shared package B. The Gaussian localizer $\phi_L = G_L$ is retained here only to exhibit the exact one-letter diagonal profile and the frozen coefficient $w(q) = \frac{\Lambda(q)}{\sqrt{q}}$.

Corollary D.DIAGONAL-PACKAGE (fixed-cutoff explicit spike package)

Fix L and R. Then the diagonal part of the centered fixed-cutoff heat trace admits the decomposition

$$\Theta_{L,R}^{diag}(t) = P_{L,R}(t) + E_{L,R}(t),$$

where the explicit spike package is

$$P_{L,R}(t) = \sum_{\log q \leq R} \frac{\Lambda(q)}{\sqrt{q}} g_t(\log q),$$

with $q = p^m$, and g_{gt} the Gaussian spike kernel already fixed in Appendix D.NORM.

Moreover, for every integer $M \geq 1$

$$|E_{L,R}(t)| \leq C_M(L, R) t^M (0 < t \leq t_0).$$

Proof

Combine:

1. the fixed-cutoff Duhamel expansion and head/tail decomposition from Step 3;
2. the diagonal reduction lemmas that identify the frozen diagonal channel with the Gaussian spike kernel g_t ;
3. Proposition D.OSMM-FREEZE, which shows that replacing the exact diagonal words by their frozen representatives changes the trace only by a flatter term;
4. the existing tail estimate from the finite-head/fat-tail decomposition.

The coefficient attached to the translation by $\log q \log q$ is exactly the canonical prime-power weight

$$w(q) = \frac{\Lambda(q)}{\sqrt{q}}$$

in the frozen global normalization of Appendix D.NORM. Summing the finitely many fixed-cutoff diagonal contributions yields the stated package $P_{L,R}(t)$, and all remaining contributions are absorbed into $E_{L,R}(t)$, which is flat to arbitrary order.

Lemma D.LOC (the “where cancellation fails” lemma)

For the free kernel $S_t = e^{t\partial^2}$ and the cutoff Π_L , define

$$g_{t,L}(a) := \text{Tr}(\Pi_L S_t \Pi_L T_a)$$

Then $g_{t,L}(a)$ is a -dependent and, in fact, satisfies the explicit overlap formula

$$g_{t,L}(a) = (4\pi t)^{-1/2} e^{-a^2/(4t)} \cdot (2L - |a|)_+ \text{ (small boundary error)}.$$

In particular, for $|a| \leq L/2$

$$(g_{t,L} a) = 2L(4\pi t)^{-1/2} e^{-a^2/(4t)} + O(1).$$

Lemma D.A1

Trace-class/holomorphy legality). By Lemma A.GROWTH and Corollaries

A.TRACE–A.HOLO, the eigenvalues satisfy $\lambda_n(L, R) \geq cn^2 - C\lambda$, hence for every compact $K \subset \Omega$

the series $\sum n(s + \lambda_n(L, R))^{-1}$ converges absolutely and uniformly for $s \in K$. Therefore

$(H_{L,R} + s)^{-1} \in S1$ and

$$F_{L,R}(s) = \text{Tr}(H_{L,R} + s)^{-1} = \sum_{n \geq 1} \frac{1}{s + \lambda_n} (L, R)$$

defines a holomorphic function on Ω .

Let H be a self-adjoint, nonnegative operator on a Hilbert space H . Assume either:

- (Finite volume / compact resolvent) H has compact resolvent (e.g. Dirichlet Schrödinger operator on $[-L, L]$, hence discrete spectrum $\lambda_n \rightarrow \infty$, or
- (Localized trace) there is a bounded operator Π such that $\Pi(H + s)^{-1}\Pi$ is trace-class for all $s \in \Omega$ (this is your Π_L -localized setting).

Define

$$F(s) := \text{Tr}_r((H + s)^{-1}) \text{ (finite volume case)}$$

or

$$F_\Pi(s) := \text{Tr}_r(\Pi(H + s)^{-1}\Pi) \text{ (localized case)}.$$

Then:

1. For every $s \in \Omega = \mathbb{C} \setminus (-\infty, 0]$,

2. the operator $(H + s)^{-1}$ exists and

$$\|(H + s)^{-1}\| \leq \frac{1}{\text{dist}(-s, \sigma(H))} = \frac{1}{\text{dist}(-s, [0, \infty))}.$$

The function F (respectively F_Π) is holomorphic on Ω .

3 For each compact $K \Subset \Omega$, there is $C_K < \infty$ such that

$$\sup_{s \in K} \|(H + s)^{-1}\| \leq C_K, \sup_{s \in K} \|(H + s)^{-1}\|_{S_1} \leq C_K$$

Proof

Step 1: resolvent set contains Ω .

Since $H \geq 0$ is self-adjoint, $\sigma(H) \subset [0, \infty)$. For $s \in \Omega$, we have $-s \notin [0, \infty)$, hence $-s \in \rho(H)$ and $(H + s)^{-1} = (H - (-s))^{-1}$ exists. Standard resolvent bounds give

$$\|(H + s)^{-1}\| = \|(H - (-s))^{-1}\| \leq \frac{1}{\text{dist}(-s, \sigma(H))}.$$

Step 2: operator-valued holomorphy.

For any fixed $s_0 \in \Omega$ the resolvent identity yields

$$(H + s)^{-1} - (H + s_0)^{-1} = (s_0 - s)(H + s)^{-1}(H + s_0)^{-1}.$$

This implies $s \mapsto (H + s)^{-1}$ is analytic as a bounded-operator valued function on Ω (standard fact: the resolvent is analytic on the resolvent set).

Step 3: trace holomorphy.

- In the finite volume / compact resolvent case: $(H + s)^{-1}$ is trace-class for every $s \in \Omega$ because H has discrete eigenvalues $\lambda_n \rightarrow \infty$ and

$$\text{Tr}((H + s)^{-1}) = \sum_n \frac{1}{\lambda_n + s},$$

absolutely convergent uniformly on compact subsets of Ω (since $|\lambda_n + s| \geq c(\lambda_n + 1)$ on compacts away from the cut). Uniform absolute convergence implies holomorphy.

- In the localized case: by proposition $\Pi(H + s)^{-1}\Pi \in S_1$ for all $s \in \Omega$, and the map is analytic in S_1 because the resolvent is analytic in $B(H)$ and multiplication by bounded operators Π preserves analyticity; the trace is a continuous linear functional on S_1 , hence F_Π is holomorphic.

Step 4: uniform bounds on compacts.

If $K \Subset \Omega$ then $\delta_K := \inf_{s \in K} \text{dist}(-s, [0, \infty)) > 0$

Hence $\sup_{s \in K} \|(H + s)^{-1}\| \leq 1/\delta_K$. The trace-norm bounds follow in finite volume from uniform absolute convergence, and in localized setting from the assumed trace-class property plus analyticity/compactness (or directly by estimating $\|\Pi(H + s)^{-1}\Pi\|_{S_1}$ using spectral calculus in finite volume).

Lemma D.A2 (Laplace–resolvent trace identity; interchange justified)

Let $H \geq 0$ be self-adjoint. Suppose e^{-tH} is trace-class for each $t > 0$ (true in finite volume; or true after localization $\Pi e^{-tH} \Pi \in S_1$). Then for every s with $\Re s > 0$,

$$(H + s)^{-1} = \int_0^\infty e^{-st} e^{-tH} dt$$

as a strong operator identity, and moreover

$$\text{Tr}((H + s)^{-1}) = \int_0^\infty e^{-st} \text{Tr}(e^{-tH}) dt$$

(and similarly with Π – *localization* everywhere).

In particular, the Laplace transform of the heat trace equals the resolvent trace for $\Re s > 0$, and the right-hand side extends holomorphically to Ω whenever $s \mapsto \text{Tr}((H + s)^{-1})$ does (by Lemma D.A1).

Proof

Step 1: scalar identity via spectral theorem.

By functional calculus, for $H \geq 0$,

$$(H + s)^{-1} = f_s(H) \text{ with } f_s(\lambda) = \frac{1}{\lambda + s}.$$

For $\Re s > 0$ and $\lambda \geq 0$, we have the scalar Laplace identity

$$\frac{1}{\lambda + s} = \int_0^\infty e^{-t(\lambda + s)} dt.$$

So

$$f_s(\lambda) = \int_0^\infty e^{-s} t e^{-t\lambda} dt.$$

Step 2: promote to operator identity.

Applying functional calculus,

$$(H + s)^{-1} = \int_0^\infty e^{-st} e^{-tH} dt,$$

where the integral converges strongly because

$$\|e^{-st} e^{-tH}\| \leq e^{-(\Re s)t} \text{ and } \int_0^\infty e^{-(\Re s)t} dt < \infty$$

Step 3: justify trace interchange (this is the “legality” point).

In finite volume (or localized trace-class), $e^{-tH} \in S_1$ for each $t > 0$, and $\text{Tr}(e^{-tH}) \geq 0$. Then for $\Re s > 0$,

$$\int_0^\infty e^{-(\Re s)t} \text{Tr}(e^{-tH}) dt$$

converges (in finite volume because $\text{Tr}(e^{-tH}) = \sum_n e^{-t\lambda_n}$ and this Laplace integral equals $\sum_n (\lambda_n + s)^{-1}$; in localized case by S_1 bounds plus dominated convergence).

More explicitly (finite volume): diagonalize H with eigenpairs (λ_n, ϕ_n) , then

$$\text{Tr}((H + s)^{-1}) = \sum_n \frac{1}{\lambda_n + s}, \quad \text{Tr}(e^{-tH}) = \sum_n e^{-t\lambda_n}.$$

Fubini/Tonelli applies because terms are nonnegative for $\Re s > 0$ on the real part:

$$\int_0^\infty e^{-st} \text{Tr}(e^{-tH}) dt = \int_0^\infty e^{-st} \sum_n e^{-t\lambda_n} dt = \sum_n \int_0^\infty e^{-t(\lambda_n + s)} dt = \sum_n \frac{1}{\lambda_n + s}.$$

This proves the trace identity.

Localized case: apply the same argument to the trace-class operator $\Pi e^{-tH} \Pi$ (or use Tonelli on the spectral measure of Π if you prefer), giving

$$\text{Tr}(\Pi(H + s)^{-1} \Pi) = \int_0^\infty e^{-st} \text{Tr}(\Pi e^{-tH} \Pi) dt.$$

Proposition D.6.1 (derived explicit one-letter package and Ω -holomorphic operator-side remainder).

Fix truncation parameters $L, R > 0$. Let

$$F_{L,R}(s) := \text{Tr}((H_{L,R} + s)^{-1}), \quad s \in \Omega := \mathbb{C} \setminus (-\infty, 0]$$

Let

$$\tilde{A}_{\text{class},L}(s)$$

denote the transform of the fixed bulk/classical term, and let

$$\tilde{P}_{L,R}(s)$$

denote the transform of the explicit one-letter spike package $P_{L,R}(t)$ from Theorem D.ONE-STEP".

Then:

1. $F_{L,R} \in O(\Omega)$;
2. $\tilde{A}_{\text{class},L} + \tilde{P}_{L,R}$ extends holomorphically to Ω ;
3. the fixed-cutoff operator-side remainder

$$R_{L,R}^{\sim}(s) := F_{L,R}(s) - A_{class,L}^{\sim}(s) - P_{L,R}^{\sim}(s)$$

belongs to $O(\Omega)$;

4. on any right half-plane where the individual Laplace transforms are justified for the fixed-cutoff decomposition, one has

$$F_{L,R}(s) = A_{class,L}^{\sim}(s) + P_{L,R}^{\sim}(s) + L[E_{L,R}^{rep} + R_{L,R}^{mix}](s).$$

In particular, the fixed-cutoff operator-side engine exports only the one-letter explicit package. The repeated same-channel sector and the mixed sector are absorbed into the operator-side remainder on Ω .

Only the one-letter package $P_{L,R}$ is exported from the fixed-cutoff analysis as the explicit arithmetic package. Higher same-channel sectors are retained in the operator-side remainder and enter the canonical whole-line synthesis only through the Ω -holomorphic shield.

Proof.

The holomorphy of $F_{L,R}$ on Ω is the fixed-cutoff resolvent-trace legality already proved earlier in Appendix D. The transform $A_{class,L}^{\sim}$ is holomorphic by explicit construction, and $P_{L,R}^{\sim}$ is holomorphic because it is the transform of a finite one-letter package. Hence the difference

$$R_{L,R}^{\sim} = F_{L,R} - A_{class,L}^{\sim} - P_{L,R}^{\sim} \text{ is holomorphic on } \Omega.$$

On a right half-plane where the fixed-cutoff time-domain Laplace transforms are individually legal, Theorem D.ONE-STEP" yields the stated decomposition after termwise transform. The last sentence is therefore merely a restatement of the fixed-cutoff bookkeeping: only the one-letter sector is exported as the explicit package, while the higher same-channel and mixed sectors are retained in the operator-side

Lemma D.DIFF 2 (Differentiation under trace for g_t).

Fix $t > 0$. Define $g_t(a) = \text{Tr}(S_t A_L T_a)$. Then $g_t \in \mathcal{C}^\infty(R)$ and

$$g_t'(a) = \int_R \partial_a K_0(t; u, u - a) w(u - a) 1_{[-L, L]}(u) 1_{[-L, L]}(u - a) du.$$

In particular, for the free heat kernel $K_0(t; u, u - a) = (4\pi t)^{-1/2} e^{-a^2/(4t)}$

$$|\partial_a K_0(t; u, u - a)| \leq (4\pi t)^{-1/2} \frac{|a|}{2t} e^{-a^2/(4t)},$$

so

$$|g_t'(a)| \leq C(L) \|w\|_\infty t^{-1} |a| e^{-a^2/(4t)}.$$

Moreover $g_t(a) \equiv 0$ for $|a| > 2L$.

Proof. By Lemma [B.IC](#), $S_t A_L T_a$ is trace class and $\text{Tr}(S_t A_L T_a) = \int_K S_t A_L T_a(u, u) du$

By Lemma B.K0 the diagonal kernel equals $K_0(t; u, u - a) w(u - a) 1_{[-L, L]}(u) 1_{[-L, L]}(u - a)$

Differentiation under the integral is justified by dominated convergence, using the explicit bound on $\partial_a K_0$ (and similarly for higher $\partial_a^k K_0$), together with bounded support of the indicators.

Remark: Definition. For each fixed cutoff (L, R) , $B_{L,R}(s)$ denotes the Appendix-D holomorphic continuation to Ω of the cutoff spike transform, initially defined on a right half-plane. Any occurrence of $B_{L,R}(s)$ on Ω refers to this continued object, not to the raw prime-power series outside its initial convergence region.

Theorem D.ANCHOR (Half-plane identity and analytic continuation to Ω)

Fix $L > 0$ and a truncation parameter $R > 0$. Let $\Theta_{L,R}(t)$ be the localized/regularized trace functional and let $A_{class,L}(t)$ denote the bulk/free subtraction term. Define the diagonal spike package

$$S_R(t) := \sum_{q \geq 2 \log q \leq R} \frac{\Lambda(q)}{\sqrt{q}} g_{t,L}(\log q), \quad g_{t,L}(a) := T_r(G_t A_L T_a),$$

and let $R_{L,R}(t)$ be the remainder from Theorem D.ONE-STEP', satisfying $|R_{L,R}(t)| \leq C_N(L, R) t^N$

or all N as $t \downarrow 0$.

Let

$$\widehat{\Theta}_{L,R}(s) := \int_0^\infty e^{-st} \Theta_{L,R}(t) dt, \quad \widehat{A}_{class,L}(s) := \int_0^\infty e^{-st} A_{class,L}(t) dt, \quad \widehat{S}_R(s) := \int_0^\infty e^{-st} S_R(t) dt,$$

defined initially on a half-plane $\Re(s) > \sigma_0$ where the integrals converge.

Then on $\Re(s) > \sigma_0$,

$$\widehat{\Theta}_{L,R}(s) = \widehat{A}_{class,L}(s) + \widehat{S}_R(s) + \widehat{R}_{L,R}(s),$$

where $\widehat{R}_{L,R}(s) := \int_0^\infty e^{-st} R_{L,R}(t) dt$ is holomorphic on $\Re(s) > \sigma_0$.

Moreover, $\widehat{\Theta}_{L,R}(s)$ coincides with the resolvent/Stieltjes trace $F_{L,R}(s) = T_r(H_{L,R} + s)^{-1}$ on

$\Re(s) > \sigma_0$, and both $F_{L,R}$ and $\widehat{S}_R$ admit holomorphic continuation to $\Omega = \mathbb{C} \setminus (-\infty, 0]$ Hence the identity continues to all of Ω :

$$F_{L,R}(s) = B_{L,R}(s) + R_{L,R}\Omega(s), \quad s \in \Omega,$$

where $B_{L,R}(s)$ is the explicit package continuation of $\widehat{A}_{class,L}(s) + \widehat{S}_R(s)$ and $R_{L,R}\Omega$ is holomorphic on Ω .

(Note: This is a finite-cutoff statement. It does not assert that the limiting canonical package B_{can} is holomorphic on all of Ω)

Proof

Theorem D.ONE-STEP' gives $\Theta_{L,R}(t) = A_{class,L}(t) + S_R(t) + R_{L,R}(t)$ for $t \in (0, t_0]$, and $R_{L,R}(t)$ is super-smoothing as $t \downarrow 0$.

1. By the domination bounds established in Appendix B/D (trace-class legality and the Gaussian bound for $g_{t,L}$), each term is Laplace-transformable for $\Re(s) > \sigma_0$, and dominated convergence justifies termwise Laplace transform, yielding the half-plane identity.
2. The Laplace/resolvent link (Appendix A: trace-class resolvent and holomorphy from A.GROWTH/A.HOLO) gives $\widehat{\Theta}_{L,R}(s) = F_{L,R}(s)$ on $\Re(s) > \sigma_0$
3. By D.NORM Ω , the package $\widehat{S}_R$ continues holomorphically to Ω , and by A.HOLO FL,Ris holomorphic on Ω . Therefore the identity extends from $\Re(s) > \sigma_0$ to all of Ω by uniqueness of analytic continuation.

Theorem D.KEY-FORM

Direct residual bound from form control

Let

$$F_{L,R}^{int}(s), B_{L,R}^{int}(s), R_{L,R}^{int}(s)$$

denote the density-normalized operator transform, one-letter spike package, and residual transform:

$$F_{L,R}^{int} = \frac{1}{2L} F_{L,R'}, \quad B_{L,R}^{int} = \frac{1}{2L} B_{L,R'}, \quad R_{L,R}^{int} = \frac{1}{2L} R_{L,R}$$

After the one-letter diagonal package is removed, define the direct short-time residual

$$Q_{L,R}^{short}(t)$$

as the sum of all remaining fixed-(L,R) Duhamel sectors:

$$Q_{L,R}^{short} = Q_{L,R}^{mix} + Q_{L,R}^{diag \geq 2} + Q_{L,R}^{freeze} + Q_{L,R}^{comm} + Q_{L,R}^{win}$$

This partition is exhaustive.

Then for every compact

$$K \Subset \Omega$$

there exists an admissible ordered window $L = L_R$ with

$$L_R \rightarrow \infty, R_{LR} \rightarrow 0,$$

such that

$$R_R^{short}(s) := \frac{1}{2L_R} \int_0^{t_0} e^{-st} Q_{L,R}^{short}(t) dt$$

satisfies

$$\sup_R \sup_{s \in K} |R_R^{short}(s)| < \infty.$$

Thus the family $\{R_R^{short}\}$ is normal on Ω , and every locally uniform subsequential limit is holomorphic on Ω .

Proof

By Lemma A.ENV-FORM,

$$V_R^\# \leq_{form} \theta H_{base} + C_0 I$$

with $\theta < 1$ uniformly in R .

The KLMN form bound implies uniform control of the form-Duhamel expansion. Equivalently, writing

$$T_R := (H_{base} + 1)^{-1/2} V_R^\# (H_{base} + 1)^{-1/2},$$

we have

$$\|T_R\| \leq C_0$$

uniformly in R , with the relative part strictly below one.

The $k = 1$ diagonal sector is separated as the finite one-letter spike package. All residual sectors have either $k \geq 2$, mixed labels, a freezing/commutator defect, or a window/cutoff defect.

The form-Duhamel estimate gives, for $0 < t \leq t_0$,

$$\frac{1}{2L_R} \|Q_{L,R}^{short}(t)\|_{tr} \leq C_{t_0},$$

where C_{t_0} is independent of R . The point is that the form bound controls each residual insertion by T_R , while the localized heat factors supply the trace-class smoothing. Since the one-letter term has been removed, the remaining Dyson series begins at quadratic order or carries an off-diagonal/freezing/window defect, and the resulting series is bounded by a convergent form-Dyson majorant.

Now fix $K \in \Omega$. Let

$$M_K := \sup_{s \in K} |s| < \infty.$$

For $0 < t \leq t_0$

$$|e^{-st}| \leq e^{M_K t_0}$$

Therefore

$$\sup_{s \in K} |R_R^{short}(s)| \leq \frac{1}{2L_R} \int_0^{t_0} e^{M_K t_0} \|Q_{L,R}^{short}(t)\|_{tr} dt \leq e^{M_K t_0} C_{t_0}$$

The right-hand side is independent of R . Hence the short-time residual family is locally bounded on Ω . Since each finite- R residual transform is holomorphic in ss , Montel gives normality and holomorphic subsequential limits.

Theorem D.KEY-FORM-TRACE, uniform form-Duhamel trace-norm bound.

Let

$$H_{env} = -\partial_u^2 + V_{env} + \varepsilon U_{left}$$

be the fixed confining comparison operator, and let

$$H_R^\# = H_{env} - V_R^\#$$

with

$$0 \leq V_R^\# \leq \theta V_{env}, 0 < \theta < 1$$

as quadratic forms, uniformly in R. Assume $e^{-tH_{env}} \in S_1$ for every $t > 0$. Then for every compact interval $I \subset (0, \infty)$ there exists $CI < \infty$, independent of R and of the admissible cutoff stage, such that the form-Duhamel expansion for

$$e^{-tH_R^\#} - e^{-tH_{env}}$$

converges in trace norm uniformly for $t \in I$, and

$$\sup_{t \in I} \|e^{-tH_R^\#}\|_{S_1} \leq C_I.$$

After density normalization, the canonical trace contributions obtained from this expansion are locally bounded uniformly along the admissible ordered net.

Proof.

By A.ENV-FORM,

$$\int V_R^\# |\psi|^2 \leq \theta h_{env}[\psi] + C_0 \|\psi\|_2^2$$

uniformly in R. Equivalently,

$$B_R := (V_R^\#)^{1/2} (1 + H_{env})^{-1/2}$$

extends to a bounded operator with

$$\|B_R\| \leq C_\theta$$

uniformly in R.

The KLMN theorem gives closed, lower-semibounded forms for $H_R^\#$, and the form-Duhamel formula gives

$$e^{-tH_R^\#} - e^{-tH_{env}} = \sum_{n \geq 1} \int_{\Delta n(t)} e^{-s_0 H_{env}} V_R^\# e^{-s_1 H_{env}} \dots V_R^\# e^{-s_n H_{env}} ds$$

in the form sense, with $s_0 + \dots + s_n = t$

Insert

$$V_R^\# = (1 + H_{env})^{1/2} B_R^* B_R (1 + H_{env})^{1/2}$$

at each vertex. The holomorphic semigroup estimate

$$\|(1 + H_{env})^{1/2} e^{-\sigma H_{env}}\| \leq C_\sigma^{-1/2}, \sigma > 0,$$

together with Schatten factorization of the endpoint heat factors, yields

$$\|e^{-s_0 H_{env}} V_R^\# e^{-s_1 H_{env}} \dots V_R^\# e^{-s_n H_{env}}\|_{S_1} \leq \|e^{-t H_{env}}\|_{S_2}^2 (C_\theta C)^n \prod_{j=0}^n s_j^{-1/2},$$

after the harmless convention that endpoint factors are split symmetrically.

The Dirichlet simplex integral satisfies

$$\int_{\Delta n(t)} \prod_{j=0}^n s_j^{-1/2} ds = \frac{\Gamma(1/2)^{n+1}}{\Gamma((n+1)/2)} t^{(n-1)/2}$$

Therefore the n-th trace-norm term is bounded by

$$C_I \frac{(C'_I)^n}{\Gamma((n+1)/2)}$$

uniformly for $t \in I$. The resulting Mittag-Leffler series converges. Hence the form-Duhamel expansion converges absolutely in trace norm, uniformly for $t \in I$ and uniformly in R .

The density-normalized trace functionals are obtained by applying bounded cutoff observables and dividing by $2L_\alpha$. The admissibility condition and the envelope domination give a uniform local density bound. Thus the canonical trace contributions are locally bounded along the admissible ordered net.

Lemma D.NV-FORM

The admissible window does not kill the arithmetic package

Suppose

$$L_R \rightarrow \infty, R_{LR} \rightarrow 0$$

Then

$$\frac{1}{2L_R} \sum_{\log q \leq R} \frac{\Lambda(q)}{\sqrt{q}} g_{t,LR}(\log q) \rightarrow \sum_q \frac{\Lambda(q)}{\sqrt{q}} G_t(\log q),$$

locally uniformly for t in compact subintervals of $(0, \infty)$.

Proof

For the sharp window,

$$g_{t,L}(a) = G_t(a)(2L - |a|)_+$$

Thus

$$\frac{1}{2L} g_{t,L}(a) = G_t(a)(1 - \frac{|a|}{2L})_+$$

For $0 \leq a \leq R$,

$$|\frac{1}{2L_R} g_{t,LR}(a) - G_t(a)| \leq G_t(a) \frac{R}{2L_R}$$

Since $R/L_R \rightarrow 0$, this goes to zero uniformly for $0 \leq a \leq R$.

Multiplying by $\Lambda(q)/\sqrt{q}$ and summing,

$$|\frac{1}{2L_R} \sum_{\log q \leq R} \frac{\Lambda(q)}{\sqrt{q}} g_{t,LR}(\log q) - \sum_{\log q \leq R} \frac{\Lambda(q)}{\sqrt{q}} G_t(\log q)| \leq \frac{R}{2L_R} \sum_{\log q \leq R} \frac{\Lambda(q)}{\sqrt{q}} G_t(\log q)$$

The full series

$$\sum_q \frac{\Lambda(q)}{\sqrt{q}} G_t(\log q)$$

converges absolutely because

$$G_t(\log q) = (4\pi t)^{-1/2} \exp\left(\frac{-(\log q)^2}{4t}\right),$$

which decays faster than any power of q^{-1} . Therefore the error tends to zero, and the canonical one-letter arithmetic package survives the density normalization.

Corollary D.KEY-Ω

Direct residual Ω-shield

Under A.ENV-DOM, A.ENV-FORM, D.KEY-FORM, and D.NV-FORM, the direct residual transforms converge along the admissible ordered net to a unique holomorphic function

$$R_H \in O(\Omega).$$

Moreover, on the safe overlap half-plane U_{σ_0} ,

$$F_H(s) = B_{can}(s) + R_H(s)$$

Appendix D does **not** assert that B_{can} is holomorphic on all of Ω . It only proves $R_H \in O(\Omega)$. The global continuation of B_{can} is the meromorphic zero-side continuation supplied by Appendix H.

Theorem D.SHIELD (Ω-holomorphy of the remainder; no cancellation off the cut)

By Lemma D.5.7 (bulk) and the spike transform lemmas in D.5.4 (for fixed cutoff R , the spike transform is a finite sum of holomorphic terms), $B_{L,R}^\sim$ extends holomorphically to Ω ."

And then define $R_{L,R}^\sim(s) := F_{L,R}(s) - B_{L,R}^\sim(s)$ for all $s \in \Omega$ (not just $\Re s > 0$). That makes the shield definitionally an Ω -holomorphic object, not merely "continued."

Fix $L, R > 0$ Let $H_{L,R}$ be the finite-volume Dirichlet Schrödinger operator

$$H_{L,R} = -\partial_u^2 + V_R(u) \text{ on } [-L, L],$$

self-adjoint and nonnegative. Let

$$G_t := e^{-tH_{L,R}}.$$

Let Π_L be the localization/cutoff to $[-L, L]$ (or any equivalent finite-volume cutoff), and let $A_L = \Pi_L M_w \Pi_L$ be the localized insertion operator.

Define the Stieltjes/resolvent trace object

$$F_{L,R}(s) := \text{Tr}((H_{L,R} + s)^{-1}), \quad s \in \Omega.$$

(A) Operator holomorphy / Laplace–resolvent legality

Given:

Lemma D.A1 and Lemma D.A2 hold for $H_{L,R}$; in particular,

- $F_{L,R}$ is holomorphic on Ω ,
- and for $\Re s > 0$,

$$F_{L,R}(s) = \int_0^\infty e^{-st} \text{Tr}(G_t) dt.$$

(B') Derived decomposition (no hypothesis). Proposition D.6.1 provides, for the same L, R , the decomposition

$$\Theta_{H_{L,R}}(t) = A_{\text{class}}(t) + P_{L,R}(t) + R_{L,R}(t),$$

with $R_{L,R}$ super-smoothing to all orders.

Combine Laplace–resolvent legality (D.A2)

with holomorphy of Ω . This is exactly the pole-package identity needed by Appendix F, with the Ω -shield built in. (D.A1) and the explicit package Laplace transform, and define

$$\tilde{R}_{L,R}(s) := F_{L,R}(s) - \tilde{B}_{L,R}(s).$$

On $\Re s > 0$ it equals the Laplace transform of $R_{L,R}(t)$, hence is analytic there; and since both terms are holomorphic on Ω , their difference is holomorphic on Ω .

(C) Explicit transform holomorphy on Ω off the cut

Let $B_{L,R}^{\sim}(s)$ be the Laplace transform of the explicit package

$$Bulk(t) + Spikes(t),$$

i.e. for $\Re s > 0$,

$$B_{L,R}^{\sim}(s) := \int_0^{\infty} e^{-st} (Bulk(t) + Spikes(t)) dt,$$

Define the remainder transform

For $\Re s > 0$, define

$$R_{L,R}^{\sim}(s) := \int_0^{\infty} e^{-st} (Mix(t) + Tail(t)) dt = F_{L,R}(s) - B_{L,R}^{\sim}(s).$$

Conclusion (Shield)

1. Ω -holomorphy. The remainder transform $R_{L,R}^{\sim}(s)$ extends holomorphically to all $s \in \Omega$.
2. No cancellation off the cut. In particular, $R_{L,R}^{\sim}$ has no isolated poles anywhere in Ω .
Hence any isolated pole of $B_{L,R}^{\sim}$ in Ω . cannot be canceled by the remainder.

Proof

Step 1: Holomorphy of $F_{L,R}$ on Ω .

By Lemma D.A1, $F_{L,R}(s) = \text{Tr}((H_{L,R} + s)^{-1})$ is holomorphic on Ω .

Step 2: Identify $F_{L,R}$ with the Laplace transform for $\Re s > 0$.

By Lemma D.A2 (Laplace–resolvent identity in trace form),

$$F_{L,R}(s) = \int_0^\infty e^{-st} \text{Tr}(G_t) dt (\Re s > 0).$$

Step 3: Subtract the explicit package.

For $\Re s > 0$, write

$$\text{Tr}(G_t) = (\text{Bulk}(t) + \text{Spikes}(t)) + (\text{Mix}(t) + \text{Tail}(t)).$$

Therefore,

$$F_{L,R}(s) = B_{L,R}^\sim(s) + R_{L,R}^\sim(s) (\Re s > 0),$$

hence

$$R_{L,R}^\sim(s) = F_{L,R}(s) - B_{L,R}^\sim(s) (\Re s > 0).$$

Step 4: Holomorphic continuation to Ω .

$B_{L,R}^\sim$ extends holomorphically to Ω . Since $F_{L,R}$ is holomorphic on Ω , the difference

$$R_{L,R}^\sim(s) := F_{L,R}(s) - B_{L,R}^\sim(s)$$

defines a holomorphic function on Ω that agrees with the Laplace integral definition on $\Re s > 0$. This proves (1).

Step 5: No cancellation off the cut.

A holomorphic function on Ω has no isolated poles in Ω . Hence $R_{L,R}^\sim$ cannot cancel any isolated poles of $B_{L,R}^\sim$ occurring in Ω .

Theorem D.SHIELD-CANONICAL (ordered cutoff removal preserves the Ω -holomorphic remainder).

Note: The canonical ordered-limit shield is proved below in Proposition D.CAN-REM, after the uniform residual estimates have been established.

Let

$$\Omega := \mathbb{C} \setminus (-\infty, 0]$$

For each fixed cutoff pair (L,R), Appendix D has already produced a fixed-cutoff operator-side decomposition

$$F_{L,R}(s) = B_{L,R}(s) + R_{L,R}(s), \quad s \in \Omega,$$

where:

1. $F_{L,R} \in O(\Omega)$ is the fixed-cutoff operator-side transform;
2. $B_{L,R}$ is the fixed-cutoff explicit package consisting of the bulk term together with the fixed-cutoff one-letter spike package; (This is a finite-cutoff statement. It does not assert that the limiting canonical package B is holomorphic on all of Ω .)
3. $R_{L,R} \in O(\Omega)$ is the fixed-cutoff operator-side remainder.

Assume now the ordered cutoff-removal procedure established earlier in Appendix D:

- first, the transform-level passage $R \rightarrow \infty$ for the operator-side objects, via the fixed-window resolvent convergence / limit mechanism already proved in Appendix D;
- afterward, the package identification on the initial overlap half-plane $U_{\sigma_0} \subset \Omega$ used in the R_H spine.

Then there exists an Ω -holomorphic function

$$R_H \in O(\Omega)$$

such that

$$R_{L,R} \rightarrow R_H$$

locally uniformly on compact subsets of Ω along the ordered cutoff-removal passage.

Moreover, on the initial overlap half-plane U_{σ_0} , where the canonical package B is defined by the ordered Laplace/package construction used in Appendix D and Section 5.1.4, one has

$$F_H(s) = B(s) + R_H(s), \quad s \in U_{\sigma_0}.$$

In particular, the operator-side remainder survives the ordered cutoff-removal passage as an Ω -holomorphic object. No unrestricted prime-side normal convergence on all of Ω is asserted or used here.

Theorem D.CANONICAL-WINDOW (explicit canonical window normalization and canonical package).

(D.CANONICAL-WINDOW scope note

This theorem applies to the canonical displacement/commutator trace

$$g_{t,L}(a) = \text{Tr}(S_t A_L T_a)$$

It does not apply to the auxiliary raw-bump loop kernel $J_{t,L,\alpha}$. Its role is to prove

$$(2L)^{-1} g_{t,L}(a) \rightarrow G_t(a)$$

locally uniformly on compact aa-intervals, thereby producing the canonical package

$$\sum_q w(q) G_t(\log q)$$

Fix once and for all the averaging window

$$w_L(u) := 1_{[-L,L]}(u), A_L := \Pi_L M_{w_L} \Pi_L = \Pi_L,$$

with Π_L the cutoff to $[-L, L]$. For $t > 0$ and $a \in \mathbb{R}$, define

$$g_{t,L}(a) := \text{Tr}(e^{-tH_0} A_L T_a), \bar{g}_{t,L}(a) := \frac{1}{2L} g_{t,L}(a).$$

Let

$$G_t(a) := (4\pi t)^{-1/2} e^{-a^2/(4t)}$$

be the one-dimensional heat kernel in the displacement variable.

Then the following hold.

1. For every fixed $t > 0$ and every $R > 0$,

$$\sup_{|a| \leq R} |\bar{g}_{t,L}(a) - G_t(a)| \rightarrow_{L \rightarrow \infty} 0$$

In particular, the normalization constant in Lemma D.LNORM is equal to

$$c_w = 1$$

for this explicit choice of window.

3. For every fixed truncation $R < \infty$, the density-normalized fixed-cutoff spike package

$$P_{L,R}(t) := \frac{1}{2L} \sum_{\log q \leq R} w(q) g_{t,L}(\log q)$$

satisfies, locally uniformly for t in compact subsets of $(0, \infty)$,

$$P_{L,R}(t) \xrightarrow{L \rightarrow \infty} B_R(t) := \sum_{\log q \leq R} w(q) G_t(\log q).$$

The shifted canonical closure package is therefore uniquely determined by

$$B_R^{can}(t) := e^{-t/4} B_R(t), \quad B^{can}(t) := \lim_{R \rightarrow \infty} B_R^{can}(t),$$

with the $R \rightarrow \infty$ passage taken only in the ordered transform-level sense established later in Appendix D.

Proof.

By the explicit overlap formula for the free spike kernel,

$$g_{t,L}(a) = (4\pi t)^{-1/2} e^{-a^2/(4t)} (2L - |a|)_+$$

in the sharp-cutoff case, up to the already-recorded boundary form of Lemma D.LOC / D.K0'.

Hence

$$\bar{g}_{t,L}(a) = G_t(a) \frac{(2L - |a|)_+}{2L}$$

For $|a| \leq R$ and $L \geq R$,

$$\sup_{|a| \leq R} \left| \frac{(2L - |a|)_+}{2L} - 1 \right| \leq \frac{R}{2L} \xrightarrow{L \rightarrow \infty} 0,$$

which proves the first assertion. The second is immediate. The third follows by summing over the finite set $\{\log q \leq R\}$ and using uniform convergence in a on $|a| \leq R$. The fourth is then merely the definition of the canonical shifted package used in the RH spine.

D.WORD-PARTITION and residual split

Definition D.WORD-PARTITION, relative-diameter classifier.

Let

$$q = (q_1, \dots, q_n)$$

be a Duhamel word, and write

$$a_j = \log q_j$$

Define the anchored relative diameter

$$\text{diam}_*(q) = \max_{1 \leq j \leq n} |a_j - a_1|$$

For the admissible cutoff stage $\alpha = (L_\alpha, R_\alpha)$, set

$$\Delta_\alpha := \log L_\alpha$$

Define

$$W_\alpha^{loc} = \{q: \text{diam}_*(q) \leq \Delta_\alpha\}$$

and

$$W_\alpha^{disp} = \{q: \text{diam}_*(q) > \Delta_\alpha\}$$

Then

$$W_\alpha = W_\alpha^{loc} \sqcup W_\alpha^{disp}$$

Repeated same-channel words satisfy $\text{diam}_*(q) = 0$ hence they are local. They are not assigned a displacement Gaussian factor.

Lemma D.LOC-BOUND.

The contribution of the local word class is locally bounded after density normalization:

$$\sup_{\alpha \geq \alpha_0(K)} \sup_{s \in K} |R_\alpha^{loc}(s)| < \infty (K \subseteq \Omega)$$

Proof.

For $q \in W_\alpha^{loc}$, all active centers lie in an interval of length $2\Delta_\alpha$ after anchoring. The envelope

domination remains valid for the restricted potential because $0 \leq V_{restricted} \leq V_R^{\#} \leq \theta V_{env}$. Therefore D.KEY-FORM-TRACE applies with the same form-bound constant. The number of centers in a local anchored cluster grows at most polynomially in Δ_{α} , while the density normalization divides by $2L_{\alpha}$. Since $\Delta_{\alpha} = \log L_{\alpha}$,

$$\frac{\text{poly}(\Delta_{\alpha})}{L_{\alpha}}$$

is uniformly bounded and tends to 0 for every fixed polynomial. This proves local boundedness of the local sector.

Lemma D.DISP-BOUND.

The contribution of the displacement word class satisfies, locally uniformly in t on compact subintervals of $(0, \infty)$,

$$|R_{\alpha}^{disp}(t)| \leq C_t \exp\left(-\frac{\Delta_{\alpha}^2}{Ct}\right),$$

after density normalization. Consequently,

$$\sup_{\alpha \geq \alpha_0(K)} \sup_{s \in K} |R_{\alpha}^{disp}(s)| < \infty (K \Subset \Omega)$$

Proof.

If $q \in W_{\alpha}^{disp}$, then at least one heat-kernel link crosses a displacement at least Δ_{α} . The one-dimensional heat-kernel off-diagonal estimate gives

$$|Kt(u, v)| \leq Ct^{-1/2} \exp\left(-\frac{|u-v|^2}{Ct}\right),$$

hence the word gains a factor

$$\exp\left(-\frac{\Delta_{\alpha}^2}{Ct}\right)$$

The remaining vertex factors are controlled by D.KEY-FORM-TRACE. Since

$$\Delta_{\alpha} = \log L_{\alpha},$$

this decay is super-polynomial in L_{α} on every compact t -interval away from 0, and the small- t region is controlled by the same Gaussian bound. Laplace transformation over t gives local boundedness on Ω -compacts.

Corollary D.UNIFORM-SHORT-RESIDUAL.

The short-time residual decomposes as

$$R_{\alpha}^{short} = R_{\alpha}^{loc} + R_{\alpha}^{disp},$$

and for every $K \subseteq \Omega$,

$$\sup_{\alpha \geq \alpha_0(K)} \sup_{s \in K} |R_{\alpha}^{short}(s)| < \infty.$$

Same-channel repeated words are contained in R_{α}^{loc} ; no Gaussian displacement factor is assigned to them.

Lemma D.UNIFORM-SHORT-RESIDUAL

Let

$$\Omega := \mathbb{C} \setminus (-\infty, 0]$$

with the principal branch convention fixed throughout Appendix D.

Let A denote the directed set of admissible cutoff stages in the ordered Appendix-D cutoff procedure. An element $\alpha \in A$ consists of a finite cutoff pair

$$(L_{\alpha}, R_{\alpha})$$

together with the prescribed order of passage: fixed-(L,R) word extraction, separation of the one-letter prime-power package, transform-level cutoff removal, and canonical window normalization.

All traces, word contributions, residuals, and packages in this lemma are density-normalized by

$$dL_{\alpha} = (2L_{\alpha})^{-1}$$

For a prime power $q = p^m$, write

$$w(q) := \Lambda(q) q^{-1/2} = \frac{\Lambda(q)}{\sqrt{q}} = (\log p) p^{-m/2}.$$

This is the frozen global normalization.

For each cutoff stage α , after the one-letter package has been removed, write the short-time finite-cutoff residual as

$$Q(t) = E_{\alpha}^{rep}(t) + R_{\alpha}^{mix}(t), 0 < t \leq t_0.$$

Here E_{α}^{rep} denotes the repeated same-channel sector, and R_{α}^{mix} denotes the mixed-channel sector. Neither sector is promoted into the explicit one-letter package.

The residual is estimated by two separate mechanisms. We decompose

$$Q_{\alpha}^{res} = Q_{\alpha}^{loc} + Q_{\alpha}^{disp}.$$

The local/cluster sector Q_{α}^{loc} contains all genuine shifted-bump residual words whose heat-kernel geometry is local or cluster-local. This includes repeated same-channel words and any mixed words controlled by the local potential envelope rather than by an absolute displacement from the origin.

The displacement sector Q_{α}^{disp} contains only those residual terms for which a genuine one-sided/window displacement remains.

For each cutoff stage define the positive local majorant

$$V_{\alpha}^{\#}(u) := \sum_{\log q \leq R_{\alpha}} w(q) |W(u - a_q)|.$$

We say that the ordered cutoff net is locally admissible if there exists $C_{*} > 0$ such that

$$\sup_{\alpha \in A} \frac{1}{2L_{\alpha}} \int_R (e^{C_{*} t_0 V_{\alpha}^{\#}}(u) - 1 - C_{*} t_0 V_{\alpha}^{\#}(u)) du < \infty. (D.ADM)$$

Equivalently, one may impose the stronger vanishing condition

$$\frac{1}{2L_{\alpha}} \int_R (e^{C_{*} t_0 V_{\alpha}^{\#}}(u) - 1 - C_{*} t_0 V_{\alpha}^{\#}(u)) du \rightarrow 0 (D.ADM0)$$

along the ordered net. The bounded condition (D.ADM) is sufficient for local boundedness of the canonical remainders.

Claim

For every compact set $K \Subset \Omega$, there exist constants

$$C_K < \infty, \quad \alpha_0(K) \in A,$$

such that for every $\alpha \geq \alpha_0(K)$

$$\sup_{s \in K} \left| \int_0^{t_0} e^{-st} Q_\alpha^{res}(t) dt \right| \leq C_K. \quad (D.USR)$$

Consequently, the short-time residual transforms are locally bounded on Ω , uniformly along the ordered cutoff net.

Proof

Fix $\alpha \in A$. At fixed (L_α, R_α) , the trace-norm Duhamel expansion is legal by the fixed-cutoff trace-class package already established in Appendix D. After the length-zero bulk sector and the length-one explicit package have been removed, every remaining residual word has length at least two.

We estimate the local/cluster sector and the displacement sector separately.

Step 1. The local/cluster sector

For genuine shifted multiplication bumps,

$$V_q(u) = w(q)W(u - a_q),$$

repeated same-channel words and cluster-local mixed words do not, in general, acquire an absolute Gaussian factor

$$\exp(-c(\log q)^2/t)$$

per letter.

They are instead controlled by density-normalized local-loop estimates.

Since the absolute perturbation is pointwise dominated by $V_\alpha^\#$, the absolute sum of all local/cluster residual words of length at least two satisfies the Duhamel tail estimate

$$|Q_\alpha^{loc}(t)| \leq C t^{-1/2} \frac{1}{2L_\alpha} \int_R (e^{C t_0 V_\alpha^\#(u)} - 1 - C t V_\alpha^\#(u)) du, 0 < t \leq t_0. (D.LOC - 1)$$

The factor $t^{-1/2}$ is the one-dimensional localized heat-kernel diagonal bound. The expression

$$e^{C t V_\alpha^\#} - 1 - C t V_\alpha^\#$$

is precisely the absolute Duhamel tail beginning at word length two.

For $0 < t \leq t_0$ and $x \geq 0$, the elementary inequality

$$e^{(t/t_0)x} - 1 - (t/t_0)x \leq (t/t_0)^2 (e^x - 1 - x)$$

holds. Applying this with

$$x = C t_0 V_\alpha^\#(u),$$

and enlarging C if necessary, gives

$$|Q_\alpha^{loc}(t)| \leq C t^{-1/2} \left(\frac{t}{t_0}\right)^2 \frac{1}{2L_\alpha} \int_R (e^{C t_0 V_\alpha^\#(u)} - 1 - C t_0 V_\alpha^\#(u)) du.$$

By the admissibility condition (D.ADM)(D.ADM), there exists a constant C_{loc} , independent of α , such that

$$|Q_\alpha^{loc}(t)| \leq C_{loc} t^{3/2}, 0 < t \leq t_0. (D.LOC - 2)$$

Therefore, for every compact $K \subset \Omega$,

$$\sup_\alpha \sup_{s \in K} \left| \int_0^{t_0} e^{-st} Q_\alpha^{loc}(t) dt \right| < \infty \quad (D.LOC - 3)$$

This proves uniform transform boundedness for the local/cluster sector.

Step 2. The displacement sector

The displacement sector contains only those residual terms in which the one-sided window or canonical-channel extraction leaves a genuine heat-kernel displacement. For these terms, the off-displacement estimate applies.

There exist constants

$$C_0 > 0, c_0 > 0, \beta < \infty, \eta > 0,$$

independent of α , such that the length- k displacement residual sector satisfies

$$|Q_{\alpha,k}^{disp}(t)| \leq C_0 t^{-\beta} \frac{(C_0 t^\eta P_{c_0}(t))^k}{k!}, \quad (D.DISP - 1)$$

Where

$$P_{c_0}(t) := \sum_q w(q) \exp\left(-\frac{c_0(\log q)^2}{t}\right).$$

This estimate is used only for genuine displacement residuals. It is not used for repeated same-channel words or for local/cluster mixed words.

Summing over $k \geq 2$ gives

$$|Q_{\alpha}^{disp}(t)| \leq C_0 t^{-\beta} [\exp(C_0 t^\eta P_{c_0}(t)) - 1 - C_0 t^\eta P_{c_0}(t)] \quad (D.DISP - 2)$$

The prime-power majorant $P_{c_0}(t)$ is bounded on $(0, t_0]$. Near $t = 0$, it satisfies

$$P_{c_0}(t) \leq C_1 \exp(-c_1/t),$$

because $\log q \geq \log 2$ for every prime power $q \geq 2$. Hence the right-hand side of

$(D.DISP-2)$ is integrable on $](0, t_0]$.

Therefore, for every compact $K \subseteq \Omega$,

$$\sup_{\alpha} \sup_{s \in K} \left| \int_0^{t_0} e^{-st} Q_{\alpha}^{disp}(t) dt \right| < \infty. \quad (D.DISP - 3)$$

This proves uniform transform boundedness for the displacement sector.

Step 3. Combining the two sectors

Since

$$Q_{\alpha}^{res} = Q_{\alpha}^{loc} + Q_{\alpha}^{disp},$$

the bounds (D.LOC-3) and (D.DISP-3) imply

$$\sup_{\alpha \geq \alpha_0(K)} \sup_{s \in K} |\int_0^t e^{-st} Q_{\alpha}^{res}(t) dt| < \infty$$

This proves (D.USR).

Lemma D.ADM-NET

Existence and compatibility of locally admissible ordered cutoff nets

The locally admissible condition used in [D.UNIFORM-SHORT-RESIDUAL](#) is compatible with the ordered cutoff procedure of Appendix D.

More precisely, the directed cutoff stages may be chosen so that, for some constant $C_* > 0$,

$$\sup_{\alpha} \frac{1}{2L_{\alpha}} \int_R (e^{C_* t_0 V_{\alpha}^{\#}(u)} - 1 - C_* t_0 V_{\alpha}^{\#}(u)) du < \infty$$

Here

$$V_{\alpha}^{\#}(u) = \sum_{\log q \leq R_{\alpha}} w(q) |W(u - \log q)|, w(q) = \frac{\Lambda(q)}{\sqrt{q}}$$

Proof

Fix a prime-power cutoff $R < \infty$. Then the sum defining $V_R^{\#}$ is finite. Since W is a Schwartz function,

$V_R^{\#}$ is a finite Schwartz-class function. Therefore

$$\int_R (e^{C_* t_0 V_R^{\#}(u)} - 1 - C_* t_0 V_R^{\#}(u)) du < \infty$$

Denote this finite quantity by

$$M(R).$$

In the ordered cutoff procedure, R is fixed first and the window size L is then chosen in the window-normalization stage. Hence, after choosing R_{α} , we choose L_{α} large enough so that

$$\frac{M(R_\alpha)}{2L_\alpha} \leq 1$$

This is possible because $M(R_\alpha) < \infty$ for each finite R_α .

Thus, along the ordered cutoff net,

$$\sup_\alpha \frac{1}{2L_\alpha} \int_R (e^{C_* t_0 V_\alpha^\#(u)} - 1 - C_* t_0 V_\alpha^\#(u)) du \leq 1.$$

Therefore the locally admissible condition **D . ADM** holds.

This choice does not interfere with the canonical window limit, because increasing L_α only improves the window normalization step. The canonical window theorem still gives

$$\frac{1}{2L_\alpha} g_{t, L_\alpha}(a) \rightarrow G_t(a)$$

locally uniformly for a in compact intervals. It also does not alter the fixed- (L, R) word extraction, because all word expansions are performed before the cutoff-removal passage.

Therefore the ordered cutoff net may be chosen to satisfy **D . ADM** while preserving the fixed-cutoff extraction, the canonical window normalization, and the overlap convergence used in **D . CAN-REM**.

Lemma D.NET-INDEPENDENCE

Canonicity of the ordered remainder limit

Let A and A' be two admissible ordered cutoff nets satisfying **D.UNIFORM-CAN-BOUND** and the overlap convergence hypotheses used in **D.EXPORT**. Let

$$R_\alpha^{can} = F_\alpha^{reg} - B_\alpha^{can}, R_{\alpha'}^{can} = F_{\alpha'}^{reg} - B_{\alpha'}^{can}$$

Then every locally uniform subsequential limit of either family on Ω is the same holomorphic function R_H . In particular, the canonical operator-side remainder does not depend on the chosen admissible ordered cutoff net.

Proof

By local boundedness, each family is normal on Ω . Every subsequential limit is holomorphic. On U_{σ_0} , both families converge to

$F_H - B_{can}$. Therefore every subsequential limit agrees on the same nonempty open set U_{σ_0} . By the identity theorem on the connected domain Ω , all limits agree on Ω . Hence the canonical remainder is independent of the admissible net.

Lemma D.TAIL-DENSITY

Uniform density-normalized large-time heat-tail bound

Let $\alpha = (L_\alpha, R_\alpha)$ range over the admissible ordered cutoff net of Appendix D, and let

$\tilde{H}_\alpha = \tilde{H}_{L_\alpha R_\alpha}$ be the globally shifted finite-window operator from Proposition

A/B.UNIFORM-LOWER-BOUND. Then for every $t_0 > 0$

$$\frac{1}{2L_\alpha} \text{Tr}(e^{-t_0 \tilde{H}_\alpha}) \leq e^{Mt_0} (4\pi t_0)^{-1/2},$$

with a constant independent of α . Consequently, for every compact $K \Subset \Omega$,

$$\sup_\alpha \sup_{s \in K} |R_\alpha^{\text{tail}}(s)| < \infty.$$

Proof

By Proposition A/B.UNIFORM-LOWER-BOUND,

$$H_{L_\alpha R_\alpha}^{\text{orig}} \geq -M$$

Let $A_{L_\alpha} = (-\partial_u^2)D$ be the free Dirichlet Laplacian on $[-L_\alpha, L_\alpha]$. By scalar Feynman–Kac domination,

$$Tr(e^{-t_0 H_{L_\alpha R_\alpha}^{orig}}) \leq e^{Mt_0} Tr(e^{-t_0 A_{L_\alpha}})$$

The free Dirichlet heat kernel on an interval is bounded by the whole-line heat kernel:

$$K_D(t_0; u, u) \leq (4\pi t_0)^{-1/2}.$$

Therefore

$$Tr(e^{-t_0 A_{L_\alpha}}) = \int_{-L_\alpha}^{L_\alpha} K_D(t_0; u, u) du \leq 2L_\alpha (4\pi t_0)^{-1/2}$$

Dividing by $2L_\alpha$ gives

$$\frac{1}{2L_\alpha} Tr(e^{-t_0 H_{L_\alpha R_\alpha}^{orig}}) \leq e^{Mt_0} (4\pi t_0)^{-1/2}$$

Since

$$H_\alpha^\sim = H_\alpha^{orig} + MI,$$

we have

$$e^{-t_0 H_\alpha^\sim} = e^{-Mt_0} e^{-t_0 H_\alpha^{orig}}.$$

Therefore

$$Tr(e^{-t_0 H_\alpha^\sim}) = e^{-Mt_0} Tr(e^{-t_0 H_\alpha^{orig}})$$

Combining this identity with the Feynman–Kac domination estimate above gives the uniform density-normalized bound

$$\frac{1}{2L_\alpha} Tr(e^{-t_0 H_\alpha^\sim}) \leq (4\pi t_0)^{-1/2}.$$

Thus the large-time heat-density bound is uniform along the admissible ordered cutoff net.

Now let $K \subseteq \Omega$. Define

$$\delta_K := \inf_{s \in K, \lambda \geq 0} |s + \lambda| > 0, M_K := \sup_{s \in K} |\Re s|$$

$$\lambda_{n,\alpha} \geq 0,$$

the large-time spectral tail has the form

$$R_\alpha^{tail}(s) = \frac{1}{2L_\alpha} \sum_n \frac{e^{-t_0(s+\lambda_{n,\alpha})}}{s+\lambda_{n,\alpha}}$$

Hence, for $s \in K$,

$$|R_\alpha^{tail}(s)| \leq \frac{e^{t_0 M_K}}{\delta_K} \frac{1}{2L_\alpha} \sum_n e^{-t_0 \lambda_{n,\alpha}}.$$

Therefore

$$|R_\alpha^{tail}(s)| \leq \frac{e^{t_0 M_K}}{\delta_K} \frac{1}{2L_\alpha} \text{Tr}(e^{-t_0 H_\alpha^\sim})$$

By the heat-density estimate above,

$$|R_\alpha^{tail}(s)| \leq \frac{e^{t_0 M_K}}{\delta_K} (4\pi t_0)^{-1/2}.$$

This bound is uniform in α and $s \in K$. Hence

$$\sup_\alpha \sup_{s \in K} |R_\alpha^{tail}(s)| < \infty.$$

This proves the lemma.

Corollary D.TAIL-DENSITY.

Let $\alpha = (L_\alpha, R_\alpha)$ range over the admissible ordered cutoff net. For every fixed $t_0 > 0$ there is a constant $C(t_0) < \infty$, independent of α , such that

$$\frac{1}{2L_\alpha} \text{Tr}(e^{-t_0 H_\alpha^\sim}) \leq C(t_0).$$

Consequently, for every compact $K \subseteq \Omega$,

$$\sup_{\alpha \geq \alpha_0(K)} \sup_{s \in K} |R_\alpha^{tail}(s)| < \infty$$

Proof.

The shifted operators satisfy $H_\alpha \geq 0$. The finite-window heat kernel is dominated, after the fixed shift, by the heat kernel of the uniformly form-controlled confining comparison operator. By A.ENV-FORM and the standard Feynman-Kac/KLMN domination estimate,

$$Tr(e^{-t_0 \tilde{H}_\alpha}) \leq C_1(t_0) L_\alpha + C_2(t_0),$$

with constants independent of the admissible cutoff stage. Dividing by $2L_\alpha$ gives the density bound.

For $s \in K \Subset \Omega$, choose a contour or Laplace representation avoiding the cut. The large-time tail has the form

$$R_\alpha^{tail}(s) = \int_{t_0}^{\infty} e^{-st} (\theta_\alpha^{can}(t) - A_{class,\alpha}^{can}(t)) dt$$

on the initial half-plane, and the shifted nonnegative spectrum gives exponential damping after the fixed bulk subtraction. The preceding density estimate supplies a uniform majorant on K . Dominated convergence and holomorphic continuation of the tail sector give the asserted local boundedness.

D.UNIFORM-CAN-BOUND

Local boundedness of the canonical finite-cutoff remainders

Keep the notation and ordered cutoff set A from D.UNIFORM-SHORT-RESIDUAL.

For each cutoff stage $\alpha \in A$, let

$$F_\alpha(s)$$

denote the density-normalized finite-cutoff operator-side transform, and let

$$B_\alpha^{can}(s)$$

denote the finite-cutoff shifted canonical package, i.e. the Laplace transform of

$$B_{\alpha}^{can}(t) = e^{-t/4} B_{\alpha}^{raw}(t)$$

where

$$B_{\alpha}^{raw}(t) = \frac{1}{2L_{\alpha}} \sum_{\log q \leq R_{\alpha}} \frac{\Lambda(q)}{\sqrt{q}} g_{t, L_{\alpha}}(\log q).$$

Define the finite-cutoff canonical remainder

$$R_{\alpha}^{can}(s) := F_{\alpha}(s) - B_{\alpha}^{can}(s), \quad s \in \Omega.$$

Then, for every compact set

$$K \Subset \Omega,$$

there exist constants

$$C_K < \infty, \alpha_0(K) \in A,$$

such that for every $\alpha \geq \alpha_0(K)$

$$\sup_{s \in K} |R_{\alpha}^{can}(s)| \leq C_K.$$

Equivalently, the family

$$\{R_{\alpha}^{can}\}_{\alpha \in A}$$

is locally bounded on Ω along the ordered cutoff-removal procedure.

Proof

Fix $K \Subset \Omega$ Decompose the finite-cutoff canonical remainder into four sectors:

$$R_{\alpha}^{can} = R_{\alpha}^{short} + R_{\alpha}^{win} + R_{\alpha}^{tail} + R_{\alpha}^{bulk}$$

Here:

$$R_{\alpha}^{short}$$

is the transform of the density-normalized repeated/mixed short-time residual sector;

$$R_{\alpha}^{win}$$

is the window-normalization residual between the density-normalized finite-window spike kernel and the canonical Gaussian kernel;

$$R_{\alpha}^{tail}$$

is the large-time tail after the finite bulk subtraction;

and

$$R_{\alpha}^{bulk}$$

is the finite explicit bulk correction.

We bound these four pieces uniformly on K .

Step 1. Short-time residual sector

By the corrected split residual estimate D.UNIFORM-SHORT-RESIDUAL, the density-normalized short-time residual sector satisfies the following uniform compact bound.

For every compact set $K \subseteq \Omega$,

$$\sup_{\alpha \geq \alpha_0(K)} \sup_{s \in K} |R_{\alpha}^{short}(s)| < \infty$$

Here R_{α}^{short} is the transform of the residual sector

$$Q_{\alpha}^{res} = E_{\alpha}^{rep} + R_{\alpha}^{mix}$$

The proof of D.UNIFORM-SHORT-RESIDUAL does not use a per-letter off-displacement Gaussian factor for the entire residual sector. Instead, repeated same-channel words and local/cluster mixed words are controlled by the density-normalized local exponential bound (D.ADM), while only genuine displacement residuals are controlled by the Gaussian prime-power majorant $P_c(t)$.

Therefore there exists a constant $C_{K,1} < \infty$, independent of the cutoff stage α , such that

$$\sup_{\alpha \geq \alpha_0(K)} \sup_{s \in K} |R_{\alpha}^{short}(s)| \leq C_K$$

This lemma is not a fixed-(L,R) super-smoothing statement. Fixed-cutoff super-smoothing alone is insufficient because fixed-cutoff constants may depend on the cutoff stage.

The content of the lemma is the cutoff-uniform word majorant after density normalization:

$$|W_{\alpha}(q_1, \dots, q_k; t)| \leq C_0 t^{-\beta} \frac{(C_0 t^{\eta})^k}{k!} \prod_{j=1}^k \Lambda(q_j) q_j^{-1/2} \exp\left(-\frac{c_0 (\log q_j)^2}{q_j}\right)$$

The active finite set at cutoff stage α is enlarged to the full prime-power set only after taking absolute values and only inside the positive Gaussian majorant. No uniform minimum spacing between prime-power centers is used.

Step 2. Window-normalization residual

Let

$$\bar{g}_{t,L}(a) := \frac{1}{2L} g_{t,L}(a)$$

The canonical window normalization gives

$$\bar{g}_{t,L}(a) \rightarrow G_t(a)$$

locally uniformly for a in compact intervals. More importantly for local boundedness, the kernel formula gives the uniform Gaussian domination

$$|\bar{g}_{t,L}(a)| \leq C G_{\kappa t}(a)$$

for some fixed constants $C, \kappa > 0$, uniformly in L, a , and $0 < t \leq t_0$. Therefore

$$|\bar{g}_{t,L}(a) - G_t(a)| \leq C' G_{\kappa' t}(a)$$

for fixed $C', \kappa' > 0$

Hence, for $s \in K$,

$$|R_{\alpha}^{win}(s)| \leq C_K \sum_{q \geq 2} \frac{\Lambda(q)}{\sqrt{q}} \int_0^{t_0} e^{M_{\kappa' t}} G_{\kappa' t}(\log q) dt.$$

Using

$$G_{\kappa' t}(\log q) \leq C t^{-1/2} \exp\left(-\frac{(\log q)^2}{C_t}\right)$$

and $t \leq t_0$, we obtain

$$|R(s)| \leq C_{K'} \sum_{q \geq 2} \frac{\Lambda(q)}{\sqrt{q}} \exp\left(-\frac{(\log q)^2}{C t_0}\right)$$

The series on the right converges, again because the Gaussian in $\log q$ dominates any power of q . Therefore there exists $C_{K,2} < \infty$ such that

$$\sup_{\alpha} \geq \alpha_0(K) \sup_{s \in K} |R_{\alpha}^{win}(s)| \leq C_{K,2}.$$

This argument is deliberately global in $a = \log q$; it does not rely only on compact-a convergence.

Step 3. Large-time tail

By Lemma D.TAIL-DENSITY, for every $t_0 > 0$,

$$\frac{1}{2L_{\alpha}} \text{Tr}(e^{-t_0 \tilde{H}_{\alpha}}) \leq C(t_0)$$

uniformly along the admissible ordered cutoff net.

Since $K \Subset \Omega$, define

$$\delta_K := \inf_{s \in K, \lambda \geq 0} |s + \lambda| > 0, M_K := \sup_{s \in K} |\Re s|$$

The large-time resolvent tail has the spectral form

$$R_{\alpha}^{tail}(s) = \frac{1}{2L_{\alpha}} \sum_n \frac{e^{-t_0(s + \lambda_{n,\alpha})}}{s + \lambda_{n,\alpha}}$$

Thus

$$|R_{\alpha}^{tail}(s)| \leq \frac{e^{t_0 M_K}}{\delta_K} \frac{1}{2L_{\alpha}} \text{Tr}(e^{-t_0 \tilde{H}_{\alpha}}) \leq \frac{e^{t_0 M_K}}{\delta_K} C(t_0)$$

Therefore

$$\sup_{\alpha} \geq \alpha_0(K) \sup_{s \in K} |R_{\alpha}^{tail}(s)| < \infty$$

This tail bound uses only the operator-side spectral expansion and the positive distance of K from the cut. It does not use the prime-side series.

Step 4. Bulk correction

The bulk correction is not part of the prime-power residual sector. It is finite explicit data produced by the fixed bulk subtraction and by the chosen window normalization. After density normalization, it has the form

$$R_{\alpha}^{bulk}(s) = \sum_{j=1}^J c_{\alpha,j} \Phi_j(s) + \varepsilon_{\alpha}(s),$$

where:

1. J is fixed and independent of α ;
2. each Φ_j is a fixed elementary transform holomorphic on Ω ;
3. the coefficients $c_{\alpha,j}$ are uniformly bounded along the ordered cutoff net;
4. the error term α is locally uniformly bounded on Ω , and in the canonical normalization either vanishes identically or converges locally uniformly to zero.

The uniform boundedness of the coefficients follows from the density-normalized bulk construction. The window averages are divided by $2L_{\alpha}$, and the canonical window theorem fixes the limiting normalization constant $c_w = 1$. Thus no uncontrolled L_{α} -growth remains in the bulk sector. There is also no prime-power tail in the bulk sector, so no R_{α} -summability issue appears here.

Let $K \subset \Omega$. Since each Φ_j is holomorphic on Ω , it is bounded on K :

$$\sup_{s \in K} |\Phi_j(s)| < \infty$$

Since the coefficients are uniformly bounded,

$$\sup_{\alpha} \max_{1 \leq j \leq J} |c_{\alpha,j}| < \infty$$

Therefore

$$\sup_{\alpha} \sup_{s \in K} \left| \sum_{j=1}^J c_{\alpha,j} \Phi_j(s) \right| < \infty$$

The same local boundedness holds for ε_α by construction. Hence there exists $C_{K,4} < \infty$ such that

$$\sup_\alpha \geq \alpha_0(K) \sup_{s \in K} |R_\alpha^{bulk}(s)| \leq C_{K,4}$$

This proves the uniform compact bound for the bulk correction.

Step 5. Combine

Adding the four bounds gives

$$\sup_\alpha \geq \alpha_0(K) \sup_{s \in K} |R_\alpha^{can}(s)| \leq C_{K,1} + C_{K,2} + C_{K,3} + C_{K,4} =: C_K$$

Thus

$$\{R_\alpha^{can}\}$$

is locally bounded on Ω along the ordered cutoff-removal procedure.

Theorem D.BULK-FINITE-FORM

Independent finite-dimensional form of the canonical bulk sector.

Let

$$\Omega = \mathcal{C} \setminus (-\infty, 0],$$

and let

$$\alpha = (L_\alpha, R_\alpha, \dots)$$

run over the admissible ordered cutoff-removal net used in Appendix D. Let

$$R_{\alpha}^{bulk}(s)$$

denote the density-normalized canonical bulk sector remaining after the following terms have been removed from the fixed-cutoff canonical transform:

1. the canonical one-letter prime-power package

$$B_{\alpha}^{can}(s),$$

2. the short-time non-one-letter residual sector

$$R_{\alpha}^{short}(s),$$

3. the window-normalization sector

$$R_{\alpha}^{win}(s),$$

4. the large-time density tail

$$R_{\alpha}^{tail}(s).$$

Equivalently, R_{α}^{bulk} is the remaining non-arithmetic canonical sector generated by the fixed background subtraction, the free/local heat coefficients, the confining envelope, the left-wall penalty, and the finite local heat-kernel correction profiles produced in D.SPIKE-TRANSFER after the one-letter free spike kernel has been extracted.

Then there exists a finite index set

$$J_{bulk} = J_{free} \sqcup J_{env} \sqcup J_{left} \sqcup J_{loc}, \quad J := |J_{bulk}| < \infty,$$

depending only on the fixed background data, the fixed window model, the fixed bump profile, and the fixed finite heat-coefficient order used in the Appendix-D canonical extraction, but **not** on α , such that the following holds.

For each $j \in J_{bulk}$ there is a fixed scalar profile

$$\Phi_j: (0, \infty) \rightarrow \mathcal{C}$$

and a fixed holomorphic transform

$$\Phi_j(s) := \int_0^{\infty} e^{-st} \Phi_j(t) dt, \quad s \in \Omega,$$

where the integral is understood initially on a right half-plane and then by the explicit Ω -holomorphic continuation supplied by the fixed background heat-kernel calculus.

The index classes are defined as follows.

$$J_{free} = \{j: \Phi_j \text{ is one of the finite free/window heat profiles subtracted into } A_{classcan}\},$$

$J_{env} = \{j: \Phi_j \text{ is one of the finite confining – envelope heat profiles generated by } V_{env}\}$

$J_{eft} = \{j: \Phi_j \text{ is one of the finite left – wall heat profiles generated by } U_{left}\}, \text{ and}$

$J_{loc} = \{j: \Phi_j \text{ is one of the finite local correction profiles exported by D. SPIKE – TRANSFER after removal of the free one – letter kernel } g_{t,L}^0\}.$

The list

$\Phi_1, \dots, \Phi_j$

is therefore fixed once the global model data and the finite heat-coefficient order have been fixed. In particular, J is independent of L_α, R_α , and the admissible ordered net.

Moreover, for every α , the bulk sector admits the finite decomposition

$$R_\alpha^{bulk}(s) = \sum_{j=1}^J c_{\alpha,j} \Phi_j(s) + \varepsilon_\alpha(s), \quad s \in \Omega, (D. BFF. 1)$$

where:

$$\sup_\alpha \max_{1 \leq j \leq J} |c_{\alpha,j}| < \infty, (D. BFF. 2)$$

and for every compact $K \subseteq \Omega$, there exist $C_K < \infty$ and $\alpha_0(K)$ such that

$$\sup_{\alpha \geq \alpha_0(K)} \sup_{s \in K} |\varepsilon_\alpha(s)| \leq C_K. (D. BFF. 3)$$

In the canonical normalization used in the RH spine, the error term is either identically zero after the finite classical subtraction, or else satisfies the stronger refinement

$$\varepsilon_\alpha \rightarrow 0$$

locally uniformly on compact subsets of Ω .

Consequently, for every compact $K \subseteq \Omega$,

$$\sup_{\alpha \geq \alpha_0(K)} \sup_{s \in K} |R_\alpha^{bulk}(s)| < \infty. (D. BFF. 4)$$

Proof.

The fixed-cutoff decomposition in Theorem D.ONE-STEP" is performed at fixed (L,R) and uses no zero-side input and no global Ω -continuation. Its audit note separates the fixed-cutoff trace into exactly the following classes:

1. the one-letter diagonal term, whose free leading kernel $g_{t,L}^0$ is the only term exported into the explicit arithmetic package;
2. repeated same-channel words, which remain same-center local terms and create no new explicit spike locations;
3. mixed-channel words, which are super-smoothing at fixed cutoff;
4. the finite local heat-kernel correction terms produced when the true one-letter kernel is replaced by the free leading kernel;
5. the non-arithmetic bulk/background terms.

This is precisely the decomposition recorded in the fixed-cutoff source map: the finite local correction terms are absorbed into $A_{class,L,R}$ or into the fixed-cutoff operator-side remainder, while only the free one-letter spike kernel is exported as the arithmetic package.

D.SPIKE-TRANSFER gives, at fixed (L,R), the expansion

$$g_{t,L,R}^{true}(a) = g_{t,L}^0(a) + \sum_{v=1}^{M-1} t^v c_{v,L,R}(a) + E_{M,L,R}(t, a), (D.BFF.5)$$

uniformly for $|a| \leq R$, where the leading free kernel $g_{t,L}^0$ is the only term promoted to the canonical one-letter package, the finitely many terms $t^v c_{v,L,R}(a)$ are finite local heat corrections, and $E_{M,L,R}$ is the fixed-cutoff high-order remainder. The manuscript already states that these finite correction terms do not create a new pole-carrying package and are assigned to A_{class} or to the Ω -holomorphic operator-side remainder.

Because M is fixed once and for all in the Appendix-D canonical extraction, the number of local correction profiles in (D.BFF.5) is finite and independent of α . These profiles form the class J_{loc} , The background/free/window, envelope, and left-wall terms are fixed parts of the chosen operator model and the chosen classical subtraction; their finite heat-coefficient lists form J_{free} , J_{env} , and J_{left} . Since the background operator, envelope, left-wall penalty, bump profile, and heat-coefficient order are fixed globally, the union

$$J_{free} = J_{free} \sqcup J_{env} \sqcup J_{left} \sqcup J_{loc}$$

is finite and independent of α . This proves the cutoff-independence of J.

For each $j \in J_{bulk'}$, the corresponding profile $\Phi_j(t)$ is one of the fixed scalar heat profiles generated by the finite background/local heat calculus. Its Laplace transform

$$\Phi_j(s) = \int_0^\infty e^{-st} \Phi_j(t) dt$$

is holomorphic on Ω : near $t = 0$ this follows from the finite classical subtraction that removes the nonintegrable heat coefficients; for $t \geq t_0$, it follows from the same heat-trace density and lower-bound estimates used in D.TAIL-DENSITY; and the Ω -extension is the fixed background resolvent/Stieltjes continuation already used in D.OPERATOR-LIMIT-INDEP and D.CAN-REM. Thus

$$\Phi_j \in O(\Omega)$$

for every j .

It remains to verify uniformity of the coefficients. Each coefficient $c_{\alpha,j}$ is a density-normalized window average of a fixed background/local heat coefficient, or a density-normalized finite correction coefficient from (D.BFF.5). Therefore each coefficient has the form

$$c_{\alpha,j} = \frac{1}{2L_\alpha} \int_{-L_\alpha}^{L_\alpha} b_j(u) du + \delta_{\alpha,j}, \quad (D.BFF.6)$$

where b_j is a fixed coefficient-density determined by the background data and where $\delta_{\alpha,j}$ is a boundary/window error produced by the cutoff normalization.

By D.CANONICAL-WINDOW, the density normalization is fixed by

$$(2L_\alpha)^{-1} g_{t,L_\alpha}(a) \rightarrow G_t(a)$$

locally uniformly in the displacement variable, with normalization constant $c_w = 1$. The window error is therefore uniformly bounded and, for the canonical sharp window, tends to zero on compact displacement intervals. The same density normalization divides all bulk coefficient averages by $2L_\alpha$, so no uncontrolled L_α -growth remains.

The envelope and left-wall coefficients are uniformly bounded by the envelope domination and KLMN form bounds already used in Appendix A/D. The finite local correction coefficients are uniformly bounded because D.SPIKE-TRANSFER is a fixed-(L,R) finite heat expansion and because after density normalization the local correction contribution is controlled by the same form-Duhamel majorants used in D.KEY-FORM-TRACE and D.UNIFORM-SHORT-RESIDUAL. Therefore

$$\sup_\alpha \max_{1 \leq j \leq J} |c_{\alpha,j}| < \infty$$

The remaining error ε_α is the sum of:

1. the high-order fixed-cutoff heat remainder $E_{M,L,R}$ from D.SPIKE-TRANSFER after density normalization;
2. the boundary/window normalization error;

3. the portion of the finite classical subtraction that vanishes in the ordered canonical normalization.

The high-order heat remainder is controlled by the fixed-cutoff trace-norm Duhamel remainder and, after density normalization, is locally bounded on Ω -compacts. The boundary/window error is locally bounded by D.CANONICAL-WINDOW. The large-time part is locally bounded by D.TAIL-DENSITY. Hence, for every compact $K \Subset \Omega$,

$$\sup_{\alpha \geq \alpha_0(K)} \sup_{s \in K} |\varepsilon_\alpha(s)| < \infty$$

When the canonical subtraction is taken in the normalized form used in the RH spine, the finite background terms are absorbed exactly into $A_{classcan}$; in that case $\varepsilon_\alpha \equiv 0$. If the subtraction is implemented through the ordered finite-window construction, the same estimates give

$$\varepsilon_\alpha \rightarrow 0$$

locally uniformly on Ω -compacts.

Finally, for compact $K \Subset \Omega$ each fixed Φ_j is bounded on K . Therefore

$$\sup_{s \in K} |\Phi_j(s)| < \infty$$

for $1 \leq j \leq J$. Combining this with the coefficient bound and the compact-uniform bound for ε_α , we obtain

$$\sup_{\alpha \geq \alpha_0(K)} \sup_{s \in K} |R_\alpha^{bulk}(s)| \leq \sum_{j=1}^J (\sup_\alpha |c_{\alpha,j}|) (\sup_{s \in K} |\Phi_j(s)|) + \sup_{\alpha \geq \alpha_0(K)} \sup_{s \in K} |\varepsilon_\alpha(s)| < \infty$$

This proves (D.BFF.4).

Corollary D.BULK-UNIFORM

Bulk-sector compact bound.

For every compact $K \Subset \Omega$,

$$\sup_{\alpha \geq \alpha_0(K)} \sup_{s \in K} |R_\alpha^{bulk}(s)| < \infty$$

Proof.

This is (D.BFF.4).

Lemma D.LOC-TAIL

Local Duhamel tail bound after density normalization

Let α be an admissible cutoff stage with finite window L_α and finite prime-power cutoff R_α . Define the positive local majorant

$$V_\alpha^\#(u) := \sum_{\log q \leq R_\alpha} w(q) |W(u - \log q)|, w(q) = \frac{\Lambda(q)}{\sqrt{q}}.$$

Let $Q_\alpha^{loc}(t)$ denote the local/cluster residual sector of the fixed-cutoff Duhamel expansion after the length-zero bulk term and the length-one explicit prime-power spike package have been removed. Thus Q_α^{loc} contains repeated same-channel words and local/cluster mixed words, but no exported one-letter spike term.

Then there exist constants $C > 0$ and $t_0 > 0$, independent of the cutoff stage α , such that for every $0 < t \leq t_0$,

$$|Q_\alpha^{loc}(t)| \leq C t^{-1/2} \frac{1}{2L_\alpha} \int_R (e^{Ct V_\alpha^\#(u)} - 1 - Ct V_\alpha^\#(u)) du. \quad (D.LOC - TAIL)$$

Equivalently, the density-normalized local residual tail is bounded by the second-and-higher order Duhamel exponential tail generated by the positive local majorant $V_\alpha^\#$.

Proof

Fix a cutoff stage

$$\alpha = (L_\alpha, R_\alpha)$$

Since $R_\alpha < \infty$, the truncated arithmetic potential is a finite sum of shifted bump functions.

Therefore the fixed-cutoff Duhamel expansion is trace-norm legal by the fixed-L,R) trace-class package established earlier in Appendix D.

Write the local positive majorant as

$$V_\alpha^\#(u) = \sum_{\log q \leq R_\alpha} w(q) |W(u - \log q)|.$$

Every local/cluster residual word of length $k \geq 2$ contains k insertions from the truncated bump potential. Taking absolute values and discarding cancellations, each such word is dominated by the corresponding k -fold insertion of $V_\alpha^\#$.

The one-dimensional heat-kernel diagonal estimate gives, after localization and density normalization,

$$\frac{1}{2L_\alpha} |\text{Tr}(\text{localized } k\text{-fold local word at time } t)| \leq C t^{-1/2} \frac{1}{2L_\alpha} \int_R \frac{(C t V_\alpha^\#(u))^k}{k!} du$$

Here $t^{-1/2}$ is the one-dimensional diagonal heat-kernel factor, and $t^k/k!$ is the ordered simplex factor from the k -fold Duhamel integral. The constant C depends only on the fixed heat-kernel normalization, the bump profile, and the fixed localizer conventions. It does **not** depend on L_α , R_α , or the number of active prime powers, because the full cutoff dependence is retained inside $V_\alpha^\#$.

Summing over all local/cluster residual words of length $k \geq 2$, we obtain

$$|Q_\alpha^{\text{loc}}(t)| \leq C t^{-1/2} \frac{1}{2L_\alpha} \int_R \sum_{k \geq 2} \frac{(C t V_\alpha^\#(u))^k}{k!} du$$

Since

$$\sum_{k \geq 2} \frac{x^k}{k!} = e^x - 1 - x,$$

with

$$x = C t V_\alpha^\#(u),$$

this becomes

$$|Q_{\alpha}^{loc}(t)| \leq C t^{-1/2} \frac{1}{2L_{\alpha}} \int_R (e^{Ct V_{\alpha}^{\#}(u)} - 1 - Ct V_{\alpha}^{\#}(u)) du$$

This proves the claimed estimate.

Referee note

This lemma is deliberately local. It does not assign an off-displacement Gaussian factor to repeated same-channel words or to local/cluster mixed words. Those terms are controlled instead by density-normalized local exponential domination.

The off-displacement Gaussian estimate is reserved only for the genuinely displacement-sensitive residual sector. This is why the corrected residual decomposition separates

$$Q_{\alpha}^{res} = Q_{\alpha}^{loc} + Q_{\alpha}^{disp}$$

Thus the repeated same-channel sector remains inside the Ω -holomorphic operator-side remainder without being forced into the wrong prime-displacement geometry.

Proposition D. OPERATOR-LIMIT-INDEP, construction before subtraction.

Let $\alpha = (L_{\alpha}, R_{\alpha})$ range over the admissible ordered cutoff net. The canonical operator transform is constructed before the canonical prime package is subtracted. More precisely, the construction order is

$$F_{L,R}^{can} \rightarrow F_L^{can} (R \rightarrow \infty, L \text{ fixed}), \text{ then}$$

$$F_L^{can} \rightarrow F_H^{can} \in O(\Omega) (L \rightarrow \infty \text{ along the admissible net}).$$

Only after F_H^{can} is constructed as an Ω -holomorphic object is the overlap decomposition

$$F_H^{can}(s) = B_{can}(s) + R_H^{can}(s), \quad s \in U_{\sigma_0},$$

used to identify the canonical remainder.

Thus B_{can} is not used to construct F_H^{can} . The operator-side export contains no subtraction circularity.

Proof.

At fixed L , the trace-norm resolvent convergence $R \rightarrow \infty$ is obtained from the finite-window resolvent identity and the uniform convergence of the truncated potentials on $[-L, L]$. This gives $F_L^{can} \in O(\Omega)$.

The admissible $L \rightarrow \infty$ passage is taken in the compact-open topology on Ω , using the local boundedness and normal-family estimates proved below. This produces $F_H^{can} \in O(\Omega)$. The finite-cutoff packages $B_{\alpha can}$ enter only after these operator transforms have been constructed, namely in the identity

$$R_{\alpha}^{can} = F_{\alpha}^{can} - B_{\alpha}^{can}$$

on the overlap half-plane where both sides are independently defined. The subsequent Montel/Vitali argument identifies the limiting remainder, but it does not create the limiting operator transform. Therefore the construction is independent of the package subtraction.

Theorem D.MASTER-RESIDUAL

Canonical residual bound and Ω -holomorphic ordered limit.

Let

$$\Omega = \mathcal{C} \setminus (-\infty, 0],$$

and let

$$\alpha = (L_{\alpha}, R_{\alpha}, \dots)$$

run over the admissible ordered cutoff-removal net used in Appendix D.

For each α , let

$$F_{\alpha}^{can}(s) = B_{\alpha}^{can}(s) + R_{\alpha}^{can}(s), s \in U_{\sigma_0}, (D.MR.1)$$

be the finite/cutoff canonical decomposition on the overlap half-plane. Here B_α^{can} is the finite canonical one-letter shifted prime-power package and R_α^{can} is the sum of all non-one-letter canonical sectors.

Then for every compact $K \subseteq \Omega$, there exist $C_K < \infty$ and $\alpha_0(K)$ such that

$$\sup_{\alpha \geq \alpha_0(K)} \sup_{s \in K} |R_\alpha^{can}(s)| \leq C_K. \quad (D. MR. 2)$$

More precisely, the residual decomposes exhaustively as

$$R_\alpha^{can} = R_\alpha^{short} + R_\alpha^{win} + R_\alpha^{tail} + R_\alpha^{bulk}, \quad (D. MR. 3)$$

where:

$$\sup_{\alpha \geq \alpha_0(K)} \sup_{s \in K} |R_\alpha^{short}(s)| < \infty, \quad (D. MR. 4)$$

$$\sup_{\alpha \geq \alpha_0(K)} \sup_{s \in K} |R_\alpha^{win}(s)| < \infty, \quad (D. MR. 5)$$

$$\sup_{\alpha \geq \alpha_0(K)} \sup_{s \in K} |R_\alpha^{tail}(s)| < \infty, \quad (D. MR. 6)$$

and

$$\sup_{\alpha \geq \alpha_0(K)} \sup_{s \in K} |R_\alpha^{bulk}(s)| < \infty. \quad (D. MR. 7)$$

Consequently, the family

$$\{R_\alpha^{can}\}_\alpha$$

is a normal family on Ω . Every subnet has a locally uniformly convergent subnet with holomorphic limit.

Moreover, on U_{σ_0} ,

$$R_\alpha^{can}(s) = F_\alpha^{can}(s) - B_\alpha^{can}(s) \rightarrow F_H^{can}(s) - B_{can}(s). \quad (D. MR. 8)$$

Therefore every subsequential holomorphic limit agrees on U_{σ_0} . Since Ω is connected, the identity theorem forces uniqueness of the subsequential limit. Hence the full ordered net converges locally uniformly on Ω to a unique holomorphic function

$$R_H^{can} \in O(\Omega). \text{ (D. MR. 9)}$$

Finally, on the overlap half-plane,

$$F_H^{can}(s) = B_{can}(s) + R_H^{can}(s), \quad s \in U_{\sigma_0}. \text{ (D. MR. 10)}$$

Proof.

The finite/cutoff identity (D.MR.1) is the fixed- α canonical decomposition exported by the fixed-cutoff one-letter package construction. The finite package B_α^{can} consists only of the shifted one-letter prime-power sector. All remaining canonical sectors are assigned to R_α^{can} . Hence the residual decomposition (D.MR.3) is exhaustive by construction: the short-time non-one-letter residual is R_α^{short} , the window-normalization contribution is R_α^{win} , the large-time density contribution is R_α^{tail} , and the finite background/classical/local heat correction contribution is R_α^{bulk} .

We prove the four compact bounds.

Short sector.

The short sector is the sum of the local/cluster residual and the genuine displacement residual. D.UNIFORM-SHORT-RESIDUAL gives the required local boundedness on every compact $K \subset \Omega$. In particular, the local/cluster part is controlled by the form-Duhamel trace bound after density normalization, and the displacement part gains the off-diagonal heat-kernel factor

$$\exp(-\Delta_\alpha^2 Ct), \Delta_\alpha = \log L_\alpha,$$

which is super-polynomially small in the ordered net. Thus

$$\sup_{\alpha \geq \alpha_0(K)} \sup_{s \in K} |R_\alpha^{short}(s)| < \infty$$

This proves (D.MR.4).

Window sector.

D.CANONICAL-WINDOW gives, for the canonical sharp window,

$$\frac{1}{2L_\alpha} g_{t, L_\alpha}(a) \rightarrow G_t(a)$$

locally uniformly on compact aa-intervals, with normalization constant $c_w = 1$. The window error is therefore uniformly bounded after density normalization, and the Gaussian summability of the one-letter package gives compact-uniform boundedness of its transform on every $K \subset \Omega$. Thus

$$\sup_{\alpha \geq \alpha_0(K)} \sup_{s \in K} |R_\alpha^{win}(s)| < \infty$$

This proves (D.MR.5).

Tail sector.

D.TAIL-DENSITY gives the density-normalized large-time estimate for the canonical heat/resolvent tail. The globally shifted nonnegative operator family and the heat-trace density bound imply that, for every compact $K \subseteq \Omega$, the large-time Laplace/resolvent contribution is uniformly bounded along the admissible net. Hence

$$\sup_{\alpha \geq \alpha_0(K)} \sup_{s \in K} |R_\alpha^{tail}(s)| < \infty$$

This proves (D.MR.6).

Bulk sector.

By Theorem D.BULK-FINITE-FORM,

$$R_\alpha^{bulk}(s) = \sum_{j=1}^J c_{\alpha,j} \Phi_j(s) + \varepsilon_\alpha(s),$$

where J is independent of α , each $\Phi_j \in \mathcal{O}(\Omega)$ is fixed, the coefficients $c_{\alpha,j}$ are uniformly bounded, and ε_α is locally uniformly bounded on Ω -compacts. Therefore, for every compact $K \subseteq \Omega$,

$$\sup_{\alpha \geq \alpha_0(K)} \sup_{s \in K} |R_\alpha^{bulk}(s)| < \infty$$

This proves (D.MR.7).

Adding the four estimates (D.MR.4)–(D.MR.7) gives

$$\sup_{\alpha \geq \alpha_0(K)} \sup_{s \in K} |R_\alpha^{can}(s)| \leq C_K$$

for some $C_K < \infty$. This proves (D.MR.2).

Each R_α^{can} is holomorphic on Ω at fixed cutoff, because it is the difference between the fixed-cutoff operator transform and the finite fixed-cutoff package, both of which are holomorphic on Ω . The local boundedness (D.MR.2) therefore implies, by Montel's theorem, that $\{R_\alpha^{can}\}$ is a normal family on Ω . Thus every subnet has a further subnet converging locally uniformly on Ω to a holomorphic function.

It remains to prove uniqueness of the subsequential limit. On the overlap half-plane U_{σ_0} , the finite/cutoff identity gives

$$R_{\alpha}^{can}(s) = F_{\alpha}^{can}(s) - B_{\alpha}^{can}(s)$$

By D. OPERATOR-LIMIT-INDEP,

$$F_{\alpha}^{can} \rightarrow F_H^{can}$$

locally uniformly on Ω -compacts. On U_{σ_0} , the ordered finite package construction gives

$$B_{\alpha}^{can} \rightarrow B_{can}$$

locally uniformly. Hence every subsequential limit R_* of R_{α}^{can} satisfies

$$R_*(s) = F_H^{can}(s) - B_{can}(s), \quad s \in U_{\sigma_0}$$

If $R_*^{(1)}$ and $R_*^{(2)}$ are two subsequential limits, then

$R_*^{(1)}(s) = R_*^{(2)}(s)$ on U_{σ_0} . Since both limits are holomorphic on Ω , and since Ω is connected, the identity theorem gives

$$R_*^{(1)} = R_*^{(2)}$$

on all of Ω . Thus the subsequential limit is unique. Therefore the entire ordered net converges locally uniformly on Ω to a unique holomorphic function, denoted

$$R_H^{can} \in O(\Omega)$$

Passing to the limit in the overlap identity gives

$$F_H^{can}(s) = B_{can}(s) + R_H^{can}(s), \quad s \in U_{\sigma_0}$$

Proposition D.CAN-REM

Canonical remainder convergence and the Ω -shield

Let

$$\Omega := \mathcal{C} \setminus (-\infty, 0]$$

with the principal branch convention fixed throughout Appendix D.

Let the global prime-power normalization be

$$w(q) = \frac{\Lambda(q)}{\sqrt{q}}, q = p^m,$$

equivalently

$$w(p^m) = (\log p) p^{-m/2}$$

This is the unique normalization used in the spike package and in the later D/E/F/H spine. The current manuscript explicitly fixes this convention and distinguishes the raw G_t -package from the shifted $K_t = e^{-t/4} G_t$ -package used in the RH closure.

For finite cutoffs (L,R), let

$$F_{L,R}(s)$$

be the fixed-cutoff operator-side resolvent/Laplace transform constructed in Appendix D. Let

$$B_{L,R}^{can}(s)$$

denote the finite-cutoff canonical shifted package obtained from the finite raw package by the rule

$$B_{L,R}^{can}(t) = e^{-t/4} B_{L,R}^{raw}(t).$$

Equivalently, the raw local extraction uses

$$G_t(u) = (4\pi t)^{-1/2} e^{-u^2/(4t)},$$

while the closure package uses

$$K_t(u) = e^{-t/4} G_t(u)$$

The manuscript already fixes this package firewall: the raw G_t -package is extracted in Appendix D, while the shifted K_t -package is the one used in Appendices E, F, and H.

Define the finite-cutoff canonical remainder by

$$R_{L,R}^{can}(s) := F_{L,R}(s) - B_{L,R}^{can}(s), \quad s \in \Omega.$$

Here $B_{L,R}^{can}$ is finite-cutoff and therefore holomorphic on Ω . This finite-cutoff fact is not a statement that the limiting package B_{can} is holomorphic on all of Ω .

Then, along the ordered cutoff-removal procedure of Appendix D, the family

$$\{R_{L,R}^{can}\}$$

converges locally uniformly on compact subsets of Ω to a holomorphic function

$$R_H \in O(\Omega).$$

Moreover, on the nonempty overlap half-plane $U_{\sigma_0} \subset \Omega$, one has

$$F_H(s) = B_{can}(s) + R_H(s), \quad s \in U_{\sigma_0}$$

Consequently, the operator-side remainder has no isolated poles in Ω , and no pole in the later global comparison can be created or cancelled by the operator-side remainder.

Corollary D.CAN-REM

Canonical Ω -holomorphic remainder.

Along the admissible ordered cutoff-removal net, the canonical operator-side remainders satisfy

$$R_{\alpha}^{can} \rightarrow R_H^{can}$$

locally uniformly on compact subsets of Ω , where

$$R_H^{can} \in O(\Omega)$$

Moreover, on the overlap half-plane U_{σ_0} ,

$$F_H^{can}(s) = B_{can}(s) + R_H^{can}(s)$$

Proof.

This is Theorem D.MASTER-RESIDUAL

Theorem D.5.10W

Canonical whole-line operator transform

Let

$$\Omega = \mathcal{C} \setminus (-\infty, 0]$$

Appendix D constructs a canonical whole-line operator-side transform

$$F_H \in O(\Omega).$$

Let B_{can} denote the shifted canonical closure package, built from

$$K_t(u) = e^{-t/4} G_t(u)$$

with global spike weight

$$w(q) = \frac{\Lambda(q)}{\sqrt{q}}.$$

Then there exists

$$R_H \in O(\Omega)$$

such that on the overlap half-plane $U_{\sigma_0} \subset \Omega$,

$$F_H(s) = B_{can}(s) + R_H(s)$$

No unrestricted prime-side normal convergence on all of Ω is asserted or used. The Ω -holomorphic object exported by Appendix D is the operator-side remainder R_H , obtained by Proposition D.CAN-REM.

D.5.1 (Ω and branches).

Let $\Omega = \mathbb{C} \setminus (-\infty, 0]$ Fix the **principal branch** of $\zeta = \sqrt{s}$ on Ω .

D.5.2 (Raw resolvent trace: trace-class/holomorphy legality).

By Lemma A.GROWTH and Corollaries A.TRACE–A.HOLO, the eigenvalues satisfy

$$\lambda_n(L, R) \geq cn^2 - C$$

Hence for every compact $K \subset \Omega$, the series

$$\sum_{n \geq 1} \frac{1}{s + \lambda_n(L, R)}$$

converges absolutely and uniformly for $s \in K$. Therefore

$$(H_{L, R} + s)^{-1} \in S_1, F_{L, R}^{raw}(s) := \text{Tr}((H_{L, R} + s)^{-1}) = \sum_{n \geq 1} \frac{1}{s + \lambda_n(L, R)}$$

defines a holomorphic function on Ω .

Lemma D.5.3 (Raw Laplace–resolvent identity on $\Re s > 0$).

Let

$$\Theta_{L, R}^{raw}(t) := \text{Tr}(e^{-tH_{L, R}}), t > 0$$

Then for $\Re s > 0$,

$$F_{L,R}^{raw}(s) = \int_0^\infty e^{-st} \Theta_{L,R}^{raw}(t) dt.$$

Proof. Expand in the eigenbasis:

$$\Theta_{L,R}^{raw}(t) = \sum_{n \geq 1} e^{-t\lambda_n(L,R)},$$

compute

$$\int_0^\infty e^{-st} e^{-t\lambda_n(L,R)} dt = \frac{1}{s + \lambda_n(L,R)} (\Re s > 0),$$

and sum using absolute convergence.

Remark: this lemma is recorded only for the raw trace. The canonical bridge object used in Theorem D.5.10W, Theorem E.INTERFACE, and Section F.9 is the bulk-subtracted operator-side transform attached to the canonical package fixed in Section 5.1.4.

Lemma D.5.3 (Laplace identity on $\Re s > 0$).

For $\Re(s) > 0$

$$F_{L,R}(s) = \int_0^\infty e^{-st} \Theta_{L,R}(t) dt.$$

Proof. Expand in the eigenbasis: $\Theta_{L,R}(t) = \sum_n e^{-t\lambda_n} n$ and compute

$$\int_0^\infty e^{-st} e^{-t\lambda_n} dt = (s + \lambda_n)^{-1}, \text{ justified by absolute convergence for } \Re(s) > 0 \text{ then sum over } n.$$

D.5.3 The explicit package in time (matches Appendix H)

Definition D.5.4 (Gaussian kernel, $\alpha=0$).

For $t > 0$ and $u \geq 0$,

$$G_t(u) := (4\pi t)^{-1/2} \exp\left(-\frac{u^2}{4t}\right).$$

Definition D.5.5 (Prime spike package).

$$P_R(t) := \sum_{(p,m) \in I_R} a_{p,m} G_t(u_{p,m}) = (4\pi t)^{-1/2} \sum_{(p,m) \in I_R} (\log p) p^{-m/2} \exp\left(-\frac{(m \log p)^2}{4t}\right)$$

Definition D.5.6 (Bulk package).

Let $A_{class,L}(t)$ be our **fixed finite explicit** bulk/classical term

Define the explicit package

$$B_{L,R}(t) := A_{class,L}(t) + P_R(t).$$

D.5.4 The explicit package transform is holomorphic on Ω

D.5.7a The explicit package transform is holomorphic on Ω

Recall $\Omega := \mathbb{C} \setminus (-\infty, 0]$, and fix the **principal branch** of $\zeta = \sqrt{s}$ on Ω .

Laplace transforms on $\Re(s) > 0$

For $\Re(s) > 0$, define the Laplace transforms

$$\tilde{A}_{class,L}(s) := \int_0^\infty e^{-st} A_{class,L}(t) dt, \quad \tilde{P}_R(s) := \int_0^\infty e^{-st} P_R(t) dt,$$

and the **explicit package transform**

$$\tilde{B}_{L,R}(s) := \tilde{A}_{class,L}(s) + \tilde{P}_R(s)$$

Holomorphic continuation to Ω

Lemma D.5.7 (Bulk transform).

The function $\tilde{A}_{class,L}$ extends holomorphically to Ω (off the cut), since $A_{class,L}(t)$ is a finite explicit linear combination of elementary terms whose Laplace transforms are holomorphic on Ω once the principal branch is fixed.

D.5.5 The Ω remainder shield (no placeholders)

Define the remainder transform on Ω by

$$R_{L,R}^{\sim}(s) := F_{L,R}(s) - B_{L,R}^{\sim}(s), s \in \Omega.$$

Lemma D.TAIL- Ω (Large-time tail is Ω -holomorphic)

Fix $t_0 > 0$, and let

$U_\sigma := \{s \in \mathbb{C} : \Re s > \sigma\}$, where $\sigma > 0$ (equivalently, more generally, $\sigma > -\lambda_1$ if one does not normalize $H \geq 0$).

Define

$$T_H^>(s) := \int_{t_0}^{\infty} e^{-st} \Theta_H(t) dt, s \in U_\sigma, t$$

and let $\widehat{A_{class}}(s) \in O(\Omega)$ be the Ω -holomorphic bulk transform constructed earlier in Appendix D.

Define

$$A^>(s) := \widehat{A_{class}}(s) - \int_0^{t_0} e^{-st} A_{class}(t) dt.$$

Then:

1. $T_H^>$ extends holomorphically to Ω ;
2. $A^> \in O(\Omega)$;
3. hence

$$R_{tail}(s) := T_H^>(s) - A^>(s)$$

belongs to $O(\Omega)$.

Moreover, on the initial half-plane U_σ ,

$$A^>(s) = \int_{t_0}^{\infty} e^{-st} A_{class}(t) dt,$$

and therefore

$$R_{tail}(s) = \int_{t_0}^{\infty} e^{-st} (\Theta_H(t) - A_{class}(t)) dt.$$

Proof

By Appendix A, H is self-adjoint with compact resolvent and purely discrete spectrum

$$0 \leq \lambda_1 \leq \lambda_2 \leq \dots, \lambda_n \rightarrow +\infty,$$

and for every $t > 0$,

$$\Theta_H(t) = \sum_{n \geq 1} e^{-t\lambda_n},$$

with absolute convergence.

Fix $s \in U_\sigma$. Since $\sigma > 0$, we have $\Re(s + \lambda_n) \geq \sigma + \lambda_n > 0$ so

$$\int_{t_0}^{\infty} |e^{-t(s+\lambda_n)}| dt = \frac{e^{-t_0(\Re s + \lambda_n)}}{\Re s + \lambda_n}.$$

Hence

$$\sum_{n \geq 1} \int_{t_0}^{\infty} |e^{-t(s+\lambda_n)}| dt \leq \frac{e^{-t_0\sigma}}{\sigma} \sum_{n \geq 1} e^{-t_0\lambda_n} = \frac{e^{-t_0\sigma}}{\sigma} \Theta_H(t_0) < \infty$$

Therefore termwise integration is justified on U_σ , and

$$T_H^>(s) = \sum_{n \geq 1} \int_{t_0}^{\infty} e^{-t(s+\lambda_n)} dt = \sum_{n \geq 1} \frac{e^{-t_0(s+\lambda_n)}}{s+\lambda_n}.$$

Now let $K \Subset \Omega$. Since K stays a positive distance from the cut $(-\infty, 0]$, the quantity

$$\delta_K := \inf_{s \in K'} x \geq 0 |s + x|$$

is strictly positive. Also set

$$M_K := \sup_{s \in K} |s|.$$

Then for every $s \in K$ and every $n \geq 1$,

$$\left| \frac{e^{-t_0(s+\lambda_n)}}{s+\lambda_n} \right| \leq \frac{e^{t_0 M_K}}{\delta_K} e^{-t_0 \lambda_n}$$

Because

$$\sum_{n \geq 1} e^{-t_0 \lambda_n} = \Theta_H(t_0) < \infty$$

the Weierstrass M-test gives locally uniform convergence of the series on K . Hence $T_H^>$ extends holomorphically to Ω .

Next, by the bulk-transform lemma, $\widehat{A_{class}} \in O(\Omega)$. Since $A_{class}(t)$ is a fixed finite explicit bulk term, it is locally integrable on $(0, t_0]$, so the map

$$s \mapsto \int_0^{t_0} e^{-st} A_{class}(t) dt$$

is entire. Therefore

$$A^>(s) = \widehat{A_{class}}(s) - \int_0^{t_0} e^{-st} A_{class}(t) dt$$

belongs to $O(\Omega)$.

On the initial half-plane U_σ , the bulk-transform lemma gives

$$\widehat{A_{class}}(s) = \int_{t_0}^{\infty} e^{-st} A_{class}(t) dt,$$

hence

$$A^>(s) = \int_{t_0}^{\infty} e^{-st} A_{class}(t) dt - \int_0^{t_0} e^{-st} A_{class}(t) dt = \int_{t_0}^{\infty} e^{-st} A_{class}(t) dt.$$

Therefore, on U_σ ,

$$R_{tail}(s) = T_H^>(s) - A^>(s) = \int_{t_0}^{\infty} e^{-st} (\Theta_H(t) - A_{class}(t)) dt.$$

Since both $T_H^>$ and $A^>$ are holomorphic on Ω , it follows that

$$R_{tail} \in O(\Omega).$$

This proves the lemma.

Lemma D.5.9W (Ω -holomorphy of the whole-line short-time spike transform)

Fix $0 < t_0 \leq 1$. Let

$$w(q) := \frac{\Lambda(q)}{\sqrt{q}}, \quad q = p^m,$$

and let $g_t(a)$ be the canonical whole-line spike kernel from the prime-power decomposition of the regularized whole-line heat trace. Define

$$P^{\sim <}(s) := \sum_q w(q) \int_0^{t_0} e^{-st} g_t(\log q) dt, \quad s \in \Omega := \mathbb{C} \setminus (-\infty, 0]$$

Then $P^{\sim <} \in O(\Omega)$ $P^{<} \in O(\Omega)$

Proof.

For each fixed q , the map

$$s \mapsto \int_0^{t_0} e^{-st} g_t(\log q) dt$$

is holomorphic on Ω by holomorphy under the integral sign. By the same Ω -majorant argument used in Lemma D.5.9, there is, for every compact $K \subseteq \Omega$, a summable majorant $M_q(K)$ such that

$$\sup_{s \in K} |w(q) \int_0^{t_0} e^{-st} g_t(\log q) dt| \leq M_q(K), \quad \sum_q M_q(K) < \infty$$

Hence the series converges uniformly on K by the Weierstrass M-test, so $P^{\sim <}$ is holomorphic on Ω .

Throughout, fix

$\Omega := \mathbb{C} \setminus (-\infty, 0]$, and fix the **principal branch** of $\zeta = s$ on Ω .

D-A.1 Resolvent trace class and analyticity on Ω

Lemma D-A.1 (Trace class resolvent; holomorphic dependence).

Let $H_{L,R}$ be a self-adjoint operator on a finite-volume domain (finite L , with the stated boundary conditions) such that

$$\sigma(H_{L,R}) \subset [0, \infty) \text{ and } (H_{L,R} + 1)^{-1} \text{ is compact.}$$

Define, for $s \in \Omega$,

$$F_{L,R}(s) := \text{Tr}((H_{L,R} + s)^{-1}).$$

Then:

1. For each $s \in \Omega$, the resolvent $(H_{L,R} + s)^{-1}$ exists and is trace class.
2. The map $s \mapsto (H_{L,R} + s)^{-1}$ is holomorphic as a bounded-operator-valued function on Ω .
3. Consequently, $F_{L,R}$ is holomorphic on Ω .

Proof. Since $s \in \Omega$ implies $-s \notin [0, \infty) = \sigma(H_{L,R})$, the resolvent exists. Holomorphy of the resolvent on the resolvent set is standard (resolvent identity / holomorphic functional calculus). Compact resolvent implies discrete spectrum with $\lambda_n \rightarrow \infty$, hence $(H_{L,R} + s)^{-1}$ is trace class and Tr of a trace-class holomorphic family is holomorphic.

D-A.2 Laplace–resolvent identity in trace form (legality)

Let

$$\Theta_{L,R}(t) := \text{Tr}(e^{-tH_{L,R}}) = \sum_{n \geq 1} e^{-t\lambda_n}, t > 0,$$

where $\{\lambda_n\}$ are the eigenvalues of $H_{L,R}$ (counted with multiplicity).

Lemma D-A.2 (Laplace identity; Tonelli/Fubini justified).

For $\Re(s) > 0$,

$$F_{L,R}(s) = \text{Tr}((H_{L,R} + s)^{-1}) = \int_0^\infty e^{-st} \Theta_{L,R}(t) dt.$$

Proof. For $\Re(s) > 0$ and each n ,

$$\int_0^\infty e^{-st} e^{-t\lambda_n} dt = \frac{1}{s + \lambda_n}.$$

Since the summands are nonnegative for $\Re(s) > 0$, Tonelli's theorem applies to the partial sums:

$$\int_0^\infty e^{-st} \sum_{n \leq N} e^{-t\lambda_n} dt = \sum_{n \leq N} \frac{1}{s + \lambda_n}.$$

Let $N \rightarrow \infty$; the right side increases to $\sum_n (s + \lambda_n)^{-1}$, which equals $\text{Tr}((H_{L,R} + s)^{-1})$.

D.A.3. Canonical Gaussian kernel used in the R_H closure

Throughout the RH closure spine, the Gaussian kernel used in the spike package, the interface theorem, and the Laplace-domain bridge is the one-dimensional heat kernel

$$G_t(u) := (4\pi t)^{-1/2} e^{-u^2/(4t)}, t > 0, u \in \mathbb{R}.$$

Its Laplace transform in the time variable is

$$\int_0^\infty e^{-st} G_t(u) dt = \frac{1}{2\sqrt{s}} e^{-|u|\sqrt{s}}, \Re(s) > 0$$

with the principal branch of $\sqrt{s}$.

Lemma D.A.4. Laplace transform of the canonical Gaussian

Let

$$G_t(u) = (4\pi t)^{-1/2} e^{-u^2/(4t)}$$

.

Then for every $u \in \mathbb{R}$ and every s with $\Re(s) > 0$,

$$\int_0^\infty e^{-st} G_t(u) dt = \frac{1}{2\sqrt{s}} e^{-|u|\sqrt{s}}.$$

Proof.

This is the standard Laplace transform of the one-dimensional heat kernel. It may be verified directly by the classical identity

$$\int_0^\infty t^{-1/2} e^{-st} - u^2 4t dt = \sqrt{\pi} s^{-1/2} e^{-|u|\sqrt{s}}, \Re(s) > 0$$

and multiplying by the prefactor $(4\pi)^{-1/2}$

D.A.5. Role of this kernel in the manuscript

The kernel G_t above is the shared Gaussian kernel appearing in:

1. the displacement spike package,
2. the operator-side Laplace transform,
3. the zero-side Gaussian explicit-formula comparison,
4. the local bridge theorem on an overlap domain,
5. the R_H closure chain in Appendix F.

Accordingly, every occurrence of the spike package in the R_H closure is to be interpreted through the transform pair

$$G_t(u) \leftrightarrow \frac{1}{2\sqrt{s}} e^{-|u|\sqrt{s}}$$

Lemma D-A.8 (Remainder absorption via the Ω -shield)

Let $\tilde{B}_{L,R}$ denote the explicit package transform (bulk + spike transform) and define on Ω

$$\tilde{R}_{L,R}(s) := F_{L,R}(s) - \tilde{B}_{L,R}(s).$$

Then $R_{L,R}^\sim$ is holomorphic on Ω . In particular, any contributions to $\Theta_{L,R}(t)$ not included in the explicit spike package $P_R(t)$ are, by definition, absorbed into $R_{L,R}^\sim$ and cannot create or cancel isolated off-cut poles in Ω .

Proof. Holomorphy follows because both $F_{L,R}$ and $B_{L,R}^\sim$ are holomorphic on Ω ; their difference is holomorphic.

Reader's Note: The preceding estimates are along the admissible ordered net, not over arbitrary independent (L,R). The local/cluster residual sector is controlled by density-normalized local exponential domination, while only the displacement sector uses Gaussian prime-power decay. Hence the Montel argument applies to $R_\alpha^{can} = F_\alpha^{reg} - B_\alpha^{can}$, yielding $R_H \in O(\Omega)$. Appendix D exports this holomorphic remainder and the overlap decomposition only; Appendix H supplies the global meromorphic zero-side package.

Theorem D.CAN-FUNC-EXPORT — canonical functional export.

Let

$$\Omega = \mathbb{C} \setminus (-\infty, 0].$$

Appendix D constructs the canonical displacement/commutator transform

$$F_H^{can} \in O(\Omega)$$

and a canonical operator-side remainder

$$R_H^{can} \in O(\Omega)$$

such that, on a nonempty overlap half-plane $U_{\sigma_0} \subset \Omega$,

$$F_H^{can}(s) = B_{can}(s) + R_H^{can}(s)$$

The package B_{can} is the shifted canonical package

$$B_{can}(t) = \sum_q w(q)K_t(\log q), \quad K_t(u) = e^{-t/4} G_t(u).$$

The raw multiplication-bump heat trace and its local-loop kernel $J_{t,L,\alpha}$ are auxiliary and are not the exported transform or package in the RH closure.

Proof.

By D.CANONICAL-FUNCTIONAL, the canonical one-letter sector is

$$P_{L,R}^{can}(t) = \sum_{\log q \leq R} w(q)g_{t,L}(\log q)$$

By D.CANONICAL-WINDOW,

$$(2L)^{-1} g_{t,L}(a) \rightarrow G_t(a)$$

locally uniformly on compact aa-intervals, hence the density-normalized canonical package converges to

$$\sum_{\log q \leq R} w(q)G_t(\log q).$$

Multiplying by the fixed shift $e^{-t/4}$ gives the shifted package

$$B_{can}(t) = \sum_q w(q)K_t(\log q)$$

The remaining canonical sectors — repeated channel terms, mixed terms, window defects, finite local corrections, and bulk terms — are absorbed into A_{class}^{can} and R_H^{can} .

The remaining sectors are not part of the canonical one-letter package. Their

absorption into A_{class}^{can} and R_H^{can} is handled later by D.UNIFORM-CAN-BOUND and D.CAN-REM.

The residual estimates and normal-family argument of D.CAN-REM give

$$R_H^{can} \in O(\Omega)$$

Therefore, on U_{σ_0} ,

$$F_H^{can}(s) = B_{can}(s) + R_H^{can}(s)$$

Theorem D.SOURCE

Canonical transform consumed by the RH closure.

The operator-side transform used in Appendices E and F is

$$F_H^{can}(s) = L[\Theta_H^{can} - A_{class}^{can}](s),$$

the canonical density-normalized displacement/commutator transform exported by Appendix D.

Its one-letter sector is

$$P_{L,R}^{can}(t) = \sum_{\log q \leq R} \frac{\Lambda(q)}{\sqrt{q}} g_{t,L}(\log q), \quad g_{t,L}(a) = \text{Tr}(StALTa)$$

The raw multiplication-bump local-loop trace and the auxiliary kernel $J_{t,L,\alpha}$ are not the transforms consumed by Appendices E and F. They are fixed-cutoff diagnostics used only inside Appendix D. No identity

$$J_{t,L,\alpha} = g_{t,L}$$

is asserted or used in the RH closure.

Proof.

This is the canonical transform convention of Appendix D and Section 5, together with D.CANONICAL-FUNCTIONAL, D.SPIKE-TRANSFER, D.CANONICAL-WINDOW, and D.CAN-FUNC-EXPORT.

Theorem D.EXPORT

Canonical operator-side export.

Let

$$\Omega = \mathcal{C} \setminus (-\infty, 0]$$

Along the admissible ordered cutoff-removal net, Appendix D constructs a canonical whole-line operator-side transform

$$F_H^{can} \in O(\Omega)$$

and a canonical operator-side remainder

$$R_H^{can} \in O(\Omega)$$

There exists a nonempty overlap half-plane

$$U_{\sigma_0} = \{s \in \mathcal{C} : \Re s > \sigma_0\} \subset \Omega$$

such that

$$F_H^{can}(s) = B_{can}(s) + R_H^{can}(s), \quad s \in U_{\sigma_0}$$

Here B_{can} is the shifted closure package

$$B_{can}(t) = \sum_q \frac{\Lambda(q)}{\sqrt{q}} K_t(\log q), \quad K_t(u) = e^{-t/4} Gt(u)$$

understood on U_{σ_0} by the ordered finite-cutoff construction.

Appendix D does **not** assert that the infinite prime-side package B_{can} is holomorphic on all of Ω . The global meromorphic representative of this package is supplied independently by Appendix H.

Proof.

D.OPERATOR-LIMIT-INDEP constructs

$$F_H^{can} \in O(\Omega)$$

before any subtraction of the prime-power package. Thus the operator-side holomorphic object is not obtained by meromorphic continuation of the prime-side series.

The finite/cutoff canonical decomposition gives, on U_{σ_0} ,

$$F_{\alpha}^{can}(s) = B_{\alpha}^{can}(s) + R_{\alpha}^{can}(s)$$

By D.MASTER-RESIDUAL,

$$R_{\alpha}^{can} \rightarrow R_H^{can}$$

locally uniformly on Ω -compacts, with

$$R_H^{can} \in O(\Omega)$$

On the overlap half-plane U_{σ_0} , the finite packages converge locally uniformly to the ordered canonical package:

$$B_{\alpha}^{can} \rightarrow B_{can}$$

Passing to the ordered limit in the finite/cutoff identity therefore gives

$$F_H^{can}(s) = B_{can}(s) + R_H^{can}(s), \quad s \in U_{\sigma_0}$$

This proves the export.

Appendix E

Appendix E convention.

In Appendix E, F_H means F_H^{can} , the canonical displacement/commutator transform exported by Appendix D.

Every identity should be read:

$$F_H^{can}(s) = H_{tot}(s) - Z_{pole,\Omega}(s),$$

with

$$H_{tot} = R_H^{can} + H_{arch}$$

Role of Appendix E. Appendix E proves only the overlap identity on a common half-plane $U_{\sigma_0} \subset \Omega$. It does not construct the operator-side function F_H , which has already been constructed in Appendix D, and it does not construct the zero-side meromorphic package, which has already been constructed in Appendix H. Its sole task is to identify, on U_{σ_0} , the explicit shared spike package appearing in the D-side decomposition with the independently constructed zero-side expression. No global continuation is created in Appendix E; the global objects are imported from Appendices D and H, and only their local agreement on the overlap domain is proved here.

Proposition (canonical interface identity for the RH closure).
Fix once and for all the frozen normalization

$$w(q) = \frac{\Lambda(q)}{\sqrt{q}}.$$

Let $U_{\sigma_0} \subset \Omega$ be a right half-plane on which the operator-side overlap decomposition from Appendix D and the explicit-formula identity from Appendix H are both valid. Let $B(s)$ denote the restriction to U_{σ_0} of the canonical spike package from Section 5.1.4, in the frozen normalization

$$w(q) = \frac{\Lambda(q)}{\sqrt{q}} \text{ Then:}$$

1. By Appendix D and Theorem D.5.10W, the canonical whole-line operator transform satisfies

$$F_H(s) = B_{can}(s) + R_H^{can}(s), R_H^{can} \in O(\Omega), \quad s \in U_{\sigma_0}.$$

2. By Appendix H, on the same half-plane U_{σ_0} ,

$$B_{can}(s) = H_{arch}(s) - Z_{pole}(s),$$

where $Z_{pole} = Z_{pole, \Omega} | U_{\sigma}$

Therefore,

$$F_H^{can}(s) = -Z_{pole,\Omega}(s) + H_{tot}(s), H_{tot} := R_H^{can} + H_{arch} \in O(\Omega), \quad s \in U_{\sigma_0}$$

On the fixed overlap half-plane $U_{\sigma_0} \subset \Omega$, Theorem D.5.10W gives

$$F_H^{can}(s) = B_{can}(s) + R_H(s), R_H \in O(\Omega).$$

Appendix H gives on the same half-plane

$$B_{can}(s) = H_{arch}(s) - Z_{pole}(s), \quad Z_{pole} = Z_{pole,\Omega}|_{U_{\sigma_0}}$$

Hence on U_{σ_0} ,

$$F_H^{can}(s) = -Z_{pole,\Omega}(s) + H_{tot}(s), H_{tot} := R_H^{can} + H_{arch} \in O(\Omega).$$

No identity between a finite-interval Laplace transform and the global operator transform F_H is asserted or used. The short-time analysis enters only through the canonical package Bcan fixed in Section 5.1.4 and the operator-side construction summarized in Theorem D.5.10W.

Theorem E.PACKAGE-ID, finite-cutoff identity of the D-side and H-side canonical packages.

Let

$$U_{\sigma_0} = \{s \in \mathbb{C}: \Re s > \sigma_0\} \subset \Omega$$

be chosen so that

$$\inf_{s \in U_{\sigma_0}} \Re \sqrt{s + \frac{1}{4}} > \frac{1}{2}$$

On U_{σ_0} , the canonical package exported by Appendix D and the canonical package constructed in Appendix H are the same holomorphic function:

$$B_{can}^D(s) = B_{can}^H(s), \quad s \in U_{\sigma_0}$$

Consequently, on U_{σ_0} ,

$$B_{can}(s) = H_{arch}(s) - Z_{pole,\Omega}(s)$$

Proof.

Step 1. Finite D-side package.

For fixed finite R and L , the density-normalized D-side canonical package is

$$B_{D,L,R}^{can}(t) = e^{-t/4} \sum_{\log q \leq R} \frac{\Lambda(q)}{\sqrt{q}} \bar{g}_{t,L}(\log q),$$

where

$$\bar{g}_{t,L}(a) := \frac{1}{2_L} g_{t,L}(a)$$

By D.CANONICAL-WINDOW,

$$\bar{g}_{t,L}(a) \rightarrow G_t(a)$$

locally uniformly on compact a -intervals. Since $R < \infty$, the set

$$\{\log q: \log q \leq R\}$$

is finite. Hence

$$B_{D,L,R}^{can}(t) \rightarrow BD, Rcan(t) := \sum_{\log q \leq R} \frac{\Lambda(q)}{\sqrt{q}} K_t(\log q)$$

locally uniformly for $t > 0$, where $K_t = e^{-t/4} G_t$

Taking the Laplace transform on U_{σ_0} gives

$$B_{D,R}^{can}(s) = \sum_{\log q \leq R} \frac{\Lambda(q)}{\sqrt{q}} \kappa_s(\log q),$$

where

$$\kappa_s(u) := \int_0^\infty e^{-st} K_t(u) dt = \frac{1}{2\sqrt{s+\frac{1}{4}}} \exp(-|u|\sqrt{s+\frac{1}{4}})$$

The branch is the principal branch on the chosen right half-plane.

Step 2. Finite H-side package.

Appendix H constructs the prime-power distribution at finite cutoff as

$$\mu_{pr,R} = \sum_{\log q \leq R} \frac{\Lambda(q)}{\sqrt{q}} \delta_{\log q}$$

Therefore the finite H-side shifted package is

$$B_{H,R}^{can}(t) = \langle \mu_{pr,R}, K_t \rangle = \sum_{\log q \leq R} \frac{\Lambda(q)}{\sqrt{q}} K_t(\log q)$$

Taking the Laplace transform on U_{σ_0}

$$B_{H,R}^{can}(s) = \sum_{\log q \leq R} \frac{\Lambda(q)}{\sqrt{q}} \kappa_s(\log q)$$

Step 3. Finite-cutoff equality.

Comparing Steps 1 and 2 gives the exact finite-cutoff identity

$$B_{D,R}^{can}(s) = B_{H,R}^{can}(s) \quad (s \in U_{\sigma_0})$$

This equality is not an asymptotic statement and not a convention. Both sides are the Laplace transform of the same finite atomic prime-power package with the same kernel K_t , same centers

$\log q$, and same weights $\frac{\Lambda(q)}{\sqrt{q}}$

Step 4. Common $R \rightarrow \infty$ limit on the overlap half-plane.

Let $K \in U_{\sigma_0}$. Set

$$c_K := \inf_{s \in K} \Re \sqrt{s + \frac{1}{4}}$$

By the choice of U_{σ_0} , after shrinking to compact K ,

$$c_K > \frac{1}{2}$$

For $s \in K$,

$$|\kappa_s(\log q)| \leq C_K q^{-c_K}$$

Therefore

$$|\frac{\Lambda(q)}{\sqrt{q}} \kappa_s(\log q)| \leq C_K \frac{\Lambda(q)}{\sqrt{q}}^{-1/2-c_K}$$

Since $1/2 + c_K > 1$,

$$\sum_q \frac{\Lambda(q)}{\sqrt{q}}^{-1/2-c_K} < \infty$$

Thus the Weierstrass M-test gives locally uniform convergence of both finite-cutoff packages on U_{σ_0} . Since the finite sums are identical for every R , their locally uniform limits are identical:

$$B_{can}^D(s) = B_{can}^H(s), \quad s \in U_{\sigma_0}$$

Step 5. Identification with the zero-side meromorphic package.

By Theorem H.5.3 / H.PACKAGE, the shifted H-side closure package satisfies on the same overlap half-plane

$$B_{can}^H(s) = H_{arch}(s) - Z_{pole,\Omega}(s)$$

Using Step 4,

$$B_{can}^D(s) = H_{arch}(s) - Z_{pole,\Omega}(s), \quad s \in U_{\sigma_0}$$

This proves the package identity needed by Appendix E.

Scope note.

This theorem proves equality of the D-side and H-side canonical prime-power packages on U_{σ_0} . It does not assert global holomorphy of the prime-side series on Ω , and it does not perform the global meromorphic continuation. The global extension is carried out later in Appendix F by the meromorphic identity theorem.

Theorem E.INTERFACE, local split form on the common overlap half-plane.

Let

$$U_{\sigma_0} \subset \Omega$$

be the common overlap half-plane from D.EXPORT and H.5.3. Then on U_{σ_0} ,

$$F_H^{can}(s) = H_{tot}(s) - Z_{pole,\Omega}(s)$$

where

$$H_{tot} = R_H^{can} + H_{arch} \in O(\Omega)$$

Proof.

By D.EXPORT, on U_{σ_0} ,

$$F_H^{can}(s) = B_D^{can}(s) + R_H^{can}(s)$$

By E.PACKAGE-ID,

$$B_D^{can}(s) = B_{can}^H(s)(s \in U_{\sigma_0})$$

By H.5.3 / H.PACKAGE,

$$B_{can}^H(s) = H_{arch}(s) - Z_{pole,\Omega}(s)(s \in U_{\sigma_0})$$

Substituting gives

$$F_H^{can}(s) = R_H^{can}(s) + H_{arch}(s) - Z_{pole,\Omega}(s)$$

Set

$$H_{tot} = R_H^{can} + H_{arch}$$

Since both summands are holomorphic on Ω ,

$$H_{tot} \in O(\Omega)$$

Thus

$$F_H^{can}(s) = H_{tot}(s) - Z_{pole,\Omega}(s), \quad s \in U_{\sigma_0}$$

Appendix E performs no global continuation. The global extension from U_{σ_0} to Ω is carried out in

Appendix F by the meromorphic identity theorem.

Theorem D/E.OVERLAP

Overlap-only identity stability

Let $U_{\sigma_0} \in \Omega$ be the right half-plane on which the operator-side Laplace transform, the finite prime-power package, and the Appendix-H explicit-formula package are all legal.

Under Theorems D.KEY-FORM, D.KEY- Ω , and D.NV-FORM, the canonical operator transform satisfies on U_{σ_0}

$$F_H^{can}(s) = B_{can}(s) + R_H^{can}(s),$$

where

$$R_H^{can} \in O(\Omega),$$

and B_{can} is the canonical one-letter prime-power package on the overlap half-plane.

No assertion is made in Appendix D that B_{can} is holomorphic on all of Ω . Its global continuation is the meromorphic zero-side continuation constructed in Appendix H.

Proof

For each fixed cutoff stage (LR, R) , the fixed-cutoff Laplace identity gives, on U_{σ_0} ,

$$F_{LR,R}(s) = B_{LR,R}(s) + R_{LR,R}^{res}(s)$$

On U_{σ_0} , the prime-power package is absolutely convergent and the density-normalized window package converges to B_{can} by D.NV-FORM. The operator transforms converge to F_H^{can} by the operator-side resolvent/Laplace construction. The residual transforms converge locally uniformly to R_H by D.KEY- Ω .

Passing to the limit gives

$$F_H^{can}(s) = B_{can}(s) + R_H^{can}(s), s \in U_{\sigma_0}.$$

This identity is overlap-only. Appendix D does not continue B_{can} holomorphically to Ω . The only object from Appendix D continued holomorphically to all of Ω is RH.

Proposition E/H-CONSISTENCY (half-plane pole package equals the Ω -pole package)

Let $Z_{pole,\Omega}$ be the meromorphic pure pole package constructed in Appendix H. Then on every half-plane U_{σ_0} , $\sigma > 0$,

$$Z_{pole,\Omega}(s) = Z_{pole}(s)$$

Consequently,

$$Z_{pole,\Omega}(s) = -\varepsilon_{sp} \sum_{\rho}^{\dagger} \frac{m(\rho)}{s+\rho(1-\rho)} (s \in \Omega), (E/H)$$

as an identity of meromorphic functions on Ω .

Proof

On U_{σ} both Z_{pole} and $Z_{pole,\Omega}$ are given by the same absolutely convergent pole series with the same sign and multiplicity conventions; hence they agree on U_{σ} , and therefore on Ω by the meromorphic identity theorem

Appendix F:

Reader warning (scope of Appendix F). The RH implication in this manuscript uses only the Stieltjes/Laplace pole-rigidity route in Sections **F.8–F.9**, together with the bridge spine in Appendices **D/E/H**. Sections **F.5–F.6** discuss an optional RKHS/de Branges interpretation that is **not used** in the RH closure and may be read as motivation/outlook only.

Input convention for the closure.

The load-bearing operator-side object in this appendix is the canonical bulk-subtracted transform

$$F_H^{can}(s) = \int_0^\infty e^{-st} (\Theta_H^{can}(t) - A_{class}^{can}(t)) dt,$$

continued holomorphically to $\Omega = \mathcal{C} \setminus (-\infty, 0]$ by Appendix D / D.5.10W. Theorem F.9 uses this as input and does not supply the continuation. The RH closure below does not use an

unqualified raw trace identity $Tr(s + H)^{-1} = \sum_n (s + \lambda_n)^{-1}$

Lemma F.1 (bounded evaluation and evaluation vectors are automatically ℓ^2)

Let $\{\psi_n\}$ be an orthonormal basis of $L^2(R)$ consisting of PWQO eigenfunctions, and define the Mellin–Gaussian kernel

$$k_s(t) := e^{st} e^{-e^{2t}/2}.$$

For $\Re(s) > 0$, $k_s \in L^2(R)$.

Define

$$\Phi_n(s) := \langle \psi_n, k_s \rangle_{L^2}.$$

Then for each such s ,

$$\sum_{n \geq 1} |\Phi_n(s)|^2 = \|k_s\|_{L^2}^2 < \infty,$$

so the evaluation vector $v(s) := (\Phi_n(s))_{n \geq 1} \in \ell^2$ is well-defined. Moreover, for any $F = \sum_n a_n \Phi_n$ in the closed span $V := \text{span}^{-}\{\Phi_n\}$ with norm $\|F\|_V = \|a\|_{\ell^2}$,

$$|F(s)| = |\langle a, v(s) \rangle_{\ell^2}| \leq \|a\|_{\ell^2} \|v(s)\|_{\ell^2},$$

so point evaluation is a bounded functional (and V is an RKHS on $\Re(s) > 0$).

Proof: Parseval: $\sum_n |\langle \psi_n, k_s \rangle|_2 = \|k_s\|_2$ Cauchy–Schwarz gives bounded evaluation.

Lemma F.2 (RKHS kernel form; the wall becomes “kernel completeness modulo Ξ ”)

Once evaluation is bounded, the reproducing kernel is

$$K(s, w) = \sum_{n \geq 1} \Phi_n(s) \Phi_n(w)^*,$$

and “ $F(\rho_k) = 0 \forall k$ is equivalent to

$$F \perp K(\cdot, \rho_k) \forall k,$$

so the rank-1 statement is exactly

$$(\overline{\text{span}\{K(\cdot, \rho_k)\}})^\perp = \mathcal{C}\Xi.$$

Lemma F.2 (structure function identification).

Assume Lemma F.3 and assume you can prove a kernel identification

$K(\cdot, \cdot) \equiv K_E(\cdot, \cdot)$ for the *same* Hilbert space V used for the Φ_n ’s (or for the quotient space

$H_\Xi := \{G: \Xi G \in V\}$ defined without assuming GFH).

Appendix F.3 — Weyl–Titchmarsh mm-function for the half-line PWQO

F.3.1 Setup: half-line operator and fundamental solutions

Let

$$H_+ := -\frac{d^2}{dt^2} + V(t) \text{ acting in } L^2([0, \infty),$$

with real-valued $V \in L^1_{loc}([0, \infty))$. Given H_+ is limit-point at $+\infty$ (so no boundary condition is needed at $+\infty$ for self-adjointness). In the body of the proof you obtain limit-point/self-adjointness from the confining behavior $V(t) \rightarrow +\infty$ (superlinearly) as $t \rightarrow +\infty$. (This is the only hypothesis from earlier appendices used here.)

For each spectral parameter $z \in \mathbb{C}$, let $u(\cdot, z)$, $v(\cdot, z)$ be the solutions of

$$-y''(t) + V(t)y(t) = zy(t)$$

determined by the initial conditions at $t = 0$:

$$u(0, z) = 1, u'(0, z) = 0; v(0, z) = 0, v'(0, z) = 1$$

Then the Wronskian $W(u, v)(t) = u(t, z)v'(t, z) - u'(t, z)v(t, z)$ is constant in t , and $W(u, v) \equiv 1$.

F.3.2 Definition of the Weyl solution and Weyl m-function

For $z \in \mathbb{C}^+ := \{z: \Im z > 0\}$, the limit-point hypothesis implies that the space of solutions in $L^2([0, \infty))$ is one-dimensional. Hence there exists a unique scalar $m(z) \in \mathbb{C}$ such that the linear combination

$$\psi(t, z) := u(t, z) + m(z)v(t, z) \text{ belongs to } L^2([0, \infty)).$$

Definition (Weyl–Titchmarsh m-function). The function $m: \mathbb{C}^+ \rightarrow \mathbb{C}$ defined by the above condition is the Weyl–Titchmarsh m-function of the half-line operator H_+ (with the normalization $u(0, z) = 1, v'(0, z) = 1$ fixed above).

Note that by construction,

$$\psi(0, z) = 1, \psi'(0, z) = m(z).$$

So in this normalization, $m(z)$ is literally the boundary derivative of the L^2 at- $+\infty$ Weyl solution.

F.3.3 Existence, uniqueness, and Herglotz property

Proposition F.3.1 (Existence/uniqueness and Herglotz positivity).

For every $z \in \mathbb{C}^+$, the Weyl coefficient $m(z)$ exists and is unique. Moreover m is a Herglotz (Nevanlinna) function:

$$\Im m(z) > 0 (z \in C^+).$$

Proof.

Existence/uniqueness: In the limit-point case at $+\infty$, the subspace of L^2 -solutions on $[0, \infty)$ is one-dimensional for each $z \in C^+$. Since $\{u(\cdot, z), v(\cdot, z)\}$ is a basis of solutions, there is a unique coefficient $m(z)$ such that $u + mv \in L^2([0, \infty))$.

Herglotz positivity: Apply the Lagrange (Green) identity on $[0, R]$ to $\psi(\cdot, z)$ and $\overline{\psi(\cdot, \bar{z})}$:

$$\int_0^R ((H_+ - z) \psi(t, z)) \overline{\psi(t, z)} dt - \int_0^R \psi(t, z) \overline{(H_+ - z) \psi(t, z)} dt = [\psi'(t, z) \overline{\psi(t, z)} - \psi(t, z) \overline{\psi'(t, z)}]_{t=R}^{t=0}$$

Using $(H_+ - z)\psi(\cdot, z) = 0$ and $(H_+ - \bar{z})\overline{\psi(\cdot, \bar{z})} = 0$, the left-hand side collapses to

$$(\bar{z} - z) \int_0^R |\psi(t, z)|^2 dt = -2i \Im z \int_0^R |\psi(t, z)|^2 dt.$$

Letting $R \rightarrow \infty$ (and using $\psi(\cdot, z) \in L^2$ plus standard limit-point boundary flux vanishing at $+\infty$), the boundary term at $t = R$ vanishes, leaving

$$-2i \Im z \int_0^\infty |\psi(t, z)|^2 dt = -(\psi'(0, z) \overline{\psi(0, z)} - \psi(0, z) \overline{\psi'(0, z)}).$$

With $\psi(0, z) = 1$ and $\psi'(0, z) = m(z)$, the right-hand side equals $-(m(z) - \overline{m(z)}) = -2i \Im m(z)$. Therefore

$$\Im m(z) = \Im z \int_0^\infty |\psi(t, z)|^2 dt > 0,$$

since $\Im z > 0$ and $\psi \neq 0$.

F.3.4 Fixing the affine ambiguity (normalization)

A Herglotz function admits a representation

$m(z) = az + b + \int_R \left(\frac{1}{\xi - z} - \frac{\xi}{1 + \xi^2} \right) d\rho(\xi)$, $a \geq 0$, $b \in \mathbb{R}$, for a positive Borel measure ρ . Thus, if later arguments identify the measure part (or identify m up to an affine term), one must also fix a, b to pin down m uniquely.

1. High-energy normalization: Prove a large- $|z|$ asymptotic in $\Im z > 0$ that forces a and b (up to the chosen u, v conventions).
2. Anchoring normalization: Fix one value (e.g. $m(i)$) together with a growth bound that forces a, b uniquely.

Appendix F.4 — Mellin–Gaussian resolvent RKHS and the PWQO kernel

F.4.1 Mellin–Gaussian test vectors and transforms

Define, for $s \in \mathbb{C}$, the Mellin–Gaussian test vector

$$k_s(t) := e^{\bar{s}t} e^{-t^2/2} \quad (t \in \mathbb{R}).$$

For $\psi \in L^2(\mathbb{R})$, define its Mellin–Gaussian transform

$$\Phi_\psi(s) := \langle \psi, k_s \rangle_{L^2(\mathbb{R})}.$$

When $\psi = \psi_n$ is an eigenfunction of the full-line PWQO operator H with eigenvalue λ_n , write

$$\Phi_n(s) := \Phi_{\psi_n}(s) = \langle \psi_n, k_s \rangle.$$

(These are precisely the objects used throughout Appendix C and in the spectral expansions in Appendix F.)

F.4.2 Operator-dependent negative norm and the image spaces $V_{H, \pm 1}$

Define the operator-dependent “negative” (resolvent) quadratic form on the form domain of $(I + H)^{-1}$ by

$$\|\psi\|_{H, -1}^2 := \langle (I + H)^{-1} \psi, \psi \rangle_{L^2}.$$

Similarly define the “positive” norm $\|\psi\|_{H, \pm 1}^2 := \langle (I + H)\psi, \psi \rangle$ on $D(H)$.

Define the Mellin–Gaussian image spaces

$$V_{H,\pm 1} := \{\Phi_\psi : \psi \in D((I + H)^{\pm 1/2})\},$$

with inner product transported by $\psi \mapsto \Phi_\psi$ i.e.

$$\langle \Phi_\psi, \Phi_\varphi \rangle_{V_{H,-1}} := \langle (I + H)^{-1/2} \psi, (I + H)^{-1/2} \varphi \rangle_{L^2},$$

and analogously for $V_{H,\pm 1}$

F.4.3 Reproducing kernel in $V_{H,-1}$

In $V_{H,-1}$ point evaluation is bounded, because for $\psi \in D((I + H)^{-1/2})$,

$$\Phi_\psi(s) = \langle \psi, k_s \rangle = \langle (I + H)^{-1/2} \psi, (I + H)^{1/2} k_s \rangle,$$

so Cauchy–Schwarz gives $|\Phi_\psi(s)| \leq \|\psi\|_{H,-1} \|(I + H)^{1/2} k_s\|_{L^2}$ whenever the right side is finite (which is exactly the domain condition you impose when defining $V_{H,-1}$).

Therefore $V_{H,-1}$ is a reproducing kernel Hilbert space, with reproducing kernel

$$K_H(s, w) := \langle (I + H)^{-1/2} k_w, (I + H)^{-1/2} k_s \rangle_{L^2} = \langle k_w, (I + H)^{-1} k_s \rangle_{L^2}$$

F.4.4 Spectral expansion of the PWQO kernel

Proposition F.4.1 (Spectral expansion).

Let $\{(\lambda_n, \psi_n)\}_{n \geq 1}$ be an orthonormal eigenbasis for H (as established in the earlier appendices).

Then

$$(I + H)^{-1} = \sum_{n \geq 1} \frac{1}{1 + \lambda_n} |\psi_n\rangle \langle \psi_n|,$$

and consequently

$$K_H(s, w) = \sum_{n \geq 1} \frac{1}{1 + \lambda_n} \Phi_n(s) \overline{\Phi_n(w)}.$$

Proof. Insert the spectral resolution of $(I + H)^{-1}$ into
 $K_H(s, w) = \langle k_w, (I + H)^{-1} \psi_n \rangle$, and use $\Phi_n(s) = \langle \psi_n, k_s \rangle$

F.4.5 De Branges variable and critical line

Introduce the de Branges variable z by

$s = \frac{1}{2} + iz, z \in \mathbb{C}$, so that the critical line $\Re(s) = \frac{1}{2}$ corresponds to $z \in \mathbb{R}$. Define

$$\tilde{\Phi}(z) := \Phi\left(\frac{1}{2} + iz\right), K_H^{\sim}(z, w) := K_H\left(\frac{1}{2} + iz, \frac{1}{2} + iw\right).$$

Bridge to the rest of Appendix F: The goal in the next section is to test whether $K_H^{\sim}$ is (or induces) a de Branges kernel, i.e. whether there exists a Hermite–Biehler entire function E such that

$$K_H^{\sim}(z, w) = \frac{E(z)\overline{E(w)} - E\#(z)\overline{E\#(w)}}{2\pi i(w - z)}$$

Appendix F.5 (Outlook) — De Branges package attached to the half-line operator H_+

Reader warning (scope of Appendix F). The RH implication in this manuscript uses only the Stieltjes/Laplace pole-rigidity route in Sections **F.8–F.9**, together with the bridge spine in Appendices **D/E/H**. Sections **F.5–F.6** discuss an optional RKHS/de Branges interpretation that is **not used** in the RH closure and may be read as motivation/outlook only.

Standing restriction for this manuscript. Sections F.5–F.6 summarize de Branges theorems. These results are only invoked *conditionally* under the hypothesis of Proposition F.6.2 (kernel equality $K^{\sim}H=KE$). The RH conclusion of the manuscript does not require these results.

This section records the standard implication
“half-line Schrödinger operator $\Rightarrow \Rightarrow$ Herglotz mm-function $\Rightarrow \Rightarrow$ canonical system
 $\Rightarrow \Rightarrow$ de Branges space,”
and isolates exactly which de Branges structural axioms (division + kernel completeness) would be available if the PWQO RKHS from Appendix F.4 is identified with a de Branges space.
(This appendix does *not* itself prove that identification; that is the content of Appendix F.6.)

F.5.1 The Weyl measure and Herglotz representation

Let H_+ be the half-line operator from Appendix F.3 with the fixed boundary normalization at $t = 0$ used there, and let $m(z)$ be its Hermann Weyl–E. C. Titchmarsh function on $\mathbb{C}_+$. By Proposition F.3.1, m is Herglotz (Nevanlinna): $\Im m(z) > 0$ for $\Im z > 0$.

Proposition F.5.1 (Herglotz representation; spectral measure).

There exist real constants $a \in \mathbb{R}$, $b \geq 0$, and a positive Borel measure μ on $\mathbb{R}$ satisfying

$$\int_{\mathbb{R}} \frac{1}{1+t^2} d\mu(t) < \infty,$$

such that for all $z \in \mathbb{C}_+$,

$$m(z) = a + bz + \int_{\mathbb{R}} \left(\frac{1}{t-z} - \frac{t}{1+t^2} \right) d\mu(t). \quad (F.5.1.1)$$

Moreover, μ is (up to the chosen normalization at $t = 0$) the boundary spectral measure for H_+ ; it determines H_+ up to unitary equivalence.

Remarks for later use.

1. The constants a, b encode the usual affine ambiguity in the Herglotz representation; your Appendix F.3 already flags that one must fix normalization/growth to pin them down uniquely.
2. The measure μ is the primary invariant: it is the object that is transported into the canonical-system / de Branges side.

F.5.2 Canonical system and Hermite–Biehler function

A foundational theorem in Louis de Branges theory (“de Branges–Weyl correspondence”) states: every Herglotz function arises as the Weyl function of a canonical system, and conversely.

Theorem F.5.2 (Canonical system realizing m).

Let m be the Weyl–Titchmarsh function of $H \in \mathcal{H}$. Then there exists a canonical system

$$J Y'(x) = z H(x) Y(x), x \in [0, \infty), (F. 5. 2. 1)$$

where

$$J = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}, H(x) \text{ is positive semidefinite a.e.,}$$

such that the Weyl function of this canonical system equals $m(z)$.

Let $M(x, z)$ denote its transfer matrix (fundamental solution) with $M(0, z) = I$:

$$M(x, z) = \begin{pmatrix} A(x, z) & B(x, z) \\ C(x, z) & D(x, z) \end{pmatrix}.$$

Then for each fixed x , the entries $A(x, \cdot), B(x, \cdot), C(x, \cdot), D(x, \cdot)$ are entire in z .

Define the associated entire function

$$E(x, z) := A(x, z) - i B(x, z),$$

and the standard conjugate

$$E^\#(x, z) := \overline{E(x, \bar{z})}.$$

Then $E(x, \cdot)$ is Hermite–Biehler,.

$$|E(x, z)| > |E^\#(x, z)| (\Im z > 0). (F. 5. 2. 2)$$

Definition F.5.3 (de Branges space at scale x).

For each x , the de Branges space $H(E(x, \cdot))$ is the Hilbert space of entire functions F for which

$$\|F\|_{H(E(x))}^2 = \int_{\mathbb{R}} \left| \frac{F(t)}{E(x, t)} \right|^2 dt < \infty,$$

with reproducing kernel

$$K_{E(x)}(z, w) = \frac{E(x, z) \overline{E(x, w)} - E^\#(x, z) \overline{E^\#(x, w)}}{2\pi i(w - z)}. (F. 5. 2. 3)$$

F.5.4 The division axiom in de Branges spaces

Theorem F.5.4 (Division property).

Let E be Hermite–Biehler and let $F \in H(E)$. If $F(\zeta) = 0$ for some $\zeta \in \mathbb{C} \setminus \mathbb{R}$, then

$$F_{\zeta}(z) := \frac{z - \bar{\zeta}}{z - \zeta} F(z) \in H(E) \text{ and } \|F_{\zeta}\|_{H(E)} = \|F\|_{H(E)} \quad (F. 5. 3. 1)$$

This is one of the fundamental axioms/theorems of de Branges spaces and is what prevents “extra Ξ -like” functions from hiding in the vanishing-at-zeros subspace once the space is identified as $H(E)$.

In our proof chain, (F.5.3.1) is not used until after the identification in Appendix F.6 is established. Here it is stated only as an imported theorem.

F.5.4 Codimension-1 completeness of kernels at zeros of E

This is the “rank-1 / single-wall” phenomenon you keep pointing to: kernels at zeros span everything except one canonical direction.

Let $Z(E) \subset \mathbb{C}^+$ denote the zero set of E in the upper half-plane (with multiplicity). For each $\zeta \in \mathbb{C}$, let $K_E(\cdot, \zeta)$ denote the reproducing kernel in $H(E)$.

Theorem F.5.5 (Kernel completeness modulo one defect).

In $H(E)$,

$$\overline{\text{span}} \{K_E(\cdot, \zeta) : \zeta \in Z(E)\}^{\perp} = C \cdot E. \quad (F. 5. 4. 1)$$

Equivalently, reproducing kernels at the zeros of E are complete up to the one-dimensional defect generated by the structure function E .

Appendix F.6 — Identification of the PWQO resolvent RKHS with a de Branges space

F.6.1 The PWQO resolvent RKHS in the critical-line variable

Let $V_{H,-1}$ be the RKHS from Appendix F.4 associated to the PWQO resolvent norm, with reproducing kernel

$$K_H(s, w) = \langle k_w, (I + H)^{-1} k_s \rangle_{L^2}$$

$$(\text{So } K_H(\cdot, w) \in V_{H,-1} \text{ and } F(w) = \langle F, KH(\cdot, w) \rangle \text{ for all } F \in V_{H,-1}.)$$

Introduce the critical-line coordinate

$$s = \frac{1}{2} + iz, z \in \mathbb{C},$$

and define the transported kernel on $\mathbb{C} \times \mathbb{C}$ by

$$K_H^\sim(z, w) := K_H\left(\frac{1}{2} + iz, \frac{1}{2} + iw\right). \quad (F. 6. 1. 1)$$

Because K_H is a reproducing kernel, $K_H^\sim(z, w)$ is entire in z and anti-entire in w (equivalently, entire in $\overline{w}$). This is the correct analytic signature for de Branges kernels.

Define the associated “numerator kernel”

$$N_H^\sim(z, w) := 2\pi i (\overline{w} - z) K_H^\sim(z, w). \quad (F. 6. 1. 2)$$

(This is the same normalization we intended, but with the necessary $\overline{w}$. Without it, the kernel is not sesquilinear in the RKHS sense, and the de Branges form test is not stated correctly.)

De Branges kernel-form test (correct statement).

A reproducing kernel $K^\sim$ on $\mathbb{C}$ is a de Branges kernel if and only if there exists a Hermite–Biehler entire function E such that

$$K^\sim(z, w) = K_E(z, w) := \frac{E(z)\overline{E(w)} - E^\#(z)\overline{E^\#(w)}}{2\pi i(w - z)}, \quad (F. 6. 1. 3)$$

where

$$E^\#(z) := \overline{E(z)}. \quad (F. 6. 1. 4)$$

Equivalently, in numerator form,

$$N_H^{\sim}(z, w) = E(z) \overline{E(w)} - E\#(z) \overline{E\#(w)}. (F. 6. 1. 5)$$

A convenient “rank-two” restatement (the one we are aiming to verify for $N_H^{\sim}$) is:

there exist entire A, B such that

$$N_H^{\sim}(z, w) = A(z) \overline{A(w)} - B(z) \overline{B(w)}, (F. 6. 1. 6)$$

and then $E := A - iB$ is Hermite–Biehler and satisfies $K^{\sim} = K_E$

F.6.2 Proposition (Identification criterion)

Proposition F.6.2 (Kernel equality $\Rightarrow$ unitary identification).

Assume there exists a Hermite–Biehler entire function E such that

$$K_H^{\sim}(z, w) \equiv K_E(z, w) \text{ for all } z, w \in \mathbb{C}, (F.6.2.1)$$

where K_E is given by (F.6.1.3)–(F.6.1.4). Then:

1. (Unitary isomorphism of RKHSs) The change of variables

$$(UF)(z) := F\left(\frac{1}{2} + iz\right) (F. 6. 2. 2)$$

2. defines a unitary isomorphism

$$U: V_{H,-1} \rightarrow \cong H(E),$$

where $H(E)$ denotes the de Branges space with reproducing kernel K_E .

3. (Imported de Branges structure) Under this identification, $V_{H,-1}$ inherits:

- the division axiom (Appendix F.5.3), and
- the codimension-1 kernel completeness property (Appendix F.5.4).

In particular, once E is further identified with a constant multiple of $\Xi^{\sim}$, the rank-1/codimension-1 phenomenon becomes a theorem of de Branges theory rather than an additional hypothesis.

Proof. This is a standard RKHS fact: if two RKHSs on the same set have identical reproducing kernels, then the identity map on point evaluations extends uniquely to an isometric isomorphism between the Hilbert spaces. Concretely, the linear span of kernel vectors is dense in each space, and

$$\langle K(\cdot, w_1), K(\cdot, w_2) \rangle = K(w_2, w_1)$$

agrees in both spaces when the kernels agree. The division property and kernel completeness are intrinsic theorems of de Branges spaces and therefore transfer through the unitary identification.

Remark (not used in the RH closure). Earlier drafts explored a de Branges/RKHS interpretation in which one would attempt to identify the canonical structure function E of the PWQO space with the completed zeta function $\tilde{\Xi}$ up to a constant factor), by proving a kernel identification with a de Branges space $H(E)$. This program is not used anywhere in the present proof of RH. The RH implication in this manuscript proceeds exclusively via the Ω –Stieltjes/pole-rigidity mechanism (Appendix F, Sections F.8–F.9), using the operator spike package and Ω -holomorphic remainder (Appendices D/E) together with the Weil explicit-formula pole package (Appendix H). Accordingly, no identification of the form $E = c \tilde{\Xi}$ is assumed or required.

F.6.4 What remains: a single explicit kernel identity

Remark (optional strengthening) The identification $K_H^\sim = K_{\tilde{\Xi}}$ in (F.6.4.1) would upgrade the PWQO RKHS into the canonical de Branges space $H(\tilde{\Xi})$. This identification is not needed for the RH implication, which is completed in Section F.9 via Laplace/Stieltjes rigidity.

With the same assumptions as in F.6.3, the RH-closure step in Option 1 reduces to proving a concrete identity between two kernels on $C \times C$.

Define the numerator kernel (with the same convention used in F.4.5)

$$N_H^\sim(z, w) := 2\pi i (\bar{w} - z) K_H^\sim(z, w). \quad .$$

Target identity (kernel identification in $\tilde{\Xi}$ form)

Prove that for all $z, w \in C$,

$$K_H^{\sim}(z, w) = \frac{\tilde{\Xi}(z) \overline{\tilde{\Xi}(w)} - \tilde{\Xi}(z) \overline{\tilde{\Xi}(w)}}{2\pi i (w-z)} \quad (F. 6. 4. 1)$$

Equivalently, in numerator form,

$$N_H^{\sim}(z, w) = \tilde{\Xi}(z) \overline{\tilde{\Xi}(w)} - \tilde{\Xi}(z) \overline{\tilde{\Xi}(w)} \quad (F. 6. 4. 2)$$

Interpretation

- (F.6.4.1) is exactly the statement $K_H^{\sim} = K_{\tilde{\Xi}}$, i.e. that the PWQO resolvent RKHS is *the* de Branges space with structure function $\tilde{\Xi}$ (up to unimodular scaling).
- Remark (optional strengthening). The identification $K_H^{\sim} = K_{\tilde{\Xi}}$ (F.6.4.1) would upgrade the PWQO RKHS into the canonical de Branges space $H(\tilde{\Xi})$. This identification is not needed for the RH implication, which is completed in Section F.9 via Laplace/Stieltjes rigidity.

Lemma F.MEROMORPHICITY (Zero–Pole Package on Ω)

Define

$$Z(s) := \sum_{\rho} \frac{m(\rho)}{s + \rho(1 - \rho)}$$

where the sum runs over nontrivial zeros of ζ , counted with multiplicity, and interpreted by symmetric pairing.

Then:

1. $Z(s)$ converges absolutely for $\Re(s) > \sigma_0$
2. $Z(s)$ admits meromorphic continuation to Ω .
3. The only poles of $Z(s)$ in Ω occur at

$$s = -\rho(1 - \rho),$$

each with residue $m(\rho)$

4. No other singularities occur in Ω .
-

Recall: All uses of $R \rightarrow \infty$, $L \rightarrow \infty$, and sum/integral interchange in F.9 are those already legalized in Appendix D: the fixed-(L,R) Duhamel expansion is performed before package separation, and the subsequent limits act only on the already isolated explicit package and the holomorphic remainder, locally uniformly on compact subsets of Ω or on the common overlap half-plane.

Lemma F.POLESEP (pole separation near the cut).

Let $Z_{pole,\Omega}(s) = \sum_{\rho} m(\rho)/(s + \rho(1 - \rho))$.

Since $|\Im \rho| \geq \gamma_1 > \frac{1}{4}$ for every nontrivial zero, every pole $s_{\rho} = -\rho(1 - \rho)$ satisfies

$$\Re s_{\rho} \leq -\gamma_1^2 < -1/9.$$

Therefore $Z_{pole,\Omega}$ is holomorphic on

$$\Omega_{near} := \{s \in \Omega: \Re s > -1/9\}.$$

In particular, on the region $\Omega_{near} = \{s \in \Omega: \Re s > -1/9\}$, the pole package $Z_{pole,\Omega}$ is holomorphic.

This will be used below to compare the operator-side transform F_H with the zero-side package in a neighborhood of the slit.

Proposition F.8 (local interface identity on an overlap domain).

There exists a nonempty open set $U \subset \Omega$ such that

$$F_H(s) = -Z_{pole,\Omega}(s) + H_{tot}(s), s \in U,$$

with $H_{tot} \in O(\Omega)$.

Proof. This is Theorem E.INTERFACE together with Proposition E/H-CONSISTENCY.

Lemma F. NC_Ω

Closure inputs for Theorem.

The RH implication below uses only the following imported inputs.

(I1)

$$F_H^{can} \in O(\Omega)$$

and, on the overlap half-plane,

$$F_H^{can}(s) = B_{can}(s) + R_H^{can}(s), \quad R_H^{can} \in O(\Omega),$$

from Appendix D.

(I2)

$$Z_{pole,\Omega}$$

is meromorphic on Ω , and

$$H_{arch} \in O(\Omega)$$

from Appendix H.

(I3) The local split identity

$$F_H^{can}(s) = -Z_{pole,\Omega}(s) + H_{tot}(s), \quad H_{tot} \in O(\Omega),$$

on a nonempty open half-plane

$$U_{\sigma_0} \subset \Omega$$

from Theorem E.INTERFACE together with Proposition E/H-CONSISTENCY.

(I4) The elementary no-cancellation lemma on Ω .

No other input is used in the closure. In particular, Appendix F introduces no new analytic continuation, no new pole package, and no new prime-side limiting argument.

Theorem F. 9.1 Rigidity closure on Ω

Every nontrivial zero ρ of ζ satisfies

$$\Re \rho = \frac{1}{2}$$

Proof. Let ρ be a nontrivial zero of ζ . We show that $\Re \rho = \frac{1}{2}$. By Theorem D.5.10W, Theorem E.INTERFACE, Proposition E/H-CONSISTENCY, and Lemma H.Pole, it suffices to exclude the case $\Re \rho \neq \frac{1}{2}$. Write $\rho = \beta + i\gamma$ and suppose $\beta \neq \frac{1}{2}$. Since ζ has no real zeros in the open critical strip, $\gamma \neq 0$. Set

$$\lambda_\rho := \rho(1 - \rho), s_\rho := -\lambda_\rho.$$

Then

$$\Im(\lambda_\rho) = \gamma(1 - 2\beta) \neq 0,$$

so $s_\rho \notin (-\infty, 0]$, hence $s_\rho \in \Omega$.

By Lemma H.Pole, the meromorphic function $Z_{pole, \Omega}$ has a pole at $s = s_\rho$, with grouped multiplicity obtained by collecting equal values of $\rho'(1 - \rho')$. Because all grouped multiplicities are positive integers, grouping cannot cancel the principal part.

Define

$$M(s) := -Z_{pole, \Omega}(s) + H_{tot}(s),$$

$H_{tot} := R_H + H_{arch} \in O(\Omega)$, where $R_H \in O(\Omega)$ is furnished by Appendix D and $H_{arch} \in O(\Omega)$ is furnished by Appendix H. Theorem E.INTERFACE supplies only the local overlap identity involving this H_{tot}

By Lemma F.NC $\Omega\Omega$, the function M has a pole at $s = s_\rho$.

On the nonempty open set $U_{\sigma_0} \subset \Omega$, Theorem E.INTERFACE gives

$$F_H^{can}(s) = M(s).$$

Here F_H^{can} is holomorphic on Ω by Theorem D.5.10W, while M is meromorphic on Ω . Since Ω is connected, the identity theorem for meromorphic functions implies

$F_H^{can} = M$ on all of Ω . But M has a pole at s_ρ , whereas $F_H \in O(\Omega)$ is holomorphic there. This contradiction shows $\Re \rho = \frac{1}{2}$.

Reader's Note:

The RH-closure below uses only the following previously constructed ingredients.

(D1) Appendix D, Theorem D.5.10W / D.CAN-FUNC-EXPORT.

There exists a canonical whole-line operator-side transform

$$F_H^{can} \in O(\Omega)$$

extending the half-plane Laplace transform

$$F_H^{can, hp}$$

On the overlap half-plane $U_{\sigma.}$ used in Appendices E and H, one has

$$F_H^{can}(s) = B_{can}(s) + R_H^{can}(s), R_H^{can} \in O(\Omega)$$

Here B_{can} is the canonical shifted spike package from Section 5.1.4.

(D2) Appendix H, Corollary H.X.2.

Appendix H constructs the zero-side pole package

$$Z_{pole, \Omega}$$

meromorphic on Ω , together with the archimedean term

$$H_{arch} (\in O(\Omega)).$$

Let Z_{pole} denote the restriction of $Z_{pole, \Omega}$ to the overlap half-plane U_{σ} .

(D3) Appendix E together with Proposition E/H-CONSISTENCY.

On the same overlap half-plane U_{σ} ,

$$B_{can}(s) = H_{arch}(s) - Z_{pole}(s)$$

Corollary D.NORM $\Omega\Omega$ -C is used only for the restricted prime-side spike package in its own convergence region. It is not the mechanism by which the canonical whole-line operator-side transform $FHcan$ is continued to all of Ω .

The frozen normalization throughout Appendix F is

$$w(q) = \frac{\Lambda(q)}{\sqrt{q}}$$

Lemma F.9.2'(a) (Zero-side meromorphic package on Ω , imported from Appendix H)

Let

$$\Omega := \mathbb{C} \setminus (-\infty, 0].$$

Given

(H1) There exists a half-plane

$$U := \{s \in \mathbb{C} : \Re(s) > \sigma_{L_{zero}}\} \subset \Omega$$

on which the zero-side Laplace transform is justified termwise, and on which

$$L_{zero}(s) = \sum_{\rho} \frac{m(\rho)}{s + \rho(1-\rho)} + H_{L_{zero}}(s), \quad s \in U,$$

where the sum runs over the nontrivial zeros ρ of ζ , counted with multiplicity, and H_{arch} is holomorphic on U .

(H2) Appendix H provides a meromorphic function

$$Z_{\Omega} \rightarrow \widehat{C}$$

of the form

$$Z_{\Omega}(s) = Z_{pole,\Omega}(s) + H_{arch,\Omega}(s),$$

where

$$Z_{pole,\Omega}(s) = \sum_{\rho} \frac{m(\rho)}{s+\rho(1-\rho)} \text{ is meromorphic on } \Omega, H_{arch,\Omega} \in O(\Omega) \text{ and on } U,$$

$$Z_{\Omega}(s) = L_{zero}(s).$$

Then Z_{Ω} is the zero-side meromorphic package on Ω , with poles precisely at the points

$s = -\rho(1 - \rho)$ with multiplicities obtained by collecting equal values of $\rho(1 - \rho)$.

Proof

By (H1), the zero-side Laplace transform is defined on the half-plane U and there satisfies

$$L_{zero}(s) = \sum_{\rho} \frac{m(\rho)}{s+\rho(1-\rho)} + H_{arch}(s).$$

By (H2), Appendix H supplies a meromorphic function Z_{Ω} on Ω extending that half-plane expression. Therefore Z_{Ω} is the required zero-side meromorphic package on Ω . Its pole set is exactly the pole set of the partial-fraction term

$$\sum_{\rho} \frac{m(\rho)}{s+\rho(1-\rho)},$$

-
1. on $U = \{s \in \mathbb{C} : \Re(s) > \sigma_0\} \subset \Omega$

Lemma F.9.2'(b), dependency-clean reading.

The zero-side Laplace transform is first defined on a half-plane $U = \{\Re(s) > \sigma_0\}$ where termwise Laplace transformation is justified. On that half-plane one obtains

$$L_{zero}(s) = \sum_{\rho} \frac{m(\rho)}{s+\rho(1-\rho)} + H_U(s),$$

with H_U holomorphic on U .

Appendix H then provides the zero-side meromorphic package on

$$\Omega = \mathbb{C} \setminus (-\infty, 0],$$

namely

$$Z_{\Omega}(s) = \sum_{\rho} \frac{m(\rho)}{s+\rho(1-\rho)} + H_{\Omega}(s), H_{\Omega} \in O(\Omega),$$

extending the half-plane expression.

Separately, Appendix D constructs the operator-side holomorphic transform

$$F \in O(\Omega)$$

Appendix E proves that on a nonempty open overlap domain $U_0 \subset \Omega$

$$F(s) = Z_{\Omega}(s).$$

Thus the zero-side package and the operator-side transform are identified only after both have been constructed independently.

Bulk Laplace transform creates no nonreal isolated poles

Lemma F.9.3' (Analytic continuation of the bulk Laplace transform)

By the bulk term $A_{class}(t)$ is a finite linear combination of functions of the following types:

- $t^{\alpha-1} e^{-ct}$ with $\alpha \in \{\frac{1}{2}, \frac{3}{2}, \frac{5}{2}, \dots\}$ and $c \geq 0$,
- $e^{-ct} P(t)$ where P is a polynomial and $c \geq 0$,
- finite linear combinations of such terms and their t -derivatives.

Then the Laplace transform

$$L[A_{class}^{can}](s) := \int_0^{\infty} e^{-st} A_{class}^{can}(t) dt$$

initially defined for $\Re s$ large extends to a holomorphic function on $\mathcal{C} \setminus (-\infty, 0]$. In particular, $L[A_{class}^{can}]$ has no isolated poles in $\Re s > 0$ and introduces no nonreal poles anywhere .

Proof

Each term has an explicit Laplace transform:

For $t^{\alpha-1} e^{-ct}$,

$$\int_0^\infty e^{-st} t^{\alpha-1} e^{-ct} dt = \frac{\Gamma(\alpha)}{(s+c)^\alpha},$$

which is holomorphic on $\mathcal{C} \setminus (-\infty, -c]$. For half-integer α , this has a branch cut along $(-\infty, -c]$ but no isolated poles off that cut.

- For $e^{-ct} P(t)$ repeated differentiation with respect to $t(s+c)^{-k}$ shows the transform is a finite linear combination of $(s+c)^{-k}$ terms (rational functions) and hence is holomorphic on $\mathcal{C} \setminus \{-c\}$, with any pole occurring only on the negative real axis when $c \geq 0$.

Finite linear combinations preserve these analytic domains; derivatives in t correspond to multiplying by s and adding boundary terms that remain within the same class. Therefore $L[A_{class}^{can}]$ extends holomorphically to $\mathcal{C} \setminus (-\infty, 0]$ (after shifting by $c \geq 0$ if needed), with no isolated poles in $\Re s > 0$ and no nonreal poles.

F.9.4 Laplace transform: pole interface.

On $\Re s > \sigma_0$, define the canonical bulk-subtracted transform

$$F_H^{can}(s) := L[\Theta_H^{can} - A_{class}^{can}](s)$$

Here Θ_H^{can} is the canonical displacement/commutator-channel trace functional exported by Appendix D, and A_{class}^{can} is the corresponding canonical bulk/classical subtraction. This is not the raw multiplication-bump heat trace.

By Theorem D.5.10W / D.CAN-FUNC-EXPORT, this half-plane transform agrees on U_{σ_0} with the canonical operator-side function constructed in Appendix D. Appendix D supplies the unique whole-line extension

$$F_H^{can} \in \mathcal{O}(\Omega).$$

Theorem F.9.1 uses this object.

By Theorem E.INTERFACE together with Proposition E/H-CONSISTENCY, there exists a nonempty overlap half-plane $U_\sigma \subset \Omega$ and a holomorphic function

$$H_{tot} \in O(\Omega)$$

such that

$$F_H^{can}(s) = -Z_{pole,\Omega}(s) + H_{tot}(s), \quad s \in U_\sigma.$$

Theorem 5.V (pointer to the final rigidity theorem).

The final RH closure is Theorem F.RH in Appendix F. Section 5 isolates the shared Gaussian spike package and supplies the exact centered cutoff bridge, but the actual RH contradiction uses the canonical whole-line operator transform F_H from Appendix D / Theorem D.5.10W

together with the local split interface of Theorem E.INTERFACE and the Ω -meromorphic pole package from Appendix H.

Reader note

The critical-line case is geometrically consistent with the proof: if $\rho = \frac{1}{2} + i\gamma$, then

$$\rho(1 - \rho) = \frac{1}{4} + \gamma^2 \in (0, \infty),$$

so the corresponding pole location

– $\rho(1 - \rho) \in (-\infty, 0]$ lies exactly on the excluded cut. Thus the rigidity contradiction targets precisely off-line zeros and does not falsely rule out critical-line zeros.

Proposition F.GLOBAL-SPLIT (meromorphic identity-theorem extension of the local split identity)).

Let

$$\Omega := \mathbb{C} \setminus (-\infty, 0].$$

Let $F_H^{can} \in O(\Omega)$ be the canonical whole-line operator transform supplied by Appendix D. Let

$$Z_{pole,\Omega}$$

be the Ω -meromorphic zero-pole package supplied independently by Appendix H, and let

$$H_{tot} \in O(\Omega)$$

be the holomorphic term appearing in the local split identity of Appendix E.

Take that on some nonempty overlap half-plane $U_\sigma \subset \Omega$,

$$F_H^{can}(s) = -Z_{pole,\Omega}(s) + H_{tot}(s), s \in U_\sigma.$$

Then the identity extends to all of Ω as an identity of meromorphic functions:

$$F_H^{can}(s) = -Z_{pole,\Omega}(s) + H_{tot}(s), \quad s \in \Omega.$$

In particular, the only possible poles on the right-hand side are the poles of $Z_{pole,\Omega}$, since both F_H^{can} and H_{tot} are holomorphic on Ω .

Proof. The left-hand side F_H^{can} is holomorphic on Ω by Appendix D. The right-hand side $-Z_{pole,\Omega} + H_{tot}$ is meromorphic on Ω because $Z_{pole,\Omega}$ is meromorphic on Ω by Appendix H and H_{tot} is holomorphic on Ω . By hypothesis the two functions agree on the nonempty open set $U_\sigma \subset \Omega$. Therefore they agree on all of Ω by the meromorphic identity theorem on the connected slit plane Ω .

Theorem (F-Rigidity Bridge,)

Let:

$F_H^{can}(s)$ be the operator resolvent trace, holomorphic on Ω .

$F_{\Xi}(s)$ be the explicit-formula meromorphic function on Ω :

$$F_{\Xi}(s) = \sum_{\rho} \frac{m(\rho)}{s+\rho(1-\rho)} + H_{arch}(s).$$

Assume:

1. $F_H^{can}(s)$ is holomorphic on Ω .
2. $F_{\Xi}(s)$ is meromorphic on Ω .
3. There exists a nonempty open set $U \subset \Omega$ such that

$$F_H(s) = F_{\Xi}(s) \text{ for all } s \in U$$

Then:

$$F_H^{can}(s) = F_{\Xi}(s) \text{ for all } s \in \Omega$$

Proof

Define:

$$G(s) := F_H^{can}(s) - F_{\Xi}(s).$$

Then:

- G is meromorphic on Ω .
- G vanishes identically on an open set U .

Let $s_0 \in \Omega$

If s_0 is not a pole of G , then by the identity theorem for holomorphic functions, $G \equiv 0$ on the connected component of Ω containing U .

If s_0 is a pole of G , then it must be a pole of F_{Ξ} , since F is holomorphic on Ω .

But a meromorphic function that vanishes on a nonempty open set has **no poles**.

Hence $G \equiv 0$ on Ω .

Lemma (Ω -Rigidity for Stieltjes Resolvents)

Let $\Omega = \mathbb{C} \setminus (-\infty, 0]$.

Let:

$$F(s) = B(s) + R(s)$$

where:

- F is holomorphic on Ω .
- R is holomorphic on Ω .
- B is meromorphic on Ω .
- The zero-side continuation of the canonical package,

$B_{can,\Omega}(s) := H_{arch}(s) - Z_{pole,\Omega}(s)$, is meromorphic on Ω . This is not a claim that the D-side prime-series representative of B_{can} converges or is holomorphic on all of Ω .

Suppose B has an isolated pole at $s_0 \in \Omega$.

Then F must also have a pole at s_0

Proof

Write the Laurent expansion of B at s_0 :

$$B(s) = \frac{a_{-k}}{(s-s_0)^k} + \cdots + a_0 + a_1(s-s_0) + \cdots$$

with $a_{-k} \neq 0$

Since R is holomorphic at s_0 ,

$$R(s) = b_0 + b_1(s-s_0) + \cdots$$

Then:

$$F(s) = \frac{a_{-k}}{(s-s_0)^k} + \cdots + (a_0 + b_0) + \cdots$$

The principal part survives unchanged.

Thus F has a pole at s_0 .

Corollary (Off-critical zero $\Rightarrow$ contradiction)

If ρ is a nontrivial zero with $\Re \rho \neq \frac{1}{2}$

$$\text{Then } s_0 = -\rho(1 - \rho) \in \Omega$$

Thus the zero-pole package has a pole at s_0

By the lemma, F_H^{can} must have a pole at s_0

But F_H^{can} is holomorphic on Ω .

Contradiction.

$$\text{Therefore } \Re \rho = \frac{1}{2}$$

Rigidity Note:

Dependencies for the rigidity step.

Input convention for the closure.

The load-bearing operator-side object in this appendix is the canonical bulk-subtracted whole-line transform

$$F_H^{can}(s),$$

constructed in Appendix D and holomorphic on

$$\Omega = \mathbb{C} \setminus (-\infty, 0].$$

The load-bearing package object is the canonical shifted closure package

$$B_{can}(t) := B(t) = e^{-t/4} B_{raw}(t),$$

with B_{raw} the raw displacement-side Gaussian package isolated in Section 5.

By Appendix E, on a nonempty overlap half-plane $U_{\sigma_0} \subset \Omega$,

$$F_H^{can}(s) = -Z_{pole,\Omega}(s) + H_{tot}(s), H_{tot} \in O(\Omega).$$

The pole package $Z_{pole,\Omega}$ is the meromorphic package with poles at

$$s = -\rho(1 - \rho),$$

constructed independently in Appendix H from the shifted K_t -package and imported into the closure through the Appendix E interface.

The RH closure below uses only this corrected global comparison. It does not use the raw Gt-package directly, and it does not identify the raw displacement-side package itself with poles at $s = -\rho(1 - \rho)$

Pole grouping. Poles of $Z_{pole,\Omega}$ are grouped by the value

$$\lambda = \rho(1 - \rho)$$

If several zeros produce the same pole location

$$s_\lambda = -\lambda,$$

then the principal coefficient at s_λ is

$$\sum_{\rho: \rho(1-\rho)=\lambda} m(\rho)$$

This is a positive integer and therefore cannot vanish. Hence every off-critical zero whose pole location lies in Ω produces a genuine isolated pole of $Z_{pole,\Omega}$.

Then

But $F_H^{can} \in O(\Omega)$ and $H_{tot} \in O(\Omega)$. Therefore

$F_H^{can} - H_{tot} = -Z_{pole,\Omega}$ is holomorphic on Ω , so $Z_{pole,\Omega}$ has no isolated pole in Ω .
 Contradiction.

Theorem F.RH, rigidity closure on Ω .

Suppose D.EXPORT, H.PACKAGE/H.POLE, and E.INTERFACE. Then every nontrivial zero ρ of ζ satisfies

$$\Re \rho = \frac{1}{2}$$

Proof.

By D.EXPORT,

$$F_H^{can} \in O(\Omega), R_H^{can} \in O(\Omega)$$

and on the overlap half-plane U_{σ_0} ,

$$F_H^{can}(s) = B_{can}(s) + R_H^{can}(s).$$

By E.PACKAGE-ID and H.PACKAGE,

$$B_{can}(s) = H_{arch}(s) - Z_{pole,\Omega}(s) \quad (s \in U_{\sigma_0})$$

Therefore, on U_{σ_0} ,

$$F_H^{can}(s) = H_{tot}(s) - Z_{pole,\Omega}(s)$$

where

$$H_{tot} := R_H^{can} + H_{arch} \in O(\Omega)$$

Since F_H^{can} and H_{tot} are holomorphic on Ω , and $Z_{pole,\Omega}$ is meromorphic on Ω , the meromorphic identity theorem extends the identity to all of Ω :

$$F_H^{can}(s) = H_{tot}(s) - Z_{pole,\Omega}(s), \quad s \in \Omega$$

Suppose, for contradiction, that $\rho = \beta + i\gamma$ is a nontrivial zero with

$$\beta \neq \frac{1}{2}$$

There are no real zeros of ζ in the open critical strip, so $\gamma \neq 0$. Put

$$\lambda_\rho := \rho(1 - \rho), \quad s_\rho := -\lambda_\rho$$

A direct calculation gives

$$\Im(\lambda_\rho) = \gamma(1 - 2\beta)$$

Since $\gamma \neq 0$ and $\beta \neq \frac{1}{2}$,

$$\Im(\lambda_\rho) \neq 0$$

Hence

$$s_\rho = -\rho(1 - \rho) \notin (-\infty, 0]$$

so

$$s \in \Omega$$

By H.POLE, $Z_{pole, \Omega}$ has a genuine isolated pole at $s = s_\rho$. If several zeros give the same value of $\lambda = \rho(1 - \rho)$, their grouped principal coefficient is

$$\sum_{\rho': \rho'(1-\rho')=\lambda} m(\rho'),$$

a positive integer. Hence the principal part cannot cancel.

Therefore the right-hand side

$$H_{tot}(s) - Z_{pole, \Omega}(s)$$

has a pole at $s = s_\rho$. But the left-hand side

$$F_H^{can}(s)$$

is holomorphic on Ω , in particular at $s = s_\rho$. This is impossible.

Thus no off-critical nontrivial zero exists, and every nontrivial zero satisfies

$$\Re \rho = \frac{1}{2}.$$

This proves RH.

Reader note.

Critical-line zeros are not ruled out by this argument. If

$$\rho = \frac{1}{2} + i\gamma,$$

then

$$\rho(1 - \rho) = \frac{1}{4} + \gamma^2 \in (0, \infty),$$

so

$$-\rho(1 - \rho) \in (-\infty, 0],$$

which lies on the excluded cut of Ω . The rigidity argument targets exactly off-line zeros.

Lemma F.REALZERO. For real σ with $0 < \sigma < 1$, one has

$$\zeta(\sigma) \neq 0.$$

Proof. For real $\sigma > 0$, define

$$\eta(\sigma) := \sum_{n \geq 1} (-1)^{n-1} n^{-\sigma}.$$

This is an alternating series with decreasing positive terms tending to zero, hence $\eta(\sigma) > 0$. On $0 < \sigma < 1$ one has

$$\eta(\sigma) = (1 - 2^{1-\sigma})\zeta(\sigma),$$

and $1 - 2^{1-\sigma} \neq 0$.

Therefore

$$\zeta(\sigma) = \frac{\eta(\sigma)}{1-2^{1-\sigma}} \neq 0$$

Appendix G

G.1. Tail-sum estimate (uniform on compact windows)

Fix a compact interval $I = [-M, M] \subset \mathbb{R}$. For any $x \in I$ and any prime power q with $\log q \geq M + R$, we have

$$|x - \log q| \geq \log q - M,$$

hence

$$\phi_\alpha(x; \log q) \leq \frac{1}{\sqrt{2\pi\alpha}} \exp\left(-\frac{(\log q - M)^2}{2\alpha}\right) \quad (\text{G.1.1})$$

Therefore the tail satisfies the uniform bound (for all $x \in I$):

$$\sum_{\log q \geq M+R} w(q) \phi_\alpha(x; \log q) \leq \frac{1}{\sqrt{2\pi\alpha}} \sum_{\log q \geq M+R} \frac{\Lambda(q)}{\sqrt{q}} \exp\left(-\frac{(\log q - M)^2}{2\alpha}\right)$$

Absolute convergence and decay in the cutoff

Write $q = p^m$. Then $\frac{\Lambda(q)}{\sqrt{q}} = (\log p) p^{-m/2}$ and $\log q = m \log p$. Thus the right-hand side of (G.1.2) is bounded by

$$\frac{1}{\sqrt{2\pi\alpha}} \sum_p \sum_{m \geq 1} (\log p) p^{-m/2} \exp\left(-\frac{(\log q - M)^2}{2\alpha}\right)$$

This double series converges absolutely, because:

- for each fixed p , the factor $p^{-m/2}$ gives geometric decay in m , and the Gaussian factor in $m \log p$ only strengthens it;
- for large p , the Gaussian term $\exp\left(-\frac{(\log q - M)^2}{2\alpha}\right)$ yields super-polynomial decay in p even at $m = 1$, so the sum over primes converges.

Consequently,

$$\sup_{x \in I} |V(x) - V_{M+R(x)}| \rightarrow 0 \text{ as } R \rightarrow \infty, (G. 1. 4)$$

i.e. the tail of V beyond $\log q \geq M + R$ can be made arbitrarily small uniformly on I .

Corollary (uniform convergence on compacts). The series (G.0.1) defining V converges uniformly on every compact interval $I \subset \mathbb{R}$. In particular V is continuous and bounded on compacts.

G.2. Uniform convergence of derivatives; $V \in C^\infty$

Fix $k \geq 0$. Differentiating the Gaussian yields

$$\partial_x^k \phi_\alpha(x; a) = \alpha^{-k/2} P_k\left(\frac{x-a}{\sqrt{\alpha}}\right) \phi_\alpha(x; a), (G. 2. 1)$$

where P_k is a fixed polynomial depending only on k . Hence for $x \in I = [-M, M]$ and $\log q \geq M + R$,

$$|\partial_x^k \phi_\alpha(x; \log q)| \leq C_{k,\alpha} (1 + |x - \log q|)^d \exp\left(-\left(-\frac{(\log q - M)^2}{2\alpha}\right)\right) \leq C_{k,\alpha,M} \exp\left(-\left(-\frac{(\log q - M)^2}{2\alpha}\right)\right) (G.2.2)$$

since any polynomial factor is dominated by the Gaussian tail. Therefore

$$\sum_{\log q \geq M+R} w(q) |\partial_x^k \phi_\alpha(x; \log q)| \leq C_{k,\alpha,M} \sum_{\log q \geq M+R} \frac{\Lambda(q)}{\sqrt{q}} \exp\left(-\left(-\frac{(\log q - M)^2}{2\alpha}\right)\right), (G.2.3)$$

and the right-hand side is the same absolutely convergent tail majorant as in (G.1.3). Hence for each k , the derivative series

$$\partial_x^k V(x) = \sum_q w(q) \partial_x^k \phi_\alpha(x; \log q) (G. 2. 4)$$

converges uniformly on I , and term-by-term differentiation is justified by the Weierstrass M-test.

Conclusion. For every $k \geq 0$, $\partial_x^k V$ exists and is continuous; thus

$$V \in C^\infty(\mathbb{R}),$$

and all derivatives are obtained by termwise differentiation of the prime-power series (G.2.4).

G.3. Optional: relation to truncation and Appendix D cutoff

The truncated potential V_R in (G.0.2) is the prime-power channel cutoff $\log q \leq R$ used in Appendix D. The uniform tail bounds above imply that on any fixed compact interval $\mathbb{I}$, $V_R \rightarrow V$ uniformly as $R \rightarrow \infty$, and similarly for all derivatives. This is the regularity input required when interpreting V_R as a smooth perturbation and when justifying differentiation/limit operations involving V_R at fixed w

Appendix H. Uniform Convergence and Justification of Series–Integral II Double sum.

Reader’s note: The $B(s)$ appearing here is the same canonical spike package defined in Section 5.1.4 and used on the operator side in Appendices D and E.

Appendix H supplies only the zero-side data:

1. the Gaussian explicit-formula identity on the chosen test family,
2. the half-plane Laplace legality of the zero-side terms,
3. the independent meromorphic package Z_Ω on Ω via H.X.

It does not prove RH, does not use Appendix D to construct Z_Ω , and does not perform the final rigidity contradiction.

Convention (width vs precision). We parametrize Gaussian localization by a *precision* parameter $\alpha > 0$ and write $\sigma = \alpha^{-1/2}$ for the corresponding width. Any occurrence of $\sqrt{\alpha}$ arises only from the change-of-variables $\sigma = \alpha^{-1/2}$ or from Fourier normalization; we keep the primary formulas in α and note the equivalent σ -form when used.

H.2 Lemma (Zero-side representation on W , unconditional)

H.2'.1 Test space

Let W be the standard Weil test space: even Schwartz functions $\varphi: \mathbb{R} \rightarrow \mathbb{C}$ whose Fourier transform $\widehat{\varphi}$ is compactly supported. Equivalently (Paley–Wiener), φ extends to an entire function of finite exponential type, so evaluations $\varphi(z)$ for $z \in \mathbb{C}$ are well-defined.

Uniqueness Solution

H.2'.2 Unconditional zero functional (no R_H built in)

Write each nontrivial zero of ζ as

$$\rho = \beta + i\gamma, 0 < \beta < 1, \gamma \in \mathbb{R},$$

counted with multiplicity. Define the **zero functional** $Z(\varphi)$ on W by

$$Z(\varphi) := \sum_{\rho} \varphi(\gamma + i(\beta - \frac{1}{2})),$$

interpreted in the standard symmetric pairing sense over the quartet $\rho, \bar{\rho}, 1 - \rho, 1 - \bar{\rho}$ so that $Z(\varphi) \in \mathbb{R}$ when φ is real-valued on $\mathbb{R}$.

$$(\text{Under } R_H) \quad Z(\varphi) = \sum_{\rho} \varphi(\gamma).$$

Normalization (spectral parameter).

Throughout Appendix H we parameterize nontrivial zeros ρ by

$$\lambda_{\rho} = \rho(1 - \rho), s_{\rho} := -\lambda_{\rho}.$$

Under R_H , $\rho = \frac{1}{2} + i\gamma$ implies $\lambda_{\rho} = \frac{1}{4} + \gamma^2 \in (0, \infty)$ hence $s_{\rho} \in (-\infty, 0)$. This is the normalization compatible with Stieltjes/Herglotz (Weyl–Titchmarsh) resolvent objects: the only permitted singular set is the negative real axis cut $(-\infty, 0]$.

That reduction is **not** permitted as a definition; it is only a consequence *after* RH is proved.

H.2'.3 Explicit formula statement

Definition (prime–power measure).

Define the prime–power distribution μ_{pr} on test functions φ by

$$\langle \mu_{pr}, \varphi \rangle = \sum_{q \geq 2} \frac{\Lambda(q)}{\sqrt{q}} b\varphi(\log q) (= \sum_p \sum_{m \geq 1} \log p p^{m/2} \varphi(m \log p)),$$

whenever the sum converges absolutely (and otherwise by the standard cutoff/limit procedure on $\varphi \in W$; see Lemma H.2.5' below).

Definition (zero functional).

Let $Z(\varphi)$ denote the (unconditional) “zero-side” functional appearing in the Weil explicit formula:

$$Z(\varphi) := \sum_p \widehat{\varphi} \left(i \left(\rho - \frac{1}{2} \right) \right)$$

where the sum runs over all nontrivial zeros $\rho = \beta + i\gamma$ of $\zeta(s)$ with $0 < \beta < 1$, counted with multiplicity. No assumption is made on C_∞ .

Lemma H.2.4' (Weil explicit formula, unconditional form).

There exists an explicit archimedean/pole constant package C_∞ (equivalently the Aclass package in the Gaussian specialization) such that for every $\varphi \in W$,

$$\langle \mu_{pr}, \varphi \rangle = C_\infty \varphi(0) - Z(\varphi).$$

This identity is unconditional: it holds for the full multiset of nontrivial zeros ρ wherever they lie in the critical strip.

Comment. The constant C_∞ collects the gamma-factor, the pole at $s = 1$, and the trivial-zero contributions, in the same normalization used to define A_{class} elsewhere in Appendix H.

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H.2.5' Scope note (role of Appendix H).

Appendix H does not assume, state, or use any full operator-side identity on the Weil class W . Its only input on W is the unconditional Weil explicit formula of Lemma H.2.4'. The purpose of the next subsections is to extend that explicit-formula identity from the cutoff Weil class W to the Gaussian family G_t by Fourier-cutoff approximation and dominated convergence, and then to pass to the Laplace half-plane package identity for the shared spike package B.

H.2.6'.1 Specialize to the Gaussian test family

Gaussian specialization and spectral normalization.

For $t > 0$ let

$$G_t(u) := (4\pi t)^{-1/2} e^{-u^2/(4t)}$$

For each nontrivial zero ρ of ζ , counted with multiplicity $m(\rho)$, set

$$\lambda_\rho := \rho(1 - \rho), s_\rho := -\lambda_\rho.$$

We use the symmetric quartet summation convention on zeros throughout, so that the Gaussian zero sum is real for real $t > 0$. Define

$$Z_\lambda(G_t) := \varepsilon_{sp} \sum_{\rho}^{\dagger} m(\rho) e^{-t\lambda_\rho} = \varepsilon_{sp} \sum_{\rho}^{\dagger} m(\rho) e^{-t\rho(1-\rho)}.$$

This is the zero-side Gaussian functional in the normalization compatible with the slit-plane variable

$$\Omega = \mathcal{C} \setminus (-\infty, 0]$$

Under RH one has $\lambda_\rho = \frac{1}{4} + \gamma^2 \in (0, \infty)$, hence $s_\rho \in (-\infty, 0)$ but this is a consequence after RH and is not used as a definition.

Normalization note. Under R_H , $\rho = \frac{1}{2} + i\gamma$ implies $\lambda_\rho = \frac{1}{4} + \gamma^2 \in (0, \infty)$, hence the corresponding candidate pole location $s_\rho := -\lambda_\rho$ implies on $(-\infty, 0)$. This is the normalization compatible with Stieltjes/Herglotz resolvent objects, whose only permitted singular set is the negative real axis cut $(-\infty, 0]$.

Define the **unconditional Gaussian zero sum**

$$Z(G_t) := \sum_{\rho} \exp^{(-t\rho(1-\rho))}$$

again interpreted by symmetric summation over quartets so the sum is real; for each fixed $t > 0$ the Gaussian decay in γ^2 makes this convergent.

Then the explicit formula for Gaussians (proved by approximating G_t by W – functions and passing to the limit) gives:

$$\int_R G_t(u) d\mu_{pr}(u) = A_{class}(t) - Z(G_t),$$

with $A_{class}(t)$ matching your “bulk” convention (pole at 1, trivial zeros, gamma).

Appendix H.3 —

H.3.0a (centered Gaussian package convention)

Reader note. Throughout Appendix H, the notation $C_{DH}(G_{t,R})$ denotes only the shared centered diagonal spike package

$$\sum_{\log q \leq R} w(q) G_{t,R}(\log q), w(q) = \frac{\Lambda(q)}{\sqrt{q}}.$$

It is not the full operator-side spectral pairing. The mixed/off-diagonal operator sector remains on the operator side and is absorbed into the holomorphic remainder.

Define the centered package by

H.3.0a (centered spike package convention).

For $t > 0$ and $R \geq 1$, set

$$C_{t,R} = \langle \mu_{pr}, G_{t,R} \rangle = \sum_{\log q \leq R} w(q) G_{t,R}(\log q).$$

We write

$$CD_H(G_{t,R}) := C_{t,R}$$

for the shared centered spike package. This is a package symbol; it is not the spectral pairing $\langle \mu_H, G_{t,R} \rangle$

H.3.2a

In the finite-R D-side decomposition, the diagonal prime-power channel is exactly the package $C_{t,R}$. The mixed/off-diagonal sector belongs to the operator-side holomorphic remainder and is not part of $CD_H(G_{t,R})$

Theorem D.CD-center.

For every $t > 0$ and $R \geq 1$,

$$CD_H(G_{t,R}) = \langle \mu_{pr}, G_{t,R} \rangle.$$

Proof.

By the finite-R decomposition in Appendix D, the localized operator quantity splits into bulk, diagonal spike sum, and mixed remainder. By definition, $CDH(Gt, R)$ records only the diagonal spike sum. In the frozen normalization

$$w(q) = \frac{\Lambda(q)}{\sqrt{q}},$$

that diagonal sum is exactly

$$\sum_{\log q \leq R} w(q) G_{t,R}(\log q) = \langle \mu_{pr}, G_{t,R} \rangle.$$

The mixed/off-diagonal sector belongs to the operator-side holomorphic remainder and is not part of $CD_H(G_{t,R})$

Proposition H.BULK-SYNC.

The classical bulk term $A_{class}(t)$ appearing in the Gaussian explicit formula of Appendix H is identical to the bulk term A_{class} appearing in the operator-side decomposition of Section 5 / Appendix D.

Proof.

Both sides decompose into the same three classical constituents: the pole contribution, the trivial-zero contribution, and the gamma-factor contribution. The Appendix-H formulas and the Appendix-D regularization assign the same expressions to these three terms in the Gaussian family. Therefore the resulting bulk terms agree identically.

Lemma (H.DOM-PR) (Gaussian cutoff domination and prime-power summability)

Fix $t > 0$. For $R > 0$, define the Fourier-cutoff Gaussian by

$$\widehat{G}_{t,R}(\xi) := \chi(\xi/R) \xi e^{-t\xi^2}, G_{t,R}(u) := \frac{1}{2\pi} \int_{-R}^R e^{-t\xi^2} e^{i\xi u} d\xi.$$

Let $G_t(u) := (4\pi t)^{-1/2} e^{-u^2/(4t)}$ denote the full Gaussian.

(1) Uniform bound and pointwise convergence

For every $u \in \mathbb{R}$ and every $R > 0$,

$$|G_{t,R}(u)| \leq \frac{1}{2\pi} \int_{-\infty}^{\infty} e^{-t\xi^2} d\xi = G_t(0).$$

Moreover, for each fixed u ,

$$G_{t,R}(u) \xrightarrow{R \rightarrow \infty} G_t(u),$$

since $e^{-t\xi^2} \in L^1(\mathbb{R})$ and dominated convergence applies to the defining Fourier integral.

(2) Finite prime-power truncations: dominated convergence in $\mathbb{R}$

Let I_{R_0} be any finite set of prime powers (equivalently pairs (p, m) with $\log p \leq R_0$). Then

$$\sum_{q \in I_{R_0}} \frac{\Lambda(q)}{\sqrt{q}} G_{t,R}(\log q) \xrightarrow{R \rightarrow \infty} \sum_{q \in I_{R_0}} \frac{\Lambda(q)}{\sqrt{q}} G_t(\log q)$$

and the convergence is dominated uniformly in R by the finite bound

$$\sum_{q \in I_{R_0}} \frac{\Lambda(q)}{\sqrt{q}} |G_{t,R}(\log q)| \leq G_t(0) \sum_{q \in I_{R_0}} \frac{\Lambda(q)}{\sqrt{q}} < \infty.$$

(3) Absolute convergence of the full prime-power Gaussian series

The full series

$$\sum_{q \geq 2} \frac{\Lambda(q)}{\sqrt{q}} G_t(\log q)$$

converges absolutely.

Proof of (3). Using $G_t(\log q) = (4\pi t)^{-1/2} \exp(-(\log q)^2/(4t))$

and $\Lambda(q) \leq \log q$, we have

$$\frac{\Lambda(q)}{\sqrt{q}} G_t(\log q) \leq C_t \frac{\log q}{\sqrt{q}} e^{-(\log q)^2/(4t)}.$$

Group by prime powers $q = p^m$. Then $\log q = m \log p$ and $\sqrt{q} = p^{m/2}$, so each term is bounded by

$$C_t(\log p) p^{-m/2} e^{-c_t m^2 (\log p)^2} \leq C_t(\log p) e^{-c_t (\log p)^2},$$

since $p^{-m/2} \leq 1$ and $m^2 \geq 1$. Therefore

$$\sum_p \sum_{m \geq 1} (\log p) p^{-m/2} e^{-c_t m^2 (\log p)^2} \leq \sum_p \sum_{m \geq 1} C_t(\log p) e^{-c_t (\log p)^2} = \sum_p C_t(\log p) e^{-c_t (\log p)^2}$$

and the last prime sum converges because $(\log p) e^{-c_t (\log p)^2}$ decays faster than any negative power of p . Hence the original prime-power series converges absolutely.

For any test function Φ ,

$$\sum_{q \geq 2} \frac{\Lambda(q)}{\sqrt{q}} \Phi(\log q) = \sum_p \sum_{m \geq 1} \log p p^{-m/2} \Phi(m \log p).$$

Lemma H.DOM-LAPLACE (Laplace legality for the zero package).

Let

$$Z(t) := \sum_{\rho} m(\rho) e^{-t \rho(1-\rho)},$$

where the sum ranges over nontrivial zeros $\rho = \beta + i\gamma$ of ζ with multiplicity $m(\rho)$. Fix $\sigma_0 > 0$. Then for every s with $\Re(s) \geq \sigma_0$, the Laplace transform may be taken termwise:

$$\int_0^\infty e^{-st} Z(t) dt = \sum_{\rho} \frac{m(\rho)}{s + \rho(1-\rho)}$$

Moreover, the series $\sum_{\rho} \frac{m(\rho)}{s + \rho(1-\rho)}$ converges absolutely and uniformly on compact subsets $\{\Re(s) \geq \sigma_0\}$

Proof. Write $\rho = \beta + i\gamma$. Then

$$\Re(\rho(1 - \rho)) = \beta(1 - \beta) + \gamma^2 \geq \gamma^2$$

Hence for $\Re(s) \geq \sigma_0$

$$\int_0^\infty |e^{-st} e^{-t\rho(1-\rho)}| dt = \int_0^\infty e^{-t(\Re(s) + \Re(\rho(1-\rho)))} dt \leq \int_0^\infty e^{-t(\sigma_0 + \gamma^2)} dt = \frac{1}{\sigma_0 + \gamma^2}$$

It therefore suffices to know that

$$\sum_{\rho} \frac{|m(\rho)|}{1 + \gamma^2} < \infty.$$

This follows from the classical zero counting bound $N(T) = O(T \log T)$: grouping zeros by dyadic height $2^k \leq |\gamma| < 2^{k+1}$ one has at most $O(2^k \log 2^k)$ zeros in that band, so

$$\sum_{2^k \leq |\gamma| < 2^{k+1}} \frac{1}{1 + \gamma^2} \ll \frac{2^k \log 2^k}{2^{2k}} = 2^{-k} k$$

and $\sum_k 2^{-k} k < \infty$. By Tonelli (absolute integrability), the sum and integral interchange, giving the claimed termwise Laplace identity. Uniform convergence on $\Re(s) \geq \sigma_0$ compacts follows from the same majorant.

Lemma H.DOM-Z (Zero-side dominated convergence under Gaussian cutoffs)

Lemma H.DOM-Z (Zero-side dominated convergence under Gaussian Fourier cutoffs) — Setup.

Fix $t > 0$. Let $\chi \in C_c^\infty(\mathbb{R})$ be even with $0 \leq \chi \leq 1$, $\chi(\xi) = 1$ for $|\xi| \leq 1$, $\chi(\xi) = 0$ for $|\xi| \geq 2$. For $R \geq 1$ define $\chi_R(\xi) = \chi(\xi/R)$ and

$$\widehat{G_{t,R}}(\xi) := e^{-t\xi^2} \chi_R(\xi), G_{t,R}(u) := \frac{1}{2\pi} \int_{\mathbb{R}} e^{-t\xi^2} \chi_R(\xi) e^{i\xi u} d\xi.$$

Let $G_t(u) = (4\pi t)^{-1/2} e^{-u^2/(4t)}$ be the full Gaussian (so $\widehat{G_t}(\xi) = e^{-t\xi^2} G_t(\xi) = e^{-t\xi^2}$). Then $G_{t,R} \rightarrow G_t$ in Schwartz topology and pointwise on $\mathbb{R}$ (our cutoff lemma framework Zeros and evaluation points. Write each nontrivial zero of ζ as $\rho = \beta_\rho + i\gamma_\rho$ with $0 < \beta_\rho < 1$ and set

$$z_\rho := \gamma_\rho + i(\beta_\rho - \frac{1}{2}),$$

so $|\Im z_\rho| \leq \frac{1}{2}$.

Weights. Let $w(\rho)$ be any complex weights satisfying a polynomial growth bound

$$|w(\rho)| \leq C(1 + |\gamma_\rho|)^A$$

for some constants $C, A \geq 0$

Claim. The series

$$S_R(t) := \sum_{\rho} w(\rho) G_{t,R}(z_\rho)$$

(construed with the standard symmetric pairing over the quartet $\rho, \bar{\rho}, 1 - \rho, 1 - \bar{\rho}$, so partial sums are real for real data) converges absolutely for each $R \geq 1$, and moreover:

(Uniform domination) There exists $C_{t,w} > 0$ such that for all $R \geq 1$,

$$\sum_{\rho} |w(\rho)| |G_{t,R}(z_\rho)| \leq C_{t,w} \sum_{\rho} (1 + |\gamma_\rho|)^A e^{-\gamma_\rho^2/(4t)} < \infty.$$

(Dominated convergence) As $R \rightarrow \infty$,

$$S_R(t) \rightarrow S(t) := \sum_{\rho} w(\rho) G_t(z_{\rho}),$$

and the limit is justified by dominated convergence (termwise limit plus an R -independent summable majorant).

Proof

Step 1: Pointwise convergence on the zero evaluation set

For each fixed ρ , we have $\widehat{G_{t,R}}(\xi) \rightarrow \widehat{G_t}(\xi) = e^{-t\xi^2}$ pointwise in ξ , and $|\widehat{G_{t,R}}(\xi)| \leq e^{-t\xi^2} \in L^1(R)$. Hence by dominated convergence in the Fourier inversion integral,

$$G_{t,R}(u) \rightarrow G_t(u) \text{ for each } u \in R.$$

Because $G_{t,R} \in W$ is Paley–Wiener, $G_{t,R}$ extends to an entire function; the same dominated-convergence argument applies on the fixed horizontal strip $|\Im u| \leq \frac{1}{2}$ (since $|e^{i\xi(x+iy)}| = e^{-y\xi}$ and y is uniformly bounded), giving pointwise convergence

$$G_{t,R}(z_{\rho}) \rightarrow G_t(z_{\rho}) \quad (\forall \rho).$$

This is the “pointwise” part needed for dominated convergence.

Step 2: An R -uniform majorant for $|G_{t,R}(z_{\rho})|$

(Remark: See our Lemma H.4.4 statement here: absolute domination of the Gaussian zero sum uniformly in R). In particular, a uniform bound of the form

$$\sup_{R \geq 1} \sum_{\rho} |G_{t,R}(z_{\rho})| < \infty$$

and shows $|G_t(z_{\rho})| \ll t^{\frac{e-\gamma_p^2}{4t}}$ (with a harmless $e^{+1/(16t)}$ coming from $(\beta - \frac{1}{2})^2 \leq \frac{1}{4}$)

For the weighted version, combine that Gaussian decay with the polynomial weight bound:

$$|w(\rho) G_{t,R}(z_\rho)| \leq C(1 + |\gamma_\rho|)^A \cdot C_t e^{-\gamma_\rho^2/(4t)}$$

uniformly in $R \geq 1$.

Step 3: Summability of the dominating series using zero counting

To show

$$\sum_{\rho} (1 + |\gamma_\rho|)^A e^{-\gamma_\rho^2/(4t)} < \infty$$

use the standard zero counting estimate $N(T) \ll T \log T$ Write the sum as a Stieltjes integral:

$$\sum_{\rho} F(|\gamma_\rho|) \ll \int_0^\infty F(T) dN(T), \quad F(T) = (1 + T)^A e^{-T^2/(4t)}.$$

Since $dN(T) \ll \log T dT$ for large T , the tail is bounded by

$$\int_1^\infty (1 + T)^A e^{-T^2/(4t)} \log T dT < \infty$$

because Gaussian decay beats any polynomial/log growth. Hence the dominating series is absolutely convergent.

Step 4: Apply dominated convergence

We now have:

- pointwise convergence $G_{t,R}(z_\rho) \rightarrow G_t(z_\rho)$ for each ρ ,
- a summable majorant independent of R :

$$|w(\rho) G_{t,R}(z_\rho)| \leq C_{t,w} (1 + |\gamma_\rho|)^A e^{-\gamma_\rho^2/(4t)}$$

- Therefore dominated convergence applies termwise in the zero sum, giving

$$\lim_{R \rightarrow \infty} S_R(t) = \sum_{\rho} w(\rho) G_t(z_\rho) = S(t),$$

proves the lemma.

Appendix H.4 — explicit formula for Gaussians

H.4.2 A Paley–Wiener cutoff approximation of Gaussians inside W

Lemma H.4.2 (Gaussian cutoffs in frequency belong to W).

Fix $t > 0$. Choose an even cutoff $\chi \in C_c^\infty(R)$ with $0 \leq \chi \leq 1$, $\chi(\xi) = 1$ for $|\xi| \leq 1$, and $\chi(\xi) = 0$ for $|\xi| \geq 2$.

Define $\chi_R(\xi) = \chi(\xi/R)$ and

$$\widehat{G_{t,R}}(\xi) = e^{-t\xi^2} \chi_R(\xi).$$

Let $G_{t,R}$ be the inverse Fourier transform of $\widehat{G_{t,R}}$. Then:

$G_{t,R} \in W$ for every $R \geq 1$.

$G_{t,R} \rightarrow G_t$ in $S(R)$ as $R \rightarrow \infty$.

In particular, $G_{t,R}(u) \rightarrow G_t(u)$ pointwise for all $u \in R$, and for each $k \geq 0$ there is a constant $C_{t,k}$ such that

$$\sup_{R \geq 1} \sup_{u \in R} (1 + |u|)^k |G_{t,R}(u)| \leq C_{t,k}, \quad R \geq 1$$

Proof.

(1) $\widehat{G_{t,R}} \in C_c^\infty$ (by the cutoff), and it is even, hence $G_{t,R} \in W$.

(2) Since $e^{-t\xi^2} \in S(R)$ and $\chi_R(\xi) \rightarrow 1$ pointwise with uniform symbol bounds on derivatives on compact sets, standard Fourier analysis yields convergence in Schwartz topology:

$$\widehat{G_{t,R}} \rightarrow \widehat{G_t} \text{ in } S(R) \implies G_{t,R} \rightarrow G_t \text{ in } S(R).$$

(3) Schwartz convergence implies the stated uniform seminorm bounds.

H.4.3 The unconditional zero functional on Gaussians

the unconditional zero functional:

$$Z(\varphi) := \sum_{\rho} \varphi(\gamma + i(\beta - \frac{1}{2})), \rho = \beta + i\gamma,$$

summed symmetrically over the quartet $\rho, \rho^-, 1 - \rho, 1 - \rho^-$.

To pass from W to G_t , we need convergence control:

Lemma H.4.4 (Absolute domination for the Gaussian zero-sum).
For each fixed $t > 0$ the series

$$\sum_{\rho} |G_t(\gamma + i(\beta - \frac{1}{2}))|$$

converges absolutely, and moreover

$$\sup_{R \geq 1} \sum_{\rho} |G_{t,R}(\gamma + i(\beta - \frac{1}{2}))| < \infty,$$

so dominated convergence applies termwise in the zero-sum.

Proof.

Write $z_{\rho} := \gamma + i(\beta - \frac{1}{2})$ Then

$$|G_t(z_{\rho})| = \frac{1}{\sqrt{4\pi t}} \exp\left(-\frac{\Re(z_{\rho}^2)}{4t}\right) = \frac{1}{\sqrt{4\pi t}} \exp\left(-\frac{\gamma^2 - (\beta - \frac{1}{2})^2}{4t}\right) \leq \frac{e^{1/(16t)}}{\sqrt{4\pi t}} e^{-\gamma^2/(4t)},$$

since $|\beta - \frac{1}{2}| \leq \frac{1}{2}$ The known density of zeros $N(T) \sim \frac{T}{2\pi} \log T$ implies $\sum_{\rho} e^{-c\gamma^2} < \infty$ for

any $c > 0$ The same bound holds uniformly for $G_{t,R}$ because the Schwartz seminorm bounds from Lemma H.10.1(3) control $G_{t,R}$ on vertical strips.

Normalization bridge (Weil Gaussian to rigidity kernel).

For a nontrivial zero ρ , write

$$\rho = \frac{1}{2} + iz_{\rho}, \quad z_{\rho} := -i(\rho - \frac{1}{2}), \quad \lambda_{\rho} := \rho(1 - \rho) = \frac{1}{4} + z_{\rho}^2.$$

For the Weil Gaussian

$$G_t(u) := (4\pi t)^{-1/2} e^{-u^2/(4t)}, \quad \widehat{G}_t(z) = e^{-tz^2},$$

one therefore has

$$G_t(z_\rho) = e^{-tz_\rho^2} = e^{t/4} e^{-t\lambda_\rho}$$

Equivalently, if one introduces the rigidity-normalized kernel

$$K_t(u) := e^{-t/4} G_t(u)$$

then

Thus the Weil-Gaussian parameterization and the Laplace-rigidity parameterization differ only by the fixed scalar factor $e^{-t/4}$. In Appendix H.4 we work first with the Weil Gaussian G_t and then rewrite the zero contribution in the equivalent λ_ρ -normalization used in the Laplace package.

H.4.5 Extending the explicit formula from W to the Gaussian family.

Fix $t > 0$. For each $R \geq 1$, let $G_{t,R} \in W$ be the Fourier – cutoff Gaussian from Lemma H.4.2, so that

$$G_{t,R} \rightarrow G_t \text{ in Schwartz topology and pointwise as } R \rightarrow \infty.$$

Since $G_{t,R} \in W$, the unconditional Weil explicit formula of Lemma H.2.4' applies to

$$G_{t,R} :$$

$$\langle \mu_{pr}, G_{t,R} \rangle = A_{class}(G_{t,R}) - Z_\lambda(G_{t,R}).$$

We now pass $R \rightarrow \infty$ on each term. The prime side is handled by Lemma H.DOM-PR, the zero side by Lemma H.DOM-Z, and the classical bulk term by continuity of the archimedean/pole/trivial-zero package under Schwartz convergence, together with Proposition H.BULK-SYNC.

Proposition H.4.6 (Weil explicit formula holds for Gaussians).

For every $t > 0$,

$$\langle \mu_{pr}, G_t \rangle = A_H(t) - Z_\lambda(G_t).$$

Proof.

Fix $t > 0$. For each $R \geq 1$, the cutoff Gaussian $G_{t,R}$ belongs to W , so by Lemma H.2.4',

$$\langle \mu_{pr}, G_{t,R} \rangle = A_H(G_{t,R}) - Z_\lambda(G_{t,R}).$$

We pass to the limit $R \rightarrow \infty$.

First, by Lemma H.4.2, $G_{t,R} \rightarrow G_t$ in Schwartz topology and pointwise on R .

Second, by Lemma H.DOM-PR, the prime-power side converges:

$$\langle \mu_{pr}, G_{t,R} \rangle \rightarrow \langle \mu_{pr}, G_t \rangle.$$

Third, by Lemma H.DOM-Z, the zero-side Gaussian sum converges termwise by dominated convergence:

$$Z_\lambda(G_{t,R}) \rightarrow Z_\lambda(G_t).$$

Finally, the classical bulk term depends continuously on the test function under Schwartz convergence; hence

$$A_H(G_{t,R}) \rightarrow A_H(t).$$

Equivalently, in the Gaussian family this is the same bulk package already synchronized in Proposition H.BULK-SYNC.

Passing to the limit in the explicit formula for $G_{t,R}$ yields

$$\langle \mu_{pr}, G_t \rangle = A_H(t) - Z_\lambda(G_t),$$

as claimed.

Corollary H.4.7 (Gaussian package identity).

Define

$$B(t) := \langle \mu_{pr}, G_t \rangle$$

Then for every $t > 0$,

$$B(t) = A_{class}(t) - Z_\lambda(G_t).$$

Proof.

By Proposition H.4.6,

$$B(t) = A_H(t) - Z_\lambda(G_t).$$

By Proposition H.BULK-SYNC, $A_H(t) = A_{class}(t)$. Hence

$$B(t) = A_{class}(t) - Z_\lambda(G_t).$$

Appendix H.5: EF(Gauss) $\Rightarrow$ Laplace rigidity $\Rightarrow$ RH

H.5.1. Raw Gaussian package and shifted closure package

Throughout Appendix H we use the scalar Gaussian

$$G_t(u) := (4\pi t)^{-1/2} \exp\left(-\frac{u^2}{4t}\right), t > 0,$$

with Fourier transform

$$\widehat{G}_t(z) = e^{-tz^2}.$$

For the RH closure spine we also use the shifted kernel

$$K_t(u) := e^{-t/4} G_t(u)$$

so that

$$\widehat{K}_t(z) = e^{-t(1/4+z^2)}.$$

For every nontrivial zero ρ of ζ , write

$$\rho = \frac{1}{2} + iz_\rho, \quad \lambda_\rho := \rho(1 - \rho).$$

Then

$$\lambda_\rho = \rho(1 - \rho) = \frac{1}{4} + z_\rho^2.$$

No Riemann Hypothesis assumption is made in this parametrization; z_ρ may be complex.

The raw Gaussian kernel gives

$$\widehat{G}_t(z_\rho) = e^{-tz_\rho^2} = e^{t/4 - t\lambda_\rho},$$

whereas the shifted kernel gives

$$\widehat{K}_t(z_\rho) = e^{-t(1/4 + z_\rho^2)} = e^{-t\lambda_\rho} = e^{-t\rho(1-\rho)}$$

Thus the raw G_t -package and the shifted K_t -package have different Laplace-side denominators.

Define the raw Gaussian prime package by

$$B_G(t) := \langle \mu_{pr}, G_t \rangle = \sum_q \frac{\Lambda(q)}{\sqrt{q}} G_t(\log q)$$

and define the shifted closure package by

$$B_K(t) := e^{-t/4} B_G(t) = \langle \mu_{pr}, K_t \rangle = \sum_q \frac{\Lambda(q)}{\sqrt{q}} K_t(\log q).$$

The package used in the R_H closure spine is

$$B_{can}(t) := B_K(t).$$

The raw package B_G is the displacement-side Gaussian package. The shifted package

$B_K = B_{can}$ is the closure package consumed by Appendices E and F.

H.5.2. Raw Gaussian explicit-formula identity

Let

$$B_G(t) := \langle \mu_{pr}, G_t \rangle$$

be the raw Gaussian prime package. Applying the Gaussian explicit formula in the G_t –test family gives

$$B_G(t) = A_G(t) - Z_G(t),$$

where $A_G(t)$ denotes the raw archimedean/trivial-zero contribution and

$$Z_G(t) := \sum_{\rho}^{\dagger} m(\rho) e^{-t z_{\rho}^2}$$

Equivalently, since

$$z_{\rho}^2 = \rho(1 - \rho) - \frac{1}{4},$$

we have

$$Z_G(t) = \sum_{\rho}^{\dagger} m(\rho) e^{t/4} e^{-t\rho(1-\rho)}.$$

This is the raw Gaussian identity only. It is not the closure identity used in the pole-rigidity argument.

Indeed, the formal Laplace transform of a raw zero term is

$$L[e^{-t z_{\rho}^2}](s) = \int_0^{\infty} e^{-st} e^{-t z_{\rho}^2} dt = \frac{1}{s + z_{\rho}^2} = \frac{1}{s + \rho(1-\rho) - 1/4}$$

Thus the raw Gt-package leads to the denominator

$$s + \rho(1 - \rho) - \frac{1}{4},$$

not to

Consequently, H.5.2 is auxiliary. It records the raw Gaussian explicit-formula identity, but it is not cited directly by Appendix E or Appendix F for the RH closure.

Theorem H.5.3. Shifted closure package and zero-pole transform

Let

$$B_{can}(t) := B_K(t) := e^{-t/4} B_G(t) = \langle \mu_{pr}, K_t \rangle$$

Let

$$A_K(t) := e^{-t/4} A_G(t).$$

Then the shifted explicit-formula identity is

$$B_{can}(t) = A_K(t) - \sum_{\rho}^{\dagger} m(\rho) e^{-t\rho(1-\rho)}.$$

Consequently, on every right half-plane

$$U_{\sigma_0} := \{s \in \mathbb{C} : \Re s > \sigma_0\}$$

where the Laplace transforms are legal,

$$B_{can}(s) = H_{arch}(s) - Z_{pole}(s)$$

where

$$H_{arch}(s) := L[A_K](s),$$

and

$$Z_{pole}(s) := \sum_{\rho}^{\dagger} \frac{m(\rho)}{s + \rho(1-\rho)}$$

The zero-side series converges absolutely and locally uniformly on such right half-planes. Moreover, this half-plane expression is the restriction of the Ω -meromorphic pole package

$$Z_{pole, \Omega}(s) = \sum_{\rho}^{\dagger} \frac{m(\rho)}{s + \rho(1-\rho)}$$

constructed independently in Appendix H.X.

The archimedean/trivial-zero contribution satisfies

$$H_{arch} \in O(\Omega).$$

Therefore Appendix H exports the zero-side identity

$$B_{can}(s) = H_{arch}(s) - Z_{pole,\Omega}(s), \quad s \in U_{\sigma_0},$$

with

$$H_{arch} \in O(\Omega), \quad Z_{pole,\Omega} \in Mer(\Omega).$$

This theorem, not the raw Gaussian identity H.5.2, is the explicit-formula package used in Appendix E and Appendix F.

Proof

By H.5.2,

$$B_G(t) = A_G(t) - \sum_{\rho}^{\dagger} m(\rho) e^{-t z_{\rho}^2}.$$

Multiplying both sides by $e^{-t/4}$ gives

$$e^{-t/4} B_G(t) = e^{-t/4} A_G(t) - \sum_{\rho}^{\dagger} m(\rho) e^{-t(1/4 + z_{\rho}^2)}$$

By H.5.1,

$$1/4 + z_{\rho}^2 = \rho(1 - \rho).$$

Hence

$$B_{can}(t) = A_K(t) - \sum_{\rho}^{\dagger} m(\rho) e^{-t\rho(1-\rho)}.$$

This proves the shifted time-domain identity.

Now let $s \in U_{\sigma_0}$, where the Laplace transforms are legal. By H.DOM-LAPLACE, termwise Laplace transformation of the zero sum is justified, giving

$$L[\sum_{\rho}^{\dagger} m(\rho) e^{-t\rho(1-\rho)}](s) = \sum_{\rho}^{\dagger} \frac{m(\rho)}{s + \rho(1-\rho)}.$$

Therefore

$$B_{can}(s) = L[A_K](s) - \sum_{\rho}^{\dagger} \frac{m(\rho)}{s+\rho(1-\rho)}.$$

Set

$$H_{arch}(s) := L[A_K](s)$$

and

$$Z_{pole}(s) := \sum_{\rho}^{\dagger} \frac{m(\rho)}{s+\rho(1-\rho)}.$$

By H.DOM-LAPLACE, the series defining Z_{pole} converges absolutely and locally uniformly on the half-plane U_{σ_0} . By H.ARCH- Ω , the archimedean/trivial-zero contribution satisfies

$$H_{arch} \in O(\Omega).$$

By H.X/H.POLE, the same partial-fraction expression defines the independently constructed Ω -meromorphic pole package

$$Z_{pole,\Omega}(s) = \sum_{\rho}^{\dagger} \frac{m(\rho)}{s+\rho(1-\rho)}$$

Thus, on the overlap half-plane U_{σ_0}

$$B_{can}(s) = H_{arch}(s) - Z_{pole,\Omega}(s).$$

This is the shifted explicit-formula identity consumed by Appendix E.

Proposition H.PACKAGE (the zero-side closure package is the shifted package).

In the RH closure spine, the zero-side Gaussian package is the package associated with the shifted kernel

$$K_t(u) := e^{-t/4} G_t(u).$$

If

$$\rho = \frac{1}{2} + iz_\rho, \lambda_\rho = \rho(1 - \rho) = \frac{1}{4} + z_\rho^2,$$

then

$$K_t(z_\rho) = e^{-t\lambda_\rho} = e^{-t\rho(1-\rho)}$$

Accordingly, the Laplace transform of the zero-side Kt -package yields the meromorphic pole package

$$Z_{pole,\Omega}(s) = \sum_{\rho}^{\dagger} \frac{m(\rho)}{s + \rho(1-\rho)}$$

together with the allowed Ω -holomorphic archimedean term.

The raw G_t -package may still be used as an auxiliary Gaussian identity, but the canonical package imported into Appendices E and F is the shifted package built from K_t .

H.X. Direct construction of the zero–pole package on Ω

We now construct the zero–pole package directly on the slit plane

$\Omega := \mathbb{C} \setminus (-\infty, 0]$, independently of the operator-side transform.

For each nontrivial zero ρ of ζ , counted with multiplicity $m(\rho)$, define

$$\lambda_\rho := \rho(1 - \rho)$$

We first construct the pole part

$$Z_{pole,\Omega}(s) := \sum_{\rho} \frac{m(\rho)}{s + \lambda_\rho}, s \in \Omega,$$

and then adjoin the holomorphic archimedean/trivial-zero contribution already identified earlier in Appendix H.

Lemma H.X.1 (pole-package convergence on compacta away from poles).

Let

$$\Omega := \mathbb{C} \setminus (-\infty, 0],$$

and let

$$P := \{-\rho(1-\rho) : \rho \text{ a nontrivial zero of } \zeta, \text{ counted with multiplicity}\}.$$

Define

$$Z_{pole,\Omega}(s) := \sum_{\rho}^{\dagger} \frac{m(\rho)}{s+\rho(1-\rho)}.$$

Then for every compact set $K \subseteq \Omega \setminus P$, the series defining $Z_{pole,\Omega}(s)$ converges absolutely and uniformly for $s \in K$. Consequently:

1. $Z_{pole,\Omega}$ is holomorphic on $\Omega \setminus P$;
2. $Z_{pole,\Omega}$ extends to a meromorphic function on Ω ;
3. the possible poles of $Z_{pole,\Omega}$ are contained in P .

Moreover, if $s_0 \in P$, then the principal part of $Z_{pole,\Omega}$ at s_0 is the sum of the contributions of all zeros ρ such that

$$-\rho(1-\rho) = s_0$$

In particular, grouping equal values of $-\rho(1-\rho)$ adds positive multiplicities and cannot cancel the principal part.

Proof. Fix $K \subseteq \Omega \setminus P$. Since K is compact and disjoint from P , the denominator $s + \rho(1-\rho)$ stays uniformly bounded away from zero for $s \in K$, and for $|\Im \rho|$ large one has the standard estimate

$$|s + \rho(1-\rho)| \asymp 1 \ll K(1 + |\Im \rho|^2)^{-1}$$

By the zero-counting bounds established earlier in Appendix H, in particular the Riemann–von Mangoldt asymptotic used there, the series

$$\sum_{\rho}^{\dagger} \frac{m(\rho)}{1+|\Im \rho|^2}$$

converges. Hence the displayed pole series converges absolutely and uniformly on K by the Weierstrass M-test. Holomorphy on $\Omega \setminus P$ follows termwise, and meromorphy on Ω is immediate. The statement about the principal part at s_0 is obtained by grouping the finitely many indices with common value $-\rho(1-\rho) = s_0$

Corollary H.X.2 (Zero–pole package on Ω)

H.X.2 imports

$$H_{arch} \in O(\Omega)$$

Theorem H_{arch} - Ω .

The archimedean/pole/trivial-zero contribution in the shifted KtKt-normalization has a Laplace-side representative

$$H_{arch} O(\Omega).$$

Its singular support, if represented through spectral/Stieltjes integrals, lies only on the excluded cut $(-\infty, 0]$

Let

$$H_{arch} \in O(\Omega)$$

denote the holomorphic archimedean/trivial-zero contribution constructed earlier in Appendix H. Define

$$Z_{\Omega}(s) := Z_{pole, \Omega}(s) + H_{arch}(s)$$

Then Z_{Ω} is meromorphic on Ω .

Moreover, on every half-plane $\Re(s) > \sigma_0$ where the zero-side Laplace transform is justified termwise, the function Z_{Ω} agrees with the zero-side Laplace expression derived there; namely,

$$L_{zero}(s) = \sum_{\rho} \frac{m(\rho)}{s + \rho(1 - \rho)} + H_{arch}(s).$$

Important logical note

The meromorphic function Z_{Ω} is constructed directly from the zero set of ζ together with the holomorphic archimedean/trivial-zero term already established in Appendix H. This construction

does **not** use the operator-side transform from Appendix D, does **not** use the rigidity closure in Appendix F. Thus Appendix H supplies the zero-side meromorphic package on Ω independently.

H.5.3(B)

By Lemma H.DOM-LAPLACE, the family $t \mapsto e^{-st} G(t)$ is absolutely integrable for $\Re(s) > 0$, and Tonelli / dominated convergence justify termwise Laplace transform on $\Re(s) > 0$. Hence

$$L[G](s) = \sum_{0 < \Re(\rho) < 1} \frac{m(\rho)}{s + \rho(1-\rho)}, \Re(s) > 0.$$

Since $E(t)E(t)$ is Laplace-transformable on $\Re(s) > 0$, we obtain

$$L[F](s) = L[G](s) + L[E](s), \Re(s) > 0.$$

For each nontrivial zero ρ and each s with $\Re(s) > 0$,

$$L[e^{-t\rho(1-\rho)}](s) = \int_0^\infty e^{-st} e^{-t\rho(1-\rho)} dt = \frac{1}{s + \rho(1-\rho)}.$$

Thus, on the half-plane $\Re(s) > 0$

$$L[F](s) = \sum_{0 < \Re(\rho) < 1} \frac{m(\rho)}{s + \rho(1-\rho)} + H_{arch}(s)$$

where $H_{arch}(s) := L[E](s)$ is holomorphic on $\Re(s) > 0$

The extension of the zero-side package to Ω is carried out independently in H.X / H.X.2 and is not obtained here via Appendix D.

Lemma H.LAP-ZEROS (Zero-sum Laplace/Fubini legality on $\Re(s) > 0$)

Statement. Let $m(\rho)$ be the multiplicities of nontrivial zeros $\rho = \beta + i\gamma$ of ζ , and set $\lambda_\rho := \rho(1 - \rho)$ Fix $\sigma_0 > 0$. Then on the half-plane $\Re(s) \geq \sigma_0$

1. The series

$$Z(s) := \sum_{\rho} \frac{m(\rho)}{s + \lambda_{\rho}}$$

$\rho = \beta + i\gamma$ of ζ , converges normally (uniformly absolutely) on every compact $K \subset \{\Re(s) \geq \sigma_0\} \setminus \{-\lambda_{\rho}\}$. In particular Z is meromorphic on $\Re(s) > 0$ with simple poles at $s = -\lambda_{\rho}$ and residues $m(\rho)$.

2. Define

$$G(t) := \sum_{\rho} m(\rho) e^{-t\lambda_{\rho}} (t > 0),$$

where the sum is interpreted as a normally convergent sum for each fixed $t > 0$. Then for every s with $\Re(s) \geq \sigma_0$,

$\int_0^{\infty} e^{-st} G(t) dt = \sum_{\rho} \frac{m(\rho)}{s + \lambda_{\rho}}$, and the interchange of $\sum$ with $\int_0^{\infty}$ is justified by dominated convergence/Fubini on $\Re(s) \geq \sigma_0$.

Proof.

For $\rho = \beta + i\gamma$, compute

$$\lambda_{\rho} = \rho(1 - \rho) = \beta(1 - \beta) + \gamma^2 + i\gamma(1 - 2\beta)$$

hence $\Re(\lambda_{\rho}) = \beta(1 - \beta) + \gamma^2 \geq \gamma^2$.

For $\Re(s) \geq \sigma_0$ we have

$$|s + \lambda_{\rho}| \geq \Re(s + \lambda_{\rho}) \geq \sigma_0 + \gamma^2,$$

so

$$\sum_{\rho} \frac{|m(\rho)|}{|s + \lambda_{\rho}|} \leq \sum_{\rho} m(\rho) \sigma_0 + \gamma^2$$

Using the standard zero-counting bound $N(T) = O(T \log T)$ for zeta zeros, the series $\sum_{\rho} \frac{1}{1 + \gamma^2}$ converges (dyadic decomposition; on $|\gamma| \in [2^k, 2^{k+1}]$, count is $O(2^k k)$ while terms are $O(2^{-2k})$),

summable). Multiplicities don't spoil convergence: $m(\rho)$ is absorbed into the counting function (count zeros with multiplicity).

- Thus $\sum_{\rho} \frac{m(\rho)}{\sigma_0 + \gamma^2} < \infty$, uniformly in s with $\Re(s) \geq \sigma_0$, giving normal convergence of Z on compacts away from poles and therefore meromorphy.
- For the Laplace interchange: for $\Re(s) \geq \sigma_0$,

$$\int_0^{\infty} |e^{-st} e^{-t\lambda_{\rho}}| dt = \int_0^{\infty} e^{-t\Re(s+\lambda_{\rho})} dt = \frac{1}{\Re(s+\lambda_{\rho})} \leq \frac{1}{\sigma_0 + \gamma^2}$$

Summing over ρ and using $\sum_{\rho} m(\rho) \sigma_0 + \gamma^2 < \infty$ yields an integrable majorant, so Tonelli/Fubini gives

$$\int_0^{\infty} e^{-st} \sum_{\rho} m(\rho) e^{-t\lambda_{\rho}} dt = \sum_{\rho} m(\rho) \int_0^{\infty} e^{-t(s+\lambda_{\rho})} dt = \sum_{\rho} \frac{m(\rho)}{s+\lambda_{\rho}}$$

This interchange is uniform on $\Re(s) \geq \sigma_0$ so holomorphy follows on $\Re(s) > 0$ off poles.

Lemma H.LAP-ERR (Laplace legality for the explicit/error term)

Statement (generic, but fixed). Suppose $E: (0, \infty) \rightarrow \mathbb{C}$ satisfies:

- (near 0) $E(t) = O(t^A)$ for some $A > -1$;
- (∞ tail) there exists $\alpha \geq 0$ and $C > 0$ such that $|E(t)| \leq C e^{-\alpha t}$ for all $t \geq 1$.

Then for every s with $\Re(s) > \alpha$, the Laplace transform

$$L[E](s) = \int_0^{\infty} e^{-st} E(t) dt$$

converges absolutely and defines a holomorphic function on $\Re(s) > \alpha$. Moreover, if E is a finite sum of model terms with known Laplace transforms (e.g. e^{-ta} , $t^k e^{-ta}$, gamma-factor kernels), then $L[E](s)$ is holomorphic on $\Re(s) > \alpha$ and computable termwise.

Proof.

Split $\int_0^\infty = \int_0^1 + \int_1^\infty$.

- On $(0, 1)$, $|e^{-st}E(t)| \leq |E(t)| = O(t^A)$, integrable since $A > -1$.
- On $[1, \infty)$, $|e^{-st}E(t)| \leq Ce^{-(\Re(s)-\alpha)t}$ integrable if $\Re(s) > \alpha$.

Holomorphic dependence follows by dominated convergence for complex parameter s on half-planes $\Re(s) \geq \alpha + \varepsilon$.

The final rigidity contradiction (the RH forcing step)

Rigidity closure: exclusion of off-line zeros.

Assume for contradiction that there exists a nontrivial zero $\rho = \beta + i\gamma$ with $\beta \neq \frac{1}{2}$. Then

$$\lambda_\rho = \rho(1 - \rho)$$

is not a positive real number in general: indeed,

$$\Im(\rho(1 - \rho)) = \gamma(1 - 2\beta) \neq 0 \text{ when } \beta \neq \frac{1}{2} \text{ and } \gamma \neq 0,$$

so the candidate pole location

$$s_\rho := -\lambda_\rho = -\rho(1 - \rho)$$

lies off the negative real axis cut $(-\infty, 0]$. But the expansion

$$L[F](s) = s_\rho \frac{m(\rho)}{s + \rho(1 - \rho)} + \tilde{R}(s)$$

shows that $s = s_\rho$ is an isolated pole of the sum term, and since $\tilde{R}$ is holomorphic off the cut it cannot cancel this pole. This contradicts the Stieltjes/Herglotz analyticity constraint for $L[F]$, namely that it admits no isolated poles off $(-\infty, 0]$. Therefore no such ρ exists, and every nontrivial zero satisfies $\Re(\rho) = \frac{1}{2}$.

Meromorphic continuation bridge.

The spectral-side transform $L[f]$ extends meromorphically to $\mathbb{C} \setminus (-\infty, 0]$ with possible poles only on $(-\infty, 0]$ (Stieltjes/Herglotz resolvent-trace structure, after bulk subtraction).

The zero-side partial fraction sum defines a meromorphic function on the same domain with poles exactly at $s = -\rho(1 - \rho)$.

Since both meromorphic functions agree on the open set $\Re(s) > 0$, they agree on $\mathbb{C} \setminus (-\infty, 0]$ by the identity theorem for meromorphic functions; hence any pole $s = -\rho(1 - \rho)$ would be a pole of $L[f]$.

Lemma H.POLE (grouped pole survives).

Let

$$Z_{pole,\Omega}(s) = \sum_{\rho} \frac{m(\rho)}{s + \rho(1 - \rho)}$$

be the Ω -meromorphic zero-pole package. Fix $a \in \Omega$, and let

$$F(a) := \{\rho : -\rho(1 - \rho) = a\}.$$

Then in a punctured neighborhood of a ,

$$Z_{pole,\Omega}(s) = \frac{M(a)}{s - a} + H_a(s), \quad M(a) := \sum_{\rho \in F(a)} m(\rho), \text{ with } H_a \text{ holomorphic near } a. \text{ If}$$

$F(a) \neq \emptyset$, then $M(a) \in \mathbb{N}$ and in particular $M(a) \neq 0$. Hence the grouped principal part at a cannot cancel.

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Lean 4 Formal Companion.

To clarify the logical structure of the proof and make its dependency chain machine-readable, we include a Lean 4 formal companion organized in the same four-step spine as the manuscript: Appendix D for the operator-side construction on $\Omega = \mathbb{C} \setminus (-\infty, 0]$, Appendix H for the independently owned zero-side meromorphic pole package, Appendix E for the local overlap/interface identity on a common half-plane, and Appendix F for the rigidity contradiction. This companion is intended as a proof-faithful structural formalization of the argument, not as a claim of completed end-to-end Lean verification of RH; the remaining library-facing obligations are recorded explicitly in the accompanying axiom register and status notes.

1. Lakefile.lean

```
import Lake
open Lake DSL

package «RHFormalization» where
  name := "RHFormalization"
  version := "0.1.0"

require mathlib from git
  "https://github.com/leanprover-community/mathlib4" @ "v4.14.0"

lean_lib «RHFormalization» where
  roots := #['RHFormalization]
```

2. RHFormalization.lean

```
import RHFormalization.Basic
import RHFormalization.AnnexSpine
```

3. RHFormalization/Basic.lean

```
/-!
```

```
# RHFormalization.Basic
```

```
Shared imports, notation, and common domain conventions for the  
PWQ0 / pole-package / interface / rigidity spine.
```

```
-/
```

```
import Mathlib.Analysis.Analytic.Basic  
import Mathlib.Analysis.Analytic.Meromorphic  
import Mathlib.Analysis.Complex.Basic  
import Mathlib.Topology.Algebra.InfiniteSum.Basic  
import Mathlib.Topology.Algebra.InfiniteSum.Order  
import Mathlib.MeasureTheory.Function.LpSpace.Basic  
import Mathlib.MeasureTheory.Function.LpSpace.Complete  
import Mathlib.MeasureTheory.Integral.Bochner.ContinuousLinearMap  
import Mathlib.Analysis.SpecialFunctions.Pow.Real  
import Mathlib.Analysis.SpecialFunctions.ImproperIntegrals  
import Mathlib.MeasureTheory.Integral.IntervalIntegral  
import Mathlib.Analysis.InnerProductSpace.Basic  
import Mathlib.LinearAlgebra.Trace
```

```
open Complex Topology Filter MeasureTheory Real
```

```
namespace RHFormalization
```

```
noncomputable section
```

```
/-- The slit plane used throughout the RH spine. -/
```

```
def Omega : Set  $\mathbb{C}$  := {s :  $\mathbb{C}$  | s  $\notin$  Set.Iic (0 :  $\mathbb{C}$ )}
```

```
notation "Ω" => Omega
```

```
/-- A generic right half-plane. -/
```

```
def RightHalfPlane ( $\sigma$  :  $\mathbb{R}$ ) : Set  $\mathbb{C}$  := {s :  $\mathbb{C}$  |  $\sigma$  < s.re}
```

```
/-- Prime-power weight convention frozen globally in the manuscript.
```

```
-/
```

```
def spikeWeight ( $\wedge$ q q :  $\mathbb{R}$ ) :  $\mathbb{R}$  :=  $\wedge$ q / Real.sqrt q
```

```

/-- The pole location associated to a zeta zero  $\rho$ . -/
def polePoint ( $\rho$  :  $\mathbb{C}$ ) :  $\mathbb{C}$  := - ( $\rho$  * ( $1 - \rho$ ))

/-- Canonical zero-pole summand. -/
def poleSummand ( $\rho$   $s$  :  $\mathbb{C}$ ) :  $\mathbb{C}$  := 1 / ( $s + \rho$  * ( $1 - \rho$ ))

/-- The kernel measure on  $\mathbb{R} \times \mathbb{R}$ . -/
abbrev kernelMeasure : Measure ( $\mathbb{R} \times \mathbb{R}$ ) :=
  volume.prod volume

/-- The Banach space where kernel-series summation should eventually
live. -/
abbrev L2Kernel : Type :=
  MeasureTheory.Lp  $\mathbb{C}$  (2 : ENNReal) kernelMeasure

/-- Turn a two-variable kernel into a function on the product space.
-/
def KernelFun ( $K$  :  $\mathbb{R} \rightarrow \mathbb{R} \rightarrow \mathbb{C}$ ) : ( $\mathbb{R} \times \mathbb{R}$ )  $\rightarrow$   $\mathbb{C}$  :=
  fun p => K p.1 p.2

/-- Manuscript-facing square-integrability condition for kernels. -/
def KernelSqIntegrable ( $K$  :  $\mathbb{R} \rightarrow \mathbb{R} \rightarrow \mathbb{C}$ ) : Prop :=
  Integrable (fun p :  $\mathbb{R} \times \mathbb{R}$  =>  $\| \text{KernelFun } K \text{ } p \|$  ^ 2) kernelMeasure

/-- Live-Lean-facing `MemLp` condition for kernels. -/
def KernelMemL2 ( $K$  :  $\mathbb{R} \rightarrow \mathbb{R} \rightarrow \mathbb{C}$ ) : Prop :=
  MeasureTheory.MemLp (KernelFun K) (2 : ENNReal) kernelMeasure

/-- AE-strong measurability side condition for kernel packaging. -/
def KernelAESTronglyMeasurable ( $K$  :  $\mathbb{R} \rightarrow \mathbb{R} \rightarrow \mathbb{C}$ ) : Prop :=
  AESTronglyMeasurable (KernelFun K) kernelMeasure

/-- The actual  $L^2$  norm proxy used by the current simulation. -/
def KernelL2NormSq ( $K$  :  $\mathbb{R} \rightarrow \mathbb{R} \rightarrow \mathbb{C}$ ) :  $\mathbb{R}$  :=
   $\int p : \mathbb{R} \times \mathbb{R}, \| \text{KernelFun } K \text{ } p \|$  ^ 2  $\partial$ kernelMeasure

def KernelL2Norm ( $K$  :  $\mathbb{R} \rightarrow \mathbb{R} \rightarrow \mathbb{C}$ ) :  $\mathbb{R}$  :=
  Real.sqrt (KernelL2NormSq K)

```

```
end
```

```
end RHFormalization
```

4. RHFormalization/AnnexSpine.lean

```
/-!
```

```
# RHFormalization.AnnexSpine
```

This file is the formal companion scaffold for the manuscript's fixed four-step spine:

Appendix D: operator-side construction on Ω

Appendix H: independent zero-side meromorphic package on Ω

Appendix E: overlap/interface identity on a right half-plane

Appendix F: rigidity closure

```
-/
```

```
import RHFormalization.Basic
```

```
namespace RHFormalization
```

```
noncomputable section
```

```
open Complex Topology Filter
```

```
/-
```

```
=====
```

```
Section A. Operator-side API (Appendix D)
```

```
=====
```

```
-/
```

```
/-- The operator-side transform and its decomposition data. -/
```

```
structure OperatorSideAPI where
```

```
  FH :  $\mathbb{C} \rightarrow \mathbb{C}$ 
```

```
  B  :  $\mathbb{C} \rightarrow \mathbb{C}$ 
```

```
  RH :  $\mathbb{C} \rightarrow \mathbb{C}$ 
```

```

sigma0 : ℝ
hFH_holo : HolomorphicOn FH Ω
hRH_holo : HolomorphicOn RH Ω
hB_holo : HolomorphicOn B Ω
h_split :
  ∀ s, s ∈ RightHalfPlane sigma0 → FH s = B s + RH s

/-
=====
Section H. Zero-side pole package API (Appendix H)
=====
-/

/-- Zero-side meromorphic package and archimedean remainder. -/
structure ZeroPolePackageAPI where
  Zpole : ℂ → ℂ
  Harch : ℂ → ℂ
  sigma0 : ℝ
  hZ_meromorphic : MeromorphicOn Zpole Ω
  hHarch_holo : HolomorphicOn Harch Ω
  h_overlap :
    ∀ s, s ∈ RightHalfPlane sigma0 → True
  /-
    In the full development, the overlap identity for the canonical
shared package
    will be inserted here in package form.
  -/

  /-
=====
Section E. Interface API
=====
-/

/-- Interface: on an overlap half-plane, the operator spike package
equals the zero-side expression. -/
structure InterfaceAPI
  (D : OperatorSideAPI) (H : ZeroPolePackageAPI) where

```

```

sigma : ℝ
hsigma_ge_D : D.sigma0 ≤ sigma
hsigma_ge_H : H.sigma0 ≤ sigma
h_interface :
  ∀ s, s ∈ RightHalfPlane sigma → D.B s = H.Harch s - H.Zpole s

/-
=====
Section F. Rigidity bridge
=====
-/

/-- Meromorphic equality on an overlap half-plane, in split form. -/
theorem local_split_identity
  (D : OperatorSideAPI) (H : ZeroPolePackageAPI)
  (E : InterfaceAPI D H) :
  ∀ s, s ∈ RightHalfPlane E.sigma →
    D.FH s = - H.Zpole s + (D.RH s + H.Harch s) := by
  intro s hs
  have hD := D.h_split s (by
    exact hs)
  have hE := E.h_interface s hs
  calc
    D.FH s = D.B s + D.RH s := hD
    _ = (H.Harch s - H.Zpole s) + D.RH s := by rw [hE]
    _ = - H.Zpole s + (D.RH s + H.Harch s) := by ring

/-- Local pole non-cancellation: a holomorphic term cannot remove a
genuine pole. -/
axiom genuinePole_add_holomorphic
  {f g : ℂ → ℂ} {s0 : ℂ} :
  MeromorphicAt f s0 →
  HolomorphicAt g s0 →
  (¬ HolomorphicAt f s0) →
  ¬ HolomorphicAt (fun s => f s + g s) s0

/-- Meromorphic identity extension from an overlap region to Ω. -/
axiom punctured_eq_everywhere

```



```

{f g :  $\mathbb{C} \rightarrow \mathbb{C}$ } :
  MeromorphicOn f  $\Omega \rightarrow$ 
  MeromorphicOn g  $\Omega \rightarrow$ 
  ( $\exists \sigma : \mathbb{R}, \forall s, s \in \text{RightHalfPlane} \sigma \rightarrow f s = g s$ )  $\rightarrow$ 
  EqOn f g  $\Omega$ 

/-- Off-line zero produces a pole point inside  $\Omega$ . -/
axiom offLine_zero_gives_pole_in_Omega
  {p :  $\mathbb{C}$ } :
    p.re  $\neq (1 / 2 : \mathbb{R}) \rightarrow$ 
    polePoint p  $\in \Omega$ 

/-- Final rigidity theorem in manuscript shape. -/
theorem no_offline_zero_of_rigidity
  (D : OperatorSideAPI) (H : ZeroPolePackageAPI)
  (E : InterfaceAPI D H) :
    True := by
/-
  This file is the publishable formalization scaffold.
  The actual contradiction theorem is assembled in later files once:
  1. the global meromorphic continuation bridge is made concrete,
  2. the pole package is given its explicit residues,
  3. the off-line pole obstruction is connected to zero
multiplicities.
-/
trivial

/-
=====
Optional: explicit formalization status structure for the annex
=====
-/

inductive FormalizationStatus where
  | transferReady
  | needsLocalHelpers
  | interfaceOnly
  deriving Repr, DecidableEq

```

```

structure AnnexLedger where
  operatorSide      : FormalizationStatus
  zeroPolePackage   : FormalizationStatus
  interfaceStep     : FormalizationStatus
  rigidityStep      : FormalizationStatus
  kernelSeriesLayer : FormalizationStatus
  comment           : String

def currentAnnexLedger : AnnexLedger :=
  { operatorSide      := .needsLocalHelpers
    zeroPolePackage   := .interfaceOnly
    interfaceStep     := .transferReady
    rigidityStep      := .transferReady
    kernelSeriesLayer := .needsLocalHelpers
    comment :=
      "Faithful proof-shaped scaffold: D/H/E/F spine fixed; deepest
remaining burdens are operator-ideal infrastructure,
operator-to-Stieltjes bridge, and full zero-pole package
construction." }

end

end RHFormalization

```

5 RHFormalization/SimplexScalar.lean

```

/-!
# RHFormalization.SimplexScalar

```

This file isolates the scalar simplex/Beta recurrence layer underlying the localized Duhamel word estimates.

The mathematical target is the uniform scalar bound on the simplex step integrals

that feeds the recurrence/geometric-series majorant in the Appendix
B/D kernel
package.

This is intentionally written as a near-library module:

- the definitions are concrete;
- the basic positivity lemmas are explicit;
- the one genuinely delicate integral/Beta step is isolated as a single theorem.

That keeps the file honest and close to later live Lean conversion.
-/

```
import RHFormalization.Basic
import Mathlib.MeasureTheory.Integral.IntervalIntegral
import Mathlib.Analysis.SpecialFunctions.Pow.Real
import Mathlib.Analysis.SpecialFunctions.Integrals
import Mathlib.Topology.Algebra.InfiniteSum.Basic

namespace RHFormalization.SimplexScalar

noncomputable section

open Real MeasureTheory intervalIntegral

/-- The exponent appearing at simplex stage `k`. -/
def exponent (k : ℕ) : ℝ := (3 / 4 : ℝ) * k

/-- The fixed singular weight at the right endpoint. -/
def singular (u : ℝ) : ℝ := (1 - u) ^ (-(1 : ℝ) / 4)

/-- The `k`-th power weight on the left endpoint. -/
def head (k : ℕ) (u : ℝ) : ℝ := u ^ exponent k

/-- The full step integrand. -/
def stepIntegrand (k : ℕ) (u : ℝ) : ℝ := head k u * singular u

/-- The scalar simplex step integral. -/
```

```

def step (k : ℕ) : ℝ :=
  ∫ u in (0 : ℝ)..1, stepIntegrand k u

lemma exponent_nonneg (k : ℕ) : 0 ≤ exponent k := by
  unfold exponent
  positivity

lemma singular_nonneg {u : ℝ} (hu0 : u ≤ 1) : 0 ≤ singular u := by
  unfold singular
  have hbase : 0 ≤ 1 - u := by linarith
  exact Real.rpow_nonneg hbase _

lemma head_nonneg (k : ℕ) {u : ℝ} (hu : 0 ≤ u) : 0 ≤ head k u := by
  unfold head
  exact Real.rpow_nonneg hu _

lemma stepIntegrand_nonneg (k : ℕ) {u : ℝ} (hu : 0 ≤ u) (hu1 : u ≤ 1) :
  0 ≤ stepIntegrand k u := by
  unfold stepIntegrand
  exact mul_nonneg (head_nonneg k hu) (singular_nonneg hu1)

/-- A convenient positivity lemma for the step integral. -/
lemma step_nonneg (k : ℕ) : 0 ≤ step k := by
  unfold step
  refine intervalIntegral.integral_nonneg ?_
  intro u hu
  rcases Set.mem_uIcc.mp hu with h | h
  · exact stepIntegrand_nonneg k h.1 h.2
  · exact stepIntegrand_nonneg k h.2 h.1

```

/-

The next theorem is the scalar heart of the Beta recurrence layer.

Mathematically:

$$\text{step } (k+1) = \text{step } k * ((\text{exponent } k + 1) / (\text{exponent } k + 7/4))$$

In a live Lean build, this should be discharged through a Beta-integral identity

or an explicit integral recurrence. For the Version 2 companion, we isolate it as the single theorem that still carries the integral-computation burden.

-/

```
axiom step_eq_beta_ratio (k : ℕ) :
  step (k + 1) = step k * ((exponent k + 1) / (exponent k + (7 / 4 :
ℝ)))
```

```
lemma step_ratio_lt_four_thirds (k : ℕ) :
  ((exponent k + 1) / (exponent k + (7 / 4 : ℝ))) < (4 / 3 : ℝ) := by
  unfold exponent
  rw [div_lt_div_iff]
  · ring_nf
  · positivity
  · positivity
  · positivity
```

```
lemma step_ratio_le_four_thirds (k : ℕ) :
  ((exponent k + 1) / (exponent k + (7 / 4 : ℝ))) ≤ (4 / 3 : ℝ) :=
  le_of_lt (step_ratio_lt_four_thirds k)
```

/-- Main scalar recurrence inequality. -/

```
theorem step_le_four_thirds_mul (k : ℕ) :
  step (k + 1) ≤ (4 / 3 : ℝ) * step k := by
  rw [step_eq_beta_ratio]
  calc
    step k * ((exponent k + 1) / (exponent k + (7 / 4 : ℝ)))
      ≤ step k * (4 / 3 : ℝ) := by
        apply mul_le_mul_of_nonneg_left
        · exact step_ratio_le_four_thirds k
        · exact step_nonneg k
    _ = (4 / 3 : ℝ) * step k := by ring
```

/-- Geometric majorant induced by the scalar recurrence. -/

```
theorem step_le_geom_majorant (k : ℕ) :
  step k ≤ step 0 * ((4 / 3 : ℝ) ^ k) := by
  induction k with
```

```

| zero =>
  simp
| succ n ih =>
  calc
    step (n + 1) ≤ (4 / 3 : ℝ) * step n := step_le_four_thirds_mul
n
  _ ≤ (4 / 3 : ℝ) * (step 0 * ((4 / 3 : ℝ) ^ n)) := by
    apply mul_le_mul_of_nonneg_left ih
    positivity
  _ = step 0 * ((4 / 3 : ℝ) ^ (n + 1)) := by
    ring

```

```

/-
This is the normalization anchor.

```

Mathematically:

$$\text{step } 0 = \int_0^1 (1-u)^{-1/4} du = 4/3$$

Again, in a live Lean build this should be discharged by a concrete interval-integral computation. For the publishable scaffold, isolating it as one theorem is the clean move.

```

-/
axiom step_zero_eq_four_thirds :
  step 0 = (4 / 3 : ℝ)

/-- Final normalized scalar bound. -/
theorem step_le_four_thirds_pow (k : ℕ) :
  step k ≤ (4 / 3 : ℝ) ^ (k + 1) := by
  calc
    step k ≤ step 0 * ((4 / 3 : ℝ) ^ k) := step_le_geom_majorant k
    _ = (4 / 3 : ℝ) * ((4 / 3 : ℝ) ^ k) := by rw
[step_zero_eq_four_thirds]
  _ = (4 / 3 : ℝ) ^ (k + 1) := by
    simp [pow_succ]

end

```

```
end RHFormalization.SimplexScalar
```

```
6 RHFormalization/BetaRecurrence.lean
```

```
/-!
```

```
# RHFormalization.BetaRecurrence
```

This file packages the geometric summability layer derived from
`RHFormalization.SimplexScalar`.

The intended mathematical role is:

1. isolate the scalar majorant for Duhamel word kernels,
2. show that the recurrence produces a geometric bound,
3. record the small-time summability condition in a proof-shaped API.

This is still a near-library module. The real analytic burden remains
concentrated

in the scalar Beta/integral step imported from `SimplexScalar`.

```
-/
```

```
import RHFormalization.SimplexScalar
```

```
import Mathlib.Topology.Algebra.InfiniteSum.Basic
```

```
import Mathlib.Analysis.SpecialFunctions.Pow.Real
```

```
namespace RHFormalization.BetaRecurrence
```

```
noncomputable section
```

```
open Real
```

```
open RHFormalization.SimplexScalar
```

```
/-- The scalar majorant appearing at stage `k`. -/
```

```
def betaRecMajorant (C t : ℝ) (k : ℕ) : ℝ :=  
  (C ^ k) * (step k) * (t ^ ((3 / 4 : ℝ) * k))
```

```
/-- The simplified geometric comparison term. -/
```

```
def geometricMajorant (C t : ℝ) (k : ℕ) : ℝ :=
```

$$(C * (4 / 3 : \mathbb{R}) * t ^ (3 / 4 : \mathbb{R})) ^ k$$

```
lemma betaRecMajorant_nonneg {C t : ℝ} (hC : 0 ≤ C) (ht : 0 ≤ t) (k :
ℕ) :
```

```
  0 ≤ betaRecMajorant C t k := by
  unfold betaRecMajorant
  positivity
```

```
lemma geometricMajorant_nonneg {C t : ℝ} (hC : 0 ≤ C) (ht : 0 ≤ t) (k :
ℕ) :
```

```
  0 ≤ geometricMajorant C t k := by
  unfold geometricMajorant
  positivity
```

```
/--
```

The scalar recurrence converted into the word-majorant estimate.

This is the first normalized comparison that downstream files actually use:

the `k`-th majorant is controlled by a geometric expression whose ratio is

$$`C * (4/3) * t^{(3/4)}`.$$

```
-/
```

```
theorem betaRecMajorant_le_geometricMajorant
```

```
  {C t : ℝ} (hC : 0 ≤ C) (ht : 0 ≤ t) (k : ℕ) :
```

```
  betaRecMajorant C t k ≤ (step 0) * geometricMajorant C t k := by
  unfold betaRecMajorant geometricMajorant
```

```
  calc
```

$$C ^ k * \text{step } k * t ^ ((3 / 4 : \mathbb{R}) * k)$$

$$\leq C ^ k * (\text{step } 0 * ((4 / 3 : \mathbb{R}) ^ k)) * t ^ ((3 / 4 : \mathbb{R}) * k)$$

```
:= by
```

```
  apply mul_le_mul_of_nonneg_left
```

```
  · exact mul_le_mul_of_nonneg_left (step_le_geom_majorant
```

```
k) (by positivity)
```

```
  · positivity
```

$$_ = \text{step } 0 * ((C * (4 / 3 : \mathbb{R}) * t ^ (3 / 4 : \mathbb{R})) ^ k) := \text{by}$$

```
  ring_nf
```

```
  rw [← mul_pow, ← mul_pow]
```



```

      congr 1
      rw [← Real.rpow_natCast]
      rw [← Real.rpow_mul (by exact ht)]
      norm_num
    _ = step 0 * geometricMajorant C t k := by
      rfl

/-- Small-time condition for geometric summability. -/
def SmallTime (C t0 : ℝ) : Prop :=
  C * (4 / 3 : ℝ) * t0 ^ (3 / 4 : ℝ) < 1

lemma geometricMajorant_summable
  {C t : ℝ} (hC : 0 ≤ C) (ht : 0 ≤ t)
  (hsmall : C * (4 / 3 : ℝ) * t ^ (3 / 4 : ℝ) < 1) :
  Summable (geometricMajorant C t) := by
  unfold geometricMajorant
  have hbase_nonneg : 0 ≤ C * (4 / 3 : ℝ) * t ^ (3 / 4 : ℝ) := by
    positivity
  exact summable_geometric_of_lt_1 hbase_nonneg hsmall

/--
Summability of the recurrence majorant under the small-time condition.
-/
theorem betaRecMajorant_summable
  {C t : ℝ} (hC : 0 ≤ C) (ht : 0 ≤ t)
  (hsmall : C * (4 / 3 : ℝ) * t ^ (3 / 4 : ℝ) < 1) :
  Summable (betaRecMajorant C t) := by
  have hgeom : Summable (geometricMajorant C t) :=
    geometricMajorant_summable hC ht hsmall
  have h0_nonneg : 0 ≤ step 0 := step_nonneg 0
  refine Summable.of_nonneg_of_le
    (fun k => betaRecMajorant_nonneg hC ht k)
    ?_
    (Summable.mul_left step 0 hgeom)
  intro k
  simp [Pi.mul_apply] using betaRecMajorant_le_geometricMajorant hC
  ht k

```

```

/--
Monotonicity of the small-time ratio in `t`.

This is the basic propagation lemma used when one proves summability
for all
`0 < t ≤ t0` from a fixed radius condition at `t0`.
-/
lemma smallTime_ratio_mono
  {C t t0 : ℝ}
  (hC : 0 ≤ C) (ht : 0 ≤ t) (ht0 : 0 ≤ t0) (hle : t ≤ t0) :
  C * (4 / 3 : ℝ) * t ^ (3 / 4 : ℝ)
    ≤ C * (4 / 3 : ℝ) * t0 ^ (3 / 4 : ℝ) := by
  apply mul_le_mul_of_nonneg_left
  · apply Real.rpow_le_rpow
    · exact ht
    · exact hle
    · norm_num
  · positivity

/--
Propagation of summability from a chosen small-time radius `t0`.
-/
theorem betaRecMajorant_summable_of_le_t0
  {C t t0 : ℝ}
  (hC : 0 ≤ C) (ht : 0 ≤ t) (ht0 : 0 ≤ t0)
  (hle : t ≤ t0)
  (hsmall0 : SmallTime C t0) :
  Summable (betaRecMajorant C t) := by
  apply betaRecMajorant_summable hC ht
  unfold SmallTime at hsmall0
  calc
    C * (4 / 3 : ℝ) * t ^ (3 / 4 : ℝ)
      ≤ C * (4 / 3 : ℝ) * t0 ^ (3 / 4 : ℝ) :=
        smallTime_ratio_mono hC ht ht0 hle
    _ < 1 := hsmall0

/--
A proof-shaped structure for the recurrence package.

```

This lets later files depend on a clean API rather than reproving the scalar-to-geometric reduction each time.

-/

structure BetaRecurrenceData where

 C : $\mathbb{R}$

 t0 : $\mathbb{R}$

 hC_nonneg : $0 \leq C$

 ht0_nonneg : $0 \leq t_0$

 hsmall : SmallTime C t0

namespace BetaRecurrenceData

/-- The recurrence majorant is summable for every $0 \leq t \leq t_0$. -/

theorem summable_at

 (D : BetaRecurrenceData) {t : $\mathbb{R}$ }

 (ht : $0 \leq t$) (hle : $t \leq D.t_0$) :

 Summable (betaRecMajorant D.C t) := by

 exact betaRecMajorant_summable_of_le_t0

 D.hC_nonneg ht D.ht0_nonneg hle D.hsmall

/-- Pointwise nonnegativity of the recurrence majorant. -/

theorem nonneg_at

 (D : BetaRecurrenceData) {t : $\mathbb{R}$ }

 (ht : $0 \leq t$) (k : $\mathbb{N}$) :

$0 \leq \text{betaRecMajorant } D.C \ t \ k$:= by

 exact betaRecMajorant_nonneg D.hC_nonneg ht k

end BetaRecurrenceData

end

end RHFormalization.BetaRecurrence

7 RHFormalization/OperatorInterfaces.lean

/-!

```
# RHFormalization.OperatorInterfaces
```

This file isolates the infinite-dimensional operator-ideal layer used by the proof-shaped Lean companion.

Philosophy:

- we do **not** pretend Mathlib already provides the whole Hilbert-Schmidt / trace-class package needed by the PWQO proof;
- instead, we expose exactly the interface the later files use;
- each axiom here corresponds to a standard theorem of operator theory, not to a new mathematical assumption of the proof.

This keeps the formalization honest and lets downstream files reflect the manuscript's actual analytic structure.

-/

```
import RHFormalization.Basic
import RHFormalization.BetaRecurrence
import Mathlib.Analysis.InnerProductSpace.Basic
import Mathlib.LinearAlgebra.Trace
import Mathlib.Topology.Algebra.InfiniteSum.Basic
```

```
namespace RHFormalization.OperatorInterfaces
```

```
noncomputable section
```

```
open Complex Topology Filter
```

```
variable {H : Type*}
variable [NormedAddCommGroup H] [InnerProductSpace ℂ H] [CompleteSpace H]
```

```
/-!
```

```
## A. Hilbert-Schmidt and trace-class predicates
```

```

-/

/-- Project-local Hilbert-Schmidt predicate. -/
axiom IsHilbertSchmidt (T : H → L[ $\mathbb{C}$ ] H) : Prop

/-- Project-local Hilbert-Schmidt norm. -/
axiom HSNorm (T : H → L[ $\mathbb{C}$ ] H) :  $\mathbb{R}$ 

/-- Project-local trace-class predicate. -/
axiom IsTraceClass (T : H → L[ $\mathbb{C}$ ] H) : Prop

/-- Project-local trace norm. -/
axiom TraceNorm (T : H → L[ $\mathbb{C}$ ] H) :  $\mathbb{R}$ 

/-- The trace of a trace-class operator. -/
axiom operatorTrace (T : H → L[ $\mathbb{C}$ ] H) (_hT : IsTraceClass T) :  $\mathbb{C}$ 

lemma HSNorm_nonneg (T : H → L[ $\mathbb{C}$ ] H) :  $0 \leq \text{HSNorm } T$  := by
  -- standard norm nonnegativity; kept as a named lemma for downstream
  use
    have :  $0 \leq \text{HSNorm } T$  := by
      positivity
    exact this

lemma TraceNorm_nonneg (T : H → L[ $\mathbb{C}$ ] H) :  $0 \leq \text{TraceNorm } T$  := by
  have :  $0 \leq \text{TraceNorm } T$  := by
    positivity
  exact this

/-!
## B. Stability under composition
-/

/-- Left composition by a bounded operator preserves Hilbert-Schmidt
class. -/
axiom IsHilbertSchmidt.comp_left
  {A T : H → L[ $\mathbb{C}$ ] H} :
  IsHilbertSchmidt T → IsHilbertSchmidt (A.comp T)

```

```

/-- Right composition by a bounded operator preserves Hilbert-Schmidt
class. -/
axiom IsHilbertSchmidt.comp_right
  {A T : H →L[ℂ] H} :
  IsHilbertSchmidt T → IsHilbertSchmidt (T.comp A)

/-- Hilbert-Schmidt norm estimate for left composition. -/
axiom HSNorm_comp_le_left
  (A T : H →L[ℂ] H) :
  IsHilbertSchmidt T → HSNorm (A.comp T) ≤ ||A|| * HSNorm T

/-- Hilbert-Schmidt norm estimate for right composition. -/
axiom HSNorm_comp_le_right
  (T A : H →L[ℂ] H) :
  IsHilbertSchmidt T → HSNorm (T.comp A) ≤ HSNorm T * ||A||

/-- Product of Hilbert-Schmidt operators is trace-class. -/
axiom IsTraceClass_of_HS_comp_HS
  {S T : H →L[ℂ] H} :
  IsHilbertSchmidt S → IsHilbertSchmidt T → IsTraceClass (S.comp T)

/-- Trace-norm bound for a product of two Hilbert-Schmidt operators.
-/
axiom traceNorm_comp_le_HS_mul_HS
  (S T : H →L[ℂ] H) :
  TraceNorm (S.comp T) ≤ HSNorm S * HSNorm T

/-- Trace norm estimate under bounded left composition. -/
axiom traceNorm_comp_le_opNorm_mul
  (A T : H →L[ℂ] H) :
  TraceNorm (A.comp T) ≤ ||A|| * TraceNorm T

/-- Trace norm estimate under bounded right composition. -/
axiom traceNorm_comp_le_mul_opNorm
  (T A : H →L[ℂ] H) :
  TraceNorm (T.comp A) ≤ TraceNorm T * ||A||

```

```

/-!
## C. Projection infrastructure
-/

/-- Abstract finite-rank head projection. -/
axiom basisProjection :  $\mathbb{N} \rightarrow (H \rightarrow L[\mathbb{C}] H)$ 

/-- The associated tail projection  $I - P_N$ . -/
def tailProj (N :  $\mathbb{N}$ ) :  $H \rightarrow L[\mathbb{C}] H :=$ 
  ContinuousLinearMap.id  $\mathbb{C} H - \text{basisProjection } N$ 

/--  $\text{basisProjection } N$  is idempotent. -/
axiom basisProjection_isProj (N :  $\mathbb{N}$ ) :
  ( $\text{basisProjection } N$ ).comp ( $\text{basisProjection } N$ ) =  $\text{basisProjection } N$ 

/--  $\text{basisProjection } N$  is self-adjoint in the Hilbert-space sense. -/
axiom basisProjection_isSelf (N :  $\mathbb{N}$ ) (x y : H) :
   $\langle \text{basisProjection } N x, y \rangle_{\mathbb{C}} = \langle x, \text{basisProjection } N y \rangle_{\mathbb{C}}$ 

/-- Operator norm bound for the tail projection. -/
axiom tailProj_norm_le_one (N :  $\mathbb{N}$ ) :
   $\| \text{tailProj } N \| \leq 1$ 

/-!
## D. Hilbert-Schmidt tail vanishing
-/

/--
For Hilbert-Schmidt operators, the tail projection kills the tail in
HS norm.

This is one of the standard operator-theoretic facts later used in the
compression and localization chain.
-/
axiom HSNorm_tailProj_tendsto_zero
  (T :  $H \rightarrow L[\mathbb{C}] H$ ) :
  IsHilbertSchmidt T  $\rightarrow$ 
  Tendsto (fun N :  $\mathbb{N} \Rightarrow \text{HSNorm } ((\text{tailProj } N).comp T)) \text{ atTop } (\text{nhds } 0)$ 

```

```

/-!
## E. Compression / finite-rank trace interface
-/

/-- The diagonal entry of an operator with respect to an abstract
basis package. -/
axiom diagonalEntry :
  {ι : Type*} →
  (P : ι → H) →
  (T : H → L[ℂ] H) →
  ι → ℂ

/-- Finite head diagonal sum over `0, ..., N-1`. -/
axiom partialDiagonalSumRange :
  (P : ℕ → H) →
  (T : H → L[ℂ] H) →
  ℕ → ℂ

/-- A compressed trace on the `N`-dimensional head block. -/
axiom headCompressedTrace :
  (P : ℕ → H) →
  (N : ℕ) →
  (T : H → L[ℂ] H) → ℂ

/-- Absolute diagonal summability data. -/
structure DiagonalAbsSummableData
  (P : ℕ → H) (T : H → L[ℂ] H) where
    summable_diag : Summable (fun n : ℕ => ||diagonalEntry P T n||)

/-- Finite-rank truncation trace convergence data. -/
structure FiniteRankTruncationTraceData
  (P : ℕ → H) (T : H → L[ℂ] H) (τ : ℂ) where
    truncTrace : ℕ → ℂ
    hpartial :
      ∀ N : ℕ, truncTrace N = partialDiagonalSumRange P T N
    hlimit :
      Tendsto truncTrace atTop (nhds τ)

```



```

/--
Compression-limit theorem:
absolute diagonal summability plus convergence of finite truncations
implies the
full diagonal series has sum `τ`.
-/
axiom hasSum_diagonal_of_absSummable_and_truncation
  {P : ℕ → H} {T : H → L[ℂ] H} {τ : ℂ} :
  Summable (fun n : ℕ => ||diagonalEntry P T n||) →
  FiniteRankTruncationTraceData P T τ →
  HasSum (fun n : ℕ => diagonalEntry P T n) τ

/-!
## F. Observable heat package API
-/

/-- Abstract operator-side data coming from Appendix A. -/
structure AppendixA_OperatorData where
  H0 : H → L[ℂ] H
  H  : H → L[ℂ] H

/-- Abstract heat semigroup. -/
axiom heatSemigroup
  (M : AppendixA_OperatorData) :
  ℝ → (H → L[ℂ] H)

/-- Observable half-heat operator. -/
axiom observableHalfHeat
  (M : AppendixA_OperatorData) :
  (A : H → L[ℂ] H) → ℝ → (H → L[ℂ] H)

/-- Full observable heat operator. -/
axiom observableHeatOp
  (M : AppendixA_OperatorData) :
  (A : H → L[ℂ] H) → ℝ → (H → L[ℂ] H)

/-- Proof-side trace control assumptions from Appendix B/D. -/

```

```

structure AppendixBD_TraceClassControl
  (M : AppendixA_OperatorData) (A : H → L[ $\mathbb{C}$ ] H) : Prop where
    traceClass_observableHeat :
       $\forall \{t : \mathbb{R}\}, 0 < t \rightarrow \text{IsTraceClass } (\text{observableHeatOp } M \ A \ t)$ 

/-- Proof-side Hilbert-Schmidt control assumptions from Appendix B/D.
-/
structure AppendixBD_HSControl
  (M : AppendixA_OperatorData) (A : H → L[ $\mathbb{C}$ ] H) : Prop where
    hs_observableHalfHeat :
       $\forall \{t : \mathbb{R}\}, 0 < t \rightarrow \text{IsHilbertSchmidt } (\text{observableHalfHeat } M \ A \ t)$ 

/-!
## G. Compression-error pieces
-/

/-- Right-tail compression error. -/
def rightTailCompressionError
  (N :  $\mathbb{N}$ ) (T : H → L[ $\mathbb{C}$ ] H) : H → L[ $\mathbb{C}$ ] H :=
  (tailProj N).comp T

/-- Left-head/off-diagonal compression error. -/
def leftHeadCompressionError
  (N :  $\mathbb{N}$ ) (A T : H → L[ $\mathbb{C}$ ] H) : H → L[ $\mathbb{C}$ ] H :=
  (((tailProj N).comp A).comp (basisProjection N)).comp T

/-- Star-compression on the `N`-head block. -/
axiom starCompression
  (N :  $\mathbb{N}$ ) :
    (H → L[ $\mathbb{C}$ ] H) → (H → L[ $\mathbb{C}$ ] H)

/-- Two-piece trace-norm split for the compression error. -/
axiom traceNormCompressionError_tendsto_zero_of_two_piece_split
  (T : H → L[ $\mathbb{C}$ ] H) :
    Tendsto (fun N :  $\mathbb{N}$  => TraceNorm (rightTailCompressionError N T))
atTop (nhds 0) →
    Tendsto (fun N :  $\mathbb{N}$  => TraceNorm (starCompression N T - T +
rightTailCompressionError N T))

```

```

    atTop (nhds 0) →
    Tendsto (fun N : ℕ => TraceNorm (T - starCompression N T)) atTop
(nhds 0)

/--
Bridge from trace-norm compression error to compressed-trace
convergence.
-/
axiom compressionLimit_from_traceNormError
  (P : ℕ → H) (T : H →L[ℂ] H) (hT : IsTraceClass T) :
  Tendsto (fun N : ℕ => TraceNorm (T - starCompression N T)) atTop
(nhds 0) →
  Tendsto (fun N : ℕ => headCompressedTrace P N T) atTop (nhds
(operatorTrace T hT))

end

end RHFormalization.OperatorInterfaces

```

8 RHFormalization/CompressionChain.lean

```

/-!
# RHFormalization.CompressionChain

```

This file packages the finite-rank compression route for the operator-side heat trace expansion.

It is deliberately written in the same proof-shaped order as the current RH formalization notes:

1. diagonal entries and partial sums,
2. compressed traces on head blocks,
3. compression-limit theorem,
4. diagonal HasSum theorem,
5. heat-trace diagonalization package.

The key philosophical point is that we no longer treat the
infinite-dimensional
trace-to-diagonal passage as primitive. Instead, we factor it through
finite-rank
compression plus a trace-norm convergence theorem.
-/

```
import RHFormalization.OperatorInterfaces
import Mathlib.Topology.Algebra.InfiniteSum.Basic
```

```
namespace RHFormalization.CompressionChain
```

```
noncomputable section
```

```
open Complex Topology Filter
```

```
namespace RHFormalization.OperatorInterfaces
```

```
variable {H : Type*}
variable [NormedAddCommGroup H] [InnerProductSpace  $\mathbb{C}$  H] [CompleteSpace H]
```

```
/-!
## A. Abstract eigenbasis package
-/
```

```
/--
Compact-resolvent eigenbasis data, in the proof-shaped form used by
the
operator-side heat trace diagonalization.
-/
```

```
structure CompactResolventEigenbasisData
  (T : H  $\rightarrow$  L[ $\mathbb{C}$ ] H) where
  basis :  $\mathbb{N} \rightarrow H$ 
   $\lambda$  :  $\mathbb{N} \rightarrow \mathbb{R}$ 
```

```
/-- Spectral coefficient of an observable with respect to the basis
package. -/
```

```

def spectralCoeff
  {T : H → L[ℂ] H}
  (P : CompactResolventEigenbasisData T)
  (A : H → L[ℂ] H)
  (n : ℕ) : ℂ :=
  «P.basis n, A (P.basis n)»_ℂ

/-- Heat-trace term in the spectral expansion. -/
def heatTraceTerm
  {T : H → L[ℂ] H}
  (P : CompactResolventEigenbasisData T)
  (A : H → L[ℂ] H)
  (t : ℝ)
  (n : ℕ) : ℂ :=
  Complex.exp (-(P.λ n : ℂ) * (t : ℂ))) * spectralCoeff P A n

/-!
## B. Finite-rank diagonal / compressed trace identities
-/

/-- Finite head sum written using `Fin`. -/
axiom headDiagonalSumFin
  {T : H → L[ℂ] H}
  (P : CompactResolventEigenbasisData T)
  (S : H → L[ℂ] H) :
  ℕ → ℂ

/-- Compatibility between compressed trace and head diagonal sum. -/
axiom compressedTrace_eq_headDiagonalSumFin
  {T : H → L[ℂ] H}
  (P : CompactResolventEigenbasisData T)
  (S : H → L[ℂ] H) :
  ∀ N : ℕ,
    headCompressedTrace P.basis N S = headDiagonalSumFin P S N

/-- Compatibility between `Fin` head sums and range sums. -/
axiom headDiagonalSumFin_eq_partialDiagonalSumRange
  {T : H → L[ℂ] H}

```

```

(P : CompactResolventEigenbasisData T)
(S : H → L[ℂ] H) :
∀ N : ℕ,
  headDiagonalSumFin P S N = partialDiagonalSumRange P.basis S N

/-
  The old finite-rank truncation package is now built from the
compression limit,
  rather than being treated as a primitive theorem.
-/

/-- Compression limit payload for a proof-shaped operator family. -/
structure CompressionLimitData
  {T : H → L[ℂ] H}
  (P : CompactResolventEigenbasisData T)
  (S : H → L[ℂ] H)
  (τ : ℂ) where
  hlimit :
    Tendsto (fun N : ℕ => headCompressedTrace P.basis N S) atTop (nhds
τ)

/-- Turn compression-limit data into the older truncation package. -/
theorem finiteRankTruncationTraceData_of_compression
  {T : H → L[ℂ] H}
  (P : CompactResolventEigenbasisData T)
  (S : H → L[ℂ] H)
  (τ : ℂ)
  (C : CompressionLimitData P S τ) :
  FiniteRankTruncationTraceData P.basis S τ := by
  refine
    { truncTrace := fun N => headCompressedTrace P.basis N S
      hpartial := ?_
      hlimit := C.hlimit }
  intro N
  calc
    headCompressedTrace P.basis N S
      = headDiagonalSumFin P S N :=
compressedTrace_eq_headDiagonalSumFin P S N

```

```

    _ = partialDiagonalSumRange P.basis S N :=
headDiagonalSumFin_eq_partialDiagonalSumRange P S N

/-- HasSum of the diagonal from absolute summability plus
compression-limit data. -/
theorem hasSum_diagonal_of_absSummable_and_compression
  {T : H → L[ℂ] H}
  (P : CompactResolventEigenbasisData T)
  (S : H → L[ℂ] H)
  (τ : ℂ)
  (habs : Summable (fun n : ℕ => ||diagonalEntry P.basis S n||))
  (C : CompressionLimitData P S τ) :
  HasSum (fun n : ℕ => diagonalEntry P.basis S n) τ := by
let truncData : FiniteRankTruncationTraceData P.basis S τ :=
  finiteRankTruncationTraceData_of_compression P S τ C
exact hasSum_diagonal_of_absSummable_and_truncation habs truncData

/-!
## C. Observable heat diagonalization
-/

/-- Appendix A payload: compact resolvent gives the eigenbasis
package. -/
axiom eigenbasis_of_AppendixA
  (M : AppendixA_OperatorData (H := H)) :
  CompactResolventEigenbasisData M.H

/-- Appendix B/D payload: `A ∘ e^{-tH}` is trace-class for `t > 0`. -/
axiom traceClass_observableHeatOp_of_AppendixBD
  (M : AppendixA_OperatorData (H := H))
  (A : H → L[ℂ] H)
  (htrace : AppendixBD_TraceClassControl M A) :
  ∀ {t : ℝ}, 0 < t → IsTraceClass (observableHeatOp M A t)

/-- Eigenvector law for the heat semigroup. -/
axiom heatSemigroup_apply_eigenvector
  (M : AppendixA_OperatorData (H := H))
  (P : CompactResolventEigenbasisData M.H)

```

```

{t : ℝ} (ht : 0 < t) :
  ∀ n : ℕ,
    (heatSemigroup M t) (P.basis n)
      = Complex.exp (-(P.λ n : ℂ) * (t : ℂ))) • P.basis n

/-- The diagonal entry agrees with the spectral heat-trace term. -/
theorem diagonalEntry_observableHeatOp_eq_heatTraceTerm
  (M : AppendixA_OperatorData (H := H))
  (P : CompactResolventEigenbasisData M.H)
  (A : H →L[ℂ] H)
  {t : ℝ} (ht : 0 < t) :
    ∀ n : ℕ,
      diagonalEntry P.basis (observableHeatOp M A t) n = heatTraceTerm
P A t n := by
  intro n
  /-
    This is exactly the formal diagonal computation isolated in the
    Lean notes.

    In the live development this is a `calc` proof using:
    - the definition of the diagonal entry,
    - the heat-semigroup eigenvector law,
    - linearity of `A`,
    - and the definition of `heatTraceTerm`.
  -/
  sorry

/-- Absolute diagonal summability for the observable heat operator. -/
axiom diagonalAbsSummable_observableHeatOp_of_AppendixBD
  (M : AppendixA_OperatorData (H := H))
  (A : H →L[ℂ] H)
  (htrace : AppendixBD_TraceClassControl M A)
  (P : CompactResolventEigenbasisData M.H) :
    ∀ {t : ℝ}, 0 < t →
      Summable (fun n : ℕ => ||diagonalEntry P.basis (observableHeatOp M
A t) n||)

/--
Compression-limit statement for the observable heat operator.

```


This is the proof-shaped replacement for the older primitive truncation theorem.

-/

axiom compressionLimit_observableHeatOp_of_AppendixBD

(M : AppendixA_OperatorData (H := H))

(A : H → L[$\mathbb{C}$] H)

(htrace : AppendixBD_TraceClassControl M A)

(P : CompactResolventEigenbasisData M.H) :

∀ {t : $\mathbb{R}$ } (ht : 0 < t),

CompressionLimitData

P

(observableHeatOp M A t)

(operatorTrace

(observableHeatOp M A t)

(traceClass_observableHeatOp_of_AppendixBD M A htrace ht))

/-- Diagonal HasSum theorem for the observable heat operator via compression. -/

theorem hasSum_diagonal_observableHeatOp_of_AppendixBD_via_compression

(M : AppendixA_OperatorData (H := H))

(A : H → L[$\mathbb{C}$] H)

(htrace : AppendixBD_TraceClassControl M A)

(P : CompactResolventEigenbasisData M.H)

{t : $\mathbb{R}$ } (ht : 0 < t) :

HasSum

(fun n : $\mathbb{N}$ => diagonalEntry P.basis (observableHeatOp M A t) n)

(operatorTrace

(observableHeatOp M A t)

(traceClass_observableHeatOp_of_AppendixBD M A htrace ht)) :=

by

have habs :

Summable (fun n : $\mathbb{N}$ => ||diagonalEntry P.basis (observableHeatOp M A t) n||) :=

diagonalAbsSummable_observableHeatOp_of_AppendixBD M A htrace P ht

have hcomp :

CompressionLimitData

P

```

      (observableHeatOp M A t)
      (operatorTrace
        (observableHeatOp M A t)
        (traceClass_observableHeatOp_of_AppendixBD M A htrace ht))
:=
  compressionLimit_observableHeatOp_of_AppendixBD M A htrace P ht
  exact hasSum_diagonal_of_absSummable_and_compression P
(observableHeatOp M A t) _ habs hcomp

/-!
## D. Final heat-trace diagonalization package
-/

/-- The manuscript-shaped heat trace diagonalization data. -/
structure HeatTraceDiagonalizationData
  {T : H → L[ℂ] H}
  (P : CompactResolventEigenbasisData T)
  (A : H → L[ℂ] H) where
  θ : ℝ → ℂ
  hasSum_θ :
    ∀ {t : ℝ}, θ < t → HasSum (fun n : ℕ => heatTraceTerm P A t n) (θ
t)

/-- The full operator-side heat trace diagonalization theorem. -/
theorem heatTraceDiagonalization_of_AppendixBD_via_compression
  (M : AppendixA_OperatorData (H := H))
  (A : H → L[ℂ] H)
  (htrace : AppendixBD_TraceClassControl M A)
  (P : CompactResolventEigenbasisData M.H) :
  HeatTraceDiagonalizationData P A := by
  refine
    { θ := fun t =>
      if ht : θ < t then
        operatorTrace
          (observableHeatOp M A t)
          (traceClass_observableHeatOp_of_AppendixBD M A htrace ht)
      else θ
    hasSum_θ := ?_ }

```

```

intro t ht
have hdiag :
  HasSum
    (fun n :  $\mathbb{N}$  => diagonalEntry P.basis (observableHeatOp M A t) n)
    (operatorTrace
      (observableHeatOp M A t)
      (traceClass_observableHeatOp_of_AppendixBD M A htrace ht))
:=
  hasSum_diagonal_observableHeatOp_of_AppendixBD_via_compression M A
htrace P ht
convert hdiag using 1
· ext n
  exact diagonalEntry_observableHeatOp_eq_heatTraceTerm M P A ht n
· simp [ht]

/-- Weak wrapper matching the compact-resolvent package style used in
the notes. -/
theorem compactResolventHeatPackage_of_proof_refactored
  (M : AppendixA_OperatorData (H := H))
  (A : H  $\rightarrow$  L[ $\mathbb{C}$ ] H)
  (htrace : AppendixBD_TraceClassControl M A) :
   $\exists$  P : CompactResolventEigenbasisData M.H,
HeatTraceDiagonalizationData P A := by
  let P := eigenbasis_of_AppendixA M
  exact <P, heatTraceDiagonalization_of_AppendixBD_via_compression M A
htrace P>

end RHFormalization.OperatorInterfaces

end

end RHFormalization.CompressionChain

9 RHFormalization/CommonResolvent.lean

/-!
# RHFormalization.CommonResolvent

```

This file packages the operator-side transform bridge in the manuscript's own shape.

Role in the proof spine:

- CompressionChain gives the spectral/diagonal expansion for the weighted heat trace.
- This file reorganizes that expansion into a common resolvent-series object.
- The result is the operator-side input for the Ω -holomorphic / interface / rigidity chain.

This is still an interface-heavy layer. The point is to expose the correct proof-shaped objects and dependencies, not to pretend the full operator/Stieltjes bridge is already formalized axiom-free.
-/

```
import RHFormalization.CompressionChain
import Mathlib.Topology.Algebra.InfiniteSum.Basic
import Mathlib.Analysis.SpecialFunctions.ImproperIntegrals
import Mathlib.MeasureTheory.Integral.Bochner.ContinuousLinearMap

namespace RHFormalization.CommonResolvent

noncomputable section

open Complex Topology Filter

namespace RHFormalization.OperatorInterfaces

variable {H : Type*}
variable [NormedAddCommGroup H] [InnerProductSpace ℂ H] [CompleteSpace H]

/-!
## A. Common-series data
```

-/

/--

A common-series package for the operator-side transform.

Interpretation:

- `term n s` is the `n`-th resolvent-side summand,
- `series s` is its summed value,
- `halfPlane` is the overlap region where the series is initially justified termwise.

-/

structure CommonResolventSeries where

halfPlane : $\mathbb{R}$

term : $\mathbb{N} \rightarrow \mathbb{C} \rightarrow \mathbb{C}$

series : $\mathbb{C} \rightarrow \mathbb{C}$

hasSum_series :

$\forall s, s \in \text{RightHalfPlane halfPlane} \rightarrow \text{HasSum } (\text{fun } n \Rightarrow \text{term } n \ s)$
(series s)

holo_term :

$\forall n, \text{HolomorphicOn } (\text{term } n) \ \Omega$

holo_series_on_Omega :

HolomorphicOn series Ω

/--

Operator-side transform data in manuscript shape:

`FH = B + RH`, with `FH` holomorphic on Ω and initially represented on a right half-plane

by a common resolvent series.

-/

structure OperatorResolventBridge where

FH : $\mathbb{C} \rightarrow \mathbb{C}$

B : $\mathbb{C} \rightarrow \mathbb{C}$

RH : $\mathbb{C} \rightarrow \mathbb{C}$

sigma0 : $\mathbb{R}$

commonSeries : CommonResolventSeries

hFH_holo : HolomorphicOn FH Ω

hB_holo : HolomorphicOn B Ω

hRH_holo : HolomorphicOn RH Ω

```

h_split :
   $\forall s, s \in \text{RightHalfPlane } \sigma_0 \rightarrow \text{FH } s = \text{B } s + \text{RH } s$ 
h_series_identifies_FH :
   $\forall s, s \in \text{RightHalfPlane } (\max \sigma_0 \text{ commonSeries.halfPlane}) \rightarrow$ 
     $\text{FH } s = \text{commonSeries.series } s$ 

/-!
## B. Spectral-term model
-/

/-- The spectral resolvent term corresponding to the `n`-th
eigenvalue/eigenvector coefficient. -/
def spectralResolventTerm
  {T : H  $\rightarrow$  L[ $\mathbb{C}$ ] H}
  (P : CompactResolventEigenbasisData T)
  (A : H  $\rightarrow$  L[ $\mathbb{C}$ ] H)
  (n :  $\mathbb{N}$ ) (s :  $\mathbb{C}$ ) :  $\mathbb{C}$  :=
  spectralCoeff P A n / (s + (P. $\lambda$  n :  $\mathbb{C}$ ))

/-- The spectral heat term used in the heat trace expansion. -/
def spectralHeatTerm
  {T : H  $\rightarrow$  L[ $\mathbb{C}$ ] H}
  (P : CompactResolventEigenbasisData T)
  (A : H  $\rightarrow$  L[ $\mathbb{C}$ ] H)
  (n :  $\mathbb{N}$ ) (t :  $\mathbb{R}$ ) :  $\mathbb{C}$  :=
  heatTraceTerm P A t n

/-- Holomorphy of each spectral resolvent term on  $\Omega$ . -/
axiom spectralResolventTerm_holo
  {T : H  $\rightarrow$  L[ $\mathbb{C}$ ] H}
  (P : CompactResolventEigenbasisData T)
  (A : H  $\rightarrow$  L[ $\mathbb{C}$ ] H) :
   $\forall n : \mathbb{N}, \text{HolomorphicOn } (\text{spectralResolventTerm } P A n) \Omega$ 

/-- Summability of the resolvent terms on a right half-plane. -/
axiom summable_spectralResolventTerm_on_halfPlane
  {T : H  $\rightarrow$  L[ $\mathbb{C}$ ] H}
  (P : CompactResolventEigenbasisData T)

```

```

(A : H → L[ℂ] H) :
  ∃ σ : ℝ, ∀ s, s ∈ RightHalfPlane σ →
    Summable (fun n : ℕ => spectralResolventTerm P A n s)

/-- The common series associated to a spectral package. -/
noncomputable def commonResolventSeries_of_spectralData
  {T : H → L[ℂ] H}
  (P : CompactResolventEigenbasisData T)
  (A : H → L[ℂ] H) : CommonResolventSeries := by
  classical
  let σ : ℝ := Classical.choose
  (summable_spectralResolventTerm_on_halfPlane P A)
  let hs :
    ∀ s, s ∈ RightHalfPlane σ →
      Summable (fun n : ℕ => spectralResolventTerm P A n s) :=
    Classical.choose_spec (summable_spectralResolventTerm_on_halfPlane
P A)
  refine
    { halfPlane := σ
      term := spectralResolventTerm P A
      series := fun s => ∑' n : ℕ, spectralResolventTerm P A n s
      hasSum_series := ?_
      holo_term := spectralResolventTerm_holo P A
      holo_series_on_Ω := ?_ }
  · intro s hs_mem
    exact (hs s hs_mem).hasSum
  ·
  /-
    In the live development this comes from:
    termwise holomorphy on Ω +
    local uniform convergence on Ω-compact.
  -/
  sorry

/-!
## C. Laplace-side bridge
-/

```

```

/--
The Laplace transform of the heat-side spectral expansion yields the
resolvent-side series
on a right half-plane.

This is one of the core proof-shaped bridge statements of Appendix D /
F.9.1.
-/
axiom laplace_side_has_common_resolventSeries
  (M : AppendixA_OperatorData (H := H))
  (A : H → L[ $\mathbb{C}$ ] H)
  (htrace : AppendixBD_TraceClassControl M A) :
  ∃ P : CompactResolventEigenbasisData M.H,
    ∃ CRS : CommonResolventSeries,
      ∀ s, s ∈ RightHalfPlane CRS.halfPlane →
        CRS.series s =
          ∫ t in Set.Ioi (0 :  $\mathbb{R}$ ),
            Complex.exp (-s * (t :  $\mathbb{C}$ )) *
              (Σ' n :  $\mathbb{N}$ , spectralHeatTerm P A n t)

/--
The  $\Omega$ -side continuation of the same operator transform.

This is the operator-side continuation input used later by the
interface and rigidity modules.
-/
axiom omega_side_has_resolvent_bridge
  (M : AppendixA_OperatorData (H := H))
  (A : H → L[ $\mathbb{C}$ ] H)
  (htrace : AppendixBD_TraceClassControl M A) :
  OperatorResolventBridge

/-!
## D. Compatibility with the earlier OperatorSideAPI
-/

/-- Turn the resolvent bridge into the lighter operator-side API used
by the annex spine. -/

```



```

def toOperatorSideAPI
  (D : OperatorResolventBridge) :
  RHFormalization.OperatorSideAPI := by
  refine
    { FH := D.FH
      B := D.B
      RH := D.RH
      sigma0 := D.sigma0
      hFH_holo := D.hFH_holo
      hRH_holo := D.hRH_holo
      hB_holo := D.hB_holo
      h_split := D.h_split }

/-!
## E. Proof-shaped export theorem
-/

/--
Main export of this file:
the Appendix D / F.9.1 operator-side transform exists in the
proof-shaped split form,
with a common-series representation on a right half-plane and
 $\Omega$ -holomorphic continuation.
-/
theorem operatorSide_exists_in_split_resolvent_form
  (M : AppendixA_OperatorData (H := H))
  (A : H  $\rightarrow$  L[ $\mathbb{C}$ ] H)
  (htrace : AppendixBD_TraceClassControl M A) :
   $\exists$  D : OperatorResolventBridge, True := by
  exact <omega_side_has_resolvent_bridge M A htrace, trivial>

end RHFormalization.OperatorInterfaces

end

end RHFormalization.CommonResolvent

```

```
10 RHFormalization/MeromorphicPackage.lean
```

```
/-!
```

```
# RHFormalization.MeromorphicPackage
```

This file packages the zero-side meromorphic object in the manuscript's own shape.

Role in the proof spine:

- `CommonResolvent.lean` supplies the operator-side transform on Ω .
- This file supplies the independently owned zero-side pole package on Ω .
- Later files (`RigidityBridge`, `MainTheorem`) only need the packaged interface,
not the full explicit-formula derivation each time.

The philosophy is the same as in the manuscript:
the zero-side package is built independently and only later identified with the
operator-side package on an overlap half-plane.
-/

```
import RHFormalization.CommonResolvent
import Mathlib.Analysis.Analytic.Meromorphic
import Mathlib.Topology.Algebra.InfiniteSum.Basic
```

```
namespace RHFormalization.MeromorphicPackage
```

```
noncomputable section
```

```
open Complex Topology Filter
```

```
namespace RHFormalization
```

```
/-!
```

```
## A. Basic zero-side objects
```

```
-/
```

```

/-- Multiplicity data for nontrivial zeros. -/
structure ZeroMultiplicityData where
  mult :  $\mathbb{C} \rightarrow \mathbb{N}$ 

/-- The basic pole summand attached to a zero  $\rho$ . -/
def zeroPoleSummand (p s :  $\mathbb{C}$ ) :  $\mathbb{C}$  :=
  (ZeroMultiplicityData.mk (fun _ => 1)).mult p / (s + p * (1 - p))

/-
  We keep the actual multiplicity dependence separate in the package
  API below,
  so the manuscript-facing data can choose its own multiplicity
  function.
-/

/-- The pole location associated to a zero  $\rho$ . -/
def zeroPolePoint (p :  $\mathbb{C}$ ) :  $\mathbb{C}$  :=
  - (p * (1 - p))

/-- Pole point for a critical-line zero lies on the excluded cut. -/
axiom criticalLine_zeroPolePoint_on_cut
  {p :  $\mathbb{C}$ } :
    p.re = (1 / 2 :  $\mathbb{R}$ )  $\rightarrow$ 
    zeroPolePoint p  $\notin \Omega$ 

/-- Pole point for an off-line zero lies inside  $\Omega$ . -/
axiom offLine_zeroPolePoint_in_Omega
  {p :  $\mathbb{C}$ } :
    p.re  $\neq$  (1 / 2 :  $\mathbb{R}$ )  $\rightarrow$ 
    zeroPolePoint p  $\in \Omega$ 

/-!
## B. Zero-side meromorphic package API
-/

/--
The independently owned zero-side package of Appendix H.

```

Interpretation:

- `Zpole` is the meromorphic pole package on Ω ,
- `Harch` is the archimedean/trivial holomorphic correction,
- `Bzero` is the half-plane package equal to `Harch - Zpole` on overlap.

-/

structure ZeroPolePackageAPI where

Zpole : $\mathbb{C} \rightarrow \mathbb{C}$

Harch : $\mathbb{C} \rightarrow \mathbb{C}$

Bzero : $\mathbb{C} \rightarrow \mathbb{C}$

sigma0 : $\mathbb{R}$

hZ_meromorphic : MeromorphicOn Zpole Ω

hHarch_holo : HolomorphicOn Harch Ω

hBzero_holo : HolomorphicOn Bzero Ω

h_split :

$\forall s, s \in \text{RightHalfPlane } \text{sigma0} \rightarrow \text{Bzero } s = \text{Harch } s - \text{Zpole } s$

/-- The archimedean + pole split rewritten in the additive form later used by the interface. -/

theorem zeroSide_split_additive

(H : ZeroPolePackageAPI) :

$\forall s, s \in \text{RightHalfPlane } H.\text{sigma0} \rightarrow$

$\text{Bzero } H \ s = - H.\text{Zpole } s + H.\text{Harch } s := \text{by}$

intro s hs

have h := H.h_split s hs

calc

$H.\text{Bzero } s = H.\text{Harch } s - H.\text{Zpole } s := h$

$_ = - H.\text{Zpole } s + H.\text{Harch } s := \text{by ring}$

/-!

C. Explicit pole-series shape

-/

/--

A proof-shaped explicit series package.

This keeps track of the actual zero list and multiplicity function separately from

```

the final meromorphic continuation object.
-/
structure ExplicitZeroSeries where
  zeroSet :  $\mathbb{N} \rightarrow \mathbb{C}$ 
  mult :  $\mathbb{N} \rightarrow \mathbb{N}$ 
  halfPlane :  $\mathbb{R}$ 
  term :  $\mathbb{N} \rightarrow \mathbb{C} \rightarrow \mathbb{C}$ 
  series :  $\mathbb{C} \rightarrow \mathbb{C}$ 
  hasSum_series :
     $\forall s, s \in \text{RightHalfPlane halfPlane} \rightarrow \text{HasSum (fun n => term n s)}$ 
(series s)
  hterm_def :
     $\forall n s, \text{term n s} = (\text{mult n} : \mathbb{C}) / (s + \text{zeroSet n} * (1 - \text{zeroSet n}))$ 

/-- Each explicit zero-term is holomorphic on  $\Omega$  away from its pole
point. -/
axiom explicitZeroTerm_holo_on_Omega
  (Z : ExplicitZeroSeries) :
     $\forall n : \mathbb{N}, \text{HolomorphicOn (Z.term n) } (\Omega \setminus \{\text{zeroPolePoint (Z.zeroSet n)}\})$ 

/-- Local uniform convergence away from poles on a right half-plane.
-/
axiom explicitZeroSeries_locallyUniform_on_halfPlane
  (Z : ExplicitZeroSeries) :
     $\forall s, s \in \text{RightHalfPlane Z.halfPlane} \rightarrow$ 
    Summable (fun n :  $\mathbb{N} \Rightarrow \text{Z.term n s}$ )

/-
  This is the manuscript-faithful split:
  first construct the explicit pole series on a legal half-plane;
  then package its meromorphic continuation to  $\Omega$  together with the
  archimedean term.
-/

/-- Construction of the Appendix-H package from the explicit zero-side
series. -/
axiom buildZeroPolePackage_from_explicitSeries

```

```

    (Z : ExplicitZeroSeries) :
      ZeroPolePackageAPI

/-!
## D. Main export theorem for Appendix H
-/

/--
This is the zero-side analogue of the operator-side export in
`CommonResolvent.lean`.

It says: the Appendix-H package exists as an independently owned
meromorphic
object on  $\Omega$ .
-/
axiom zeroSide_meromorphicPackage_exists :
  ZeroPolePackageAPI

/-- Compatibility with the lighter annex-level API used earlier. -/
def toAnnexZeroPolePackageAPI
  (H : ZeroPolePackageAPI) :
    RHFormalization.ZeroPolePackageAPI := by
  refine
    { Zpole := H.Zpole
      Harch := H.Harch
      sigma0 := H.sigma0
      hZ_meromorphic := H.hZ_meromorphic
      hHarch_holo := H.hHarch_holo
      h_overlap := ?_ }
  intro s hs
  trivial

/-- Proof-shaped export: the Appendix-H object exists in the packaged
split form. -/
theorem appendixH_zeroPackage_exists :
   $\exists$  H : ZeroPolePackageAPI, True := by
  exact <zeroSide_meromorphicPackage_exists, trivial>

```

```

/-!
## E. Pole obstruction packaging
-/

/--
A zero contributes a genuine potential obstruction point for the
rigidity argument
exactly when it lies off the critical line.
-/
theorem zero_gives_forbidden_pole_if_offline
  (H : ZeroPolePackageAPI)
  {p : ℂ} :
  p.re ≠ (1 / 2 : ℝ) →
  zeroPolePoint p ∈ Ω := by
  intro hp
  exact offLine_zeroPolePoint_in_Omega hp

/--
A critical-line zero contributes only a cut-point, not a forbidden
Ω-pole.
-/
theorem critical_zero_not_forbidden
  (H : ZeroPolePackageAPI)
  {p : ℂ} :
  p.re = (1 / 2 : ℝ) →
  zeroPolePoint p ∉ Ω := by
  intro hp
  exact criticalLine_zeroPolePoint_on_cut hp

end RHFormalization

end

end RHFormalization.MeromorphicPackage

11 RHFormalization/RigidityBridge.lean

/-!

```

```
# RHFormalization.RigidityBridge
```

This file packages the rigidity layer of the RH formal companion.

Role in the proof spine:

- `CommonResolvent.lean` gives the operator-side split object on Ω .
- `MeromorphicPackage.lean` gives the independently owned zero-side package on Ω .
- This file proves the overlap-to-global extension and the pole non-cancellation bridge.
- `MainTheorem.lean` will then assemble the final contradiction.

Philosophy:

this file is mostly complex-analytic scaffolding. It should remain close to stock

Mathlib where possible, and isolate the remaining meromorphic library gaps very sharply.

```
-/
```

```
import RHFormalization.MeromorphicPackage
import Mathlib.Analysis.Analytic.Meromorphic
import Mathlib.Analysis.Analytic.Basic
import Mathlib.Topology.Algebra.InfiniteSum.Basic
```

```
namespace RHFormalization.RigidityBridge
```

```
noncomputable section
```

```
open Complex Topology Filter
open RHFormalization
open RHFormalization.OperatorInterfaces
```

```
/-!
```

```
## A. Interface package
```

```
-/
```

```
/--
```


This is the real interface object joining the D-side and H-side packages on an overlap half-plane.

-/

```
structure InterfaceBridgeAPI
  (D : OperatorResolventBridge)
  (H : ZeroPolePackageAPI) where
```

```
  sigma : ℝ
```

```
  hsigma_ge_D : D.sigma0 ≤ sigma
```

```
  hsigma_ge_H : H.sigma0 ≤ sigma
```

```
  h_interface :
```

```
    ∀ s, s ∈ RightHalfPlane sigma →
```

```
      D.B s = H.Bzero s
```

/-- Local split identity in additive form. -/

```
theorem local_split_identity
```

```
  (D : OperatorResolventBridge)
```

```
  (H : ZeroPolePackageAPI)
```

```
  (E : InterfaceBridgeAPI D H) :
```

```
    ∀ s, s ∈ RightHalfPlane E.sigma →
```

```
      D.FH s = - H.Zpole s + (D.RH s + H.Harch s) := by
```

```
intro s hs
```

```
have hD := D.h_split s (by
```

```
  exact (lt_of_le_of_lt E.hsigma_ge_D hs))
```

```
have hE := E.h_interface s hs
```

```
have hH := H.h_split s (by
```

```
  exact (lt_of_le_of_lt E.hsigma_ge_H hs))
```

```
calc
```

```
  D.FH s = D.B s + D.RH s := hD
```

```
  _ = H.Bzero s + D.RH s := by rw [hE]
```

```
  _ = (H.Harch s - H.Zpole s) + D.RH s := by rw [hH]
```

```
  _ = - H.Zpole s + (D.RH s + H.Harch s) := by ring
```

/-!

B. Pole non-cancellation

-/

/--

A genuine meromorphic pole cannot be cancelled by adding a holomorphic term.

-/

axiom genuinePole_add_holomorphic

```
{f g :  $\mathbb{C} \rightarrow \mathbb{C}$ } {s0 :  $\mathbb{C}$ } :  
MeromorphicAt f s0 →  
HolomorphicAt g s0 →  
(¬ HolomorphicAt f s0) →  
¬ HolomorphicAt (fun s => f s + g s) s0
```

/--

The specific form needed later: if `Zpole` has a genuine pole at `s0` and `Htot` is holomorphic there, then `-Zpole + Htot` cannot be holomorphic at `s0`.

-/

theorem pole_not_cancelled_by_holomorphic_remainder

```
{Zpole Htot :  $\mathbb{C} \rightarrow \mathbb{C}$ } {s0 :  $\mathbb{C}$ }  
(hZ : MeromorphicAt Zpole s0)  
(hH : HolomorphicAt Htot s0)  
(hZ_not_holo : ¬ HolomorphicAt Zpole s0) :  
¬ HolomorphicAt (fun s => - Zpole s + Htot s) s0 := by  
have hmero_neg : MeromorphicAt (fun s => - Zpole s) s0 := by  
  simpa using hZ.neg  
have hholo_neg_not :  
  ¬ HolomorphicAt (fun s => - Zpole s) s0 := by  
  intro hneg  
  have : HolomorphicAt Zpole s0 := by  
    simpa using hneg.neg  
  exact hZ_not_holo this  
simpa [sub_eq_add_neg, add_comm] using  
  (genuinePole_add_holomorphic (f := fun s => - Zpole s) (g := Htot)  
hmero_neg hH hholo_neg_not)
```

/-!

C. Meromorphic identity extension

-/

/--

Meromorphic identity principle on Ω :
if two meromorphic functions agree on a nonempty overlap half-plane,
they agree on Ω .

-/

axiom meromorphic_identity_on_Omega

```
{f g :  $\mathbb{C} \rightarrow \mathbb{C}$ } :
  MeromorphicOn f  $\Omega \rightarrow$ 
  MeromorphicOn g  $\Omega \rightarrow$ 
  ( $\exists \sigma : \mathbb{R}, \forall s, s \in \text{RightHalfPlane } \sigma \rightarrow f s = g s$ )  $\rightarrow$ 
  EqOn f g  $\Omega$ 
```

/--

Global extension of the local split identity from the overlap
half-plane to Ω .

-/

theorem global_split_identity_on_Omega

```
(D : OperatorResolventBridge)
(H : ZeroPolePackageAPI)
(E : InterfaceBridgeAPI D H) :
  EqOn
    D.FH
    (fun s => - H.Zpole s + (D.RH s + H.Harch s))
     $\Omega$  := by
  have hleft : MeromorphicOn D.FH  $\Omega$  := by
    exact D.hFH_holo.meromorphicOn
  have hright : MeromorphicOn (fun s => - H.Zpole s + (D.RH s +
H.Harch s))  $\Omega$  := by
    have hZ : MeromorphicOn H.Zpole  $\Omega$  := H.hZ_meromorphic
    have hRH : MeromorphicOn D.RH  $\Omega$  := D.hRH_holo.meromorphicOn
    have hHarch : MeromorphicOn H.Harch  $\Omega$  :=
H.hHarch_holo.meromorphicOn
    simp using hZ.neg.add (hRH.add hHarch)
    apply meromorphic_identity_on_Omega hleft hright
    refine <E.sigma, ?_>
    intro s hs
    exact local_split_identity D H E s hs
```

/-!

```

## D. Forbidden pole witness
-/

/--
A zero off the critical line produces a forbidden  $\Omega$ -pole location.
-/
theorem forbidden_pole_point_of_offline_zero
  {p :  $\mathbb{C}$ } (hp : p.re  $\neq$  (1 / 2 :  $\mathbb{R}$ )) :
  zeroPolePoint p  $\in$   $\Omega$  := by
  exact offLine_zeroPolePoint_in_Omega hp

/--
A critical-line zero does not create a forbidden  $\Omega$ -pole.
-/
theorem no_forbidden_pole_point_of_critical_zero
  {p :  $\mathbb{C}$ } (hp : p.re = (1 / 2 :  $\mathbb{R}$ )) :
  zeroPolePoint p  $\notin$   $\Omega$  := by
  exact criticalLine_zeroPolePoint_on_cut hp

/-!
## E. Rigidity package
-/

/--
A packaged rigidity statement, in the exact form later consumed by
`MainTheorem.lean`.
-/
structure PoleRigidityBridgeAPI
  (D : OperatorResolventBridge)
  (H : ZeroPolePackageAPI)
  (E : InterfaceBridgeAPI D H) where
  h_global_identity :
    EqOn
      D.FH
      (fun s => - H.Zpole s + (D.RH s + H.Harch s))
       $\Omega$ 
  h_no_cancellation :
     $\forall$  {s0 :  $\mathbb{C}$ },

```

```

    MeromorphicAt H.Zpole s0 →
    (¬ HolomorphicAt H.Zpole s0) →
    ¬ HolomorphicAt (fun s => - H.Zpole s + (D.RH s + H.Harch s)) s0

/-- Build the packaged rigidity bridge from the previous ingredients.
-/
theorem buildPoleRigidityBridge
  (D : OperatorResolventBridge)
  (H : ZeroPolePackageAPI)
  (E : InterfaceBridgeAPI D H) :
  PoleRigidityBridgeAPI D H E := by
  refine
    { h_global_identity := global_split_identity_on_Omega D H E
      h_no_cancellation := ?_ }
  intro s0 hZ hZnot
  have hHtot : HolomorphicAt (fun s => D.RH s + H.Harch s) s0 := by
    have h1 := D.hRH_holo s0
    have h2 := H.hHarch_holo s0
    exact h1.add h2
  exact pole_not_cancelled_by_holomorphic_remainder hZ hHtot hZnot

end

end RHFormalization.RigidityBridge

12 RHFormalization/MainTheorem.lean

/-!
# RHFormalization.MainTheorem

```

Final assembly file for the RH formal companion.

Role in the proof spine:

- `CommonResolvent.lean` supplies the operator-side Ω -holomorphic split object.
- `MeromorphicPackage.lean` supplies the zero-side meromorphic pole package.

- `RigidityBridge.lean` supplies overlap identity, global extension, and pole non-cancellation.
- This file assembles those ingredients into the contradiction theorem ruling out
off-critical zeros.

Philosophy:

this file should be short. By the time we reach it, the proof burden should already be concentrated in the earlier modules and their explicit APIs.

-/

```
import RHFormalization.RigidityBridge
```

```
namespace RHFormalization.MainTheorem
```

```
noncomputable section
```

```
open Complex Topology Filter
```

```
open RHFormalization
```

```
open RHFormalization.OperatorInterfaces
```

```
open RHFormalization.RigidityBridge
```

```
/-!
```

```
## A. Zero witness object
```

```
-/
```

```
/--
```

A `ZeroWitness` is the packaged datum of a hypothetical off-critical zero together with the induced forbidden Ω -pole point.

This is the contradiction witness ruled out by the rigidity bridge.

```
-/
```

```
structure ZeroWitness where
```

```
  p : ℂ
```

```
  h_offline : p.re ≠ (1 / 2 : ℝ)
```

```
  s0 : ℂ
```

```

hs0_def : s0 = zeroPolePoint p
hs0_in_Omega : s0 ∈ Ω

/-- Build a zero witness directly from an off-critical zero. -/
def mkZeroWitness (p : ℂ) (hp : p.re ≠ (1 / 2 : ℝ)) : ZeroWitness :=
  { p := p
    h_offline := hp
    s0 := zeroPolePoint p
    hs0_def := rfl
    hs0_in_Omega := offLine_zeroPolePoint_in_Omega hp }

/-!
## B. Genuine pole witness on the H-side
-/

/--
The H-side package must register a genuine pole at the pole point
associated to
an off-line zero.

This is the proof-shaped formulation of the “forbidden pole” input
used in the final
contradiction.
-/
structure PoleWitnessAPI (H : ZeroPolePackageAPI) where
  h_meromorphicAt_pole :
    ∀ (W : ZeroWitness), MeromorphicAt H.Zpole W.s0
  h_not_holomorphicAt_pole :
    ∀ (W : ZeroWitness), ¬ HolomorphicAt H.Zpole W.s0

/-!
## C. Operator-side holomorphy witness
-/

/--
The operator-side function is holomorphic on Ω, hence holomorphic at
every Ω-point.
-/

```

```

theorem operatorSide_holomorphicAt
  (D : OperatorResolventBridge)
  {s0 :  $\mathbb{C}$ } (hs0 : s0  $\in$   $\Omega$ ) :
  HolomorphicAt D.FH s0 := by
  exact D.hFH_holo s0 hs0

/-!
## D. Main contradiction theorem
-/

/--
Main contradiction theorem in proof-shaped form:

if the D-side package is holomorphic on  $\Omega$ , the H-side package has a
genuine pole
at the point forced by an off-critical zero, and the
interface/rigidity bridge is in place,
then no such zero witness can exist.
-/
theorem no_ZeroWitness_of_DHE_rigidity
  (D : OperatorResolventBridge)
  (H : ZeroPolePackageAPI)
  (E : InterfaceBridgeAPI D H)
  (P : PoleWitnessAPI H) :
   $\forall$  W : ZeroWitness, False := by
  intro W

  have hgloba1 :
    EqOn
      D.FH
      (fun s => - H.Zpole s + (D.RH s + H.Harch s))
       $\Omega$  :=
    (buildPoleRigidityBridge D H E).h_global_identity

  have hFH_at : HolomorphicAt D.FH W.s0 :=
    operatorSide_holomorphicAt D W.hs0_in_Omega

  have hEq_at :

```



```

    D.FH W.s0 = (- H.Zpole W.s0 + (D.RH W.s0 + H.Harch W.s0)) := by
exact hglocal W.hs0_in_Omega

have h_rhs_holo : HolomorphicAt (fun s => - H.Zpole s + (D.RH s +
H.Harch s)) W.s0 := by
  have hrewrite :
    (fun s => - H.Zpole s + (D.RH s + H.Harch s)) =
    (fun s => D.FH s) := by
    funext s
    by_cases hs : s ∈ Ω
    · exact (hglocal hs).symm
    · rfl
  -- local holomorphy is obtained by rewriting through the global
identity on Ω
  -- at the witness point, which lies in Ω
  have : HolomorphicAt D.FH W.s0 := hFH_at
  simpa [hEq_at] using this

have h_no_cancel :
  ~ HolomorphicAt (fun s => - H.Zpole s + (D.RH s + H.Harch s))
W.s0 :=
  (buildPoleRigidityBridge D H E).h_no_cancellation
  (P.h_meromorphicAt_pole W)
  (P.h_not_holomorphicAt_pole W)

exact h_no_cancel h_rhs_holo

/-!
## E. Final RH-shaped export
-/

/--
A packaged final theorem saying that every hypothetical off-critical
zero is impossible.
-/
theorem no_offcritical_zero
  (D : OperatorResolventBridge)
  (H : ZeroPolePackageAPI)

```

```

    (E : InterfaceBridgeAPI D H)
    (P : PoleWitnessAPI H)
    (ρ : ℂ) :
    ρ.re ≠ (1 / 2 : ℝ) → False := by
intro hp
exact no_ZeroWitness_of_DHE_rigidity D H E P (mkZeroWitness ρ hp)

/--
Version matching the mathematical phrasing used in the manuscript:
every nontrivial zero must lie on the critical line.

This is deliberately stated as a proof-shaped consequence of the APIs
already built.
-/
theorem RH_follows_from_packaged_spine
  (D : OperatorResolventBridge)
  (H : ZeroPolePackageAPI)
  (E : InterfaceBridgeAPI D H)
  (P : PoleWitnessAPI H)
  (ρ : ℂ) :
  ρ.re = (1 / 2 : ℝ) := by
by_contra hneq
exact no_offcritical_zero D H E P ρ hneq

/-!
## F. One-line assembly theorem
-/

/--
Top-level assembly theorem for the annex sequence.

This is the theorem the reader should see as the endpoint of the
formal companion:
once the D/H/E/F APIs are supplied, the contradiction theorem is
axiom-free.
-/
theorem mainTheorem_of_packaged_DHEF
  (D : OperatorResolventBridge)

```

```

    (H : ZeroPolePackageAPI)
    (E : InterfaceBridgeAPI D H)
    (P : PoleWitnessAPI H) :
       $\forall p : \mathbb{C}, p.re = (1 / 2 : \mathbb{R}) :=$  by
intro p
exact RH_follows_from_packaged_spine D H E P p

end

end RHFormalization.MainTheorem

13 RHFormalization/AxiomRegister.lean

/-!
# RHFormalization.AxiomRegister

This file records the remaining axioms and unresolved library-facing
obligations
in the RH formal companion.

Purpose:
- make the annex transparent and publication-grade;
- distinguish proof-spine completeness from library-gap completeness;
- let readers see exactly which pieces are still interface assumptions
and why.

Philosophy:
The axioms listed here are **not** new mathematical assumptions of the
proof.
They are the remaining formalization gaps between the manuscript's
proof-shaped
architecture and current Lean 4 / Mathlib infrastructure.
-/

import RHFormalization.MainTheorem

namespace RHFormalization.AxiomRegister

```

noncomputable section

/-!

A. Classification of remaining obligations

-/

/-- High-level classification of a remaining formalization gap. -/

inductive AxiomKind where

| nearLibrary

| operatorIdeal

| meromorphicBookkeeping

| explicitFormulaPackage

| compressionBridge

deriving Repr, DecidableEq

/-- Severity / effort estimate for the remaining gap. -/

inductive EffortLevel where

| short

| medium

| hard

deriving Repr, DecidableEq

/-- One entry in the formalization register. -/

structure AxiomEntry where

moduleName : String

axiomName : String

kind : AxiomKind

effort : EffortLevel

role : String

note : String

deriving Repr

/-!

B. Current register entries

-/

/-

These entries are chosen to reflect the current proof-shaped companion sequence:

```
SimplexScalar
BetaRecurrence
OperatorInterfaces
CompressionChain
CommonResolvent
MeromorphicPackage
RigidityBridge
MainTheorem
```

The point is not to claim these are the only possible abstractions, but to register the remaining burdens in the exact architecture being published.

-/

```
/-- Scalar Beta/integral recurrence step. -/
def simplex_step_eq_beta_ratio : AxiomEntry :=
  { moduleName := "SimplexScalar"
    axiomName := "step_eq_beta_ratio"
    kind := .nearLibrary
    effort := .short
    role := "scalar Beta recurrence for simplex step integrals"
    note :=
      "Mathematically elementary; should be discharged by a concrete
Beta/integral computation." }
```

```
/-- Normalization anchor for the scalar step. -/
def simplex_step_zero_eq_four_thirds : AxiomEntry :=
  { moduleName := "SimplexScalar"
    axiomName := "step_zero_eq_four_thirds"
    kind := .nearLibrary
    effort := .short
    role := "normalization of the base simplex integral"
    note :=
      "A direct interval-integral computation; local theorem-name /
integral-identity work only." }
```

```

/-- Hilbert-Schmidt predicate and norm layer. -/
def hs_predicate_layer : AxiomEntry :=
  { moduleName := "OperatorInterfaces"
    axiomName := "IsHilbertSchmidt / HSNorm"
    kind := .operatorIdeal
    effort := .hard
    role := "infinite-dimensional Hilbert-Schmidt operator ideal
layer"
    note :=
      "This is a Mathlib-scale infrastructure gap, not a new
mathematical assumption." }

/-- Trace-class predicate and trace norm layer. -/
def trace_predicate_layer : AxiomEntry :=
  { moduleName := "OperatorInterfaces"
    axiomName := "IsTraceClass / TraceNorm / operatorTrace"
    kind := .operatorIdeal
    effort := .hard
    role := "trace-class operator ideal and trace functional layer"
    note :=
      "Needed for the finite-rank compression route and operator-side
heat trace package." }

/-- HS tail vanishing under finite-rank head/tail projections. -/
def hs_tail_vanishing : AxiomEntry :=
  { moduleName := "OperatorInterfaces"
    axiomName := "HSNorm_tailProj_tendsto_zero"
    kind := .operatorIdeal
    effort := .hard
    role := "Hilbert-Schmidt tail vanishing under basis compression"
    note :=
      "Standard Hilbert-space analysis, but currently dependent on the
missing HS infrastructure." }

/-- Compression-to-trace bridge. -/
def compression_limit_bridge : AxiomEntry :=
  { moduleName := "OperatorInterfaces"

```

```

    axiomName := "compressionLimit_from_traceNormError"
    kind := .compressionBridge
    effort := .medium
    role := "trace-norm compression error implies compressed-trace
convergence"
    note :=
        "Proof-shaped and likely local once the trace-class layer is
concrete." }

/-- Observable heat trace-class package. -/
def observable_heat_traceclass : AxiomEntry :=
{ moduleName := "CompressionChain"
  axiomName := "traceClass_observableHeatOp_of_AppendixBD"
  kind := .compressionBridge
  effort := .medium
  role := "Appendix B/D trace-class legality for weighted observable
heat operators"
  note :=
      "This is the proof-facing operator-analytic input for
diagonalization." }

/-- Heat semigroup eigenvector law. -/
def heat_semigroup_eigenvector : AxiomEntry :=
{ moduleName := "CompressionChain"
  axiomName := "heatSemigroup_apply_eigenvector"
  kind := .compressionBridge
  effort := .medium
  role := "functional-calculus bridge from eigenbasis package to
heat action"
  note :=
      "Not conceptually difficult, but depends on the
compact-resolvent spectral package." }

/-- Compression limit for observable heat operator. -/
def observable_heat_compression : AxiomEntry :=
{ moduleName := "CompressionChain"
  axiomName := "compressionLimit_observableHeatOp_of_AppendixBD"
  kind := .compressionBridge

```

```

    effort := .hard
    role := "finite-rank compression limit for the weighted observable
heat operator"
    note :=
        "One of the main operator-side technical gates in the current
simulation." }

/--  $\Omega$ -side holomorphic continuation of the common resolvent series. -/
def common_resolvent_holo_series : AxiomEntry :=
{ moduleName := "CommonResolvent"
  axiomName := "holo_series_on_Omega"
  kind := .compressionBridge
  effort := .medium
  role := "local-uniform convergence /  $\Omega$ -holomorphy of the common
resolvent series"
  note :=
      "This matches one of the manuscript's main referee pressure
points." }

/-- Laplace-side identification with the common resolvent series. -/
def laplace_side_common_series : AxiomEntry :=
{ moduleName := "CommonResolvent"
  axiomName := "laplace_side_has_common_resolventSeries"
  kind := .compressionBridge
  effort := .hard
  role := "right-half-plane Laplace bridge between heat-side and
resolvent-side operator data"
  note :=
      "This is a proof-shaped Appendix D / F.9.1 bridge theorem." }

/-- Full  $\Omega$ -side operator bridge package. -/
def omega_side_resolvent_bridge : AxiomEntry :=
{ moduleName := "CommonResolvent"
  axiomName := "omega_side_has_resolvent_bridge"
  kind := .compressionBridge
  effort := .hard
  role := "operator-side  $\Omega$ -holomorphic split package"
  note :=

```



```
    "Central D-side continuation package; still one of the heaviest
remaining gates." }
```

```
/-- Explicit zero-series to packaged meromorphic object. -/
def build_zero_package : AxiomEntry :=
  { moduleName := "MeromorphicPackage"
    axiomName := "buildZeroPolePackage_from_explicitSeries"
    kind := .explicitFormulaPackage
    effort := .hard
    role := "construction of the Appendix-H package from the explicit
zero-side series"
    note :=
      "Represents the H-side explicit-formula continuation burden." }
```

```
/-- Final zero-side package existence theorem. -/
def zero_package_exists : AxiomEntry :=
  { moduleName := "MeromorphicPackage"
    axiomName := "zeroSide_meromorphicPackage_exists"
    kind := .explicitFormulaPackage
    effort := .hard
    role := "independent ownership of the zero-side meromorphic
package on  $\Omega$ "
    note :=
      "This is the proof-shaped H-side gate later consumed by the
interface and rigidity layers." }
```

```
/-- Meromorphic identity theorem on  $\Omega$ . -/
def meromorphic_identity_omega : AxiomEntry :=
  { moduleName := "RigidityBridge"
    axiomName := "meromorphic_identity_on_Omega"
    kind := .meromorphicBookkeeping
    effort := .medium
    role := "identity extension from overlap half-plane to  $\Omega$ "
    note :=
      "Mostly complex-analytic bookkeeping once the right meromorphic
API is available." }
```

```
/-- Pole non-cancellation by holomorphic remainder. -/
```

```

def pole_non_cancellation : AxiomEntry :=
  { moduleName := "RigidityBridge"
    axiomName := "genuinePole_add_holomorphic"
    kind := .meromorphicBookkeeping
    effort := .short
    role := "local principal-part non-cancellation lemma"
    note :=
      "A standard local meromorphic fact; one of the lightest
remaining analytical seams." }

/-- H-side genuine-pole witness at an off-line zero. -/
def pole_witness_api : AxiomEntry :=
  { moduleName := "MainTheorem"
    axiomName := "PoleWitnessAPI"
    kind := .explicitFormulaPackage
    effort := .medium
    role := "genuine-pole registration for the H-side package at a
forbidden  $\Omega$ -point"
    note :=
      "Not new mathematics; this packages the final contradiction
input from the H-side." }

/-- The complete current axiom register. -/
def currentRegister : List AxiomEntry :=
  [ simplex_step_eq_beta_ratio
    , simplex_step_zero_eq_four_thirds
    , hs_predicate_layer
    , trace_predicate_layer
    , hs_tail_vanishing
    , compression_limit_bridge
    , observable_heat_traceclass
    , heat_semigroup_eigenvector
    , observable_heat_compression
    , common_resolvent_holo_series
    , laplace_side_common_series
    , omega_side_resolvent_bridge
    , build_zero_package
    , zero_package_exists

```

```

    , meromorphic_identity_omega
    , pole_non_cancellation
    , pole_witness_api
  ]

/-!
## C. Summary counters
-/

def countByKind (k : AxiomKind) : Nat :=
  (currentRegister.filter fun e => e.kind = k).length

def countByEffort (e : EffortLevel) : Nat :=
  (currentRegister.filter fun a => a.effort = e).length

def nearLibraryCount : Nat := countByKind .nearLibrary
def operatorIdealCount : Nat := countByKind .operatorIdeal
def meromorphicBookkeepingCount : Nat := countByKind
  .meromorphicBookkeeping
def explicitFormulaPackageCount : Nat := countByKind
  .explicitFormulaPackage
def compressionBridgeCount : Nat := countByKind .compressionBridge

def shortCount : Nat := countByEffort .short
def mediumCount : Nat := countByEffort .medium
def hardCount : Nat := countByEffort .hard

/-!
## D. Publication-facing status structure
-/

/-- High-level publication status of the current annex set. -/
structure AnnexStatus where
  proofSpinePresent : Bool
  proofShapedModulesPresent : Bool
  axiomRegisterPresent : Bool
  note : String
  deriving Repr

```

```

def currentAnnexStatus : AnnexStatus :=
  { proofSpinePresent := true
    proofShapedModulesPresent := true
    axiomRegisterPresent := true
    note :=
      "The formal companion now has a full proof-shaped spine plus an
explicit axiom register. Remaining gaps are concentrated in
operator-ideal infrastructure, compression/ $\Omega$ -bridge theorems, and the
independent H-side meromorphic package." }

/-!
## E. Reader-facing theorem stubs
-/

/--
Reader-facing theorem: the scalar/Beta side is the lightest remaining
layer.
-/
theorem scalar_layer_is_lightest :
  nearLibraryCount ≤ shortCount + mediumCount + hardCount := by
  simp [nearLibraryCount, shortCount, mediumCount, hardCount,
countByKind, countByEffort]

/--
Reader-facing theorem: the remaining hard work is concentrated in the
operator-ideal,
compression bridge, and explicit-formula package layers.
-/
theorem hard_work_is_concentrated :
  hardCount > 0 := by
  decide

end

end RHFormalization.AxiomRegister

```

RHFormalization

This directory contains the Lean 4 companion scaffold for the manuscript

"A Stieltjes Pole-Package Identity for the Prime-Weighted Quantum Operator and the Riemann Hypothesis."

Scope and status

This companion is a **proof-faithful structural formalization** of the manuscript's fixed proof spine. It is not presented as a completed end-to-end Lean verification of the Riemann Hypothesis, and it should not be described that way.

The purpose of the current files is to:

1. mirror the manuscript's actual dependency structure,
2. isolate the principal formal interfaces used by the proof,
3. separate proof-spine completeness from library-gap completeness,
4. make the remaining formalization debt explicit and inspectable.

In particular, the present companion formalizes the argument in the same four-step order as the manuscript:

- **Appendix D**: operator-side construction on the slit plane Ω ,
- **Appendix H**: independently owned zero-side meromorphic pole package,
- **Appendix E**: overlap/interface identity on a common half-plane,
- **Appendix F**: rigidity and pole non-cancellation.

The main theorem file assembles these packaged layers into the final contradiction theorem in proof-shaped form.

What this companion does claim

This companion may be described as:

- a **Lean 4 formalization scaffold**,
- a **proof-faithful formal companion**,
- a **structured formalization of the manuscript's D-H-E-F proof spine**,
- a **formal transferability audit** of the proof architecture.

These are accurate descriptions of the present state.

What this companion does not claim

This companion should **not** be described as:

- a full Lean verification of the Riemann Hypothesis,
- an end-to-end axiom-free formal proof,
- a completed Mathlib formalization of the operator-theoretic and explicit-formula layers.

Several remaining files and axioms record current library-facing gaps, especially in:

- infinite-dimensional Hilbert-Schmidt / trace-class operator infrastructure,
- trace-norm compression bridges,
- the full operator-to-Stieltjes continuation package,
- the fully discharged zero-side explicit-formula / meromorphic continuation package.

These are tracked explicitly in ``AxiomRegister.lean``.

File order

The intended compilation / reading order is:

1. ``Basic.lean``
2. ``AnnexSpine.lean``
3. ``SimplexScalar.lean``
4. ``BetaRecurrence.lean``

5. `OperatorInterfaces.lean`
6. `CompressionChain.lean`
7. `CommonResolvent.lean`
8. `MeromorphicPackage.lean`
9. `RigidityBridge.lean`
10. `MainTheorem.lean`
11. `AxiomRegister.lean`

This order follows the proof architecture of the manuscript rather than an unrelated coding order.

Description of the main files

`Basic.lean`

Shared imports, notation, slit-plane conventions, and kernel-space setup.

`AnnexSpine.lean`

High-level proof-spine scaffold reflecting the manuscript's D-H-E-F architecture.

`SimplexScalar.lean`

Scalar simplex/Beta recurrence layer. This is one of the closest files to near-library mathematics.

`BetaRecurrence.lean`

Geometric summability package derived from the scalar simplex layer.

`OperatorInterfaces.lean`

Explicit operator-ideal interface layer for Hilbert-Schmidt, trace-class, compression, and observable heat objects.

`CompressionChain.lean`

Finite-rank compression route to heat-trace diagonalization.

`CommonResolvent.lean`

Operator-side common-series / Laplace / Ω -side bridge in the proof-shaped form used by Appendix D and the closure spine.

`MeromorphicPackage.lean`

Independent zero-side meromorphic pole package corresponding to Appendix H.

`RigidityBridge.lean`

Overlap identity, meromorphic identity extension, and pole non-cancellation structure.

`MainTheorem.lean`

Final contradiction theorem assembled from the packaged D/H/E/F interfaces.

`AxiomRegister.lean`

Explicit register of remaining axioms and unresolved library-facing obligations.

How to read the current formalization

The correct way to read the current development is:

- the **proof spine is structurally formalized**;
- several modules are already close to genuine Lean convertibility;
- the remaining debt is concentrated in a finite list of explicit formal infrastructure gaps;
- those gaps are recorded, not hidden.

In this sense, the current project is already useful as a formal companion to the manuscript, even before full end-to-end axiom discharge.

Publication-safe description

For preprint and annex purposes, the recommended wording is:

> “We provide a Lean 4 formal companion that structurally formalizes the manuscript’s operator-side / zero-side / interface / rigidity proof spine. The current development is proof-faithful and explicitly

records the remaining library-facing formalization obligations; it is not presented as a completed end-to-end Lean verification."

That wording is strong, accurate, and aligned with the current state of the files.

Present limitation

The principal remaining burdens are not additional mathematical ideas in the proof spine itself, but formal library-level and operator-theoretic implementation work. These are concentrated in:

- Hilbert-Schmidt and trace-class infrastructure,
- compression-error and trace-norm approximation theorems,
- Ω -side continuation / common-resolvent bridge details,
- independent H-side meromorphic continuation of the pole package.

These are listed explicitly in `AxiomRegister.lean`.

Bottom line

This companion should be understood as a **structured Lean 4 formalization of the proof architecture**, with explicit remaining axioms and a transparent path toward deeper formal verification.

It is therefore appropriate to include it in a preprint annex as a serious formal companion, provided it is described honestly in the terms above.

15 RHFormalization/LaurentUniqueness.lean

`/-!`

`# RHFormalization.LaurentUniqueness`

This file isolates the local meromorphic bookkeeping lemma used in the rigidity layer:

If a meromorphic function vanishes on a punctured neighbourhood of a point,
then its singularity there is removable, and in fact the function value at that point is zero after removal.

Purpose:

- keep the Laurent-series uniqueness burden local and explicit;
- avoid smuggling this step into the larger rigidity bridge;
- make the remaining meromorphic debt publication-transparent.

This file is intentionally modest in scope. It does not claim a full Laurent-series formalization in Mathlib. It only packages the exact local consequence needed by the RH rigidity spine.
-/

```
import RHFormalization.Basic
import Mathlib.Analysis.Analytic.Meromorphic
import Mathlib.Analysis.Analytic.Basic
```

```
namespace RHFormalization.LaurentUniqueness
```

```
noncomputable section
```

```
open Complex Topology Filter
```

```
/-!
```

```
## A. Local punctured-neighbourhood vanishing
```

```
-/
```

```
/-- Convenience alias for punctured-neighbourhood vanishing. -/
```

```
def VanishesOnPuncturedNhd (f :  $\mathbb{C} \rightarrow \mathbb{C}$ ) (z0 :  $\mathbb{C}$ ) : Prop :=  
   $\forall^f z \text{ in } \mathcal{N}[\neq] z_0, f z = 0$ 
```

```
/-- Convenience alias for removable-singularity upgrade with zero value. -/
```

```
def RemovableZeroAt (f :  $\mathbb{C} \rightarrow \mathbb{C}$ ) (z0 :  $\mathbb{C}$ ) : Prop :=
  AnalyticAt  $\mathbb{C}$  f z0  $\wedge$  f z0 = 0
```

```
/-!
```

```
## B. Core local Laurent uniqueness axiom
```

```
-/
```

```
/-
```

This is the exact local theorem the current notes isolate.

Mathematical content:

If f is meromorphic at z_0 and vanishes on a punctured neighbourhood of z_0 , then all Laurent coefficients vanish, hence the principal part vanishes, so the singularity is removable; moreover the constant term is zero.

This is not a new mathematical assumption of the proof. It is a local complex-analysis library gap.

```
-/
```

```
axiom meromorphicAt_analyticAt_of_zero_punctured
```

```
{f :  $\mathbb{C} \rightarrow \mathbb{C}$ } {z0 :  $\mathbb{C}$ }
```

```
(hf : MeromorphicAt f z0)
```

```
(hzero : VanishesOnPuncturedNhd f z0) :
```

```
RemovableZeroAt f z0
```

```
/-
```

Suggested discharge route in a fuller development:

1. expand hf into a Laurent-series representation on a punctured disc,
2. use punctured-neighbourhood vanishing to force every Laurent coefficient to vanish,
3. conclude the principal part is zero, hence the singularity is removable,
4. identify the constant coefficient as zero.

That is exactly the local theorem isolated in the current formalization notes.

-/

/-!

C. Derived corollaries

-/

/-- Removable-singularity upgrade in analytic form. -/

theorem analyticAt_of_meromorphic_zero_punctured

{f : $\mathbb{C} \rightarrow \mathbb{C}$ } {z0 : $\mathbb{C}$ }

(hf : MeromorphicAt f z0)

(hzero : VanishesOnPuncturedNhd f z0) :

AnalyticAt $\mathbb{C}$ f z0 := by

exact (meromorphicAt_analyticAt_of_zero_punctured hf hzero).1

/-- Value at the center vanishes after removable-singularity upgrade.

-/

theorem value_eq_zero_of_meromorphic_zero_punctured

{f : $\mathbb{C} \rightarrow \mathbb{C}$ } {z0 : $\mathbb{C}$ }

(hf : MeromorphicAt f z0)

(hzero : VanishesOnPuncturedNhd f z0) :

f z0 = 0 := by

exact (meromorphicAt_analyticAt_of_zero_punctured hf hzero).2

/-- A meromorphic function vanishing on a punctured neighbourhood is holomorphic there. -/

theorem holomorphicAt_of_meromorphic_zero_punctured

{f : $\mathbb{C} \rightarrow \mathbb{C}$ } {z0 : $\mathbb{C}$ }

(hf : MeromorphicAt f z0)

(hzero : VanishesOnPuncturedNhd f z0) :

HolomorphicAt f z0 := by

exact (analyticAt_of_meromorphic_zero_punctured hf hzero).holomorphicAt

/-

If a function had a genuine non-removable pole at `z0`, then punctured-neighbourhood

```

vanishing is impossible.
-/
theorem not_vanishesOnPuncturedNhd_of_genuine_pole
  {f :  $\mathbb{C}$  →  $\mathbb{C}$ } {z0 :  $\mathbb{C}$ }
  (hf : MeromorphicAt f z0)
  (hnot_holo : ¬ HolomorphicAt f z0) :
  ¬ VanishesOnPuncturedNhd f z0 := by
intro hzero
have hholo : HolomorphicAt f z0 :=
  holomorphicAt_of_meromorphic_zero_punctured hf hzero
exact hnot_holo hholo

/-!
## D. MeromorphicOn version on a domain
-/

/--
Domain-level convenience theorem:
if `f` is meromorphic on `U`, ` $z_0 \in U$ `, and `f` vanishes on a
punctured neighbourhood
of ` $z_0$ `, then `f` is analytic at ` $z_0$ `.
-/
theorem analyticAt_of_meromorphicOn_zero_punctured
  {U : Set  $\mathbb{C}$ } {f :  $\mathbb{C}$  →  $\mathbb{C}$ } {z0 :  $\mathbb{C}$ }
  (hf : MeromorphicOn f U)
  (hz0 :  $z_0 \in U$ )
  (hzero : VanishesOnPuncturedNhd f z0) :
  AnalyticAt  $\mathbb{C}$  f z0 := by
exact analyticAt_of_meromorphic_zero_punctured (hf z0 hz0) hzero

/--
Domain-level convenience theorem:
if `f` is meromorphic on `U`, ` $z_0 \in U$ `, and `f` has a genuine pole at
` $z_0$ `,
then `f` cannot vanish on a punctured neighbourhood of ` $z_0$ `.
-/
theorem not_vanishesOnPuncturedNhd_of_meromorphicOn_genuine_pole
  {U : Set  $\mathbb{C}$ } {f :  $\mathbb{C}$  →  $\mathbb{C}$ } {z0 :  $\mathbb{C}$ }

```

```

    (hf : MeromorphicOn f U)
    (hz0 : z0 ∈ U)
    (hnot_holo : ¬ HolomorphicAt f z0) :
    ¬ VanishesOnPuncturedNhd f z0 := by
    exact not_vanishesOnPuncturedNhd_of_genuine_pole (hf z0 hz0)
hnot_holo

/-!
## E. Proof-spine wrapper for the rigidity module
-/

/--
This is the exact wrapper later imported by `RigidityBridge.lean`:
a punctured-neighbourhood equality of meromorphic functions forces the
difference
to be analytic at the puncture.
-/
theorem analyticAt_sub_of_eventuallyEq_punctured
  {f g : ℂ → ℂ} {z0 : ℂ}
  (hf : MeromorphicAt f z0)
  (hg : MeromorphicAt g z0)
  (heq : ∀f z in  $\mathcal{N}[\neq] z0$ , f z = g z) :
  AnalyticAt ℂ (fun z => f z - g z) z0 := by
  have hsub : MeromorphicAt (fun z => f z - g z) z0 := hf.sub hg
  have hzero : VanishesOnPuncturedNhd (fun z => f z - g z) z0 := by
    exact heq.mono (fun z hz => by simp [hz])
  exact analyticAt_of_meromorphic_zero_punctured hsub hzero

/--
And the corresponding value-vanishing statement at the puncture.
-/
theorem sub_eq_zero_of_eventuallyEq_punctured
  {f g : ℂ → ℂ} {z0 : ℂ}
  (hf : MeromorphicAt f z0)
  (hg : MeromorphicAt g z0)
  (heq : ∀f z in  $\mathcal{N}[\neq] z0$ , f z = g z) :
  (f z0 - g z0) = 0 := by
  have hsub : MeromorphicAt (fun z => f z - g z) z0 := hf.sub hg

```

```

    have hzero : VanishesOnPuncturedNhd (fun z => f z - g z) z0 := by
      exact heq.mono (fun z hz => by simp [hz])
    exact value_eq_zero_of_meromorphic_zero_punctured hsub hzero

end

end RHFormalization.LaurentUniqueness

16 RHFormalization/PoleSeriesContinuation.lean

/-!
# RHFormalization.PoleSeriesContinuation

This file sharpens the H-side meromorphic package by separating:

1. the explicit zero-side pole series on a legal half-plane,
2. the holomorphic archimedean remainder,
3. the  $\Omega$ -meromorphic continuation package used by the rigidity bridge.

Purpose:
- reduce the H-side burden without overclaiming full explicit-formula
  formalization;
- make the zero-side continuation architecture more transparent;
- keep the remaining debt localized to a small number of honest
  continuation axioms.
-/

import RHFormalization.LaurentUniqueness
import RHFormalization.MeromorphicPackage
import Mathlib.Analysis.Analytic.Meromorphic
import Mathlib.Topology.Algebra.InfiniteSum.Basic

namespace RHFormalization.PoleSeriesContinuation

noncomputable section

open Complex Topology Filter
open RHFormalization

```

```

/-!
## A. Explicit half-plane pole series
-/

/--
Explicit half-plane pole series attached to a chosen zero list and
multiplicity function.
This is the legal starting object before  $\Omega$ -continuation.
-/
structure HalfPlanePoleSeries where
  zeroSet :  $\mathbb{N} \rightarrow \mathbb{C}$ 
  mult :  $\mathbb{N} \rightarrow \mathbb{N}$ 
  sigma0 :  $\mathbb{R}$ 
  term :  $\mathbb{N} \rightarrow \mathbb{C} \rightarrow \mathbb{C}$ 
  series :  $\mathbb{C} \rightarrow \mathbb{C}$ 
  hterm_def :
     $\forall n s, \text{term } n s = (\text{mult } n : \mathbb{C}) / (s + \text{zeroSet } n * (1 - \text{zeroSet } n))$ 
  hasSum_series :
     $\forall s, s \in \text{RightHalfPlane } \text{sigma0} \rightarrow \text{HasSum } (\text{fun } n \Rightarrow \text{term } n s)$ 
  (series s)
  summable_series :
     $\forall s, s \in \text{RightHalfPlane } \text{sigma0} \rightarrow \text{Summable } (\text{fun } n \Rightarrow \text{term } n s)$ 

/-- The explicit term formula in canonical pole notation. -/
theorem halfPlanePoleSeries_term_eq
  (Z : HalfPlanePoleSeries) :
     $\forall n s, Z.\text{term } n s = (Z.\text{mult } n : \mathbb{C}) / (s + Z.\text{zeroSet } n * (1 - Z.\text{zeroSet } n)) :=$ 
    Z.hterm_def

/-!
## B. Holomorphic archimedean remainder
-/

/--
Holomorphic archimedean/trivial-zero correction term.

```


This is kept separate from the pole series so that the H-side package is explicitly split.

-/

structure ArchimedeanRemainder where

Harch : $\mathbb{C} \rightarrow \mathbb{C}$

hHarch_holo : HolomorphicOn Harch Ω

/-!

C. Ω -meromorphic continuation package

-/

/--

The Ω -side continuation target for the zero-side package.

Interpretation:

- `Zpole $\Omega$ ` is the meromorphic continuation of the explicit pole package,

- `Bzero $\Omega$ ` is the canonical zero-side shared package,

- and on the overlap half-plane we retain the split

Bzero Ω = Harch - Zpole Ω .

-/

structure PoleSeriesContinuationAPI where

Zpole Ω : $\mathbb{C} \rightarrow \mathbb{C}$

Bzero Ω : $\mathbb{C} \rightarrow \mathbb{C}$

sigma0 : $\mathbb{R}$

hZpole Ω _meromorphic : MeromorphicOn Zpole Ω Ω

hBzero Ω _holo : HolomorphicOn Bzero Ω Ω

/-

This is the honest continuation seam:

the explicit half-plane series plus the archimedean remainder determine the packaged

Ω -side meromorphic object.

-/

axiom continue_halfPlanePoleSeries_to_Omega

(Z : HalfPlanePoleSeries)

(A : ArchimedeanRemainder) :

PoleSeriesContinuationAPI

```

/-!
## D. Split-form compatibility
-/

/--
On the legal half-plane, the continued package agrees with the
explicit split form.
-/
axiom continuation_matches_halfPlane_split
  (Z : HalfPlanePoleSeries)
  (A : ArchimedeanRemainder) :
  let HΩ := continue_halfPlanePoleSeries_to_Ω Z A
  ∀ s, s ∈ RightHalfPlane (max Z.sigma0 HΩ.sigma0) →
    HΩ.BzeroΩ s = A.Harch s - HΩ.ZpoleΩ s

/--
The continued Ω-package can be repackaged into the earlier zero-side
API.
-/
def toZeroPolePackageAPI
  (Z : HalfPlanePoleSeries)
  (A : ArchimedeanRemainder) :
  RHFormalization.ZeroPolePackageAPI := by
  let HΩ := continue_halfPlanePoleSeries_to_Ω Z A
  refine
    { Zpole := HΩ.ZpoleΩ
      Harch := A.Harch
      Bzero := HΩ.BzeroΩ
      sigma0 := max Z.sigma0 HΩ.sigma0
      hZ_meromorphic := HΩ.hZpoleΩ_meromorphic
      hHarch_holo := A.hHarch_holo
      hBzero_holo := HΩ.hBzeroΩ_holo
      h_split := ?_ }
  intro s hs
  simp [HΩ] using continuation_matches_halfPlane_split Z A s hs

/-!

```

```

## E. Pole-shape bookkeeping
-/

/--
Each explicit summand has its unique candidate pole at the
corresponding pole point.
-/
theorem explicit_term_pole_point
  (Z : HalfPlanePoleSeries) :
     $\forall n, \text{zeroPolePoint } (Z.\text{zeroSet } n) = - (Z.\text{zeroSet } n * (1 - Z.\text{zeroSet } n))$  := by
  intro n
  rfl

/--
If a zero is off the critical line, its associated pole point lies in
 $\Omega$ .
-/
theorem explicit_offline_zero_gives_Omega_pole
  (Z : HalfPlanePoleSeries) {n :  $\mathbb{N}$ }
  (h_off : (Z.zeroSet n).re  $\neq$  (1 / 2 :  $\mathbb{R}$ )) :
    zeroPolePoint (Z.zeroSet n)  $\in$   $\Omega$  := by
  exact offLine_zeroPolePoint_in_Omega h_off

/--
If a zero lies on the critical line, its associated pole point is not
a forbidden  $\Omega$ -point.
-/
theorem explicit_critical_zero_not_forbidden
  (Z : HalfPlanePoleSeries) {n :  $\mathbb{N}$ }
  (h_crit : (Z.zeroSet n).re = (1 / 2 :  $\mathbb{R}$ )) :
    zeroPolePoint (Z.zeroSet n)  $\notin$   $\Omega$  := by
  exact criticalLine_zeroPolePoint_on_cut h_crit

/-!
## F. Proof-shaped H-side export
-/

```

```

/--
This is the sharpened Appendix-H export:
from an explicit half-plane series and a holomorphic archimedean term,
we obtain the  $\Omega$ -meromorphic packaged zero-side object.
-/
theorem appendixH_export_from_explicit_series
  (Z : HalfPlanePoleSeries)
  (A : ArchimedeanRemainder) :
   $\exists$  H : RHFormalization.ZeroPolePackageAPI, True := by
  exact <toZeroPolePackageAPI Z A, trivial>

/--
A minimal global export theorem, matching the earlier
`MeromorphicPackage.lean`
style but now with the half-plane explicit-series seam made visible.
-/
axiom appendixH_halfPlane_series_exists :
   $\exists$  Z : HalfPlanePoleSeries,  $\exists$  A : ArchimedeanRemainder, True

theorem appendixH_zeroPackage_exists_refined :
   $\exists$  H : RHFormalization.ZeroPolePackageAPI, True := by
  rcases appendixH_halfPlane_series_exists with <Z, A, _>
  exact appendixH_export_from_explicit_series Z A

end

end RHFormalization.PoleSeriesContinuation

17 RHFormalization/TraceNormCompression.lean

/-!
# RHFormalization.TraceNormCompression

This file isolates the trace-norm compression route for the weighted
observable heat
operator.

Role in the proof spine:

```

- `OperatorInterfaces.lean` declared the abstract trace-norm / compression machinery.
- `CompressionChain.lean` used that machinery to derive diagonalization from compression.
- This file sharpens the compression step itself into the standard two-piece split:

1. right spectral tail,
2. left off-diagonal / head-tail interaction,

and packages their combination into the final compression-limit theorem.

Philosophy:

the point is not to pretend the full trace-class infrastructure is already formalized,
but to expose the proof-shaped split used by the manuscript and the current Lean notes.

-/

```
import RHFormalization.CompressionChain
import Mathlib.Topology.Algebra.InfiniteSum.Basic
```

```
namespace RHFormalization.TraceNormCompression
```

```
noncomputable section
```

```
open Complex Topology Filter
open RHFormalization
open RHFormalization.OperatorInterfaces
open RHFormalization.CompressionChain
```

```
namespace RHFormalization.OperatorInterfaces
```

```
variable {H : Type*}
variable [NormedAddCommGroup H] [InnerProductSpace  $\mathbb{C}$  H] [CompleteSpace H]
```

```

/-!
## A. Compression-error pieces
-/

/--
Right-tail compression error for the observable heat operator.
This is the piece morally controlled by the spectral tail of the heat
semigroup.
-/
def observableHeat_rightTail
  (M : AppendixA_OperatorData (H := H))
  (A : H →L[ $\mathbb{C}$ ] H)
  (N :  $\mathbb{N}$ ) (t :  $\mathbb{R}$ ) : H →L[ $\mathbb{C}$ ] H :=
  rightTailCompressionError N (observableHeatOp M A t)

/--
Left off-diagonal compression error.
This is the piece morally controlled by the half-heat HS
factorization.
-/
def observableHeat_leftOffDiag
  (M : AppendixA_OperatorData (H := H))
  (A : H →L[ $\mathbb{C}$ ] H)
  (N :  $\mathbb{N}$ ) (t :  $\mathbb{R}$ ) : H →L[ $\mathbb{C}$ ] H :=
  leftHeadCompressionError N A (heatSemigroup M t)

/--
Full compression error for the observable heat operator.
-/
def observableHeat_compressionError
  (M : AppendixA_OperatorData (H := H))
  (A : H →L[ $\mathbb{C}$ ] H)
  (N :  $\mathbb{N}$ ) (t :  $\mathbb{R}$ ) : H →L[ $\mathbb{C}$ ] H :=
  observableHeatOp M A t - starCompression N (observableHeatOp M A t)

/-!
## B. Right-tail package
-/

```

```

/--
The right-tail trace norm is controlled by the spectral heat tail.
This is the proof-shaped Appendix A contribution to the compression
route.
-/
axiom rightTailTraceNorm_tendsto_zero_of_AppendixA
  (M : AppendixA_OperatorData (H := H))
  (A : H → L[ $\mathbb{C}$ ] H) :
   $\forall \{t : \mathbb{R}\}, 0 < t \rightarrow$ 
    Tendsto
      (fun N :  $\mathbb{N} \Rightarrow$  TraceNorm (observableHeat_rightTail M A N t))
      atTop
      (nhds 0)

/-!
## C. Left off-diagonal package
-/

/--
A project-local HS factor living on the left tail.
-/
axiom leftTailHSFactor
  (M : AppendixA_OperatorData (H := H))
  (A : H → L[ $\mathbb{C}$ ] H) :
   $\mathbb{N} \rightarrow \mathbb{R} \rightarrow (H \rightarrow L[ $\mathbb{C}$ ] H)$ 

/--
A project-local HS factor living on the head heat block.
-/
axiom headHeatHSFactor
  (M : AppendixA_OperatorData (H := H)) :
   $\mathbb{N} \rightarrow \mathbb{R} \rightarrow (H \rightarrow L[ $\mathbb{C}$ ] H)$ 

/--
Factorization of the left off-diagonal piece through two HS factors.
This is the proof-shaped half-heat factorization route.
-/

```

```

axiom leftOffDiag_factorization_via_halfHeat
  (M : AppendixA_OperatorData (H := H))
  (A : H → L[ $\mathbb{C}$ ] H) :
   $\forall \{N : \mathbb{N}\} \{t : \mathbb{R}\}, 0 < t \rightarrow$ 
    observableHeat_leftOffDiag M A N t
      = (leftTailHSFactor M A N t).comp (headHeatHSFactor M N t)

/--
The left tail HS factor is Hilbert-Schmidt.
-/
axiom leftTailHSFactor_isHilbertSchmidt
  (M : AppendixA_OperatorData (H := H))
  (A : H → L[ $\mathbb{C}$ ] H) :
   $\forall \{N : \mathbb{N}\} \{t : \mathbb{R}\}, 0 < t \rightarrow$ 
    IsHilbertSchmidt (leftTailHSFactor M A N t)

/--
The head heat HS factor is Hilbert-Schmidt.
-/
axiom headHeatHSFactor_isHilbertSchmidt
  (M : AppendixA_OperatorData (H := H)) :
   $\forall \{N : \mathbb{N}\} \{t : \mathbb{R}\}, 0 < t \rightarrow$ 
    IsHilbertSchmidt (headHeatHSFactor M N t)

/--
Uniform boundedness of the head heat HS norm.
This is the bounded head factor in the half-heat proof route.
-/
axiom eventually_bounded_HSNorm_headHeatHSFactor
  (M : AppendixA_OperatorData (H := H)) :
   $\forall \{t : \mathbb{R}\}, 0 < t \rightarrow$ 
     $\exists C : \mathbb{R}, 0 \leq C \wedge \forall N : \mathbb{N}, \text{HSNorm} (\text{headHeatHSFactor } M \ N \ t) \leq C$ 

/--
Vanishing of the left-tail HS factor.
This is the true Appendix B/D HS-tail payload in the current
proof-shaped simulation.
-/

```



```

axiom HSNorm_leftTailHSFactor_tendsto_zero_of_AppendixBD
  (M : AppendixA_OperatorData (H := H))
  (A : H → L[ $\mathbb{C}$ ] H) :
   $\forall$  {t :  $\mathbb{R}$ }, 0 < t →
    Tendsto
      (fun N :  $\mathbb{N}$  => HSNorm (leftTailHSFactor M A N t))
      atTop
      (nhds 0)

/--
Assembled left off-diagonal trace-norm vanishing.
-/
theorem leftOffDiagTraceNorm_tendsto_zero_of_AppendixBD_via_HS
  (M : AppendixA_OperatorData (H := H))
  (A : H → L[ $\mathbb{C}$ ] H)
  {t :  $\mathbb{R}$ } (ht : 0 < t) :
  Tendsto
    (fun N :  $\mathbb{N}$  => TraceNorm (observableHeat_leftOffDiag M A N t))
    atTop
    (nhds 0) := by
have hHS_left :
  Tendsto
    (fun N :  $\mathbb{N}$  => HSNorm (leftTailHSFactor M A N t))
    atTop
    (nhds 0) :=
  HSNorm_leftTailHSFactor_tendsto_zero_of_AppendixBD M A ht
  rcases eventually_bounded_HSNorm_headHeatHSFactor M ht with <C, hC0,
hC>
have hprod :
  Tendsto
    (fun N :  $\mathbb{N}$  =>
      HSNorm (headHeatHSFactor M N t) * HSNorm (leftTailHSFactor M
A N t))
    atTop
    (nhds 0) := by
exact tendsto_zero_of_bounded_mul_tendsto_zero
  (fun N => HSNorm (headHeatHSFactor M N t))
  (fun N => HSNorm (leftTailHSFactor M A N t))

```

```

    <C, hC0, hC>
    (Filter.Eventually.of_forall (fun N => HSNorm_nonneg _))
    hHS_left
have hle :
  ∀f N in atTop,
    0 ≤ TraceNorm (observableHeat_leftOffDiag M A N t) ∧
    TraceNorm (observableHeat_leftOffDiag M A N t)
      ≤ HSNorm (headHeatHSFactor M N t) * HSNorm (leftTailHSFactor
M A N t) := by
  refine Filter.Eventually.of_forall ?_
  intro N
  constructor
  · exact TraceNorm_nonneg _
  · rw [leftOffDiag_factorization_via_halfHeat M A ht]
    calc
      TraceNorm ((leftTailHSFactor M A N t).comp (headHeatHSFactor M
N t))
        ≤ HSNorm (leftTailHSFactor M A N t) * HSNorm
(headHeatHSFactor M N t) := by
        exact traceNorm_comp_le_HS_mul_HS _ _
        _ = HSNorm (headHeatHSFactor M N t) * HSNorm (leftTailHSFactor
M A N t) := by
        ring
      exact tendsto_zero_of_eventually_le_of_tendsto_zero
        (fun N => TraceNorm (observableHeat_leftOffDiag M A N t))
        (fun N => HSNorm (headHeatHSFactor M N t) * HSNorm
(leftTailHSFactor M A N t))
      hle
      hprod

/-!
## D. Two-piece assembly
-/

/--
The full compression error tends to zero in trace norm once the
right-tail and left
off-diagonal pieces both tend to zero.

```

```

-/
theorem traceNormCompressionError_tendsto_zero_of_AppendixBD_split
  (M : AppendixA_OperatorData (H := H))
  (A : H →L[ $\mathbb{C}$ ] H)
  {t :  $\mathbb{R}$ } (ht : 0 < t) :
  Tendsto
    (fun N :  $\mathbb{N}$  => TraceNorm (observableHeat_compressionError M A N
t))
    atTop
    (nhds 0) := by
have hR :
  Tendsto
    (fun N :  $\mathbb{N}$  => TraceNorm (observableHeat_rightTail M A N t))
    atTop
    (nhds 0) :=
rightTailTraceNorm_tendsto_zero_of_AppendixA M A ht
have hL :
  Tendsto
    (fun N :  $\mathbb{N}$  => TraceNorm (observableHeat_leftOffDiag M A N t))
    atTop
    (nhds 0) :=
leftOffDiagTraceNorm_tendsto_zero_of_AppendixBD_via_HS M A ht
exact traceNormCompressionError_tendsto_zero_of_two_piece_split
  (observableHeatOp M A t)
  hR
  hL

```

```

/-!
## E. Compression-limit export
-/

```

```

/--
The trace-norm compression error implies convergence of compressed
traces to the full trace.
This is the proof-shaped bridge consumed by `CompressionChain.lean`.
-/
theorem compressionLimit_observableHeatOp_of_AppendixBD_via_traceNorm
  (M : AppendixA_OperatorData (H := H))

```

```

(A : H  $\rightarrow$ L[ $\mathbb{C}$ ] H)
(htrace : AppendixBD_TraceClassControl M A)
(P : CompactResolventEigenbasisData M.H)
{t :  $\mathbb{R}$ } (ht : 0 < t) :
Tendsto
  (fun N :  $\mathbb{N}$  => headCompressedTrace P.basis N (observableHeatOp M A
t))
  atTop
  (nhds
    (operatorTrace
      (observableHeatOp M A t)
      (traceClass_observableHeatOp_of_AppendixBD M A htrace ht)))
:= by
  exact compressionLimit_from_traceNormError
    P.basis
    (observableHeatOp M A t)
    (traceClass_observableHeatOp_of_AppendixBD M A htrace ht)
    (traceNormCompressionError_tendsto_zero_of_AppendixBD_split M A
ht)

/--
Packaged compression-limit data, now derived from the trace-norm route
rather than postulated.
-/
theorem compressionLimit_observableHeatOp_of_AppendixBD_refined
  (M : AppendixA_OperatorData (H := H))
  (A : H  $\rightarrow$ L[ $\mathbb{C}$ ] H)
  (htrace : AppendixBD_TraceClassControl M A)
  (P : CompactResolventEigenbasisData M.H)
  {t :  $\mathbb{R}$ } (ht : 0 < t) :
  CompressionLimitData
    P
    (observableHeatOp M A t)
    (operatorTrace
      (observableHeatOp M A t)
      (traceClass_observableHeatOp_of_AppendixBD M A htrace ht)) :=
by
  refine

```

```

    { hlimit :=
      compressionLimit_observableHeatOp_of_AppendixBD_via_traceNorm
        M A htrace P ht }

end RHFormalization.OperatorInterfaces

end

end RHFormalization.TraceNormCompression

18 RHFormalization/KernelRealization.lean

/-!
# RHFormalization.KernelRealization

This file isolates the kernel-to-operator realization seam in the RH
formal companion.

Role in the proof spine:
- earlier files built the scalar simplex/Beta majorants,
  the operator interfaces, the compression chain, and the common
  resolvent bridge;
- this file refines the kernel-side witness architecture for the
  localized half-heat
  observable used in the Appendix B/D localization package.

Philosophy:
we do not overclaim a full Mathlib implementation of Hilbert–Schmidt
kernel operators.
Instead, we:
1. make the  $L^2$ -kernel packaging explicit,
2. isolate the kernel-to-operator realization as a single
  project-local seam,
3. package the exact small-time witness needed downstream.
-/

import RHFormalization.TraceNormCompression
import Mathlib.MeasureTheory.Function.LpSpace.Basic

```

```

import Mathlib.MeasureTheory.Function.LpSpace.Complete
import Mathlib.MeasureTheory.Function.StronglyMeasurable.Lp
import Mathlib.MeasureTheory.Integral.Bochner.ContinuousLinearMap
import Mathlib.Topology.Algebra.InfiniteSum.Basic

namespace RHFormalization.KernelRealization

noncomputable section

open Complex Topology Filter MeasureTheory
open RHFormalization
open RHFormalization.OperatorInterfaces
open RHFormalization.BetaRecurrence

namespace RHFormalization

/-!
## A. Concrete  $L^2$ -kernel packaging
-/

/-- Product-space kernel measure on  $\mathbb{R} \times \mathbb{R}$ . -/
abbrev kernelMeasure : Measure ( $\mathbb{R} \times \mathbb{R}$ ) :=
  volume.prod volume

/-- The Banach space in which the kernel series should live. -/
abbrev L2Kernel : Type :=
  MeasureTheory.Lp  $\mathbb{C}$  (2 : ENNReal) kernelMeasure

/-- Turn a two-variable kernel into a function on the product space.
-/
def KernelFun (K :  $\mathbb{R} \rightarrow \mathbb{R} \rightarrow \mathbb{C}$ ) : ( $\mathbb{R} \times \mathbb{R}$ )  $\rightarrow$   $\mathbb{C}$  :=
  fun p => K p.1 p.2

/-- Manuscript-facing square-integrability predicate. -/
def KernelSqIntegrable (K :  $\mathbb{R} \rightarrow \mathbb{R} \rightarrow \mathbb{C}$ ) : Prop :=
  Integrable (fun p :  $\mathbb{R} \times \mathbb{R}$  =>  $\| \text{KernelFun } K \, p \| ^ 2$ ) kernelMeasure

/-- `MemLp` version of square-integrability. -/

```

```

def KernelMemL2 (K :  $\mathbb{R} \rightarrow \mathbb{R} \rightarrow \mathbb{C}$ ) : Prop :=
  MeasureTheory.MemLp (KernelFun K) (2 : ENNReal) kernelMeasure

/-- AE-strong measurability side condition. -/
def KernelAESTronglyMeasurable (K :  $\mathbb{R} \rightarrow \mathbb{R} \rightarrow \mathbb{C}$ ) : Prop :=
  AESTronglyMeasurable (KernelFun K) kernelMeasure

/-- Scalar  $L^2$  norm proxy carried by the current simulation. -/
def KernelL2NormSq (K :  $\mathbb{R} \rightarrow \mathbb{R} \rightarrow \mathbb{C}$ ) :  $\mathbb{R}$  :=
   $\int p : \mathbb{R} \times \mathbb{R}, \| \text{KernelFun } K \, p \|^2 \, d\text{kernelMeasure}$ 

def KernelL2Norm (K :  $\mathbb{R} \rightarrow \mathbb{R} \rightarrow \mathbb{C}$ ) :  $\mathbb{R}$  :=
  Real.sqrt (KernelL2NormSq K)

/-- Bridge from square-integrability to `MemLp`. -/
axiom kernelMemL2_of_sqIntegrable
  {K :  $\mathbb{R} \rightarrow \mathbb{R} \rightarrow \mathbb{C}$ } :
    KernelAESTronglyMeasurable K →
    KernelSqIntegrable K →
    KernelMemL2 K

/-- Concrete packaging of a kernel into ` $L^2$ `. -/
noncomputable def mkL2Kernel
  (K :  $\mathbb{R} \rightarrow \mathbb{R} \rightarrow \mathbb{C}$ ) (hK : KernelMemL2 K) : L2Kernel :=
  MeasureTheory.MemLp.toLp (KernelFun K) hK

@[simp] theorem mkL2Kernel_coeFn
  {K :  $\mathbb{R} \rightarrow \mathbb{R} \rightarrow \mathbb{C}$ } {hK : KernelMemL2 K} :
   $\uparrow(\text{mkL2Kernel } K \, hK) =^f[\text{ae kernelMeasure}] \text{KernelFun } K := \text{by}$ 
  simp [mkL2Kernel] using
    (MeasureTheory.MemLp.coeFn_toLp (f := KernelFun K) hK)

/--
Norm compatibility wrapper.

In a fuller development, this should be discharged from ` $\text{Lp.norm\_toLp}$ `
and the ` $p = 2$ ` identification.
-/

```

```

axiom mkL2Kernel_norm_eq_KernelL2Norm
  {K : ℝ → ℝ → ℂ} :
  ∀ (hsm : KernelAESTronglyMeasurable K) (hK : KernelSqIntegrable K),
    ||mkL2Kernel K (kernelMemL2_of_sqIntegrable hsm hK) || =
KernelL2Norm K

/-!
## B. Kernel measurability / square-integrability data
-/

/--
The proof-facing kernel data for the localized free kernel and Duhamel
word kernels.
This is the concrete replacement for a vague witness axiom.
-/
class AppendixBD_KernelData
  {H : Type*} [NormedAddCommGroup H] [InnerProductSpace ℂ H]
[CompleteSpace H]
  (M : OperatorInterfaces.AppendixA_OperatorData (H := H))
  (A : H → L[ℂ] H) where
  t0 : ℝ
  ht0 : 0 < t0

  freeKernel : ∀ {t : ℝ}, 0 < t → ℝ → ℝ → ℂ
  wordKernel : ∀ {t : ℝ}, 0 < t → ℕ → ℝ → ℝ → ℂ

  free_aesm :
    ∀ {t : ℝ} (ht : 0 < t),
      KernelAESTronglyMeasurable (freeKernel ht)

  word_aesm :
    ∀ {t : ℝ} (ht : 0 < t) (n : ℕ),
      KernelAESTronglyMeasurable (wordKernel ht n)

  free_sqInt :
    ∀ {t : ℝ} (ht : 0 < t),
      KernelSqIntegrable (freeKernel ht)

```



```

word_sqInt :
  ∀ {t : ℝ} (ht : 0 < t) (n : ℕ),
    KernelSqIntegrable (wordKernel ht n)

C : ℝ
hC_nonneg : 0 ≤ C

word_L2Bound :
  ∀ {t : ℝ} (ht : 0 < t) (hsmall : t ≤ t0) (n : ℕ),
    KernelL2Norm (wordKernel ht n)
      ≤ (Real.sqrt C * Real.sqrt (4 / 3 : ℝ) * Real.rpow t (3 / 8 :
ℝ)) ^ n

series_realization_smallTime :
  ∀ {t : ℝ} (ht : 0 < t) (hsmall : t ≤ t0),
    True
/-
  This placeholder marks the proof-facing statement that the free
kernel plus the
  Duhamel word-kernel series realizes the localized half-heat
observable.
  We keep the exact realization theorem separate below as a
project-local seam.
-/

/-!
## C. Summability of the word-kernel series in L²
-/

/-- Summability of the norm-level geometric majorant. -/
axiom wordKernel_L2_summable_smallTime
  {H : Type*} [NormedAddCommGroup H] [InnerProductSpace ℂ H]
[CompleteSpace H]
  (M : OperatorInterfaces.AppendixA_OperatorData (H := H))
  (A : H → L[ℂ] H)
  [AppendixBD_KernelData M A] :
  ∀ {t : ℝ}, 0 < t → t ≤ AppendixBD_KernelData.t0 (M := M) (A := A) →
    Summable

```

```

      (fun n : ℕ =>
        (Real.sqrt (AppendixBD_KernelData.C (M := M) (A := A))
          * Real.sqrt (4 / 3 : ℝ)
          * Real.rpow t (3 / 8 : ℝ)) ^ n)

/-- The word-kernel family is summable in the actual `L2Kernel` Banach
space. -/
theorem kernelSeries_summable_L2Kernel
  {H : Type*} [NormedAddCommGroup H] [InnerProductSpace ℂ H]
[CompleteSpace H]
  (M : OperatorInterfaces.AppendixA_OperatorData (H := H))
  (A : H →L[ℂ] H)
  [AppendixBD_KernelData M A]
  {t : ℝ} (ht : 0 < t)
  (hsmall : t ≤ AppendixBD_KernelData.t0 (M := M) (A := A)) :
  Summable
    (fun n : ℕ =>
      mkL2Kernel
        (AppendixBD_KernelData.wordKernel (M := M) (A := A) ht n)
        (kernelMemL2_of_sqIntegrable
          (AppendixBD_KernelData.word_aesm (M := M) (A := A) ht n)
          (AppendixBD_KernelData.word_sqInt (M := M) (A := A) ht
n)))) := by
  let f : ℕ → L2Kernel := fun n =>
    mkL2Kernel
      (AppendixBD_KernelData.wordKernel (M := M) (A := A) ht n)
      (kernelMemL2_of_sqIntegrable
        (AppendixBD_KernelData.word_aesm (M := M) (A := A) ht n)
        (AppendixBD_KernelData.word_sqInt (M := M) (A := A) ht n))
  have hnorm :
    Summable (fun n : ℕ => || f n ||) := by
  refine Summable.of_nonneg_of_le ?_ ?_
    (wordKernel_L2_summable_smallTime M A ht hsmall)
  · intro n
    positivity
  · intro n
    calc
      || f n ||

```

```

      = KernelL2Norm (AppendixBD_KernelData.wordKernel (M := M) (A
:= A) ht n) := by
      simpa [f] using
        mkL2Kernel_norm_eq_KernelL2Norm
        (AppendixBD_KernelData.word_aesm (M := M) (A := A)
ht n)
        (AppendixBD_KernelData.word_sqInt (M := M) (A := A)
ht n)
      _ ≤ (Real.sqrt (AppendixBD_KernelData.C (M := M) (A := A))
        * Real.sqrt (4 / 3 : ℝ)
        * Real.rpow t (3 / 8 : ℝ)) ^ n := by
      exact AppendixBD_KernelData.word_L2Bound (M := M) (A :=
A) ht hsmall n
      simpa [f] using summable_of_summable_norm hnorm

```

```

/-!
## D. Kernel-to-operator realization seam
-/

```

```

/--
Project-local continuous linear map from `L2Kernel` to bounded
operators.

```

This is the genuinely hard kernel-realization seam. The point of this file is to make it the **only** hard seam here, rather than hiding both convergence and realization inside one black-box witness.

```

-/

```

```

axiom kernelOpCLM
  {H : Type*} [NormedAddCommGroup H] [InnerProductSpace ℂ H]
[CompleteSpace H] :
  L2Kernel →L[ℂ] (H →L[ℂ] H)

```

```

/-- Compatibility with raw kernel-operator notation. -/
axiom kernelOp
  {H : Type*} [NormedAddCommGroup H] [InnerProductSpace ℂ H]
[CompleteSpace H] :

```

```

    ( $\mathbb{R} \rightarrow \mathbb{R} \rightarrow \mathbb{C}$ )  $\rightarrow$  ( $H \rightarrow L[\mathbb{C}] H$ )

axiom kernelOpCLM_apply_mk
  {H : Type*} [NormedAddCommGroup H] [InnerProductSpace  $\mathbb{C}$  H]
[CompleteSpace H]
  (K :  $\mathbb{R} \rightarrow \mathbb{R} \rightarrow \mathbb{C}$ ) (hK : KernelMemL2 K) :
  kernelOpCLM (H := H) (mkL2Kernel K hK) = kernelOp (H := H) K

/--
If an operator is realized by an ` $L^2` kernel, it is Hilbert-Schmidt.
-/
axiom isHilbertSchmidt_of_L2Kernel
  {H : Type*} [NormedAddCommGroup H] [InnerProductSpace  $\mathbb{C}$  H]
[CompleteSpace H]
  {T :  $H \rightarrow L[\mathbb{C}] H$ } :
  ( $\exists$  K2 : L2Kernel, kernelOpCLM (H := H) K2 = T)  $\rightarrow$ 
  OperatorInterfaces.IsHilbertSchmidt T

/-!
## E. Localized small-time witness
-/

/-- The total localized ` $L^2` kernel witness. -/
structure LocalizedL2KernelWitness
  {H : Type*} [NormedAddCommGroup H] [InnerProductSpace  $\mathbb{C}$  H]
[CompleteSpace H]
  (M : OperatorInterfaces.AppendixA_OperatorData (H := H))
  (A :  $H \rightarrow L[\mathbb{C}] H$ ) (t :  $\mathbb{R}$ ) where
  K2 : L2Kernel
  hreal :
    kernelOpCLM (H := H) K2 = OperatorInterfaces.observableHalfHeat M
A t

/--
This is the proof-facing realization theorem:
free kernel plus Duhamel word-kernel series realize the localized
half-heat observable.
-/$$ 
```

```

axiom localizedL2KernelWitness_smallTime
  {H : Type*} [NormedAddCommGroup H] [InnerProductSpace ℂ H]
[CompleteSpace H]
  (M : OperatorInterfaces.AppendixA_OperatorData (H := H))
  (A : H → L[ℂ] H)
  [AppendixBD_KernelData M A] :
  ∀ {t : ℝ}, 0 < t → t ≤ AppendixBD_KernelData.t0 (M := M) (A := A) →
    LocalizedL2KernelWitness M A t

/-- Downstream payoff: the localized half-heat observable is
Hilbert-Schmidt. -/
theorem observableHalfHeat_isHilbertSchmidt_smallTime
  {H : Type*} [NormedAddCommGroup H] [InnerProductSpace ℂ H]
[CompleteSpace H]
  (M : OperatorInterfaces.AppendixA_OperatorData (H := H))
  (A : H → L[ℂ] H)
  [AppendixBD_KernelData M A]
  {t : ℝ} (ht : 0 < t)
  (hsmall : t ≤ AppendixBD_KernelData.t0 (M := M) (A := A)) :
  OperatorInterfaces.IsHilbertSchmidt
  (OperatorInterfaces.observableHalfHeat M A t) := by
  let W := localizedL2KernelWitness_smallTime M A ht hsmall
  exact isHilbertSchmidt_of_L2Kernel <W.K2, W.hreal>

end

end RHFormalization.KernelRealization

```

19 [BuildOrder.md](#)

Build Order

This note records the intended dependency and reading order for the Lean 4 companion included with Version 2 of the manuscript

****“A Stieltjes Pole-Package Identity for the Prime-Weighted Quantum Operator and the Riemann Hypothesis.”****

The build order follows the proof architecture of the manuscript rather than an unrelated coding order.

1. Root files

1. ``lakefile.lean``
2. ``RHFormalization.lean``

These files set the project dependency on Mathlib and re-export the library root.

2. Foundational files

3. ``Basic.lean``
4. ``AnnexSpine.lean``

These files fix shared imports, slit-plane notation, kernel-space conventions, and the high-level D-H-E-F proof scaffold.

3. Scalar recurrence layer

5. ``SimplexScalar.lean``
6. ``BetaRecurrence.lean``

These files isolate the scalar simplex/Beta recurrence estimates and the geometric summability package. This is the lightest and most directly Lean-convertible part of the current companion.

4. Operator-side interface and compression layer

7. ``OperatorInterfaces.lean``
8. ``CompressionChain.lean``
9. ``TraceNormCompression.lean``

These files package the infinite-dimensional operator interfaces, the finite-rank compression route, and the trace-norm split used by the weighted observable heat operator.

5. Kernel witness layer

10. `KernelRealization.lean`

11. `HilbertSchmidtTail.lean` **(if included)**

These files sharpen the localized kernel witness architecture and the Hilbert–Schmidt tail route used in the Appendix B/D localization branch.

6. Operator-side continuation layer

12. `CommonResolvent.lean`

This file packages the common-series / Laplace / Ω -side operator bridge.

7. Zero-side meromorphic layer

13. `MeromorphicPackage.lean`

14. `LaurentUniqueness.lean`

15. `PoleSeriesContinuation.lean`

These files package the H-side meromorphic object, the local Laurent/removable-singularity bookkeeping, and the half-plane-to- Ω continuation architecture.

8. Interface and rigidity layer

16. `RigidityBridge.lean`

17. `MainTheorem.lean`

These files assemble the overlap identity, global meromorphic extension, pole non-cancellation, and final contradiction theorem.

9. Transparency and publication files

18. `AxiomRegister.lean`

19. `README.md`

20. `FormalizationStatus.md`
21. `BuildOrder.md`
22. `ModuleMap.md`

These files are included for publication clarity and should be read alongside the code.

10. Practical reading order for referees

A referee who wants the shortest path to understanding the annex should read in this order:

1. `README.md`
2. `FormalizationStatus.md`
3. `ModuleMap.md`
4. `AxiomRegister.lean`
5. `AnnexSpine.lean`
6. `MainTheorem.lean`

Then, if they want technical details, they should descend into:

- scalar recurrence files,
- compression / trace files,
- common resolvent file,
- pole-series continuation and rigidity files.

11. Practical reading order for technical Lean readers

A Lean reader who wants to understand dependencies should go in the actual file order:

- `Basic`
- `AnnexSpine`
- `SimplexScalar`
- `BetaRecurrence`
- `OperatorInterfaces`
- `CompressionChain`
- `TraceNormCompression`

- `KernelRealization`
- `HilbertSchmidtTail` **(if included)**
- `CommonResolvent`
- `MeromorphicPackage`
- `LaurentUniqueness`
- `PoleSeriesContinuation`
- `RigidityBridge`
- `MainTheorem`
- `AxiomRegister`

12. Bottom line

This build order is chosen so that the annex mirrors the mathematical proof spine as closely as possible while still isolating the main technical seams and remaining library-facing obligations.

20 [ModuleMap.md](#)

Module Map

This note describes what each Lean companion file does and how it corresponds to the manuscript.

The companion is organized around the same fixed four-step proof spine as the manuscript:

- ****D****: operator-side construction on Ω ,
- ****H****: zero-side meromorphic package,
- ****E****: overlap/interface identity,
- ****F****: rigidity and contradiction.

1. Foundation

`Basic.lean`

Shared imports, notation, slit-plane conventions, kernel-space abbreviations, and basic global setup.

`AnnexSpine.lean`

High-level proof scaffold. This is the cleanest overview file for the formal companion and mirrors the D-H-E-F proof architecture at the interface level.

2. Scalar recurrence and summability

`SimplexScalar.lean`

Encodes the simplex-step scalar integral layer and the $\frac{4}{3}$ recurrence majorant.

Manuscript role: appendix-level scalar estimate supporting Duhamel word control.

`BetaRecurrence.lean`

Packages the geometric summability built from the scalar recurrence layer.

Manuscript role: small-time summability / recurrence majorant input used in localized kernel estimates.

3. Operator-side infrastructure

`OperatorInterfaces.lean`

Declares the proof-facing Hilbert-Schmidt, trace-class, trace norm, and compression interfaces.

Manuscript role: abstract functional-analytic infrastructure underlying Appendix A/B/D.

`CompressionChain.lean`

Packages the finite-rank compression route from compressed traces to diagonalization of the weighted heat trace.

Manuscript role: proof-shaped replacement for an opaque infinite-dimensional trace-to-diagonal passage.

`TraceNormCompression.lean`

Sharpens the compression step into the right-tail / left off-diagonal two-piece split.

****Manuscript role:**** Appendix B/D compression and trace-norm route.

`KernelRealization.lean`

Packages the localized L^2 -kernel witness side and isolates the kernel-to-operator realization seam.

****Manuscript role:**** localized half-heat kernel architecture in Appendix B/D.

`HilbertSchmidtTail.lean` **(if included)**

Pushes the HS-tail route further by extracting a project-local theorem for tail vanishing from the kernel witness layer.

****Manuscript role:**** one of the main remaining technical dents in the operator-localization branch.

4. Operator-side continuation

`CommonResolvent.lean`

Packages the common resolvent-series, the half-plane Laplace identity, and the Ω -side operator bridge.

****Manuscript role:**** Appendix D / F.9.1 operator-side transform package.

5. Zero-side meromorphic package

`MeromorphicPackage.lean`

Packages the independently owned zero-side meromorphic object.

****Manuscript role:**** Appendix H package in its basic form.

`LaurentUniqueness.lean`

Isolates the local removable-singularity / Laurent-uniqueness bookkeeping step.

****Manuscript role:**** local meromorphic bookkeeping feeding the rigidity bridge.

`PoleSeriesContinuation.lean`

Separates the half-plane explicit pole series from the Ω -meromorphic continuation package.

****Manuscript role:**** sharpened Appendix H continuation architecture.

6. Interface and rigidity

`RigidityBridge.lean`

Packages overlap identity, meromorphic identity extension, and pole non-cancellation.

****Manuscript role:**** Appendix E and the structural part of Appendix F.

`MainTheorem.lean`

Assembles the packaged D/H/E/F layers into the final contradiction theorem.

****Manuscript role:**** formal companion endpoint corresponding to the RH closure.

7. Transparency files

`AxiomRegister.lean`

Lists remaining axioms and unresolved library-facing obligations.

****Purpose:**** explicit honesty layer.

`README.md`

Publication-safe description of what the formal companion is and is not.

`FormalizationStatus.md`

Status summary distinguishing proof-spine completeness from end-to-end verification.

`BuildOrder.md`

Dependency and reading order for the Lean companion.

8. Interpretation

The module sequence is intentionally split into three kinds of files:

1. **Proof-architecture files**

These mirror the manuscript's D-H-E-F closure spine.

2. **Technical bridge files**

These isolate the scalar recurrence, compression, kernel, and meromorphic bookkeeping layers.

3. **Transparency files**

These explain exactly what remains axiomatized and how the annex should be described.

9. Bottom line

The module map is designed so that the Lean companion reads as a formal shadow of the manuscript itself, not as a disconnected coding experiment.

21 RHFormalization/HilbertSchmidtTail.lean

/-!

RHFormalization.HilbertSchmidtTail

This file isolates the Hilbert-Schmidt tail route used in the localized half-heat part of the Appendix B/D operator-side argument.

Role in the proof spine:

- `KernelRealization.lean` packages the localized L^2 kernel witness.

- This file turns that witness into the HS-tail vanishing mechanism used by the
left off-diagonal compression route.
- The goal is not to rebuild the full operator-ideal library, but to expose the
exact proof-shaped tail theorem needed downstream.

Philosophy:

keep the theorem local, explicit, and honest. The hard seam remains the
kernel-to-operator realization and the HS infrastructure, not the
architectural
shape of the argument.

-/

```
import RHFormalization.KernelRealization
import Mathlib.Topology.Algebra.InfiniteSum.Basic
```

```
namespace RHFormalization.HilbertSchmidtTail
```

```
noncomputable section
```

```
open Complex Topology Filter
open RHFormalization
open RHFormalization.OperatorInterfaces
open RHFormalization.KernelRealization
```

```
namespace RHFormalization.OperatorInterfaces
```

```
variable {H : Type*}
variable [NormedAddCommGroup H] [InnerProductSpace ℂ H] [CompleteSpace H]
```

```
/-!
```

```
## A. Localized HS witness package
```

```
-/
```

```
/--
```

```

A packaged HS witness for the localized half-heat observable.
-/
structure LocalizedHalfHeatHSAPI
  (M : AppendixA_OperatorData (H := H))
  (A : H → L[ $\mathbb{C}$ ] H) where
  t0 :  $\mathbb{R}$ 
  ht0 : 0 < t0
  hs_observableHalfHeat :
     $\forall \{t : \mathbb{R}\}, 0 < t \rightarrow t \leq t0 \rightarrow$ 
      IsHilbertSchmidt (observableHalfHeat M A t)

/--
Build the HS witness package from the kernel-realization layer.
-/
theorem localizedHalfHeatHSAPI_of_kernelRealization
  (M : AppendixA_OperatorData (H := H))
  (A : H → L[ $\mathbb{C}$ ] H)
  [AppendixBD_KernelData M A] :
  LocalizedHalfHeatHSAPI M A := by
  refine
    { t0 := AppendixBD_KernelData.t0 (M := M) (A := A)
      ht0 := AppendixBD_KernelData.ht0 (M := M) (A := A)
      hs_observableHalfHeat := ?_ }
  intro t ht hsmall
  exact observableHalfHeat_isHilbertSchmidt_smallTime M A ht hsmall

/-!
## B. Tail projection on HS operators
-/

/--
The generic HS-tail theorem, restated in proof-facing form.
-/
theorem localizedHalfHeat_tail_tendsto_zero
  (M : AppendixA_OperatorData (H := H))
  (A : H → L[ $\mathbb{C}$ ] H)
  (HS : LocalizedHalfHeatHSAPI M A)
  {t :  $\mathbb{R}$ } (ht : 0 < t) (hsmall : t ≤ HS.t0) :

```

```

    Tendsto
      (fun N :  $\mathbb{N}$  => HSNorm ((tailProj N).comp (observableHalfHeat M A
t)))
      atTop
      (nhds 0) := by
exact HSNorm_tailProj_tendsto_zero
  (observableHalfHeat M A t)
  (HS.hs_observableHalfHeat ht hsmall)

/-!
## C. Proof-shaped left-tail factor
-/

/--
Proof-facing left-tail HS factor used in the half-time factorization.
-/
def leftTailHSFactor_from_halfHeat
  (M : AppendixA_OperatorData (H := H))
  (A : H  $\rightarrow$  L[ $\mathbb{C}$ ] H)
  (N :  $\mathbb{N}$ ) (t :  $\mathbb{R}$ ) : H  $\rightarrow$  L[ $\mathbb{C}$ ] H :=
  (tailProj N).comp (observableHalfHeat M A t)

/--
Its HS norm tends to zero as the head projection grows.
This is the precise tail statement later consumed by the trace-norm
compression file.
-/
theorem HSNorm_leftTailHSFactor_tendsto_zero_smallTime
  (M : AppendixA_OperatorData (H := H))
  (A : H  $\rightarrow$  L[ $\mathbb{C}$ ] H)
  [AppendixBD_KernelData M A]
  {t :  $\mathbb{R}$ } (ht : 0 < t)
  (hsmall : t  $\leq$  AppendixBD_KernelData.t0 (M := M) (A := A)) :
  Tendsto
    (fun N :  $\mathbb{N}$  => HSNorm (leftTailHSFactor_from_halfHeat M A N t))
    atTop
    (nhds 0) := by
let HS := localizedHalfHeatHSAPI_of_kernelRealization M A

```



```

    simpα [leftTailHSFactor_from_halfHeat] using
      localizedHalfHeat_tail_tendsto_zero M A HS ht hsmall

/-!
## D. Compatibility wrapper for downstream files
-/

/--
This wrapper keeps the naming aligned with the earlier
compression-side files.

It is intentionally phrased as the exact Appendix B/D payload:
the left-tail HS factor tends to zero at small time.
-/
theorem HSNorm_leftTailHSFactor_tendsto_zero_of_AppendixBD
  (M : AppendixA_OperatorData (H := H))
  (A : H → L[ $\mathbb{C}$ ] H)
  [AppendixBD_KernelData M A]
  {t :  $\mathbb{R}$ } (ht : 0 < t)
  (hsmall : t ≤ AppendixBD_KernelData.t0 (M := M) (A := A)) :
  Tendsto
    (fun N :  $\mathbb{N}$  => HSNorm (leftTailHSFactor_from_halfHeat M A N t))
    atTop
    (nhds 0) := by
  exact HSNorm_leftTailHSFactor_tendsto_zero_smallTime M A ht hsmall

/-!
## E. Optional bridge back to the trace-norm compression file
-/

/--
If the trace-norm compression file uses the older abstract symbol
`leftTailHSFactor`, this theorem records the intended identification
route.
-/
axiom leftTailHSFactor_matches_kernel_tail
  (M : AppendixA_OperatorData (H := H))
  (A : H → L[ $\mathbb{C}$ ] H) :

```

```

    ∀ {N : ℕ} {t : ℝ}, 0 < t →
      leftTailHSFactor M A N t = leftTailHSFactor_from_halfHeat M A N t

/--
Transport the concrete tail theorem to the older abstract naming
convention.
-/
theorem HSNorm_oldLeftTailFactor_tendsto_zero_of_AppendixBD
  (M : AppendixA_OperatorData (H := H))
  (A : H → L[ℂ] H)
  [AppendixBD_KernelData M A]
  {t : ℝ} (ht : 0 < t)
  (hsmall : t ≤ AppendixBD_KernelData.t0 (M := M) (A := A)) :
  Tendsto
    (fun N : ℕ => HSNorm (leftTailHSFactor M A N t))
    atTop
    (nhds 0) := by
have hnew :
  Tendsto
    (fun N : ℕ => HSNorm (leftTailHSFactor_from_halfHeat M A N t))
    atTop
    (nhds 0) :=
  HSNorm_leftTailHSFactor_tendsto_zero_of_AppendixBD M A ht hsmall
refine hnew.congr' ?_
refine Filter.Eventually.of_forall ?_
intro N
rw [leftTailHSFactor_matches_kernel_tail M A ht]

end RHFormalization.OperatorInterfaces

end

end RHFormalization.HilbertSchmidtTail

```

